BECOMING A DOCTOR OF

INTERNATIONAL ENERGY AND NATURAL RESOURCES LAW

DOCTORAL TEXT AND ADVANCED PROFESSIONAL REFERENCE



TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

BECOMING A DOCTOR OF INTERNATIONAL ENERGY AND NATURAL RESOURCES LAW

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International Energy and Natural Resources Law

A Doctoral Text and Advanced Professional Reference on Resource Sovereignty, Energy Transactions, Investment Protection, Environmental and Climate Governance, and International Dispute Resolution

TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

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Becoming a Doctor of International Energy and Natural Resources Law

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DEDICATION

To the scholars, regulators, negotiators, advocates, and arbitrators whose rigorous work turns the world's energy and natural resources into lawful, accountable, and equitable development —

and to the communities whose lands, waters, and futures that work must always serve.

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PART I

Jurisprudential and Institutional Foundations

The conceptual architecture of the discipline: its definition and history, its sources and principles, the doctrine of permanent sovereignty over natural resources, and the institutions through which global energy governance is organised.

1 CHAPTER

The Nature, Scope, and Evolution of International Energy and Natural Resources Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Define international energy and natural resources law as a field of legal study and distinguish it from adjacent fields, including public international law, international economic law, and international environmental law.
2. Trace the historical development of resource governance from the early concession era through decolonisation, the New International Economic Order, liberalisation, and the contemporary energy transition.
3. Explain how the internationalisation of energy markets has transformed the regulation of exploration, production, transportation, and trade.
4. Analyse the interaction between public law, private law, and transnational legal relations in energy and resource projects.
5. Evaluate the significance of legal pluralism and competing regulatory authorities in global resource governance.
6. Identify the principal foundational debates that structure doctoral research in the field.

Introduction

Energy and natural resources sit at the junction of almost every question that modern legal systems are asked to answer. Who owns what lies beneath the soil and the seabed? Under what conditions may a sovereign state grant rights to explore for and extract those resources, and on what terms may it later change its mind? How should the economic rent produced by a non-renewable asset be shared between the state, the investor, the host community, and future generations? What obligations attach to the transport of oil and gas across national frontiers, to the generation and trade of electricity, or to the emissions released

when fossil fuels are burned? How should disputes between powerful states and powerful corporations be resolved, and by whom? Each of these questions is, at bottom, a legal question, and the body of law that has developed to answer them is the subject of this book.

International energy and natural resources law is a young discipline in name but an old one in substance. The legal instruments through which states have granted, regulated, taxed, and expropriated mineral and petroleum rights have a history that stretches back more than a century, and the doctrines that frame those transactions — sovereignty, contract, property, state responsibility, and the protection of aliens — are older still. What is new is the recognition that these instruments and doctrines constitute a coherent field of study, with its own sources, principles, institutions, and methods, and with a distinctive set of intellectual problems that cannot be adequately addressed from within any single traditional legal category.

This opening chapter lays the foundation for the remainder of the book. It begins by defining the field and identifying its boundaries. It then traces the historical development of resource governance, explains the internationalisation of energy markets and regulation, and examines the interplay between public law, private law, and transnational legal relations. It considers the relationship between energy law and international economic law, the phenomenon of legal pluralism, and the contemporary challenges confronting global resource governance. The chapter closes by setting out the foundational debates that animate doctoral research and that recur throughout the chapters that follow.

The approach throughout is analytical rather than descriptive. A doctoral candidate in this field must be able to do more than recite the content of treaties and statutes; he or she must be able to explain why the law takes the form that it does, to identify the tensions and contradictions within it, to evaluate competing interpretations, and to propose reasoned reforms. The chapter therefore emphasises the conceptual architecture of the field — the ideas that hold it together and the fault lines along which it is contested.

1.1 Defining Energy and Natural Resources Law

The Problem of Definition

Any attempt to define a field of law must confront a familiar difficulty: legal categories are not natural kinds. They are constructed by scholars, practitioners, courts, and legislators for particular purposes, and their boundaries shift as those purposes change. “Energy law” and “natural resources law” are no exception. Neither phrase appears as a term of art in any multilateral treaty of general application, and neither corresponds to a single body of legislation in most domestic legal systems. Instead, each describes a cluster of rules, principles, institutions, and

practices that share a common subject matter — the ownership, development, transformation, trade, and use of energy sources and natural resources — and that are united by a common set of policy concerns.

A useful working definition, adopted for the purposes of this book, is the following:

> International energy and natural resources law is the body of rules, principles, institutions, and practices — derived from public international law, international economic law, comparative domestic law, and transnational commercial practice — that governs the allocation of rights over energy sources and natural resources, the conduct of activities involving their exploration, development, transformation, transport, trade, and use, and the distribution of the benefits and burdens arising from those activities, in situations that involve more than one legal order.

Several features of this definition deserve comment. First, it is *functional* rather than formal: it defines the field by reference to its subject matter and the problems it addresses, not by reference to a particular source of law. Second, it is *pluralist*: it expressly acknowledges that the relevant rules derive from multiple sources, including domestic legislation and private contracts as well as treaties and custom. Third, it is *distributive*: it identifies the allocation of benefits and burdens as a central concern, reflecting the fact that energy and resource law is fundamentally about who gets what, when, and how. Fourth, it is *transnational*: the field is concerned with situations that involve more than one legal order, whether because a resource straddles a boundary, because an investor is foreign, because a commodity is traded internationally, or because environmental effects cross frontiers.

Energy Law and Natural Resources Law: One Field or Two?

The conjunction "energy and natural resources" requires justification. Energy law, narrowly understood, is concerned with the production, conversion, transmission, distribution, and consumption of energy, whether derived from fossil fuels, nuclear fission, or renewable sources. Natural resources law, narrowly understood, is concerned with the ownership, management, and exploitation of resources found in nature, including minerals, petroleum, water, forests, fisheries, and biological diversity. The two fields overlap substantially — petroleum and natural gas are both natural resources and energy sources — but neither is wholly contained within the other. Renewable electricity generated from wind or solar radiation is an energy source but is not usually described as a natural resource in the extractive sense; copper and lithium are natural resources but are not themselves sources of energy, although they are indispensable to its transmission and storage.

There are strong reasons for treating the two fields together. Historically, the law governing petroleum developed out of the law governing hard-rock minerals, and many of the foundational instruments — concessions, licences, royalties, and the doctrine of permanent sovereignty — are common to both. Functionally, the energy transition is collapsing the distinction: the decarbonisation of energy systems depends on the extraction of critical minerals on an unprecedented scale, on access to land and water for renewable installations, and on the use of geological formations for the storage of carbon dioxide and hydrogen. Institutionally, the same ministries, regulators, and courts often have jurisdiction over both energy and minerals. Doctrinally, the same principles of international law — sovereignty over natural resources, the prohibition of transboundary harm, the protection of foreign investment, and the rights of indigenous peoples — govern both.

This book therefore treats energy law and natural resources law as a single integrated field, while recognising that particular chapters will emphasise one or the other. The integrated approach has the further advantage of exposing the connections between subjects that are too often studied in isolation: between petroleum fiscal regimes and renewable energy incentives, for example, or between mining law and climate law.

Relationship with Adjacent Fields

The field overlaps with several established branches of law, and a doctoral candidate must be able to locate it precisely in relation to each.

Public international law. The general rules of public international law — on sources, treaties, state responsibility, jurisdiction, immunity, and dispute settlement — supply the framework within which international energy and resource law operates. The doctrine of permanent sovereignty over natural resources, discussed in Chapter 2, is a principle of general international law. The law of the sea, international watercourse law, and international environmental law are specialised branches of public international law with direct application to resource activities.

International economic law. The law of international trade, investment, finance, and taxation governs the cross-border movement of energy commodities, capital, and revenues. The World Trade Organization (WTO) agreements apply to trade in energy goods and, to a more limited extent, services; bilateral and multilateral investment treaties protect foreign investors in the energy and mining sectors; and international tax rules shape the fiscal treatment of resource projects. The relationship between energy law and international economic law is examined in section 1.5 below.

International environmental and climate law. Resource activities are among the most significant sources of environmental harm, and international

environmental law imposes obligations of prevention, precaution, assessment, and cooperation that constrain the manner in which they may be carried out. Climate change law, in particular, is reshaping energy law by setting long-term objectives for the reduction of greenhouse gas emissions that are incompatible with the indefinite expansion of fossil fuel production.

Comparative and domestic law. Most of the rules that govern energy and resource activities on a day-to-day basis are domestic: constitutional provisions on resource ownership, petroleum and mining statutes, electricity regulation, environmental legislation, and tax codes. International energy law cannot be understood without close attention to these domestic regimes and to the patterns of convergence and divergence among them, which are explored in Chapter 24.

Transnational commercial law. Energy and resource projects are structured through complex networks of private contracts — joint operating agreements, sales and purchase agreements, engineering and construction contracts, financing agreements — that are frequently governed by English or New York law and subject to international commercial arbitration. The model contracts and standard terms produced by industry associations constitute a form of transnational private ordering that interacts with, and sometimes substitutes for, state regulation.

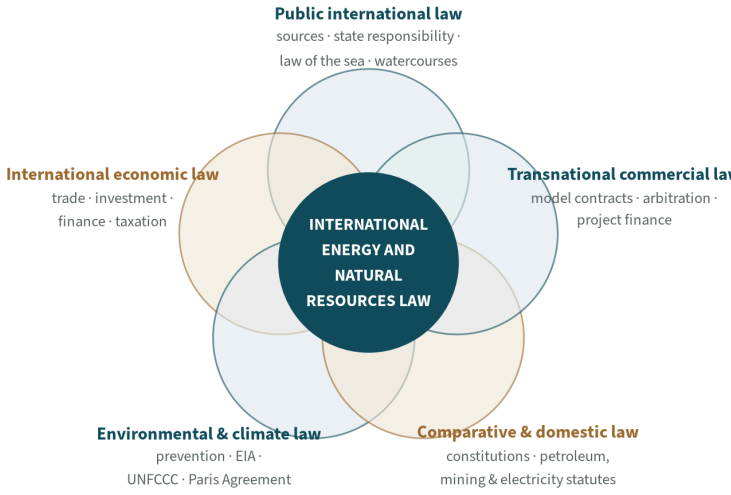


Figure 1.1 The constituent fields of international energy and natural resources law. The discipline is defined by its subject matter and draws on five overlapping bodies of law. *Source:* prepared for this edition.

The Scope of the Field

The scope of the field, as treated in this book, may be described along three axes.

The first axis is the *resource or energy source* concerned: petroleum (crude oil and natural gas), coal, hard-rock minerals including critical minerals, uranium and nuclear energy, renewable energy sources (solar, wind, hydropower, geothermal, biomass, marine), hydrogen and other energy carriers, electricity, water, and marine resources.

The second axis is the *stage of the value chain*: the allocation of rights to explore and develop resources (upstream), the transport and processing of those resources (midstream), and their sale, distribution, and consumption (downstream), together with the decommissioning of installations and the remediation of affected environments at the end of the project life cycle.

The third axis is the *legal function*: the allocation of property and access rights, the regulation of conduct, the protection of investment, the allocation of commercial risk by contract, the raising and distribution of revenue, the protection of the environment and human rights, and the resolution of disputes.

The chapters of this book are organised principally along the third axis, while particular chapters address specific resources (Chapters 5 to 12) and specific legal functions (Chapters 13 to 27).

Is There a "Lex Petrolea"?

One of the most contested questions in the definition of the field concerns the existence of a distinct transnational legal order governing the international petroleum industry — sometimes described as the *lex petrolea*. The idea has a long pedigree. In the arbitral award in *Kuwait v American Independent Oil Company (Aminoil)*, the tribunal referred to the existence of a body of practice in the oil industry that informed the interpretation of the concession, while stopping short of endorsing an autonomous legal system.[1] Proponents of the *lex petrolea* argue that the repeated use of standard contractual forms, the consistent practice of industry participants, the jurisprudence of arbitral tribunals, and the model contracts of bodies such as the Association of International Energy Negotiators (AIEN) have generated a body of transnational norms that operates partly independently of any national legal system.[2]

Critics respond that the concept is either trivially true — industry practice obviously influences the interpretation of industry contracts — or dangerously

[1] *Government of the State of Kuwait v American Independent Oil Company (Aminoil)*, Award of 24 March 1982, (1982) 21 ILM 976.

[2] See R. Doak Bishop, "International Arbitration of Petroleum Disputes: The Development of a *Lex Petrolea*" (1998) 23 *Yearbook of Commercial Arbitration* 1131; and Thomas C.C. Childs, "Update on *Lex Petrolea*: The Continuing Development of Customary Law Relating to International Oil and Gas Exploration and Production" (2011) 4 *Journal of World Energy Law and Business* 214.

overstated, in that it purports to detach contractual relations from the legal systems that ultimately give them force, and that it tends to privilege the interests of international oil companies over those of host states and affected communities. The debate is significant for doctoral researchers because it raises fundamental questions about the nature of law, the legitimacy of private ordering, and the relationship between state and non-state sources of normativity. It is revisited in Chapter 2 in the context of soft law and industry standards and in Chapter 26 in the context of arbitral practice.

1.2 Historical Development of Resource Governance

A historical perspective is indispensable to an understanding of contemporary energy and resource law. Many of the field's defining doctrines — the stabilisation clause, the doctrine of permanent sovereignty, the standard of compensation for expropriation, the treatment of production sharing contracts — were forged in specific historical controversies, and their meaning cannot be grasped without appreciating the circumstances in which they arose. This section traces five broad phases in the development of resource governance.

Phase One: The Classical Concession Era (c. 1900–1950)

The modern international petroleum industry was built on the classical concession. In 1901, William Knox D'Arcy obtained from the Shah of Persia a concession conferring the exclusive right to explore for, produce, and export petroleum over approximately three-quarters of the territory of Persia for a period of sixty years, in return for a modest initial payment and a share of annual net profits. The D'Arcy concession became the template for a generation of agreements across the Middle East, Latin America, and Asia.

The classical concession had several characteristic features. It covered vast areas, often a substantial proportion of the host state's territory. It endured for very long periods, frequently between sixty and ninety-nine years. It conferred upon the concessionaire not merely the right to extract petroleum but ownership of the petroleum once produced, together with broad managerial control over all aspects of operations. The host state's financial return was limited, typically to a fixed royalty per ton of production, and the state had little or no involvement in decisions concerning the rate of production, the pricing of crude oil, or the allocation of markets. The concessions were frequently negotiated between unequal parties, in circumstances in which the host state lacked the geological information, technical expertise, and bargaining power necessary to secure favourable terms.

The legal character of these instruments was ambiguous. From the perspective of the international oil companies and their home states, a concession was a contract that created vested rights protected by international law, and any unilateral modification or termination by the host state constituted a wrong for which

compensation was owed. From the perspective of host states, the concession was an act of sovereign authority — an administrative grant — that the state could revise in the public interest. This tension between contract and sovereignty is one of the enduring themes of the field.

During this period the industry was dominated by a small number of vertically integrated companies — later known as the "Seven Sisters" — which controlled exploration, production, refining, transportation, and marketing. Arrangements such as the Red Line Agreement of 1928, under which the partners in the Iraq Petroleum Company agreed not to compete with one another for concessions within a defined area of the former Ottoman Empire, and the Achnacarry or "As-Is" Agreement of the same year, which sought to stabilise market shares, illustrate the degree to which the international oil market was organised through private arrangements rather than public regulation.

Phase Two: Nationalisation and the Assertion of Sovereignty (1938–1960s)

The first major challenge to the concession system came from Latin America. In 1938, Mexico nationalised the holdings of foreign oil companies following a prolonged labour dispute and a confrontation with the companies over compliance with a decision of the Mexican Supreme Court. The Mexican nationalisation prompted a celebrated exchange of diplomatic notes in which the United States Secretary of State, Cordell Hull, asserted that international law required "prompt, adequate, and effective" compensation for the expropriation of foreign property — the formula that became known as the Hull standard. Mexico rejected the claim that international law imposed any such requirement, maintaining that compensation was a matter for domestic law. The dispute was eventually settled by agreement, but the doctrinal disagreement it exposed persisted for decades.

The second major challenge came in Iran. In 1951, the Iranian parliament enacted legislation nationalising the assets of the Anglo-Iranian Oil Company. The United Kingdom instituted proceedings before the International Court of Justice, but the Court held that it lacked jurisdiction because the relevant Iranian declaration accepting the Court's compulsory jurisdiction was limited to disputes concerning treaties, and the concession of 1933 was not a treaty between Iran and the United Kingdom.[3] The *Anglo-Iranian Oil Co* case is significant for its holding that a concession agreement between a state and a foreign company is not an international treaty, notwithstanding the involvement of the company's home state in its negotiation.

[3] *Anglo-Iranian Oil Co (United Kingdom v Iran)*, Preliminary Objection, Judgment, ICJ Reports 1952, p. 93.

The same period witnessed the gradual replacement of fixed royalty arrangements with profit-sharing formulas. Beginning with Venezuela in 1943 and spreading to Saudi Arabia in 1950, the so-called "fifty-fifty" principle — under which the host government received approximately half of the profits from petroleum operations through a combination of royalties and income taxes — became the industry norm. The fifty-fifty principle represented a significant shift in the distribution of resource rents, but it left the structure of the concession — and the managerial control of the companies — largely intact.

Phase Three: Decolonisation and the New International Economic Order (1960–1980)

The third phase was shaped by decolonisation and the emergence of a large number of newly independent states determined to assert control over their natural resources. Three developments were decisive.

First, in September 1960, Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela founded the Organization of the Petroleum Exporting Countries (OPEC). OPEC's initial objective was to resist reductions in posted prices unilaterally imposed by the companies, but it rapidly became a forum through which producing states coordinated their policies on pricing, taxation, and participation.

Second, the General Assembly of the United Nations adopted a series of resolutions articulating the principle of permanent sovereignty over natural resources. The most important of these, Resolution 1803 (XVII) of 14 December 1962, declared that "the right of peoples and nations to permanent sovereignty over their natural wealth and resources must be exercised in the interest of their national development and of the well-being of the people of the State concerned", while also providing that in cases of nationalisation "the owner shall be paid appropriate compensation, in accordance with the rules in force in the State taking such measures in the exercise of its sovereignty and in accordance with international law", and that foreign investment agreements "freely entered into by or between sovereign States shall be observed in good faith".[4] Resolution 1803 represented a carefully negotiated compromise between capital-exporting and capital-importing states, and it is widely regarded as reflecting customary international law in significant respects.

Third, in the early 1970s, producing states moved from profit-sharing to participation and outright nationalisation. Libya nationalised the interests of several foreign companies between 1971 and 1974; Algeria took a majority interest in French oil companies in 1971; Iraq nationalised the Iraq Petroleum Company in 1972; and the states of the Persian Gulf negotiated progressive participation

[4] UN General Assembly Resolution 1803 (XVII), "Permanent Sovereignty over Natural Resources", 14 December 1962, paras 1, 4, and 8.

agreements that culminated in full state ownership of their producing industries. The oil price increases of 1973–1974, associated with the Arab oil embargo and OPEC's assertion of control over pricing, transformed the economics of the industry and the balance of power between producing states and consuming states.

In 1974, the General Assembly adopted the Declaration on the Establishment of a New International Economic Order and the Charter of Economic Rights and Duties of States.[5] Article 2 of the Charter asserted the right of every state "to nationalize, expropriate or transfer ownership of foreign property, in which case appropriate compensation should be paid by the State adopting such measures, taking into account its relevant laws and regulations and all circumstances that the State considers pertinent", and provided that any controversy over compensation "shall be settled under the domestic law of the nationalizing State and by its tribunals", unless otherwise agreed. Unlike Resolution 1803, the Charter omitted any reference to international law as the governing standard for compensation and was adopted over the opposition or abstention of most capital-exporting states. Its status as a statement of customary international law has accordingly been doubted.

The legal consequences of the nationalisations of this period were addressed in a series of landmark arbitral awards, often referred to collectively as the "Libyan oil arbitrations" — *BP v Libya* (1973), *Texaco v Libya* (1977), and *LIAMCO v Libya* (1977) — together with the *Aminoil* award of 1982. These awards, which are examined in detail in Chapters 3 and 14, grappled with the legal status of concession contracts, the effect of stabilisation clauses, the applicable law, and the appropriate standard of compensation. Their reasoning remains foundational to the modern law of foreign investment.

[5] UN General Assembly Resolution 3201 (S-VI), "Declaration on the Establishment of a New International Economic Order", 1 May 1974; UN General Assembly Resolution 3281 (XXIX), "Charter of Economic Rights and Duties of States", 12 December 1974.



Figure 1.2 Selected milestones in the development of international energy and natural resources law, 1901–2025. *Source:* prepared for this edition.

Phase Four: Liberalisation, Investment Protection, and Market Integration (1980–2005)

The fourth phase was characterised by a reversal of the statist tendencies of the preceding period. The collapse of oil prices in the mid-1980s, the debt crisis in the developing world, the end of the Cold War, and the ascendancy of market-oriented economic policies led many states to reopen their resource sectors to foreign investment. Production sharing contracts — first used in Indonesia in the 1960s — became the dominant contractual form in many developing producer states, allowing governments to retain formal ownership of petroleum while engaging foreign companies as contractors entitled to a share of production. Mining codes across Africa and Latin America were rewritten, often with the support of international financial institutions, to attract foreign capital through security of tenure, fiscal stability, and liberal repatriation of profits.

At the same time, a dense network of bilateral investment treaties (BITs) emerged. From a few hundred in the 1980s, the number of BITs grew to more than two thousand by the early 2000s. These treaties conferred upon foreign investors substantive protections — fair and equitable treatment, full protection and security, protection against expropriation without compensation, national

and most-favoured-nation treatment — and, crucially, the right to bring claims directly against host states before international arbitral tribunals, typically under the auspices of the International Centre for Settlement of Investment Disputes (ICSID) established by the Washington Convention of 1965. The extension of investor–state arbitration transformed the enforcement of investor rights by removing the need for diplomatic protection by the investor's home state.

In the energy sector specifically, the Energy Charter Treaty (ECT), signed in Lisbon in December 1994 and in force since April 1998, established a multilateral framework for investment protection, trade, transit, and energy efficiency among the states of Europe and the former Soviet Union, together with a number of states in Asia. The ECT became the most frequently invoked investment treaty in the world, a fact that would later make it a central battleground in debates over the compatibility of investment protection with climate policy.

The liberalisation of domestic energy markets proceeded in parallel. In the electricity and gas sectors, the United Kingdom privatised its state-owned utilities in the late 1980s and 1990s; the United States introduced open access to transmission networks through Federal Energy Regulatory Commission (FERC) Order No. 888 of 1996; and the European Union adopted successive legislative packages, beginning with the first Electricity Directive of 1996, designed to create an internal energy market through the unbundling of vertically integrated utilities and the introduction of competition in generation and supply.

Phase Five: Resource Nationalism, Sustainability, and the Energy Transition (2005–present)

The fifth phase, which continues today, is marked by a complex and sometimes contradictory set of trends.

First, the commodity price boom of the 2000s prompted a revival of resource nationalism. Venezuela, Bolivia, Ecuador, and Russia, among others, increased state participation, renegotiated contracts, or expropriated foreign interests, generating a new wave of investment arbitrations, several of which resulted in awards of billions of United States dollars. Resource nationalism also took subtler forms, including increased royalties and windfall taxes, local content requirements, export restrictions on unprocessed minerals, and mandatory state participation in new projects.

Second, the international community increasingly emphasised transparency, good governance, and the equitable distribution of resource revenues. The Extractive Industries Transparency Initiative (EITI), launched in 2002 and now implemented in more than fifty countries, requires the disclosure of payments by companies to governments and of revenues received by governments from the extractive sector. Supply chain due diligence standards for conflict minerals, notably

the OECD *Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas*, sought to address the link between mineral extraction and armed conflict. The United Nations *Guiding Principles on Business and Human Rights*, endorsed by the Human Rights Council in 2011, articulated the responsibility of business enterprises, including extractive companies, to respect human rights.

Third, and most consequentially, climate change has moved from the margins to the centre of energy law. The Paris Agreement of 2015 set the objective of holding the increase in global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the increase to 1.5°C. At the twenty-eighth Conference of the Parties to the United Nations Framework Convention on Climate Change (COP28) in Dubai in 2023, the parties for the first time called in the outcome of the first global stocktake for "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner". Climate litigation has proliferated in domestic courts, regional human rights bodies, and international tribunals, culminating in advisory opinions from the International Tribunal for the Law of the Sea (2024), the Inter-American Court of Human Rights (2025), and the International Court of Justice (2025) on the obligations of states in respect of climate change. These developments are examined in Chapter 18.

Fourth, the energy transition has created new legal frontiers: the regulation of renewable energy, hydrogen, energy storage, carbon capture and storage, and offshore wind; the governance of critical minerals essential to batteries, electric vehicles, and grid infrastructure; and the management of stranded assets and the decommissioning of fossil fuel infrastructure. Simultaneously, geopolitical events — notably the Russian invasion of Ukraine in February 2022 and the resulting disruption of European gas supply — have restored energy security to the forefront of policy and have demonstrated that the transition will be shaped as much by security considerations as by environmental ones.

KEY DOCTRINAL POINT

The history of resource governance is not a linear progression from one paradigm to another. Elements of each phase coexist in contemporary law. Classical concession-style licences remain in use in many jurisdictions; the doctrine of permanent sovereignty retains its normative force; investment treaties continue to protect investors even as they are reformed or terminated; and the imperatives of the energy transition are layered on top of these older structures rather than replacing them. A doctoral analysis must therefore be historically literate: it must recognise which layer of the law a given rule belongs to and how that rule has been reinterpreted over time.

1.3 Internationalisation of Energy Markets and Resource Regulation

The Global Character of Energy Markets

Energy markets are among the most internationalised of all economic sectors. Crude oil is a global commodity, traded on international exchanges and priced by reference to benchmark grades such as Brent and West Texas Intermediate. A substantial proportion of the world's oil production crosses at least one international border before it is consumed, often passing through maritime chokepoints such as the Strait of Hormuz and the Strait of Malacca. Natural gas markets, historically regional because of the constraints of pipeline transport, have become increasingly integrated through the growth of the liquefied natural gas (LNG) trade, which permits gas to be shipped across oceans and redirected in response to price signals. Coal is traded internationally in large volumes. Electricity, while still predominantly national, is increasingly traded across borders through interconnectors and regional power pools. The supply chains for renewable energy technologies — solar panels, wind turbines, batteries — are global, with manufacturing concentrated in a small number of states.

The internationalisation of markets has legal consequences. Domestic regulatory decisions in one state — on export restrictions, subsidies, environmental standards, or sanctions — can have profound effects on prices and supplies in others. Commercial disputes arising from cross-border transactions require rules on applicable law, jurisdiction, and the enforcement of judgments and awards. Infrastructure that crosses borders, such as pipelines and interconnectors, requires intergovernmental agreements to address transit, tariffs, and the allocation of regulatory authority.

Mechanisms of Internationalisation

The internationalisation of energy and resource regulation has proceeded through several mechanisms.

Treaty-making. States have concluded treaties addressing particular aspects of energy and resource activities: investment protection (BITs and the ECT), trade (the WTO agreements and regional trade agreements), transboundary pipelines (for example, the intergovernmental agreements underlying the Baku–Tbilisi–Ceyhan pipeline and the West African Gas Pipeline), nuclear safety and liability, the law of the sea, and environmental protection.

Institutional cooperation. International organisations such as the International Energy Agency (IEA), the International Atomic Energy Agency (IAEA), the International Renewable Energy Agency (IRENA), OPEC, and the Gas Exporting Countries Forum provide forums for the coordination of national policies, the exchange of information, and, in some cases, the adoption of binding

standards. Regional bodies — the Energy Community in South-East Europe, the West African Power Pool, the Southern African Power Pool — promote regional market integration.

Regulatory convergence. States often borrow regulatory models from one another, whether through voluntary emulation, the conditions attached to development finance, or the technical assistance of international organisations. The spread of independent regulatory agencies, of unbundling requirements in electricity, and of model petroleum agreements reflects this process of regulatory transplantation, which is examined in Chapter 24.

Private ordering. The standard contracts produced by industry associations — the AIEN Model Joint Operating Agreement, the International Federation of Consulting Engineers (FIDIC) suites of construction contracts, and the master agreements used in energy trading — create transnational norms that operate across jurisdictions.

Extraterritorial regulation. Some states regulate conduct outside their territory through legislation with extraterritorial effect. Examples include anti-corruption statutes such as the United States Foreign Corrupt Practices Act of 1977 and the United Kingdom Bribery Act 2010, sanctions regimes targeting the energy sectors of particular states, and supply chain due diligence obligations such as those contained in the European Union's Corporate Sustainability Due Diligence Directive and Batteries Regulation. These are discussed in Chapters 2 and 22.

The Limits of Internationalisation

Notwithstanding these developments, energy and resource regulation remains in important respects national. There is no comprehensive multilateral energy treaty of universal application; the ECT, the most ambitious attempt to create such an instrument, never attracted the participation of major producers and consumers such as the United States, China, Saudi Arabia, or Brazil, and its membership has contracted in recent years following withdrawals by the European Union and several of its member states. Petroleum, mining, and electricity remain subject primarily to domestic legislation, and the doctrine of permanent sovereignty ensures that the fundamental decisions about whether, when, and on what terms to develop natural resources are reserved to the state in which they are located. The internationalisation of energy law is thus partial, uneven, and contested — characteristics that make it a fertile subject for doctoral research.

1.4 Public Law, Private Law, and Transnational Legal Relations

The Public–Private Divide in Energy and Resource Law

One of the distinctive features of energy and resource law is the extent to which it straddles the conventional divide between public and private law. On the public side, the state acts as sovereign: it owns or controls subsurface resources, grants licences, sets fiscal terms, regulates conduct, protects the environment, and resolves disputes through its courts. On the private side, the state frequently acts as a commercial party: it enters into contracts with investors, participates in joint ventures through national oil or mining companies, sells resources on international markets, and borrows money to finance its share of project costs.

The dual character of the state — as sovereign regulator and as contracting party — generates persistent legal difficulties. When a state that has entered into a contract with a foreign investor subsequently changes its tax laws or environmental regulations in a manner that adversely affects the investor, is it acting as a sovereign exercising its regulatory powers, or as a contracting party in breach of its contractual obligations? The answer may determine whether the investor has a remedy, which tribunal has jurisdiction, and what law applies. The distinction between acts *jure imperii* and acts *jure gestionis*, familiar from the law of state immunity, and the distinction in investment law between a mere breach of contract and a breach of treaty, are both attempts to grapple with this problem.

The State Contract

The "state contract" — a contract between a state (or a state entity) and a foreign private party — is the central legal instrument of international resource development. Petroleum concessions, production sharing contracts, mining agreements, host government agreements for pipelines, and power purchase agreements with state-owned utilities are all species of state contract.

The legal status of state contracts has been debated for more than a century. Three principal positions may be distinguished.

The first position holds that state contracts are governed exclusively by the domestic law of the host state. On this view, the state retains its sovereign power to modify the contract by legislation, and the investor's protection lies only in such guarantees as domestic law provides. This position was associated with the Calvo doctrine in Latin America and with the claims of the New International Economic Order.

The second position holds that state contracts may be "internationalised" — that is, subjected to international law or to general principles of law — by the choice of the parties, by the inclusion of stabilisation and arbitration clauses, or by the nature of the contract as an economic development agreement. This

position found its fullest expression in the award of Sole Arbitrator René-Jean Dupuy in *Texaco v Libya*, which held that the concession contracts at issue had been internationalised and that Libya's nationalisation measures were a breach of its contractual obligations under international law.[6]

The third position, now widely accepted, holds that the contract is governed by the law chosen by the parties (often the host state's law, sometimes supplemented by international law or general principles), but that the investor may also enjoy independent protection under international law — most importantly through investment treaties — which may be invoked even where the contract itself is governed by domestic law. The distinction between contract claims and treaty claims, articulated by the *ad hoc* Committee in *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentina* (*Vivendi I*), is now a cornerstone of investment arbitration: "whether there has been a breach of the BIT and whether there has been a breach of contract are different questions".[7]

Transnational Legal Relations

Beyond the public–private divide lies a further category of relations that is neither purely public nor purely private, neither purely national nor purely international. Philip Jessup, writing in 1956, coined the term "transnational law" to include "all law which regulates actions or events that transcend national frontiers", embracing both public and private international law as well as "other rules which do not wholly fit into such standard categories".[8] Energy and resource activities exemplify transnational legal relations. A single offshore gas project may involve a host government agreement with the state, a production sharing contract with the national oil company, a joint operating agreement among the co-venturers, engineering contracts with international contractors, LNG sales agreements with buyers in several countries, financing agreements with commercial banks and export credit agencies, and an investment treaty between the host state and the home states of the investors. Each relationship is governed by a different combination of national and international rules, and disputes may be resolved in different forums.

The transnational character of these relations has three important implications. First, it means that no single legal order governs the project as a whole; the applicable law must be determined relationship by relationship. Second, it creates the risk of conflicting obligations, as when an environmental obligation imposed by domestic law conflicts with a stabilisation clause in a contract, or a

[6] *Texaco Overseas Petroleum Company and California Asiatic Oil Company v Government of the Libyan Arab Republic*, Award on the Merits of 19 January 1977, (1978) 17 ILM 1; (1979) 53 ILR 389.

[7] *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentine Republic*, ICSID Case No ARB/97/3, Decision on Annulment, 3 July 2002, para 96.

[8] Philip C. Jessup, *Transnational Law* (Yale University Press 1956), p. 2.

sanctions regime imposed by one state conflicts with a contractual obligation to supply. Third, it confers significant influence upon private actors — companies, banks, industry associations, arbitrators — in the creation and interpretation of norms, raising questions of legitimacy and accountability that recur throughout this book.

1.5 The Relationship Between Energy Law and International Economic Law

Energy as an Exception in Trade Law

The relationship between energy law and international economic law is close but historically uneasy. When the General Agreement on Tariffs and Trade (GATT) was negotiated in 1947, energy products were not excluded from its scope, but in practice trade in oil and gas was largely left outside the multilateral trading system. Many major producers were not parties to the GATT; the trade in crude oil was conducted largely within vertically integrated companies; and energy policy was regarded as so closely bound up with national security that states were unwilling to subject it to multilateral discipline. The GATT security exception in Article XXI, which permits measures "relating to fissionable materials or the materials from which they are derived" and measures "taken in time of war or other emergency in international relations", reinforced this sense of energy as a domain of sovereign prerogative.

The establishment of the WTO in 1995 and the accession of major energy producers — including Saudi Arabia in 2005 and the Russian Federation in 2012 — brought energy trade more firmly within the multilateral framework. WTO disputes have since addressed export restrictions on raw materials (*China — Raw Materials* and *China — Rare Earths*), local content requirements in renewable energy programmes (*Canada — Renewable Energy/Feed-in Tariff Program* and *India — Solar Cells*), and the European Union's energy market regulation (*EU — Energy Package*). These disputes are examined in Chapter 13.

Investment Law as the Primary Interface

If trade law has historically played a limited role in energy, investment law has played a central one. Energy and mining projects are capital-intensive, long-lived, and dependent on the stability of the host state's legal and fiscal regime; they are therefore particularly vulnerable to what economists call the "obsolescing bargain" — the tendency for the host state's bargaining power to increase once the investor's capital has been sunk. Investment treaties and investor–state arbitration were designed in large part to mitigate this risk. Statistics compiled by ICSID consistently show that the oil, gas, and mining sector, together with electric power

and other energy, account for a larger share of registered cases than any other economic sector.

The obsolescing bargain theory, associated with the work of Raymond Vernon, explains much of the dynamic of resource governance. At the time of the initial investment decision, the host state competes with other states for scarce capital and expertise and must offer attractive terms. Once the investment has been made, however, the investor's assets are immobile and the host state may seek to renegotiate the terms in its favour — through increased taxation, contract renegotiation, or expropriation. Investors anticipate this risk and demand protections: stabilisation clauses, international arbitration, investment treaty coverage, and political risk insurance. The law of international energy investment can be understood, in significant part, as an attempt to manage the obsolescing bargain.

Convergence and Tension

The relationship between energy law and international economic law is characterised by both convergence and tension. Convergence occurs where economic law supports energy policy objectives — for example, where investment protection encourages the capital flows necessary for the energy transition. Tension occurs where economic law constrains the regulatory autonomy that states need to pursue energy and climate objectives — for example, where investors challenge the phase-out of coal-fired power or the withdrawal of renewable energy subsidies, or where trade rules are invoked against local content requirements designed to build domestic manufacturing capacity. The phase-out of coal-fired power in the Netherlands gave rise to ECT claims by RWE and Uniper (the latter subsequently withdrawn), and the retroactive reduction of renewable energy incentives in Spain generated dozens of ECT arbitrations. The resulting debates about regulatory chill, the right to regulate, and the reform of investment treaties are examined in Chapter 14.

1.6 Legal Pluralism and Competing Regulatory Authorities

The Concept of Legal Pluralism

Legal pluralism describes situations in which two or more legal orders coexist in the same social field. Originally developed by legal anthropologists to describe the coexistence of state law and customary law in colonial and post-colonial societies, the concept has been extended to describe the coexistence of national, international, regional, and private normative orders in a globalised world. Energy and resource governance is a paradigmatic site of legal pluralism.

At least five layers of normative authority may be identified in a typical resource project:

1. **Customary and community norms.** In many resource-rich regions, land and resources are held under customary tenure systems that predate, and

coexist with, state law. In Nigeria, for example, customary land tenure was overlaid by the Land Use Act 1978, which vested land in each state in the Governor, while section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 vests the entire property in and control of all minerals, mineral oils, and natural gas in the Government of the Federation. Communities in the Niger Delta nevertheless continue to assert claims to land and resources grounded in custom, and these claims shape the conduct of operations and the content of community development agreements.

2. **Subnational law.** In federal states, constitutional authority over resources may be divided between the federal government and the constituent units. In Canada, the provinces own and regulate most natural resources under section 92A of the Constitution Act 1867 (inserted in 1982); in Australia, the states own onshore minerals and petroleum; in the United States, mineral rights may be held privately, by the states, or by the federal government. In Nigeria, the Electricity Act 2023, enacted following a constitutional amendment, empowers states to regulate electricity markets within their territories.
3. **National law.** Constitutional provisions, petroleum and mining statutes, environmental legislation, tax law, and administrative law form the core of the regulatory framework.
4. **International and regional law.** Treaties, customary international law, and the law of regional organisations (such as the European Union, the Economic Community of West African States, and the African Union) impose obligations on states and, in some cases, confer rights on private parties.
5. **Transnational private regulation.** Industry standards, model contracts, voluntary codes of conduct, certification schemes, and the environmental and social performance standards of lenders — notably the International Finance Corporation (IFC) *Performance Standards on Environmental and Social Sustainability* and the Equator Principles adopted by commercial banks — impose requirements on companies that may exceed those of national law.

Sources of Conflict

The coexistence of multiple normative orders generates conflicts of several kinds. *Vertical* conflicts arise between different levels of authority — for example, between federal and state governments over the ownership of offshore resources, as in Nigeria's resource-control litigation, in which the Supreme Court held in *Attorney-General of the Federation v Attorney-General of Abia State* that for the purposes of revenue allocation the seaward boundary of a littoral state was the

low-water mark of its land surface.[9] *Horizontal* conflicts arise between different regimes at the same level — for example, between investment treaty obligations and environmental obligations. *Public–private* conflicts arise between state regulation and private standards or contracts. *Jurisdictional* conflicts arise between courts and tribunals asserting authority over the same dispute, as when an investor pursues parallel proceedings before domestic courts, commercial arbitration, and investment treaty arbitration.

Managing Pluralism

Legal systems have developed a range of techniques for managing these conflicts: hierarchical rules of priority (constitutional supremacy, the primacy of European Union law), principles of interpretation (systemic integration under Article 31(3)(c) of the Vienna Convention on the Law of Treaties, the presumption against conflict), doctrines of comity and deference (the margin of appreciation, the police powers doctrine in investment law), procedural mechanisms (fork-in-the-road and waiver clauses, *lis pendens*, *res judicata*), and institutional coordination (memoranda of understanding between regulators, joint commissions).

The International Law Commission's study of the fragmentation of international law, completed in 2006 under the chairmanship of Martti Koskenniemi, remains the most authoritative analysis of these techniques at the international level.[10] For energy and resource law, the key insight of the fragmentation debate is that specialised regimes — investment law, trade law, environmental law, human rights law — do not operate in isolation; they remain embedded in general international law, and their interpretation must take account of other relevant rules applicable between the parties.

> **COMPARATIVE PERSPECTIVE**
>
> **Federalism and resource authority in three jurisdictions.** In *Canada*, section 92A of the Constitution Act 1867 confirms provincial legislative authority over the exploration, development, conservation, and management of non-renewable natural resources, subject to federal paramountcy for interprovincial export. In *Nigeria*, section 44(3) of the 1999 Constitution vests ownership and control of minerals and petroleum in the Federal Government, and item 39 of the Exclusive Legislative List reserves "mines and minerals, including oil fields, oil mining, geological surveys and natural gas" to the National Assembly; the derivation principle in section 162(2) allocates not less than thirteen per cent of revenue accruing from natural resources

[9] *Attorney-General of the Federation v Attorney-General of Abia State & 35 Others* (2002) 6 NWLR (Pt 764) 542 (Supreme Court of Nigeria).

[10] International Law Commission, *Fragmentation of International Law: Difficulties Arising from the Diversification and Expansion of International Law*, Report of the Study Group finalised by Martti Koskenniemi, UN Doc A/CN.4/L.682 (13 April 2006).

to the states of origin. In the *United States*, subsurface minerals may be privately owned and severed from the surface estate, with federal and state governments owning substantial mineral estates in their own right and regulating activities on federal and state lands respectively. These differences shape the bargaining power of communities, the structure of fiscal regimes, and the incidence of resource conflicts.

1.7 Contemporary Challenges in Global Resource Governance

The field confronts a set of challenges that are without precedent in their scale and complexity. Six are of particular importance.

The Energy Transition and Stranded Assets

The first challenge is the transition from an energy system dominated by fossil fuels to one based predominantly on low-carbon sources. The legal implications are profound. Petroleum and coal licences granted for decades may become economically or legally unviable before their expiry, raising questions about compensation, decommissioning liability, and the protection of legitimate expectations. Investors in fossil fuel infrastructure may seek to rely on investment treaties to challenge phase-out measures. Producer states whose economies depend on hydrocarbon revenues face fiscal crises and must manage the social consequences of transition. The concept of "stranded assets" — assets that suffer from premature write-downs or devaluation as a result of the transition — has migrated from financial analysis into law, influencing disclosure obligations, directors' duties, and the interpretation of investment protections.

Critical Minerals and the New Extractivism

The second challenge arises from the paradox that decarbonisation requires more extraction, not less. The IEA has repeatedly projected that demand for minerals such as lithium, cobalt, nickel, copper, graphite, and rare earth elements will grow severalfold under scenarios consistent with the Paris Agreement. These minerals are geographically concentrated, and their processing is concentrated still further. States have responded with industrial policies, export restrictions, strategic stockpiles, and bilateral critical mineral agreements. Communities in mining regions confront the risk that the green transition will reproduce the environmental degradation and human rights abuses associated with earlier extractive booms. Chapter 8 addresses these issues in detail.

Energy Access and Energy Poverty

The third challenge is that a substantial proportion of humanity still lacks access to modern energy services. Hundreds of millions of people, the majority in sub-Saharan Africa, lack access to electricity, and far more lack access to clean cooking fuels. Sustainable Development Goal 7 commits states to ensure access

to affordable, reliable, sustainable, and modern energy for all by 2030. Achieving universal access while simultaneously decarbonising energy systems requires legal and regulatory frameworks that mobilise investment in decentralised and grid-based solutions, protect consumers, and allocate the costs of transition fairly. The concept of energy justice, explored in Chapter 23, provides a framework for analysing these distributional questions.

Geopolitics and Energy Security

The fourth challenge is the re-emergence of energy as an instrument of geopolitical competition. The disruption of gas supplies to Europe following the invasion of Ukraine, the use of sanctions against major energy exporters, attacks on energy infrastructure, and competition over critical mineral supply chains have demonstrated that energy security remains a central concern of state policy. The law governing sanctions, export controls, emergency powers, strategic reserves, and the protection of critical infrastructure — examined in Chapter 25 — has accordingly grown in importance.

Legitimacy of Investor–State Dispute Settlement

The fifth challenge concerns the legitimacy of the system of investor–state dispute settlement (ISDS). Critics argue that ISDS privileges foreign investors over domestic stakeholders, constrains the regulatory autonomy of states, produces inconsistent decisions, and lacks transparency and accountability. Reform efforts proceed in several forums: the United Nations Commission on International Trade Law (UNCITRAL) Working Group III, the European Union's proposal for a multilateral investment court, the modernisation of the ECT, and the renegotiation or termination of BITs by individual states. The outcome of these reforms will shape the legal environment for energy investment for decades.

Technological Change and Digitalisation

The sixth challenge arises from technological change. Hydrogen, advanced nuclear reactors, carbon capture, battery storage, offshore floating wind, and deep-sea mining all raise novel regulatory questions. Digitalisation — smart meters, distributed energy resources, automated trading, artificial intelligence in grid management — raises questions of data ownership, cybersecurity, and liability. Regulatory frameworks designed for centralised, fossil-fuel-based systems must be adapted to decentralised, digital, and decarbonised systems. Chapters 12 and 30 address these developments.

1.8 Doctoral Research Perspectives and Foundational Debates

The Nature of Doctoral Contribution

Doctoral research is distinguished from other forms of legal writing by the requirement of an original contribution to knowledge. In international energy

and natural resources law, such a contribution may take several forms: the identification and analysis of a new legal problem; a new interpretation of existing rules or cases; the application of a theoretical framework to a body of law; a comparative study revealing previously unrecognised patterns; an empirical investigation of how the law operates in practice; or a reasoned proposal for reform. Chapter 28 discusses research methodology in detail. Here, the purpose is to identify the foundational debates within which most doctoral research in the field is situated.

Debate One: Sovereignty versus Stability

The most enduring debate in the field concerns the tension between the sovereign right of states to regulate their natural resources in the public interest and the need of investors for legal and fiscal stability. The debate manifests itself in disputes over stabilisation clauses, the standard of compensation for expropriation, the scope of the fair and equitable treatment standard, and the legitimacy of contract renegotiation. Is the doctrine of permanent sovereignty compatible with binding long-term commitments to foreign investors? Can a state contract away its legislative power? How should the law balance the legitimate expectations of investors against the evolving needs of the host society?

Debate Two: Contract versus Regulation

A second debate concerns the relative roles of contract and regulation in governing resource activities. In some jurisdictions, the terms of resource development are set primarily by individually negotiated contracts; in others, primarily by general legislation and regulation applicable to all licensees. Each approach has advantages and drawbacks. Contracts offer flexibility and can be tailored to the circumstances of particular projects, but they are vulnerable to corruption, asymmetries of information, and lack of transparency. Regulation offers predictability, equality of treatment, and democratic accountability, but may be less responsive to project-specific risks. The trend in many jurisdictions, including Nigeria under the Petroleum Industry Act 2021, has been towards greater reliance on legislative frameworks, but contracts remain central.

Debate Three: Efficiency versus Equity

A third debate concerns the objectives of resource law. Economic approaches emphasise efficiency — the maximisation of resource rents, the minimisation of transaction costs, and the allocation of risk to the party best able to bear it. Equity-oriented approaches emphasise the fair distribution of benefits and burdens among states, investors, communities, and future generations, and the protection of the rights of affected persons. The "resource curse" literature, which examines why resource-rich states often experience slower growth, weaker institutions, and greater conflict than resource-poor states, has highlighted the importance of

institutional quality and has prompted reforms focused on transparency, revenue management, and accountability.

Debate Four: Development versus Sustainability

A fourth debate concerns the relationship between resource development and environmental sustainability. Developing states argue that they have a right to exploit their resources to finance development, as industrialised states did before them, and that the burden of climate mitigation should fall primarily on those states responsible for historical emissions. This argument is reflected in the principle of common but differentiated responsibilities and respective capabilities, articulated in the United Nations Framework Convention on Climate Change. At the same time, the physical limits of the carbon budget mean that a substantial proportion of known fossil fuel reserves cannot be burned if the temperature goals of the Paris Agreement are to be met. How should the right to development be reconciled with the imperative of decarbonisation? Who should bear the costs of leaving resources in the ground?

Debate Five: State-Centrism versus Transnational Governance

A fifth debate concerns the appropriate unit of analysis. Traditional approaches treat the state as the principal actor and source of law. Transnational approaches emphasise the role of corporations, international organisations, civil society, communities, and private standard-setters in creating and enforcing norms. The rise of corporate climate litigation, supply chain due diligence obligations, and private sustainability standards suggests that governance is increasingly polycentric. Doctoral research must decide how to account for this complexity without losing analytical clarity.

LEADING AUTHORITY

Texaco Overseas Petroleum Co v Libyan Arab Republic (1977)

Facts. Libya had granted deeds of concession to two American companies under its Petroleum Law of 1955. The concessions contained a stabilisation clause providing that the contractual rights expressly created by the concession should not be altered except by mutual consent, and an arbitration clause. In 1973 and 1974 Libya nationalised the companies' interests. The companies invoked arbitration; Libya refused to participate, and a sole arbitrator was appointed by the President of the International Court of Justice.

Holding. Sole Arbitrator Dupuy held that the concessions were internationalised contracts; that they were governed by principles of Libyan law common to international law and, in the absence of such common principles, by general principles of law; that the stabilisation clause was valid and binding; and that Libya had breached its obligations. Notably, he ordered *restitutio in integrum* — specific performance — as the primary remedy, although the parties subsequently settled.

Significance. The award is the high-water mark of the internationalisation theory. Its analysis of Resolution 1803 as reflecting customary law, and its treatment of the later Charter of Economic Rights and Duties of States as lacking that status, remain influential. Its remedy of restitution has rarely been followed; later tribunals, beginning with *LIAMCO* and *Aminoil*, preferred damages.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

The Kestrel Basin concession (hypothetical). The Republic of Aranda, a fictional coastal state, granted a thirty-year petroleum concession in 1998 to Northwind Energy plc, a company incorporated in a fictional state with which Aranda has a bilateral investment treaty. The concession contains (i) a choice of Arandan law "supplemented by such principles of international law as may be applicable"; (ii) a "freezing" stabilisation clause; and (iii) ICC arbitration seated in Geneva. In 2025, following discoveries of significant gas reserves and a change of government, Aranda enacts a Petroleum Revenue Act that raises the royalty rate from 8 to 15 per cent, imposes a windfall profits tax, and requires all existing concessionaires to convert to production sharing contracts within two years.

Prepare a memorandum of not more than 2,500 words that: (a) identifies each legal order potentially applicable to the dispute; (b) explains the competing characterisations of the measures as sovereign regulation or contractual breach; (c) assesses the legal effect of the stabilisation clause in the light of the doctrine of permanent sovereignty; and (d) recommends a dispute resolution strategy for Northwind, identifying the risks of parallel proceedings. *All parties, places, and instruments in this exercise are fictitious.*

CHAPTER SUMMARY

- International energy and natural resources law is a functional, pluralist, distributive, and transnational field concerned with the allocation of rights over energy sources and natural resources, the regulation of activities involving them, and the distribution of the resulting benefits and burdens.
- The field integrates energy law and natural resources law, a combination justified by shared history, doctrine, institutions, and — increasingly — the material interdependence created by the energy transition.
- Its history may be organised into five overlapping phases: the classical concession era, nationalisation, decolonisation and the New International Economic Order, liberalisation and investment protection, and the current era of resource nationalism, sustainability, and transition.
- Energy markets are highly internationalised, but energy regulation remains predominantly national; internationalisation proceeds through treaties, institutions, regulatory convergence, private ordering, and extraterritorial regulation.

- The dual character of the state as sovereign and contracting party is the source of many of the field's enduring problems, notably the status of state contracts and the distinction between contract and treaty claims.
- Legal pluralism is a defining feature of resource governance, requiring techniques of conflict management drawn from constitutional, international, and private law.
- Doctoral research is situated within five foundational debates: sovereignty versus stability, contract versus regulation, efficiency versus equity, development versus sustainability, and state-centrism versus transnational governance.

DOCTORAL RESEARCH QUESTIONS

1. To what extent does the concept of *lex petrolea* describe a genuine transnational legal order, and what are the implications of its existence or non-existence for the legitimacy of investor–state arbitration?
2. Has the energy transition rendered the integrated study of energy and natural resources law analytically indispensable, or does it obscure important differences between extractive and non-extractive energy sources?
3. How should the obsolescing bargain theory be revised to account for renewable energy projects, in which the host state's bargaining power may evolve differently from that in petroleum projects?
4. Can the techniques developed to manage the fragmentation of international law adequately address conflicts between investment protection and climate obligations, or is structural reform required?
5. What role should customary land tenure and community norms play in the allocation of resource rights in states, such as Nigeria, where the Constitution vests mineral ownership in the federal government?

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2 CHAPTER

Sources, Principles, and Institutions of International Energy Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Identify the formal sources of international energy and natural resources law and explain how each operates in the energy and resource context.
2. Analyse the content, legal status, and limits of the principle of permanent sovereignty over natural resources.
3. Apply the rules of state responsibility to conduct affecting energy investments, infrastructure, and the environment.
4. Explain the principles of jurisdiction, including the conditions under which states may regulate extraterritorially.
5. Evaluate the contribution of international organisations, soft law, industry standards, and corporate governance instruments to the field.
6. Assess the role of international courts and arbitral tribunals and analyse conflicts between domestic and international legal obligations.

Introduction

The preceding chapter described international energy and natural resources law as a pluralist field drawing upon multiple bodies of law. This chapter examines the materials from which that law is made — its sources — and the general principles and institutions through which it operates. It begins with the formal sources of international law identified in Article 38(1) of the Statute of the International Court of Justice and considers how each functions in the energy context. It then examines the principle that more than any other defines the field: permanent sovereignty over natural resources. The chapter proceeds to the law of state responsibility, the principles of jurisdiction, and the growing role of international organisations, soft law, and industry standards. It concludes with an analysis of

the institutions that adjudicate energy disputes and of the persistent problem of conflicts between domestic and international legal obligations.

A preliminary observation is in order. Students trained in a domestic legal system sometimes approach international law expecting a hierarchy of norms comparable to that of a national constitution, with clearly defined law-making institutions and a supreme court at the apex. International law does not work in that way. It is a decentralised system in which the subjects of the law — principally states — are also its makers, its interpreters, and, to a considerable extent, its enforcers. The sources of international law are therefore best understood not as a hierarchy but as a set of recognised processes through which states, and increasingly other actors, generate obligations. This decentralisation is particularly pronounced in the energy field, where there is no single governing treaty and where much of the normative work is done by contracts, standards, and the decisions of arbitral tribunals.

2.1 Treaties, Custom, and General Principles of International Law

Article 38 of the Statute of the International Court of Justice

Article 38(1) of the Statute of the International Court of Justice directs the Court, in deciding disputes in accordance with international law, to apply:

> (a) international conventions, whether general or particular, establishing rules expressly recognized by the contesting states;
> (b) international custom, as evidence of a general practice accepted as law;
> (c) the general principles of law recognized by civilized nations;
> (d) subject to the provisions of Article 59, judicial decisions and the teachings of the most highly qualified publicists of the various nations, as subsidiary means for the determination of rules of law.

Although Article 38 is formally a direction to the Court, it is universally regarded as an authoritative statement of the sources of international law. The archaic reference to "civilized nations" is today understood to refer to the community of nations generally. Article 38 does not establish a formal hierarchy among treaties, custom, and general principles, although in practice a treaty, as *lex specialis*, will usually prevail over a conflicting rule of custom as between the parties to it, subject always to the overriding force of peremptory norms (*jus cogens*).

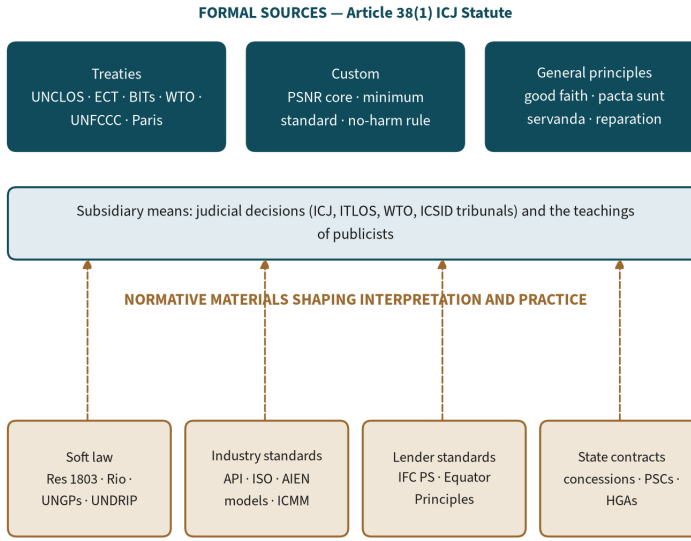


Figure 2.1 The sources of international energy and natural resources law. Formal sources under Article 38(1) of the ICJ Statute are supplemented by soft law, contracts, and transnational standards that shape practice and interpretation. *Source:* prepared for this edition.

Treaties

Treaties are the most important source of international energy law in terms of volume and specificity. The Vienna Convention on the Law of Treaties 1969 (VCLT) defines a treaty as "an international agreement concluded between States in written form and governed by international law, whether embodied in a single instrument or in two or more related instruments and whatever its particular designation". Treaties relevant to energy and natural resources may be classified as follows.

Multilateral treaties of general application that apply to energy and resource activities without being specifically directed at them. These include the United Nations Convention on the Law of the Sea 1982 (UNCLOS), the WTO agreements, the ICSID Convention 1965, the New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards 1958, the United Nations Framework Convention on Climate Change 1992 (UNFCCC), the Paris Agreement 2015, and the Convention on Biological Diversity 1992.

Multilateral treaties specific to energy. The Energy Charter Treaty 1994 is the principal example of a multilateral treaty dedicated to energy. Others include the Statute of the International Atomic Energy Agency 1956, the Agreement on an International Energy Program 1974 (which established the IEA), the Statute of the International Renewable Energy Agency 2009, and the suite of nuclear

conventions on safety, security, liability, and waste management considered in Chapter 11.

Regional treaties, including the Treaty Establishing the Energy Community 2005, the ECOWAS Energy Protocol 2003 (modelled on the ECT), and the energy provisions of the Treaty on the Functioning of the European Union, notably Article 194.

Bilateral treaties, including more than two thousand bilateral investment treaties, tax treaties, maritime boundary agreements, joint development agreements, unitisation agreements for transboundary reservoirs, and intergovernmental agreements for cross-border pipelines and interconnectors.

Treaty Interpretation

Because so much of the field turns on the interpretation of treaties — investment treaties in particular — a doctoral candidate must have a thorough command of the rules of interpretation in Articles 31 to 33 of the VCLT, which reflect customary international law. Article 31(1) provides that a treaty "shall be interpreted in good faith in accordance with the ordinary meaning to be given to the terms of the treaty in their context and in the light of its object and purpose". Article 31(3) requires account to be taken, together with the context, of subsequent agreements between the parties regarding interpretation, subsequent practice establishing agreement regarding interpretation, and "any relevant rules of international law applicable in the relations between the parties". Article 32 permits recourse to supplementary means of interpretation, including the preparatory work of the treaty and the circumstances of its conclusion, to confirm the meaning resulting from Article 31 or to determine the meaning when that interpretation leaves it ambiguous or obscure or leads to a manifestly absurd or unreasonable result.

Three interpretive issues recur in energy disputes. The first is the weight to be given to the *object and purpose* of investment treaties: tribunals that emphasise the promotion and protection of investment, often stated in treaty preambles, have tended to adopt expansive interpretations of investor protections, while others insist that the object and purpose include a balance between investment protection and the host state's right to regulate. The second is the use of *subsequent agreements and practice*, including joint interpretive statements issued by treaty parties, such as the interpretive note issued by the NAFTA Free Trade Commission in 2001 limiting the minimum standard of treatment to the customary international law minimum standard. The third is *systemic integration* under Article 31(3)(c), which permits tribunals to take account of environmental, human rights, and climate obligations in interpreting investment treaties.

Customary International Law

Customary international law arises from a general practice of states accepted as law (*opinio juris*). Both elements must be established: practice that is sufficiently widespread, representative, and consistent, and the acceptance of that practice as legally required or permitted. The International Law Commission's *Draft Conclusions on Identification of Customary International Law*, adopted in 2018, provide authoritative guidance on the evidence of each element.

Customary international law performs several functions in the energy field. It supplies rules of general application where no treaty exists, such as the rules on the protection of foreign property and the standard of compensation for expropriation, the obligation to prevent significant transboundary harm, and the rules on state responsibility. It fills gaps in treaty regimes and informs their interpretation. And in some areas — notably the minimum standard of treatment of aliens — treaty provisions expressly incorporate customary standards.

The identification of custom in the energy field is often difficult. Investment treaties are numerous and broadly similar in content, prompting debate over whether they have generated or codified customary rules. In *Barcelona Traction*, the International Court of Justice observed that, in the field of foreign investment, the law had developed through bilateral and multilateral agreements rather than general rules,[1] and this remains a contested question. The dominant view is that the proliferation of BITs has not of itself transformed their substantive standards into custom, because states conclude such treaties to obtain reciprocal benefits rather than out of a sense of legal obligation, although the treaties may serve as evidence of the content of the customary minimum standard.

General Principles of Law

General principles of law recognised by the community of nations serve primarily to fill gaps in treaty and customary law. The International Law Commission adopted *Draft Conclusions on General Principles of Law* in 2023, distinguishing between principles derived from national legal systems and principles that may be formed within the international legal system. General principles of particular importance in the energy field include good faith, *pacta sunt servanda*, the prohibition of abuse of rights, the principle of unjust enrichment, estoppel, the principle that no one may profit from his own wrong, the obligation to make reparation for injury caused by a wrongful act, and the procedural principles of due process and equality of arms.

General principles played a central role in the arbitral jurisprudence of the concession era. In the *Abu Dhabi* arbitration of 1951, Lord Asquith of Bishopstone,

[1] *Barcelona Traction, Light and Power Company, Limited (Belgium v Spain)*, Second Phase, Judgment, ICJ Reports 1970, p. 3, paras 89–90.

acting as umpire, declined to apply the law of Abu Dhabi on the ground that no settled body of legal principles existed there that could be applied to the construction of modern commercial instruments, and instead applied principles "rooted in the good sense and common practice of the generality of civilised nations — a sort of 'modern law of nature'".[2] In *Sapphire International Petroleums v National Iranian Oil Company* (1963), the arbitrator similarly applied general principles of law.[3] Such reasoning reflected attitudes towards non-Western legal systems that are now rightly regarded as unacceptable, and modern tribunals generally apply the law chosen by the parties or the law of the host state, supplemented where appropriate by international law.

Subsidiary Means: Judicial Decisions and Teachings

Judicial decisions and the teachings of highly qualified publicists are subsidiary means for the determination of rules of law. There is no formal doctrine of binding precedent in international law; Article 59 of the ICJ Statute provides that decisions of the Court have no binding force except between the parties and in respect of the particular case. Nevertheless, the decisions of the ICJ, the International Tribunal for the Law of the Sea, the WTO Appellate Body, and investment tribunals carry significant persuasive authority, and tribunals routinely cite earlier awards. In investment arbitration, a *jurisprudence constante* has emerged on many issues, although significant divergences persist. The influence of scholarly writing is also considerable: the works of authors such as Nico Schrijver on permanent sovereignty, Christoph Schreuer on the ICSID Convention, and Philippe Sands on environmental law are frequently cited by tribunals.

2.2 Permanent Sovereignty over Natural Resources

Origins and Development

The principle of permanent sovereignty over natural resources (PSNR) is the cornerstone of international resource law. It emerged in the 1950s from the efforts of newly independent and developing states to secure control over their natural wealth, which in many cases remained subject to concessions granted by colonial administrations or on terms that the host states regarded as inequitable. The principle was first articulated in General Assembly Resolution 523 (VI) of 1952 on integrated economic development and commercial agreements, and Resolution 626 (VII) of the same year, which affirmed the right of peoples "freely to use and exploit their natural wealth and resources". A Commission on Permanent Sover-

[2] *Petroleum Development (Trucial Coast) Ltd v Sheikh of Abu Dhabi*, Award of September 1951, (1952) 1 ICLQ 247.

[3] *Sapphire International Petroleums Ltd v National Iranian Oil Company*, Award of 15 March 1963, (1967) 35 ILR 136.

eignty over Natural Resources was established in 1958, and its work culminated in Resolution 1803 (XVII) of 1962.

The principle was incorporated into common Article 1(2) of the International Covenant on Civil and Political Rights and the International Covenant on Economic, Social and Cultural Rights 1966, which provides:

> All peoples may, for their own ends, freely dispose of their natural wealth and resources without prejudice to any obligations arising out of international economic co-operation, based upon the principle of mutual benefit, and international law. In no case may a people be deprived of its own means of subsistence.

Article 21 of the African Charter on Human and Peoples' Rights 1981 is more emphatic still, providing that "all peoples shall freely dispose of their wealth and natural resources", that this right "shall be exercised in the exclusive interest of the people", and that "in no case shall a people be deprived of it". The provision also obliges states parties to "undertake to eliminate all forms of foreign economic exploitation particularly that practised by international monopolies so as to enable their peoples to fully benefit from the advantages derived from their national resources".

Content of the Principle

PSNR comprises a bundle of rights and, increasingly, duties. The principal rights include:

- the right to possess, use, and dispose of natural resources;
- the right to determine freely the conditions of exploration, development, and disposition of resources;
- the right to regulate foreign investment and to supervise the activities of foreign investors;
- the right to nationalise or expropriate foreign property, subject to the payment of appropriate compensation;
- the right to settle disputes in accordance with national law, subject to agreements to the contrary;
- the right to manage resources in pursuance of national environmental and developmental policy.

Schrijver's influential study demonstrated that the principle has evolved from a primarily rights-based claim asserted by developing states against foreign interests into a more balanced concept that also encompasses duties: the duty to exercise sovereignty in the interest of national development and the well-being of the people; the duty to respect the rights and interests of indigenous peoples; the duty to observe international agreements in good faith; the duty to treat foreign

investors fairly; and the duty to manage resources sustainably and to avoid transboundary environmental harm.[4]

Legal Status

The legal status of PSNR has been considered by the International Court of Justice. In *Armed Activities on the Territory of the Congo (Democratic Republic of the Congo v Uganda)*, the Court recognised that Resolution 1803 and the related instruments reflected customary international law in respect of the principle of permanent sovereignty, although it held that the principle was not applicable to the specific situation of looting, pillage, and exploitation of natural resources by members of the army of a state militarily intervening in another state, which fell instead to be addressed under the law of belligerent occupation and international humanitarian law.[5] In its 2022 judgment on reparations in the same case, the Court awarded the Democratic Republic of the Congo a global sum in respect of damage to natural resources.

It is generally accepted that the core of PSNR — the right of states to control and regulate their natural resources, and to expropriate foreign property for a public purpose on payment of compensation — forms part of customary international law. More contested are the claims, associated with the Charter of Economic Rights and Duties of States, that compensation is to be determined solely by domestic law and that disputes are to be resolved exclusively by domestic courts. The arbitral tribunal in *Texaco v Libya* held that Resolution 1803, adopted by a large majority including both developed and developing states, reflected customary law, whereas the relevant provisions of the Charter, adopted over the opposition of the principal capital-exporting states, did not.

PSNR and Contractual Commitments

The relationship between PSNR and contractual commitments is one of the field's central problems. Can a state, in the exercise of its sovereignty, bind itself not to exercise its sovereign powers in the future — for example, by agreeing to a stabilisation clause that freezes the applicable law? The arbitral jurisprudence has generally answered this question in the affirmative, on the ground that the power to enter into binding commitments is itself an attribute of sovereignty. As the sole arbitrator in *Texaco v Libya* put it, "the State, by entering into an international agreement with a private person, has not alienated its sovereignty but has made use of it".[6] On this view, PSNR does not authorise a state to disregard its contrac-

[4] Nico Schrijver, *Sovereignty over Natural Resources: Balancing Rights and Duties* (Cambridge University Press 1997), especially chs 9–10.

[5] *Armed Activities on the Territory of the Congo (Democratic Republic of the Congo v Uganda)*, Judgment, ICJ Reports 2005, p. 168, para 244; Reparations, Judgment, ICJ Reports 2022, p. 13.

[6] *Texaco Overseas Petroleum Company v Libyan Arab Republic*, Award of 19 January 1977, (1979) 53 ILR 389, para 68.

tual obligations; it merely affirms the state's ultimate authority over its resources, which must be exercised in accordance with international law, including the obligation to pay compensation for any taking.

The *Aminoil* tribunal adopted a more nuanced position. It accepted that stabilisation clauses were valid but held that, in the absence of an express prohibition on nationalisation of particular duration, the clause in the Aminoil concession did not prohibit nationalisation as such; rather, its effect was to be taken into account in determining the compensation payable. The tribunal also emphasised that the concession had undergone significant evolution over time, through renegotiation and the acceptance by the company of changes in the fiscal regime, such that the contract had been transformed and its original stabilisation clause could not be read in isolation from that evolution. The approach of *Aminoil* — treating stabilisation clauses as relevant to compensation rather than as an absolute bar to sovereign action — has been influential in later practice and is examined further in Chapters 14 and 15.

KEY DOCTRINAL POINT

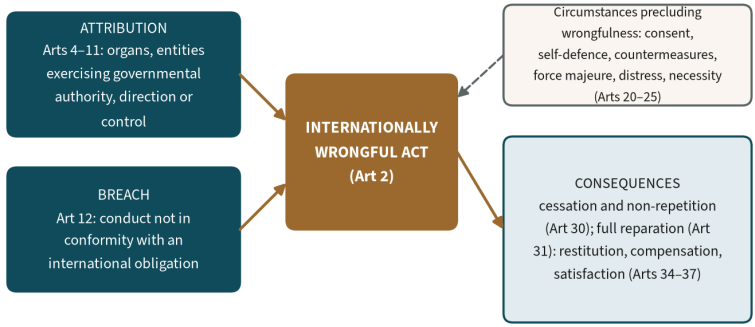
Permanent sovereignty over natural resources is not an unlimited licence to disregard commitments. In its modern form it is a principle of balanced rights and duties: the state controls its resources and may regulate and expropriate them in the public interest, but it must exercise that control in good faith, for the benefit of its people, in accordance with its international obligations, and subject to the payment of compensation where property is taken.

2.3 State Responsibility and International Legal Obligations

The Articles on State Responsibility

The law of state responsibility determines when a state has committed an internationally wrongful act and what consequences follow. Its authoritative statement is the International Law Commission's *Articles on Responsibility of States for Internationally Wrongful Acts* (ARSIWA), adopted in 2001 and commended to governments by the General Assembly. Although ARSIWA is not a treaty, many of its provisions are recognised as reflecting customary international law, and they are routinely applied by the ICJ and by investment tribunals.

Article 1 of ARSIWA states the basic principle: "Every internationally wrongful act of a State entails the international responsibility of that State." Under Article 2, an internationally wrongful act exists when conduct consisting of an action or omission (a) is attributable to the state under international law and (b) constitutes a breach of an international obligation of the state.



Chorzów Factory (1928): reparation must, as far as possible, wipe out all the consequences of the illegal act

Figure 2.2 Elements and consequences of state responsibility under the ILC Articles (2001). *Source:* prepared for this edition.

Attribution

Attribution is of particular importance in the energy sector, where states frequently act through state-owned enterprises, national oil companies, regulatory agencies, and subnational authorities. ARSIWA provides several grounds of attribution:

- **Organs of the state** (Article 4). The conduct of any state organ — legislative, executive, judicial, or other, whether central or territorial — is attributable to the state. A federal state is therefore responsible for the conduct of its constituent units, and a state is responsible for the decisions of its courts and regulators.
- **Persons or entities exercising elements of governmental authority** (Article 5). The conduct of an entity that is not a state organ but is empowered by law to exercise elements of governmental authority is attributable to the state when the entity acts in that capacity. A national oil company that is empowered to grant licences, collect royalties, or regulate operations exercises governmental authority in performing those functions.
- **Conduct directed or controlled by the state** (Article 8). The conduct of a person or group is attributable to the state if it acts on the instructions of, or under the direction or control of, the state in carrying out the conduct.
- **Conduct acknowledged and adopted by the state** (Article 11).

The attribution of the conduct of national oil companies and state-owned utilities has been extensively litigated. In *Noble Ventures v Romania*, the tribunal held that acts of a state privatisation agency were attributable to Romania because the

agency exercised governmental authority.[7] In *EDF (Services) Limited v Romania*, the tribunal examined whether the conduct of state-owned companies controlled by the state was attributable under Articles 5 and 8.[8] The general principle that emerges from the jurisprudence is that the separate legal personality of a state enterprise does not of itself shield the state from responsibility, but neither does state ownership of itself render the enterprise's commercial conduct attributable. The analysis turns on the function performed and the degree of state direction or control over the specific conduct at issue. Investment treaties sometimes contain specific provisions on the treatment of state enterprises, and the ECT, in Article 22, requires each contracting party to ensure that any state enterprise that it maintains or establishes conducts its activities in a manner consistent with the party's obligations under Part III of the treaty.

Breach of an International Obligation

A breach occurs when an act of a state is not in conformity with what is required of it by an international obligation, regardless of the origin or character of the obligation (Article 12). In the energy field, relevant obligations may arise under investment treaties, trade agreements, environmental treaties, human rights treaties, the law of the sea, and customary international law. A state may also incur responsibility for failing to exercise due diligence to prevent harm caused by private actors — for example, failing to prevent pollution from private oil operations from causing transboundary harm, or failing to provide full protection and security to a foreign investor's installations against attack.

The relationship between breach of contract and breach of international law is a recurring issue. A mere breach of a contract between a state and a foreign investor does not, of itself, constitute a breach of international law. Something more is required: an exercise of sovereign power (*puissance publique*) that goes beyond the conduct of an ordinary contracting party, a denial of justice in the resolution of the contractual dispute, or the operation of an "umbrella clause" in an investment treaty elevating contractual undertakings to treaty obligations. These issues are examined in Chapter 14.

Circumstances Precluding Wrongfulness

ARSIWA identifies circumstances that preclude the wrongfulness of conduct that would otherwise breach an international obligation: consent (Article 20), self-defence (Article 21), countermeasures (Article 22), *force majeure* (Article 23), distress (Article 24), and necessity (Article 25).

[7] *Noble Ventures, Inc v Romania*, ICSID Case No ARB/01/11, Award, 12 October 2005, paras 68–86.

[8] *EDF (Services) Limited v Romania*, ICSID Case No ARB/05/13, Award, 8 October 2009, paras 185–213.

The defence of **necessity** has been extensively litigated in the energy sector, particularly in the arbitrations arising from Argentina's economic crisis of 2001–2002, during which Argentina froze utility tariffs, converted dollar-denominated tariffs into pesos, and suspended tariff adjustment mechanisms that foreign investors in its gas and electricity sectors had been guaranteed. Article 25 permits a state to invoke necessity only if the act "is the only way for the State to safeguard an essential interest against a grave and imminent peril" and "does not seriously impair an essential interest of the State or States towards which the obligation exists, or of the international community as a whole"; necessity may not be invoked if the obligation excludes it or if the state has contributed to the situation of necessity.

The Argentine cases produced divergent results. In *CMS Gas Transmission Co v Argentina*, the tribunal rejected Argentina's necessity defence, finding that the measures were not the only way to respond to the crisis and that Argentina had contributed to the situation.[9] In *LG&E Energy Corp v Argentina*, by contrast, the tribunal accepted that a state of necessity existed for a defined period between December 2001 and April 2003, during which Argentina was exempt from liability.[10] The *ad hoc* annulment committee in *CMS* criticised the tribunal for conflating the customary necessity defence with the non-precluded measures clause in Article XI of the United States–Argentina BIT, while declining to annul the award on that ground.[11] The inconsistency of these outcomes on substantially similar facts became a prominent illustration of the legitimacy concerns surrounding investment arbitration.

Consequences: Cessation and Reparation

A state responsible for an internationally wrongful act is under an obligation to cease the act, if it is continuing, and to offer appropriate assurances and guarantees of non-repetition (Article 30). It is also under an obligation to make full reparation for the injury caused (Article 31). The classic statement of the reparation principle is that of the Permanent Court of International Justice in the *Factory at Chorzów* case:

> The essential principle contained in the actual notion of an illegal act ... is that reparation must, as far as possible, wipe out all the consequences of the illegal act

[9] *CMS Gas Transmission Company v Argentine Republic*, ICSID Case No ARB/01/8, Award, 12 May 2005, paras 315–331.

[10] *LG&E Energy Corp, LG&E Capital Corp and LG&E International Inc v Argentine Republic*, ICSID Case No ARB/02/1, Decision on Liability, 3 October 2006, paras 226–266.

[11] *CMS Gas Transmission Company v Argentine Republic*, ICSID Case No ARB/01/8, Decision of the ad hoc Committee on the Application for Annulment, 25 September 2007, paras 128–136.

> and re-establish the situation which would, in all probability, have existed if that act had not been committed.[12]

Reparation may take the form of restitution, compensation, or satisfaction, either singly or in combination (Article 34). Restitution — re-establishing the situation that existed before the wrongful act — is the primary form, but in practice it is rarely ordered in energy disputes, both because of the practical difficulty of compelling a sovereign to reverse a nationalisation and because of respect for sovereignty. Compensation, covering financially assessable damage including loss of profits insofar as established (Article 36), is the usual remedy. The valuation of compensation for energy assets — using discounted cash flow, comparable transactions, or asset-based methods — is one of the most technically demanding aspects of energy arbitration and is examined in Chapter 26.

2.4 Jurisdiction and Extraterritorial Regulation

Principles of Jurisdiction

Jurisdiction in international law refers to the authority of a state to prescribe rules (prescriptive or legislative jurisdiction), to apply those rules through its courts (adjudicative jurisdiction), and to enforce them (enforcement jurisdiction). The bases of prescriptive jurisdiction are well established:

- **Territoriality.** A state may regulate conduct occurring within its territory. The subjective territorial principle covers conduct initiated within the territory; the objective territorial principle covers conduct completed or producing its effects within it.
- **Nationality.** A state may regulate the conduct of its nationals, including corporations incorporated under its law, wherever they act.
- **Protective principle.** A state may regulate conduct abroad that threatens its security or essential governmental functions.
- **Universality.** For certain grave international crimes, any state may exercise jurisdiction.
- **Passive personality.** A state may, in limited circumstances, regulate conduct abroad that harms its nationals.

The so-called **effects doctrine**, developed in United States antitrust law, permits the regulation of conduct abroad that has substantial, direct, and foreseeable effects within the regulating state. Its application has been controversial, particularly when used to regulate the conduct of foreign companies abroad.

Enforcement jurisdiction is strictly territorial. In the *Lotus* case, the Permanent Court of International Justice held that a state "may not exercise its power in any

[12] *Factory at Chorzów (Germany v Poland)*, Merits, 1928 PCIJ Series A, No 17, p. 47.

form in the territory of another State" without a permissive rule to that effect, while affirming broad discretion as to prescriptive jurisdiction.[13]

Jurisdiction over Maritime Resources

Jurisdiction over offshore energy and resources is governed by UNCLOS, which establishes a graduated system of maritime zones. The coastal state has full sovereignty over its internal waters and territorial sea (up to 12 nautical miles from the baselines), subject to the right of innocent passage. In the exclusive economic zone (EEZ), extending to 200 nautical miles, the coastal state has sovereign rights for the purpose of exploring and exploiting, conserving, and managing natural resources, and with regard to other activities for the economic exploitation and exploration of the zone, "such as the production of energy from the water, currents and winds" (Article 56). On the continental shelf, the coastal state exercises sovereign rights for the purpose of exploring it and exploiting its natural resources (Article 77); these rights are exclusive and do not depend on occupation or express proclamation. Beyond national jurisdiction, the mineral resources of the seabed — "the Area" — are the common heritage of mankind and are administered by the International Seabed Authority. These rules are examined in Chapter 20.

Extraterritorial Regulation in the Energy Sector

Extraterritorial regulation has become increasingly important in the energy sector. Four categories may be distinguished.

Anti-corruption legislation. The United States Foreign Corrupt Practices Act 1977 prohibits the bribery of foreign officials by issuers of securities listed in the United States, domestic concerns, and persons acting within United States territory. The United Kingdom Bribery Act 2010 creates a corporate offence of failure to prevent bribery that applies to any organisation that carries on a business or part of a business in the United Kingdom, regardless of where the bribery occurs. Both statutes have been enforced extensively against energy and mining companies. The OECD Anti-Bribery Convention 1997 requires parties to criminalise the bribery of foreign public officials.

Sanctions. Economic sanctions targeting the energy sectors of particular states — including Iran, Russia, and Venezuela — are a prominent instrument of foreign policy. Some sanctions regimes have "secondary" effects, threatening penalties against foreign persons who engage in specified transactions with sanctioned parties even where there is no connection with the sanctioning state's territory. The European Union has adopted a "Blocking Statute" (Council Regulation (EC) No 2271/96) prohibiting EU persons from complying with certain extraterritorial sanctions. Sanctions are examined in Chapters 22 and 25.

[13] *SS Lotus (France v Turkey)*, 1927 PCIJ Series A, No 10, pp. 18–19.

Supply chain due diligence. A growing body of legislation imposes due diligence obligations on companies with respect to their global supply chains. Section 1502 of the United States Dodd–Frank Act 2010 required disclosure of the use of conflict minerals from the Democratic Republic of the Congo and adjoining countries. The European Union's Conflict Minerals Regulation (EU) 2017/821 imposes due diligence obligations on importers of tin, tantalum, tungsten, and gold. The EU Batteries Regulation (EU) 2023/1542 requires due diligence regarding raw materials used in batteries, and the Corporate Sustainability Due Diligence Directive (EU) 2024/1760 establishes broader human rights and environmental due diligence obligations for large companies, subject to amendments under the European Union's 2025 simplification initiatives.

Climate and environmental measures. The European Union's Carbon Border Adjustment Mechanism (CBAM), established by Regulation (EU) 2023/956, requires importers of specified carbon-intensive goods — including iron and steel, aluminium, cement, fertilisers, electricity, and hydrogen — to purchase certificates reflecting the embedded emissions of those goods, with the definitive regime applying from 2026. The EU Methane Regulation (EU) 2024/1787 imposes monitoring, reporting, and verification requirements on fossil fuel imports. Such measures regulate production processes occurring abroad by attaching conditions to market access, and their compatibility with international trade law is examined in Chapter 13.

PRACTICE NOTE

When advising a multinational energy company on extraterritorial exposure, counsel should map, for each jurisdiction in which the company is incorporated, listed, or carries on business, the anti-corruption, sanctions, export-control, and due diligence regimes that may apply to conduct anywhere in the group. Conflicts between regimes — for example, between a home-state sanction and a host-state contractual obligation to supply — should be identified at the contract negotiation stage and addressed through sanctions clauses, termination rights, and *force majeure* provisions.

2.5 International Organizations and Specialized Energy Institutions

The Fragmented Institutional Landscape

There is no single international organisation with comprehensive competence over energy. Instead, the institutional landscape is fragmented among organisations with specialised mandates, overlapping memberships, and varying legal powers. This fragmentation reflects the political sensitivity of energy, the divergent interests of producers and consumers, and the historical development of

energy governance through issue-specific responses to particular crises. Chapter 4 provides a detailed analysis of the principal energy institutions. This section introduces the main categories and their legal significance as sources and agents of international energy law.

Categories of Institutions

Consumer-oriented institutions. The International Energy Agency was established in 1974 within the framework of the OECD in response to the 1973–1974 oil crisis. Its founding treaty, the Agreement on an International Energy Program, obliges members to maintain emergency oil stocks equivalent to at least ninety days of net imports and to participate in coordinated emergency response measures.

Producer-oriented institutions. OPEC, established in 1960, and the Gas Exporting Countries Forum, established as an intergovernmental organisation in 2008, coordinate the policies of producing states. Since 2016, OPEC and a group of non-OPEC producers including the Russian Federation have coordinated production through the "OPEC+" arrangements under the Declaration of Cooperation.

Technology- and sector-specific institutions. The International Atomic Energy Agency (IAEA), established in 1957, promotes the peaceful uses of nuclear energy and administers safeguards to verify that nuclear material is not diverted to weapons. The International Renewable Energy Agency (IRENA), established in 2009 with headquarters in Abu Dhabi, promotes the adoption of renewable energy.

Treaty-based energy organisations. The Energy Charter Conference is the governing body of the ECT. The Energy Community, established by treaty in 2005, extends the European Union's internal energy market rules to contracting parties in South-East Europe and the Black Sea region, including Ukraine.

Dialogue forums. The International Energy Forum, headquartered in Riyadh, provides a forum for dialogue between producers and consumers, and administers the Joint Organisations Data Initiative (JODI) on oil and gas statistics.

General international organisations with energy competences. The United Nations and its agencies (including the United Nations Environment Programme, the UNFCCC secretariat, and UN-Energy), the World Bank Group (including the IFC and the Multilateral Investment Guarantee Agency), regional development banks, the WTO, and the OECD all exercise significant influence over energy policy.

Regional organisations. The European Union exercises extensive legislative competence over energy under Article 194 of the Treaty on the Functioning of

the European Union. In Africa, the African Union's Agenda 2063, the African Energy Commission (AFREC), the ECOWAS Regional Electricity Regulatory Authority, the West African Power Pool, and the Southern African Power Pool promote regional cooperation.

Legal Powers and Outputs

The legal powers of these institutions vary greatly. Most energy organisations lack the power to adopt decisions binding on their members, and their influence derives from information, expertise, coordination, and soft law. Exceptions include the IEA's emergency response mechanism, the IAEA's safeguards agreements (which are binding treaties between the Agency and individual states), the Energy Community's dispute settlement procedure, and, most significantly, the European Union, whose regulations and directives are binding on member states and, in the case of regulations, directly applicable.

The outputs of international organisations contribute to international energy law in several ways: they may be the forum for the negotiation of treaties; their resolutions and declarations may constitute evidence of customary international law or contribute to its formation; their standards and guidelines may be incorporated into domestic law, contracts, or investment conditions; and their data and analysis frame policy debates.

2.6 Soft Law, Industry Standards, and Corporate Governance

The Concept of Soft Law

"Soft law" refers to normative instruments that are not legally binding in themselves but that nonetheless influence the conduct of states and other actors and may acquire legal effect through incorporation, interpretation, or the formation of custom. Examples include General Assembly resolutions, declarations adopted at international conferences, codes of conduct, guidelines, recommendations of international organisations, and voluntary standards adopted by industry or multi-stakeholder initiatives. The concept is contested: some scholars reject it as a contradiction in terms, insisting on a binary distinction between law and non-law, while others regard it as an indispensable description of the graduated normativity characteristic of contemporary international governance.

Soft law is especially significant in energy and resource governance for several reasons. States are often reluctant to accept binding obligations in a field so closely tied to sovereignty and security. Many relevant actors — corporations, financial institutions, communities — are not subjects of international law in the traditional sense and cannot be parties to treaties. Technological and market conditions change rapidly, making flexible instruments preferable to treaties that are difficult to amend. And soft law can serve as a laboratory for norms that may later harden into binding law.

Principal Soft Law Instruments

The principal soft law instruments relevant to energy and natural resources include:

- **UN General Assembly Resolution 1803 (XVII) (1962)** on permanent sovereignty over natural resources, which, although a resolution, is widely regarded as reflecting customary international law.
- **The Rio Declaration on Environment and Development (1992)**, whose principles — including prevention (Principle 2), precaution (Principle 15), the polluter-pays principle (Principle 16), and environmental impact assessment (Principle 17) — have been incorporated into numerous treaties and domestic laws.
- **The UN Guiding Principles on Business and Human Rights (2011)**, establishing the "Protect, Respect and Remedy" framework.
- **The OECD Guidelines for Multinational Enterprises on Responsible Business Conduct**, revised most recently in 2023, which are backed by a system of National Contact Points that handle complaints ("specific instances") against enterprises.
- **The UN Declaration on the Rights of Indigenous Peoples (2007)**, which articulates the principle of free, prior, and informed consent.
- **The Voluntary Principles on Security and Human Rights (2000)**, a multi-stakeholder initiative addressing the use of public and private security forces by extractive companies.
- **The Natural Resource Charter**, setting out principles for the governance of natural resources for the benefit of citizens.
- **The Santiago Principles (2008)**, Generally Accepted Principles and Practices for sovereign wealth funds, many of which are funded by resource revenues.

Industry Standards and Transnational Private Regulation

Beyond intergovernmental soft law lies a dense body of industry standards and private regulation. Technical standards developed by the American Petroleum Institute (API), the International Organization for Standardization (ISO), and the International Electrotechnical Commission (IEC) govern the design, construction, and operation of energy installations and are frequently incorporated by reference into regulation and contracts. Model contracts developed by industry associations, including the AIEN Model Joint Operating Agreement and Model Unitization and Unit Operating Agreement, standardise commercial relations. The International Association of Oil & Gas Producers (IOGP) publishes guidance on health, safety, and environmental management. In mining, the International Council on Mining and Metals (ICMM) has adopted Mining Principles

that bind its member companies, and the Global Industry Standard on Tailings Management (2020) was developed following the catastrophic collapse of the Brumadinho tailings dam in Brazil in 2019.

Financial institutions exercise considerable regulatory influence through the conditions they attach to project finance. The IFC Performance Standards (2012) establish requirements on environmental and social risk management, labour, resource efficiency, community health and safety, land acquisition and involuntary resettlement, biodiversity, indigenous peoples, and cultural heritage. The Equator Principles, adopted by commercial banks, apply the IFC Performance Standards to project finance transactions above specified thresholds. Because many major energy and mining projects depend on international finance, these standards often impose more demanding requirements than domestic law.

Corporate Governance as a Source of Norms

Corporate governance instruments — codes of conduct, sustainability policies, human rights policies, and climate commitments — are increasingly treated as sources of legal obligation. In *Milieudefensie v Royal Dutch Shell*, the District Court of The Hague in 2021 relied on the UN Guiding Principles and other soft law instruments to give content to the unwritten standard of care under Dutch tort law and ordered Shell to reduce the net carbon dioxide emissions of the Shell group by 45 per cent by 2030 relative to 2019 levels.[14] On appeal, the Hague Court of Appeal in November 2024 set aside the specific reduction order, holding that there was insufficient consensus in climate science on a specific reduction percentage applicable to an individual company, while affirming that companies such as Shell have an obligation to contribute to the reduction of greenhouse gas emissions and that this obligation can be derived from the standard of care informed by human rights and soft law instruments.[15] The case illustrates how soft law can acquire legal force through domestic private law.

The Hardening of Soft Law

Soft law norms can harden into binding law through several pathways: incorporation into treaties (the Rio principles in the UNFCCC and the Convention on Biological Diversity); incorporation into domestic legislation (the UN Guiding Principles in mandatory human rights due diligence statutes); incorporation into contracts (the IFC Performance Standards in loan agreements and host government agreements); judicial reliance (as in the *Shell* litigation); and the formation of customary international law. The study of these pathways is a rich field for doctoral research.

[14] *Milieudefensie et al v Royal Dutch Shell plc*, District Court of The Hague, 26 May 2021, ECLI:NL:RBDHA:2021:5339.

[15] *Shell plc v Milieudefensie et al*, Court of Appeal of The Hague, Judgment of 12 November 2024.

2.7 International Judicial and Arbitral Institutions

The International Court of Justice

The International Court of Justice, the principal judicial organ of the United Nations, has decided numerous cases bearing on energy and natural resources, including maritime delimitation disputes involving offshore petroleum resources (*North Sea Continental Shelf*, 1969; *Land and Maritime Boundary between Cameroon and Nigeria*, 2002; *Maritime Delimitation in the Black Sea*, 2009; *Territorial and Maritime Dispute (Nicaragua v Colombia)*, 2012; *Maritime Delimitation in the Indian Ocean (Somalia v Kenya)*, 2021), disputes over shared watercourses and hydropower (*Gabčíkovo-Nagymaros Project*, 1997; *Pulp Mills on the River Uruguay*, 2010; *Dispute over the Status and Use of the Waters of the Silala*, 2022), and disputes concerning the protection of foreign investment (*Barcelona Traction*, 1970; *Elettronica Sicula SpA (ELSI)*, 1989). Its advisory opinion of 23 July 2025 on the *Obligations of States in respect of Climate Change* is of profound significance for energy law and is analysed in Chapter 18.

The Court's jurisdiction in contentious cases depends on the consent of states, given through special agreement, compromissory clauses in treaties, or optional clause declarations under Article 36(2) of its Statute. Because private parties have no standing, the Court plays a limited role in investor–state disputes, which are resolved principally through arbitration.

The International Tribunal for the Law of the Sea and Annex VII Tribunals

The International Tribunal for the Law of the Sea (ITLOS), established under Annex VI of UNCLOS and seated in Hamburg, has jurisdiction over disputes concerning the interpretation or application of the Convention. Its Special Chamber decided the maritime boundary dispute between Ghana and Côte d'Ivoire in 2017, a case driven by offshore petroleum discoveries, in which the Chamber also rejected Côte d'Ivoire's claim that Ghana had violated its sovereign rights by conducting oil activities in the disputed area.[16] Its Seabed Disputes Chamber delivered an important advisory opinion in 2011 on the responsibilities of states sponsoring activities in the Area, holding that sponsoring states have obligations of due diligence and direct obligations, including the obligation to apply a precautionary approach.[17] In 2024 the full Tribunal delivered an advisory

[16] *Dispute Concerning Delimitation of the Maritime Boundary between Ghana and Côte d'Ivoire in the Atlantic Ocean (Ghana/Côte d'Ivoire)*, ITLOS Case No 23, Judgment of the Special Chamber, 23 September 2017.

[17] *Responsibilities and Obligations of States Sponsoring Persons and Entities with respect to Activities in the Area*, Advisory Opinion, ITLOS Seabed Disputes Chamber, 1 February 2011, ITLOS Reports 2011, p. 10.

opinion holding that anthropogenic greenhouse gas emissions constitute pollution of the marine environment within the meaning of UNCLOS.

Arbitral tribunals constituted under Annex VII of UNCLOS have decided several disputes with energy dimensions, including the *South China Sea Arbitration (Philippines v China)* of 2016, which addressed the legal status of maritime features and the lawfulness of Chinese activities, including interference with Philippine petroleum exploration.

The WTO Dispute Settlement System

The WTO Dispute Settlement Body, comprising panels and the Appellate Body, has decided several disputes concerning energy goods and renewable energy policies. Since December 2019, the Appellate Body has been unable to hear appeals because the United States has blocked the appointment of new members. Many WTO members have joined the Multi-Party Interim Appeal Arbitration Arrangement (MPIA) under Article 25 of the Dispute Settlement Understanding as an interim substitute, but appeals "into the void" by members not party to the MPIA can prevent panel reports from being adopted. This crisis has weakened the enforcement of trade disciplines in the energy sector.

Investment Arbitration Institutions

Investor–state disputes in the energy sector are resolved principally through arbitration under the auspices of:

- **ICSID**, established by the Convention on the Settlement of Investment Disputes between States and Nationals of Other States 1965, with its own self-contained system of annulment rather than review by national courts, and enforcement provisions requiring contracting states to recognise awards as if they were final judgments of their own courts (Article 54);
- **the Permanent Court of Arbitration (PCA)** in The Hague, which administers many arbitrations conducted under the UNCITRAL Arbitration Rules, including the *Yukos* arbitrations and numerous ECT cases;
- **the Arbitration Institute of the Stockholm Chamber of Commerce (SCC)**, designated as one of the forums for investor–state disputes under Article 26 of the ECT;
- **the International Chamber of Commerce (ICC) International Court of Arbitration**, which administers many commercial and some investment disputes;
- **the London Court of International Arbitration (LCIA)** and regional centres, including the Singapore International Arbitration Centre, the Cairo Regional Centre for International Commercial Arbitration, and the Lagos Court of Arbitration.

The institutional framework, procedural rules, and practice of energy arbitration are examined in Chapter 26.

Regional Courts and Human Rights Bodies

Regional courts and human rights bodies have become increasingly important forums for energy-related disputes. The Court of Justice of the European Union has decided cases of enormous significance for intra-EU investment arbitration (*Achmea*, 2018; *Komstroy*, 2021; *PL Holdings*, 2021) and energy market regulation. The European Court of Human Rights, in *Verein KlimaSeniorinnen Schweiz v Switzerland* (2024), held that Article 8 of the European Convention on Human Rights encompasses a right to effective protection by state authorities from the serious adverse effects of climate change.[18] The African Commission on Human and Peoples' Rights, in *Social and Economic Rights Action Center (SERAC) and Center for Economic and Social Rights v Nigeria* (2001), found that Nigeria had violated the African Charter through the conduct of the state oil company and its failure to regulate the activities of oil companies in Ogoniland.[19] The ECOWAS Community Court of Justice, in *SERAP v Federal Republic of Nigeria* (2012), held Nigeria responsible for failing to protect the right to a satisfactory environment in the Niger Delta against the activities of oil companies.

2.8 Conflicts Between Domestic and International Legal Obligations

Monism, Dualism, and Their Limits

The relationship between international law and domestic law is traditionally described in terms of monism and dualism. Monist systems treat international law and domestic law as parts of a single legal order, such that treaties may have direct effect in domestic law upon ratification. Dualist systems treat the two as separate, such that treaties must be incorporated by legislation before they can be applied by domestic courts. Nigeria follows a dualist approach for treaties: section 12(1) of the Constitution of the Federal Republic of Nigeria 1999 provides that "no treaty between the Federation and any other country shall have the force of law except to the extent to which any such treaty has been enacted into law by the National Assembly". In *Abacha v Fawehinmi*, the Supreme Court of Nigeria held that the African Charter on Human and Peoples' Rights, having been domesticated by the African Charter on Human and Peoples' Rights (Ratification and Enforcement)

[18] *Verein KlimaSeniorinnen Schweiz and Others v Switzerland*, App No 53600/20, European Court of Human Rights (Grand Chamber), Judgment of 9 April 2024.

[19] *Social and Economic Rights Action Center (SERAC) and Center for Economic and Social Rights (CESR) v Nigeria*, Communication No 155/96, African Commission on Human and Peoples' Rights, Decision of 27 October 2001.

Act, was part of Nigerian law, although subordinate to the Constitution.[20] The United Kingdom is likewise dualist, whereas the Netherlands and many civil law states are closer to the monist model.

The International Law Perspective

From the perspective of international law, the classification of a state's domestic system is irrelevant to its international obligations. Article 27 of the VCLT provides that "a party may not invoke the provisions of its internal law as justification for its failure to perform a treaty". Article 3 of ARSIWA provides that the characterisation of an act of a state as internationally wrongful is governed by international law and is not affected by its characterisation as lawful by internal law. Accordingly, a state cannot escape liability under an investment treaty by pointing to the lawfulness of its measures under its own constitution, nor can it rely on a domestic court decision annulling a contract as a defence if the decision itself amounts to a denial of justice.

Typical Conflicts in the Energy Sector

Conflicts between domestic and international obligations arise in several recurring situations in the energy sector:

- **Legislative change and stabilisation.** A state enacts new fiscal or environmental legislation that is valid under its constitution but conflicts with stabilisation commitments in contracts or with investment treaty protections.
- **Judicial decisions.** Domestic courts annul or refuse to enforce contracts or arbitral awards, or issue anti-arbitration injunctions, in circumstances that international tribunals may regard as breaches of treaty obligations or of the New York Convention.
- **Constitutional limits on contracting.** Contracts concluded by state entities may be void under domestic law for failure to obtain required approvals, raising questions about whether the state can rely on its own law to avoid its obligations. Article 46 of the VCLT, by analogy, permits a state to invoke a violation of internal law regarding competence to conclude treaties only where the violation was manifest and concerned a rule of fundamental importance.
- **Climate and environmental obligations.** Domestic courts may order a state to adopt more stringent climate measures, as the Supreme Court of the Netherlands did in *State of the Netherlands v Urgenda Foundation* (2019),[21] potentially generating claims by investors whose investments are affected by the resulting measures.

[20] *Abacha v Fawehinmi* (2000) 6 NWLR (Pt 660) 228 (Supreme Court of Nigeria).

[21] *State of the Netherlands v Urgenda Foundation*, Supreme Court of the Netherlands, 20 December 2019, ECLI:NL:HR:2019:2007.

- **Human rights obligations and investment protection.** A state's obligations to protect the rights of indigenous peoples or the right to water may conflict with commitments to investors.

The Intra-EU Investment Arbitration Conflict

The most dramatic recent example of conflict between legal orders arose from intra-EU investment arbitration under the ECT. In *Slovak Republic v Achmea BV* (2018), the Court of Justice of the European Union held that arbitration clauses in intra-EU BITs were incompatible with EU law because they removed disputes potentially concerning the interpretation or application of EU law from the EU judicial system.[22] In *Republic of Moldova v Komstroy LLC* (2021), the Court extended this reasoning to the ECT, holding that Article 26 of the ECT must be interpreted as not applicable to disputes between a member state and an investor of another member state concerning an investment made by the latter in the first member state.[23]

Investment tribunals constituted under the ECT have overwhelmingly rejected the intra-EU objection, holding that they derive their jurisdiction from the ECT as a treaty of international law and that EU law cannot override the consent expressed in it. The result has been a direct conflict between legal orders: awards rendered against EU member states in favour of EU investors (many arising from Spain's withdrawal of renewable energy incentives) have been upheld by arbitral tribunals and enforced or recognised by courts in some non-EU jurisdictions such as the United States, Australia, and the United Kingdom, while courts within the European Union have refused enforcement and the European Commission has treated payment of such awards as potentially constituting unlawful state aid. The European Union, Euratom, and a number of member states subsequently withdrew from the ECT. This saga is a vivid case study in legal pluralism and is examined further in Chapters 14 and 26.

LEADING AUTHORITY

Factory at Chorzów (Germany v Poland), PCIJ, 1928

Facts. Following the transfer of Upper Silesia to Poland after the First World War, Poland took possession of a nitrate factory at Chorzów owned by German companies, in breach of the Geneva Convention of 1922 concerning Upper Silesia. Germany brought proceedings seeking reparation.

Holding. The Permanent Court held that it is a principle of international law that the breach of an engagement involves an obligation to make reparation in an adequate

[22] Case C-284/16, *Slowakische Republik v Achmea BV*, EU:C:2018:158 (CJEU, Grand Chamber, 6 March 2018).

[23] Case C-741/19, *République de Moldavie v Komstroy LLC*, EU:C:2021:655 (CJEU, Grand Chamber, 2 September 2021).

form, and that reparation must, as far as possible, wipe out all the consequences of the illegal act. It distinguished between lawful expropriation, which requires payment of the value of the undertaking at the time of dispossession plus interest, and unlawful expropriation, for which restitution in kind, or payment of a sum corresponding to the value that restitution would bear, is required — potentially including the value of the property at the date of the judgment rather than the date of the taking.

Significance for energy law. *Chorzów* remains the foundation of the law of reparation. Investment tribunals in energy cases — including *ADC v Hungary* (2006), *ConocoPhillips v Venezuela* (2019), and *Yukos Universal v Russia* (2014) — have relied on the distinction between lawful and unlawful expropriation to determine the valuation date and the measure of compensation, with major financial consequences.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Attributing the conduct of a national oil company (hypothetical). The fictional Federal Republic of Calvaria owns 100 per cent of Calvaria National Petroleum Company Ltd (CNPC), a company incorporated under the Calvarian Companies Act. Under the Calvarian Petroleum Act, CNPC holds all state participating interests in joint ventures and is empowered to approve work programmes and budgets on behalf of the state. In 2024, CNPC refused to approve a joint venture work programme submitted by its foreign partner, Delta Ridge Resources Inc, citing "national energy security priorities" communicated in a letter from the Minister of Petroleum. Delta Ridge alleges that the refusal was designed to force it to sell its interest at an undervalue to a company connected to senior officials.

Advise Delta Ridge on (a) whether CNPC's conduct is attributable to Calvaria under ARSIWA Articles 4, 5, and 8; (b) whether the refusal could constitute a breach of an investment treaty obligation as opposed to a mere contractual breach; and (c) what evidence should be gathered to support attribution. *All entities and facts are fictitious.*

CHAPTER SUMMARY

- The formal sources of international energy law are treaties, custom, and general principles, with judicial decisions and scholarly writings as subsidiary means of determining the law; no formal hierarchy exists, but treaties operate as *lex specialis* and peremptory norms prevail over all.
- Treaty interpretation under VCLT Articles 31–33 is central to the field, particularly in investment disputes; systemic integration under Article 31(3)(c) allows environmental and human rights obligations to inform the interpretation of investment treaties.
- Permanent sovereignty over natural resources is a principle of customary international law encompassing both rights and duties; it does not authorise a state

to disregard its commitments, but affirms the state's ultimate authority subject to international law.

- The ILC Articles on State Responsibility govern attribution, breach, defences, and reparation; the attribution of national oil company conduct and the necessity defence have generated extensive energy jurisprudence.
- Jurisdiction is primarily territorial, but extraterritorial regulation through anti-corruption, sanctions, supply chain, and carbon border measures increasingly shapes energy activities.
- International organisations, soft law, industry standards, lender requirements, and corporate governance instruments constitute a significant layer of normative authority, which may harden into binding law.
- Conflicts between domestic and international obligations are resolved, from the international perspective, by the rule that internal law cannot justify non-performance; the intra-EU arbitration saga illustrates the limits of that rule when legal orders collide.

DOCTORAL RESEARCH QUESTIONS

1. Has the proliferation of investment treaties containing similar substantive standards contributed to the formation of customary international law on the treatment of foreign energy investments? What methodology should be used to answer this question after the ILC's 2018 Draft Conclusions?
2. To what extent does the modern, duty-oriented conception of permanent sovereignty over natural resources constrain the freedom of states to grant long-term fossil fuel licences in the light of climate obligations?
3. Should the conduct of national oil companies be subject to a special rule of attribution, given their hybrid commercial and governmental functions?
4. What are the conditions under which soft law instruments such as the UN Guiding Principles acquire binding force through domestic tort law, and is this process democratically legitimate?
5. Does the intra-EU arbitration conflict demonstrate the failure of the techniques for managing fragmentation identified by the International Law Commission?

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3 CHAPTER

Resource Sovereignty, State Ownership, and Constitutional Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal foundations of resource sovereignty at the international and constitutional levels.
2. Compare the principal models of subsurface resource ownership — state ownership, the accession (landowner) system, and hybrid regimes — and their consequences for licensing and revenue.
3. Analyse the public trust doctrine and related concepts of state stewardship over natural resources.
4. Evaluate the allocation of resource authority in federal systems, with particular reference to Nigeria, Canada, Australia, and the United States.
5. Assess the rights of indigenous peoples and local communities in resource governance.
6. Analyse resource nationalism, expropriation, nationalisation, and the standards of compensation under international law.

Introduction

Chapter 2 examined permanent sovereignty over natural resources as a principle of international law. This chapter turns to the ways in which that sovereignty is organised and exercised within states. International law recognises that a state controls the natural resources within its territory and continental shelf, but it is silent on the question of how ownership and authority are to be distributed internally: whether resources belong to the state, to the owners of the surface, to subnational units, or to communities; which organs of government may grant rights to explore and exploit them; and how the revenues they generate are to be shared. These questions are answered by constitutional law, statute, and custom, and the answers vary greatly among states.

The internal organisation of resource sovereignty has profound practical consequences. It determines who an investor must negotiate with, what kind of title the investor obtains, who receives royalties, and which courts have jurisdiction over disputes. It shapes the political economy of resource-rich states, influencing the incidence of conflict, the quality of governance, and the distribution of wealth. And it determines the extent to which affected communities can participate in decisions about resource development and share in its benefits.

The chapter proceeds in eight sections. It begins with the legal foundations of resource sovereignty, then compares models of constitutional ownership of subsurface resources. It examines the public trust doctrine, federalism and subnational authority, and the position of indigenous peoples. It analyses resource nationalism and the law of expropriation and compensation, and concludes with a comparative survey of constitutional approaches.

3.1 Legal Foundations of Resource Sovereignty

Territorial Sovereignty and Resource Control

Resource sovereignty is, at its root, an incident of territorial sovereignty. A state's sovereignty over its land territory extends to the subsoil beneath it, without any fixed depth limit, and to the airspace above it. Sovereignty over the territorial sea likewise extends to the seabed and subsoil (UNCLOS, Article 2). Beyond the territorial sea, the coastal state enjoys not sovereignty but "sovereign rights" over the natural resources of the EEZ and the continental shelf — functional rights that are exclusive but limited to the purposes specified in the Convention. The distinction between sovereignty and sovereign rights is important: in the EEZ and on the continental shelf, other states enjoy freedoms of navigation, overflight, and the laying of submarine cables and pipelines, subject to the coastal state's rights.

In the classical conception, the state's territorial sovereignty entitled it to deal with its resources as it saw fit, subject only to obligations it had freely assumed and to the general rules protecting the property of aliens. The doctrine of permanent sovereignty over natural resources, discussed in Chapter 2, reinforced this conception and gave it a new political meaning in the context of decolonisation, emphasising that sovereignty over resources belongs to the people of the state and must be exercised for their development and well-being.

The People as Bearers of Sovereignty

A significant development in the modern law of resource sovereignty is the recognition that sovereignty over natural resources is vested not merely in the state as an abstract entity but in the people. Common Article 1(2) of the International Covenants refers to the right of "all peoples" to freely dispose of their natural wealth and resources. Article 21 of the African Charter on Human and Peoples' Rights similarly vests the right in "peoples" and requires that it be exercised "in the

exclusive interest of the people". Many national constitutions declare that natural resources belong to the people, with the state acting as trustee or custodian. Article 257(6) of the Constitution of Ghana 1992, for example, vests every mineral in its natural state in the President "on behalf of, and in trust for, the people of Ghana". Article 62(1)(f) of the Constitution of Kenya 2010 classifies all minerals and mineral oils as public land, and Article 62(3) provides that such public land vests in and is held by the national government in trust for the people of Kenya.

The recognition of the people as bearers of resource sovereignty has several legal implications. It supports arguments that resource revenues must be managed transparently and for the public benefit. It provides a normative basis for the participation of citizens and communities in resource decisions. And it has been invoked to challenge contracts concluded by unrepresentative or corrupt governments, raising difficult questions about the continuity of state obligations following changes of government — an issue examined in the context of odious debt and illegitimate contracts.

The Limits of Sovereignty

Resource sovereignty is subject to limits imposed by international law. These include:

- obligations assumed by treaty, including investment treaties, trade agreements, and environmental agreements;
- the customary obligation to prevent significant transboundary harm, recognised in *Trail Smelter* and affirmed by the International Court of Justice in its *Legality of the Threat or Use of Nuclear Weapons* advisory opinion and in *Pulp Mills*;
- the obligation to cooperate in the management of shared natural resources, such as transboundary watercourses and straddling petroleum deposits;
- the international minimum standard of treatment of aliens and the rules governing expropriation;
- human rights obligations, including the rights of indigenous peoples;
- obligations under the law of armed conflict regarding the exploitation of resources in occupied territory.

These limits do not negate sovereignty; they define the conditions under which it must be exercised.

3.2 Constitutional Ownership of Subsurface Resources

Three Models of Ownership

National legal systems allocate ownership of subsurface resources according to three broad models.

The regalian or domanial system (state ownership). Under this model, which has its origins in Roman law, medieval European royal prerogatives over precious metals, and the Spanish colonial *derecho minero*, subsurface minerals and petroleum belong to the state (historically, the Crown or the sovereign) irrespective of the ownership of the surface. The state grants rights to explore and exploit through licences, concessions, or contracts. The regalian system prevails in the great majority of states, including virtually all of Africa, Latin America, the Middle East, and Asia, and in Europe. In the United Kingdom, petroleum in its natural condition in strata was vested in the Crown by the Petroleum (Production) Act 1934, a vesting now continued by section 2 of the Petroleum Act 1998; coal was nationalised in 1946 and is now vested in the Coal Authority; and gold and silver have long been "royal mines" belonging to the Crown.

The accession or landowner system. Under this model, derived from the common law maxim *cuius est solum, eius est usque ad caelum et ad inferos* ("whoever owns the soil owns it up to the heavens and down to the depths"), the owner of the surface also owns the minerals beneath it, unless they have been severed by conveyance or reserved by the state. The accession system prevails in the United States (for privately owned land) and was historically important in parts of Canada. It is the principal reason why private mineral ownership, mineral leasing between private parties, and royalty interests owned by individuals are characteristic features of American oil and gas law.

The *res nullius* and claim systems. Historically, some systems treated minerals as unowned until discovered and appropriated, giving rise to "free mining" or claim-staking regimes under which the first person to discover and stake a claim acquired rights. The United States General Mining Law of 1872 retains elements of this approach for hard-rock minerals on federal public lands, permitting citizens to locate claims on valuable mineral deposits. Free entry systems have been modified in many Canadian provinces to require consultation with indigenous peoples before claims are registered.

	Regalian / state ownership	Accession / landowner	Claim / free entry
Owner of minerals	State, nation or people (e.g. Nigeria s 44(3), UK, Mexico art 27)	Surface owner unless severed (US private land)	Unowned until located and claimed (US Mining Law 1872)
Grant instrument	Licence, lease, concession, PSC, service contract	Private lease negotiated with mineral owner	Location and registration of claim; patent
Rent capture	Royalty, tax, profit oil, state participation	Bonus and royalty to owner; state taxes	Limited royalty; taxation
Regulator's role	Owner and regulator: opens acreage, sets terms	Prevent waste, protect correlative rights	Record claims, environmental control
Typical issues	Contract stability, transparency, community benefit	Rule of capture, pooling, unitisation	Conflicts with indigenous rights, environment

Figure 3.1 Models of subsurface resource ownership and their typical legal consequences. *Source:* prepared for this edition.

The Rule of Capture and Its Consequences

The accession system, applied to fugacious resources such as oil and gas that migrate through porous rock, gave rise in the United States to the "rule of capture": a landowner owns all the oil and gas produced from wells drilled on his or her land, even if some of that oil and gas migrated from beneath neighbouring land. In *Westmoreland & Cambria Natural Gas Co v De Witt* (1889), the Supreme Court of Pennsylvania analogised oil and gas to wild animals (*ferae naturae*), which belong to the person who captures them. The rule of capture encouraged a race to drill, leading to wasteful over-production, the premature loss of reservoir pressure, and the dissipation of resources.

Terence Daintith's study of the rule of capture demonstrates how this legal doctrine shaped the organisation of the world oil industry, prompting the development of conservation regulation (well spacing, production prorationing, and compulsory pooling and unitisation) in the United States, and influencing the decision of many other states to adopt state ownership and licensing systems that would avoid the inefficiencies of fragmented ownership.[1] The doctrine of correlative rights, developed to mitigate the rule of capture, recognises that each owner

[1] Terence Daintith, *Finders Keepers? How the Law of Capture Shaped the World Oil Industry* (RFF Press 2010).

of land overlying a common reservoir has a right to a fair opportunity to produce its just and equitable share of the oil and gas, and a duty not to injure the reservoir or waste its resources.

Constitutional Vesting Clauses

In states following the regalian model, ownership is typically vested in the state by a constitutional provision. Some illustrative examples follow.

Nigeria. Section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 provides:

> Notwithstanding the foregoing provisions of this section, the entire property in and control of all minerals, mineral oils and natural gas in under or upon any land in Nigeria or in, under or upon the territorial waters and the Exclusive Economic Zone of Nigeria shall vest in the Government of the Federation and shall be managed in such manner as may be prescribed by the National Assembly.

This vesting is reiterated in section 1 of the Petroleum Industry Act 2021 and section 1 of the Nigerian Minerals and Mining Act 2007. The Land Use Act 1978 separately vests all land comprised in the territory of each state in the Governor of that state, to be held in trust and administered for the use and common benefit of all Nigerians, with holders of land enjoying rights of occupancy rather than ownership. The combination of federal ownership of minerals and state control of land, overlaid on customary tenure, generates complex legal relationships that are examined further in section 3.4.

Mexico. Article 27 of the Mexican Constitution of 1917 vests in the Nation direct ownership of all minerals and hydrocarbons, declares that this ownership is inalienable and imprescriptible, and — following the constitutional reforms of 2013 and their partial reversal in 2024–2025 — regulates the participation of private parties in the hydrocarbon and electricity sectors.

Bolivia. Article 349 of the Constitution of Bolivia 2009 declares that natural resources are the property and direct domain, indivisible and imprescriptible, of the Bolivian people, and that their administration corresponds to the state in the collective interest.

Norway. Norway, which has no comparable constitutional clause, relies on statute: section 1-1 of the Petroleum Act 1996 provides that the Norwegian state has the proprietary right to subsea petroleum deposits and the exclusive right to resource management.

Consequences of the Ownership Model

The choice of ownership model has significant consequences. Under state ownership, the state is both owner and regulator; it captures the resource rent

through royalties, taxes, and production shares; and it decides which areas will be opened for exploration and on what terms. The relationship between the state and the investor is typically governed by a licence, concession, or contract, and the investor's rights are those conferred by that instrument. Under the accession system, the landowner captures a substantial portion of the rent through lease bonuses and royalties; the state's revenues derive principally from taxation; and the regulator's role is primarily to prevent waste, protect correlative rights, and safeguard the environment. Political economy research suggests that the distribution of resource rents to private landowners in the United States has helped sustain public support for oil and gas development in producing regions, whereas the concentration of rents in the state under the regalian model may contribute to the disconnection between resource extraction and local benefits that characterises many producer states.

COMPARATIVE PERSPECTIVE

Ownership and community benefit. In the United States, a farmer in North Dakota who owns the mineral estate beneath his land may receive royalties of 12.5 to 20 per cent or more of production value from a shale well, a direct private share of the resource rent. In Nigeria, a farmer in the Niger Delta whose land overlies a producing field owns neither the petroleum nor, under the Land Use Act, the land itself in the full sense; he is entitled to compensation for surface rights and crops affected by operations, and his community may benefit from the host community development trust established under Chapter 3 of the Petroleum Industry Act 2021. The contrast illustrates how the ownership model shapes the distribution of benefits and, consequently, local attitudes towards resource development.

3.3 Public Trust Doctrine and State Stewardship

Origins of the Public Trust Doctrine

The public trust doctrine holds that certain natural resources are held by the state in trust for the public and may not be alienated or managed in a manner inconsistent with the public interest. The doctrine traces its origins to Roman law, which recognised that certain things — the air, running water, the sea, and the shores of the sea — were common to all (*res communes*) under the Institutes of Justinian. It was received into English common law in relation to tidal waters and the foreshore, which the Crown held subject to the public rights of navigation and fishing, and it was developed most fully in the United States.

The leading American authority is *Illinois Central Railroad Co v Illinois* (1892), in which the United States Supreme Court held that the State of Illinois could not irrevocably convey to a private railroad company title to the submerged lands of Chicago's harbour on Lake Michigan, because those lands were held

in trust for the people for purposes of navigation, commerce, and fishing.[2] The Court reasoned that a state could no more abdicate its trust over property in which the whole people are interested than it could abdicate its police powers. The doctrine was revitalised by Joseph Sax's influential 1970 article, which proposed it as a tool for environmental protection, and was extended by the Supreme Court of California in *National Audubon Society v Superior Court* (1983) to require the state to consider the public trust in allocating water rights affecting Mono Lake.[3]

The Doctrine Beyond the United States

The public trust doctrine has been received and adapted in several other jurisdictions. The Supreme Court of India, in *M.C. Mehta v Kamal Nath* (1997), held that the public trust doctrine was part of Indian law and that the state, as trustee of natural resources, was under a legal duty to protect them; it quashed a lease of riparian forest land granted to a motel company.[4] Courts in Pakistan, the Philippines, Kenya, and South Africa have invoked the doctrine or analogous concepts. Section 3 of South Africa's Mineral and Petroleum Resources Development Act 28 of 2002 provides that "mineral and petroleum resources are the common heritage of all the people of South Africa and the State is the custodian thereof for the benefit of all South Africans", a formulation that replaced private mineral ownership with state custodianship. The Constitutional Court of South Africa, in *Agri South Africa v Minister for Minerals and Energy* (2013), held that this transition did not amount to an expropriation of private mineral rights requiring compensation, because the state had not acquired the rights for itself but had become their custodian.[5]

Stewardship and Intergenerational Equity

The public trust doctrine is closely related to the concept of intergenerational equity, which holds that each generation holds the natural and cultural resources of the planet in trust for future generations. Edith Brown Weiss's formulation of the principle of intergenerational equity — comprising conservation of options, conservation of quality, and conservation of access — has been influential in international environmental law.[6] In the context of non-renewable resources, intergenerational equity raises questions about the rate of extraction, the investment of resource revenues, and the obligation to leave future generations with comparable

[2] *Illinois Central Railroad Co v Illinois*, 146 US 387 (1892).

[3] *National Audubon Society v Superior Court of Alpine County*, 33 Cal 3d 419 (1983); see Joseph L. Sax, "The Public Trust Doctrine in Natural Resource Law: Effective Judicial Intervention" (1970) 68 *Michigan Law Review* 471.

[4] *M.C. Mehta v Kamal Nath* (1997) 1 SCC 388 (Supreme Court of India).

[5] *Agri South Africa v Minister for Minerals and Energy* [2013] ZACC 9; 2013 (4) SA 1 (CC).

[6] Edith Brown Weiss, *In Fairness to Future Generations: International Law, Common Patrimony, and Intergenerational Equity* (Transnational/United Nations University 1989).

opportunities. Sovereign wealth funds, such as Norway's Government Pension Fund Global and Nigeria's Sovereign Investment Authority, established by the Nigeria Sovereign Investment Authority (Establishment, etc.) Act 2011 with a Future Generations Fund among its constituent funds, represent institutional responses to this concern.

Climate litigants have invoked the public trust doctrine to argue that governments hold the atmosphere in trust and are obliged to protect it. In *Juliana v United States*, young plaintiffs relied in part on the public trust doctrine to claim that the federal government's promotion of fossil fuels violated their constitutional rights; the Ninth Circuit Court of Appeals held in 2020 that the plaintiffs lacked standing because the relief sought was beyond the power of an Article III court to provide, and the litigation was ultimately terminated after the Supreme Court declined review in 2025.[7] Courts in other jurisdictions have been more receptive to stewardship-based reasoning, as discussed in Chapter 18.

Legal Consequences of Stewardship

Where resources are held in trust, several legal consequences may follow. The state may be prohibited from alienating trust resources entirely or from granting rights that substantially impair the public interest. Grants of rights may be subject to implied conditions requiring their exercise consistently with the trust. Courts may review state decisions affecting trust resources with heightened scrutiny. And citizens may have standing to enforce the trust. These consequences can create tension with the security of tenure sought by investors, particularly where a court later concludes that a grant was inconsistent with the trust. Investors and their advisers should therefore assess, as part of legal due diligence, whether the host jurisdiction recognises a public trust doctrine or constitutional stewardship obligations that could affect the validity or scope of resource rights.

3.4 Federalism and Subnational Resource Authority

The Federal Dilemma

In federal states, the allocation of authority over natural resources between the federal government and the constituent units is among the most politically charged constitutional questions. Resources are often geographically concentrated, so that their ownership and revenues may disproportionately benefit particular regions. Federal ownership may be perceived by producing regions as the appropriation of their wealth by the centre; subnational ownership may produce stark inequalities between resource-rich and resource-poor regions and undermine national cohesion. The resolution of this dilemma varies widely.

[7] *Juliana v United States*, 947 F 3d 1159 (9th Cir 2020).

Nigeria: Federal Ownership, Derivation, and Resource Control

Nigeria provides a paradigmatic example of the federal dilemma. Before independence and during the First Republic, the regions enjoyed significant fiscal autonomy and the principle of derivation — the allocation of revenue to the region from which it was derived — played a major role in revenue sharing. Following the civil war and the rise of petroleum as the dominant source of public revenue, the federal government progressively centralised control. The Petroleum Act 1969 vested ownership of petroleum in the state (the Federal Military Government), and successive constitutions confirmed federal ownership. The derivation percentage fell sharply in the 1970s and 1980s.

The 1999 Constitution restored a minimum derivation component: section 162(2) provides that "the principle of derivation shall be constantly reflected in any approved formula as being not less than thirteen per cent of the revenue accruing to the Federation Account directly from any natural resources". The question of whether offshore revenues should be counted as derived from littoral states gave rise to the resource control litigation. In *Attorney-General of the Federation v Attorney-General of Abia State & 35 Others* (2002), the Supreme Court held that the seaward boundary of a littoral state for the purposes of the derivation principle was the low-water mark of its land surface (or the seaward limit of inland waters), so that revenues from offshore production on the continental shelf and in the EEZ accrued to the Federation rather than to the states. The decision was politically controversial and was effectively reversed in part by the Allocation of Revenue (Abolition of Dichotomy in the Application of the Principle of Derivation) Act 2004, which provided that the 200-metre water depth isobath would be the limit for the purposes of derivation, thereby treating shallow-water offshore production as derived from the adjoining littoral states.

The demand for "resource control" — greater control by producing states and communities over the resources found in their territories and the revenues derived from them — remains a central theme of Nigerian politics. It has been associated with the Ogoni struggle of the 1990s, the Kaiama Declaration of 1998, militancy in the Niger Delta in the 2000s, and the amnesty programme of 2009. The Petroleum Industry Act 2021 responded in part by establishing host community development trusts funded by an annual contribution of three per cent of the actual annual operating expenditure of settlors (holders of interests in petroleum prospecting licences and petroleum mining leases) in the preceding financial year (section 240(2)), although critics argue that this falls far short of meaningful resource control.

Electricity Federalism in Nigeria

In contrast to petroleum, electricity in Nigeria has undergone significant decentralisation. The Constitution (Fifth Alteration) (No. 33) Act 2023 amended the Exclusive and Concurrent Legislative Lists to permit states to legislate on electricity generation, transmission, and distribution within their territories. The Electricity Act 2023, which repealed the Electric Power Sector Reform Act 2005, provides a framework under which states may establish their own electricity markets and regulators, with the Nigerian Electricity Regulatory Commission transferring regulatory oversight of intrastate markets to state regulators once established. A number of states, including Enugu, Ekiti, Ondo, Imo, Oyo, Edo, Kogi, and Lagos, have enacted electricity laws and established regulatory commissions, and regulatory oversight has been transferred to several of them. This process of electricity federalism is examined in Chapter 9.

Canada: Provincial Ownership

In Canada, the provinces own the natural resources within their boundaries by virtue of section 109 of the Constitution Act 1867 (extended to the prairie provinces by the Natural Resources Transfer Agreements of 1930). Section 92A, added by the Constitution Act 1982 following prolonged federal–provincial conflict over energy pricing and taxation (notably the National Energy Program of 1980), confirms exclusive provincial legislative authority over exploration, development, conservation, and management of non-renewable natural resources, forestry resources, and electrical energy, and permits provinces to legislate on the export of such resources to other parts of Canada, subject to federal paramountcy. The federal government retains authority over offshore resources (confirmed by the Supreme Court of Canada in *Reference re Newfoundland Continental Shelf* (1984)), interprovincial and international pipelines, nuclear energy, and matters of national concern. Offshore resources off Newfoundland and Labrador and Nova Scotia are managed jointly through accords implemented by federal and provincial legislation.

The Supreme Court of Canada in *References re Greenhouse Gas Pollution Pricing Act* (2021) upheld federal carbon pricing legislation under the "peace, order and good government" power as a matter of national concern,[8] while in *Reference re Impact Assessment Act* (2023) it held that significant parts of the federal impact assessment regime were unconstitutional because they reached beyond federal jurisdiction into matters of provincial competence.[9]

[8] *References re Greenhouse Gas Pollution Pricing Act*, 2021 SCC 11.

[9] *Reference re Impact Assessment Act*, 2023 SCC 23.

Australia: State Ownership Onshore and Offshore Arrangements

In Australia, the states own onshore minerals and petroleum and regulate their development. Offshore, following the High Court's decision in *New South Wales v Commonwealth* (the *Seas and Submerged Lands Case*) (1975), which upheld Commonwealth sovereignty over the territorial sea and continental shelf, the Offshore Constitutional Settlement of 1979 allocated jurisdiction over the first three nautical miles of the territorial sea to the states, with the Commonwealth retaining authority beyond. Petroleum activities in Commonwealth waters are governed by the Offshore Petroleum and Greenhouse Gas Storage Act 2006, administered by the National Offshore Petroleum Titles Administrator and regulated for safety and environmental matters by the National Offshore Petroleum Safety and Environmental Management Authority (NOPSEMA).

The United States: Layered Ownership

In the United States, mineral ownership is layered among private parties, states, the federal government, and Native American tribes. The federal government owns the mineral estate under approximately 700 million acres of land and administers leasing under the Mineral Leasing Act 1920 and, for the outer continental shelf beyond state waters (generally three nautical miles), under the Outer Continental Shelf Lands Act 1953. The Submerged Lands Act 1953 confirmed state title to submerged lands within three nautical miles of the coast (three marine leagues for Texas and the Gulf coast of Florida). Regulation of oil and gas production on private and state lands is primarily a matter of state law, administered by bodies such as the Railroad Commission of Texas.

3.5 Indigenous Peoples and Resource Governance

The Encounter between Resource Development and Indigenous Rights

A large proportion of the world's remaining mineral, petroleum, and renewable energy resources lies on or beneath lands traditionally owned, occupied, or used by indigenous peoples. The encounter between resource development and indigenous rights has been one of the most significant sources of legal conflict in the field, and the recognition of indigenous rights has fundamentally altered the legal landscape for resource projects in many jurisdictions. Chapter 21 provides a detailed treatment of the international legal framework; this section focuses on the constitutional and property law dimensions.

Recognition of Indigenous Title and Rights

The recognition of indigenous land rights has proceeded through constitutional provisions, legislation, judicial decisions, and treaty arrangements. In Australia, the High Court in *Mabo v Queensland (No 2)* (1992) rejected the doctrine of *terra*

nullius and recognised native title at common law,[10] a recognition subsequently regulated by the Native Title Act 1993, which confers upon native title holders a "right to negotiate" with respect to certain future acts, including the grant of mining tenements. In *Wik Peoples v Queensland* (1996), the High Court held that pastoral leases did not necessarily extinguish native title.

In Canada, section 35 of the Constitution Act 1982 recognises and affirms existing aboriginal and treaty rights. The Supreme Court of Canada recognised aboriginal title in *Delgamuukw v British Columbia* (1997) and, for the first time, made a declaration of aboriginal title over a specific territory in *Tsilhqot'in Nation v British Columbia* (2014), holding that aboriginal title confers the right to decide how the land will be used and to enjoy its economic benefits, and that governments seeking to use titled land without consent must justify the infringement.[11] The Court has also developed a duty to consult and, where appropriate, accommodate aboriginal peoples when the Crown contemplates conduct that might adversely affect asserted or established rights (*Haida Nation v British Columbia (Minister of Forests)*, 2004). These duties have major implications for pipeline and mining projects; in *Clyde River (Hamlet) v Petroleum Geo-Services Inc* (2017), the Supreme Court quashed an authorisation for offshore seismic testing because the consultation with the Inuit had been inadequate.

In Latin America, the Inter-American Court of Human Rights has developed a substantial jurisprudence on indigenous and tribal peoples' property rights under Article 21 of the American Convention on Human Rights. In *Saramaka People v Suriname* (2007), the Court held that the state must ensure the effective participation of the Saramaka people in development or investment plans within their territory, guarantee that they receive a reasonable benefit, and ensure that no concession is issued without a prior environmental and social impact assessment; for large-scale projects with a major impact, the state has a duty to obtain their free, prior, and informed consent.[12] In *Kichwa Indigenous People of Sarayaku v Ecuador* (2012), the Court found Ecuador responsible for permitting a petroleum company to carry out seismic exploration on Sarayaku territory without prior consultation.

In Africa, the African Commission on Human and Peoples' Rights in *Centre for Minority Rights Development (Kenya) and Minority Rights Group International on behalf of Endorois Welfare Council v Kenya* (2010) found that Kenya had violated the rights of the Endorois community by evicting them from their ancestral land around Lake Bogoria without adequate consultation or compensation,

[10] *Mabo v Queensland (No 2)* (1992) 175 CLR 1 (High Court of Australia).

[11] *Tsilhqot'in Nation v British Columbia*, 2014 SCC 44, [2014] 2 SCR 257.

[12] *Saramaka People v Suriname*, Inter-American Court of Human Rights, Judgment of 28 November 2007, Series C No 172, paras 129–137.

and the African Court on Human and Peoples' Rights in *African Commission on Human and Peoples' Rights v Republic of Kenya* (the *Ogiek* case, 2017) found violations of the Ogiek people's rights to their ancestral land in the Mau Forest.

Indigenous Ownership of Subsurface Resources

A distinct question is whether indigenous peoples own the subsurface resources beneath their lands. In most regalian systems, the answer is no: the state's ownership of minerals extends beneath indigenous lands, and indigenous rights are limited to the surface and to rights of consultation, compensation, and benefit-sharing. There are exceptions. In the United States, tribes own the mineral estate beneath many reservation lands, and mineral development is governed by the Indian Mineral Leasing Act 1938 and the Indian Mineral Development Act 1982. In Australia's Northern Territory, the Aboriginal Land Rights (Northern Territory) Act 1976 confers a veto over exploration on Aboriginal land, subject to a national interest override. In Canada, the content of aboriginal title, as described in *Delgamuukw*, includes subsurface mineral rights.

The question has particular resonance in Africa, where the distinction between indigenous peoples and other local communities is often contested and where customary communities assert claims to resources beneath their lands. Under Nigerian law, such claims are foreclosed by section 44(3) of the Constitution, but communities may assert rights to compensation, environmental protection, and participation in benefit-sharing arrangements.

3.6 Resource Nationalism and Regulatory Reform

Defining Resource Nationalism

"Resource nationalism" describes the tendency of governments to assert greater control over natural resources and a larger share of the economic benefits they generate. The term is often used pejoratively by investors and commentators to describe measures perceived as opportunistic or confiscatory, but it can also describe legitimate efforts by states to correct inequitable bargains, to capture a fair share of windfall profits, and to promote national development. A more neutral analytical definition treats resource nationalism as a spectrum of state measures designed to increase state control over, or state revenue from, the resource sector, ranging from fiscal adjustments at one end to outright expropriation at the other.

Resource nationalism is cyclical. It tends to intensify during periods of high commodity prices, when the gap between the returns earned by investors and those anticipated at the time of investment widens and governments come under political pressure to capture windfall gains. It tends to recede during periods of low prices, when states compete to attract investment and are more willing to offer favourable terms. This cyclicality reflects the obsolescing bargain described in Chapter 1.

Forms of Resource Nationalism

Resource nationalism takes a variety of forms, which may be classified by increasing intensity:

1. **Fiscal measures**: increases in royalty rates, income tax rates, or special resource taxes; the introduction of windfall profit taxes or resource rent taxes; restrictions on cost recovery or deductions; and changes to the treatment of transfer pricing.
2. **Local content and beneficiation requirements**: obligations to employ nationals, procure goods and services locally, process minerals domestically before export, or transfer technology. Nigeria's Nigerian Oil and Gas Industry Content Development Act 2010 is a prominent example; Indonesia's ban on the export of unprocessed nickel ore, in force since January 2020, illustrates the beneficiation model.
3. **Export restrictions and domestic supply obligations**: requirements to supply a portion of production to the domestic market, often at below-market prices (domestic market obligations), or restrictions on exports. Nigeria's Petroleum Industry Act 2021 establishes a domestic crude oil supply obligation and a domestic gas delivery obligation.
4. **Increased state participation**: mandatory carried or paid interests for the state or national resource company in new or existing projects; requirements that a minimum percentage of shares be held by nationals. Tanzania's Natural Wealth and Resources (Permanent Sovereignty) Act 2017 and Natural Wealth and Resources Contracts (Review and Re-Negotiation of Unconscionable Terms) Act 2017 combined participation requirements with a statutory power to renegotiate "unconscionable" terms, and the Mining (Amendment) Act 2017 provided for a non-dilutable free carried interest of not less than sixteen per cent for the government in mining companies.
5. **Contract renegotiation and conversion**: unilateral or coerced renegotiation of contracts, or the mandatory conversion of existing contracts to new forms (for example, from concessions to production sharing contracts or service contracts).
6. **Expropriation and nationalisation**: the outright taking of investor assets, with or without compensation.

Drivers and Justifications

Several factors drive resource nationalism: high commodity prices and perceived windfall profits; the perception that existing contracts were negotiated on unfair terms, under duress, or as a result of corruption; fiscal pressures on governments; the political salience of resource sovereignty; the desire to build national industrial capacity; and, increasingly, concerns about energy and resource security. Governments frequently justify their measures by reference to permanent sovereignty,

the public interest, and the need to correct asymmetries of information and bargaining power at the time of contracting.

Regulatory Reform: The Nigerian Petroleum Industry Act 2021

Not all reform of resource governance is properly characterised as resource nationalism. Many states have undertaken comprehensive reforms designed to modernise their legal frameworks, improve governance, and attract investment. Nigeria's Petroleum Industry Act 2021 (PIA), the product of nearly two decades of legislative effort, is a leading example. The PIA:

- restructured the institutional framework by establishing two regulators — the Nigerian Upstream Petroleum Regulatory Commission (NUPRC) and the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA) — and providing for the incorporation of the Nigerian National Petroleum Corporation as a limited liability company, Nigerian National Petroleum Company Limited (NNPC Ltd), operating on commercial principles;
- replaced the oil prospecting licence and oil mining lease with petroleum prospecting licences and petroleum mining leases, with provisions on relinquishment, work commitments, and the conversion of existing licences;
- introduced a new fiscal regime comprising a hydrocarbon tax on upstream crude oil profits, companies income tax, and production-based royalties, with incentives for gas and frontier exploration;
- established host community development trusts;
- created frameworks for environmental remediation and decommissioning funds; and
- established the Frontier Exploration Fund and the Midstream and Downstream Gas Infrastructure Fund.

The PIA illustrates the trend towards replacing discretionary, contract-based governance with a comprehensive statutory framework. Its implementation and the conversion of existing licences to the new regime are examined in Chapters 5 and 24.

3.7 Expropriation, Nationalization, and Compensation

Terminology

"Expropriation" refers to the taking of property by the state. "Nationalisation" usually refers to large-scale takings of entire industries or sectors as part of a programme of economic or social reform. "Confiscation" refers to a taking without compensation. "Indirect expropriation" or "creeping expropriation" refers to measures that do not formally transfer title but substantially deprive the owner of the use, enjoyment, or economic value of the property. The distinction

between indirect expropriation and non-compensable regulation is one of the most difficult questions in international investment law and is examined in detail in Chapter 14. This section focuses on the general international law framework, with particular reference to its development through energy disputes.

Conditions of Lawful Expropriation

Customary international law, as reflected in Resolution 1803 and in the practice of investment treaties, permits a state to expropriate foreign property provided that the expropriation is:

1. for a public purpose;
2. non-discriminatory;
3. carried out in accordance with due process of law; and
4. accompanied by the payment of compensation.

These conditions are reflected, with variations, in virtually every investment treaty. Article 13 of the ECT, for example, prohibits expropriation of investments of investors of another contracting party except where the expropriation is "(a) for a purpose which is in the public interest; (b) not discriminatory; (c) carried out under due process of law; and (d) accompanied by the payment of prompt, adequate and effective compensation".

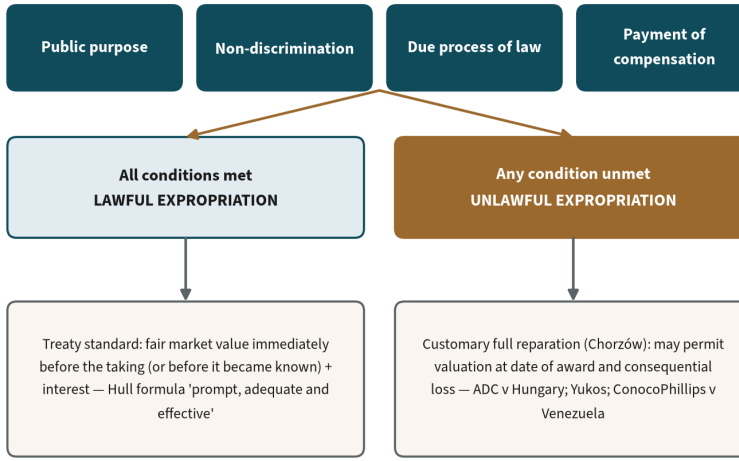


Figure 3.2 The four conditions of lawful expropriation and their consequences for compensation. *Source:* prepared for this edition.

The Standard of Compensation

The standard of compensation has been the most contested aspect of the law of expropriation. Three formulations compete.

The Hull standard — "prompt, adequate and effective" compensation — requires compensation equivalent to the full market value of the expropriated property, paid without delay and in convertible currency. It was championed by capital-exporting states and is reflected in most modern investment treaties, often with the further specification that compensation shall be equivalent to the fair market value of the investment immediately before the expropriation or before it became publicly known, whichever is earlier.

The "appropriate compensation" standard of Resolution 1803 and the Charter of Economic Rights and Duties of States permits account to be taken of a range of circumstances, potentially justifying compensation below full market value. The *Aminoil* tribunal applied this standard, holding that the "legitimate expectations" of the parties, as reflected in the evolution of their contractual relationship, informed the determination of appropriate compensation, and that compensation should take account of a reasonable rate of return rather than speculative profits.

Partial or no compensation has been asserted in the context of large-scale nationalisations forming part of fundamental social and economic reform, on the ground that full compensation would make such reform impossible. This position has not commanded general acceptance as a rule of international law, although lump-sum settlements negotiated between states have frequently provided for compensation at substantially less than full value.

In treaty-based arbitration today, the debate has largely been resolved in favour of the full market value standard for lawful expropriation, because the treaties themselves specify it. The more significant debates now concern valuation methodology, the valuation date, and the distinction between lawful and unlawful expropriation.

Lawful and Unlawful Expropriation

Following *Chorzów Factory*, tribunals distinguish between lawful expropriation, for which compensation is due under the treaty standard (typically fair market value at the date of taking), and unlawful expropriation, for which full reparation is due under customary international law. The distinction matters because full reparation may require valuation at the date of the award rather than the date of taking, allowing the claimant to benefit from subsequent increases in value — for example, increases in oil prices.

In *ADC Affiliate Ltd v Hungary* (2006), the tribunal held that because Hungary's expropriation was unlawful, the claimants were entitled to the value of their investment at the date of the award, which was considerably higher

than at the date of the taking.[13] In *Yukos Universal Ltd v Russian Federation* (2014), the tribunal found that Russia's measures against Yukos amounted to an unlawful expropriation and awarded damages of approximately USD 50 billion to the former majority shareholders, using a valuation date chosen to give the claimants the benefit of the higher value.[14] In *ConocoPhillips v Venezuela* (2019), the tribunal, having previously held that Venezuela's taking of ConocoPhillips' interests in heavy oil projects was unlawful because Venezuela had failed to negotiate compensation in good faith, awarded approximately USD 8.5 billion plus interest.[15] By contrast, in *Venezuela Holdings BV (Mobil) v Venezuela* (2014), the tribunal found that the claimants had not established that the expropriation was unlawful merely because the compensation offered was inadequate, and accordingly applied the treaty standard of compensation for lawful expropriation, valuing the investment as of the date of the taking.

The Yukos awards were set aside by the District Court of The Hague in 2016 on jurisdictional grounds relating to the provisional application of the ECT, reinstated by the Hague Court of Appeal in 2020, and subjected to further proceedings, with the Supreme Court of the Netherlands in 2021 remitting the question of alleged fraud in the arbitration to the Amsterdam Court of Appeal, which rejected Russia's fraud arguments in 2024. The saga illustrates the complexity and duration of enforcement proceedings against a sovereign that resists payment.

Compensation for Natural Resource Projects: Valuation Issues

The valuation of expropriated natural resource projects presents distinctive difficulties. The value of a petroleum or mining project depends on the volume of reserves, future commodity prices, production costs, fiscal terms, and the risk profile of the host state, each of which is uncertain. Tribunals have generally accepted the discounted cash flow (DCF) method for producing assets with an established record of profitability, while being sceptical of DCF valuations for projects at an early stage of development. In *Tethyan Copper Company Pty Limited v Pakistan* (2019), the tribunal awarded approximately USD 4 billion in damages and interest for the denial of a mining lease for the Reko Diq copper-gold deposit, valuing the project by DCF despite the fact that it had not yet entered production — a decision that attracted criticism and was followed by a negotiated

[13] *ADC Affiliate Limited and ADC & ADMC Management Limited v Republic of Hungary*, ICSID Case No ARB/03/16, Award, 2 October 2006, paras 493–499.

[14] *Yukos Universal Limited (Isle of Man) v Russian Federation*, PCA Case No AA 227, Final Award, 18 July 2014.

[15] *ConocoPhillips Petrozuata BV, ConocoPhillips Hamaca BV and ConocoPhillips Gulf of Paria BV v Bolivarian Republic of Venezuela*, ICSID Case No ARB/07/30, Award, 8 March 2019.

settlement in 2022 involving the reconstitution of the project with new participation arrangements.[16]

The Iran–United States Claims Tribunal and Lump-Sum Settlements

A substantial body of jurisprudence on the expropriation of energy assets was generated by the Iran–United States Claims Tribunal, established by the Algiers Accords of January 1981 following the Iranian revolution and the hostage crisis. Many of the claims arose from the nationalisation of the Iranian oil industry and the termination of petroleum agreements after 1979. In *Amoco International Finance Corporation v Iran* (1987), Chamber Three distinguished between lawful and unlawful expropriation and held that, where the taking was lawful, the compensation owed was the value of the undertaking as a going concern at the date of the taking, excluding the speculative future profits that would have been recoverable as damages for an unlawful taking.[17] In *Phillips Petroleum Company Iran v Iran* (1989), the Tribunal held that the claimant was entitled to compensation for the expropriation of its contractual rights under a joint structure agreement with the National Iranian Oil Company, and that the full value of those rights was to be determined using a discounted cash flow analysis informed by contemporaneous price expectations.[18] In *Starrett Housing Corporation v Iran* (1983), the Tribunal recognised that measures taken by a state can interfere with property rights to such an extent that the rights are rendered so useless that they must be deemed to have been expropriated, even though the state does not purport to have expropriated them and legal title remains with the original owner — a foundational statement of the concept of indirect expropriation.

Outside institutional adjudication, many nationalisation disputes have been resolved by lump-sum agreements between the expropriating state and the home state of the investors, under which the expropriating state pays a global sum that the home state distributes among its nationals through a domestic claims commission. The United States Foreign Claims Settlement Commission and the United Kingdom Foreign Compensation Commission administered such distributions in respect of nationalisations in Eastern Europe, Cuba, and elsewhere. The compensation provided under lump-sum agreements has frequently been considerably less than the full value of the property taken. Whether such settlements constitute state practice supporting a standard of less-than-full compensation, or merely represent negotiated compromises without *opinio juris*, remains debated;

[16] *Tethyan Copper Company Pty Limited v Islamic Republic of Pakistan*, ICSID Case No ARB/12/1, Award, 12 July 2019.

[17] *Amoco International Finance Corporation v Islamic Republic of Iran*, Partial Award No 310-56-3, 14 July 1987, 15 Iran-US CTR 189.

[18] *Phillips Petroleum Company Iran v Islamic Republic of Iran*, Award No 425-39-2, 29 June 1989, 21 Iran-US CTR 79.

the prevailing view is that they reflect pragmatic settlement rather than a belief that international law requires no more.

Due Process and Judicial Review of Expropriation

The due process condition requires that an expropriation be carried out in accordance with the procedures established by domestic law and with the fundamental international standards of procedural fairness, including reasonable advance notice, a fair hearing, and the opportunity to challenge the measure before an independent and impartial body. In *ADC v Hungary*, the tribunal held that due process "demands an actual and substantive legal procedure for a foreign investor to raise its claims against the depriving actions already taken or about to be taken against it", including a reasonable chance within a reasonable time to claim its legitimate rights and have its claims heard.[19] Domestic constitutional law frequently imposes additional requirements. Section 44(1) of the Nigerian Constitution, for instance, prohibits the compulsory acquisition of any interest in moveable or immoveable property except in the manner and for the purposes prescribed by a law that, among other things, requires the prompt payment of compensation and gives the person claiming compensation a right of access to a court or tribunal for the determination of his interest and the amount of compensation. Section 25 of the Nigerian Investment Promotion Commission Act (Cap N117, Laws of the Federation of Nigeria 2004) further provides that no enterprise shall be nationalised or expropriated by any government of the Federation, and that no person who owns the enterprise shall be compelled by law to surrender his interest to any other person, save where the acquisition is in the national interest or for a public purpose and under a law that provides for payment of fair and adequate compensation and a right of access to the courts.

3.8 Comparative Constitutional Approaches

A Typology of Constitutional Approaches

Constitutions address natural resources in strikingly different ways. A comparative survey reveals at least five approaches:

1. **Silence.** Some constitutions, including the Constitution of the United States, do not address natural resource ownership at all, leaving the matter to property law and legislation.
2. **Vesting clauses.** Many constitutions vest ownership of minerals and petroleum in the state, the nation, or the people (Nigeria, Mexico, Ghana).

[19] *ADC Affiliate Limited v Republic of Hungary*, ICSID Case No ARB/03/16, Award, 2 October 2006, para 435.

3. **Federal allocation clauses.** Federal constitutions allocate authority between levels of government (Canada's sections 92A and 109; Nigeria's Exclusive Legislative List and derivation clause).
4. **Stewardship and sustainability clauses.** Some constitutions impose duties of sustainable management. Article 112 of the Constitution of Norway (formerly Article 110b) provides that natural resources shall be managed on the basis of comprehensive long-term considerations that will safeguard this right for future generations as well. The Supreme Court of Norway in *Greenpeace Nordic and Nature and Youth v Ministry of Petroleum and Energy* (2020) held that Article 112 is a substantive right but that the threshold for judicial intervention in parliamentary decisions is high, and it declined to invalidate Arctic petroleum licences; the subsequent *Greenpeace Nordic v Norway* proceedings before the Oslo District Court in 2024 held certain field development approvals invalid for failure to assess combustion emissions.
5. **Rights of nature.** The Constitution of Ecuador 2008 (Articles 71–74) recognises the rights of nature (*Pacha Mama*) to exist, persist, maintain, and regenerate its vital cycles. The Constitutional Court of Ecuador in the *Los Cedros* case (2021) held that mining concessions in the Los Cedros protected forest violated the rights of nature.

Revenue-Sharing Clauses

Constitutions in resource-rich federations frequently prescribe revenue-sharing arrangements. Nigeria's thirteen per cent derivation principle has been noted. The Constitution of Iraq 2005, Article 112, provides that the federal government, with the producing governorates and regional governments, shall undertake the management of oil and gas extracted from present fields, with revenues distributed in a fair manner in proportion to the population distribution — a provision whose ambiguity has fuelled protracted disputes between Baghdad and the Kurdistan Regional Government, including the 2022 ruling of Iraq's Federal Supreme Court declaring the Kurdistan Region's oil and gas law unconstitutional, and the 2023 ICC arbitral award in Iraq's favour concerning exports through the Iraq–Türkiye pipeline. The Constitution of Kenya 2010 requires legislation providing for the sharing of benefits from natural resources between the national government, counties, and local communities, implemented in part by section 58 of the Petroleum Act 2019, which allocates 20 per cent of the national government's share of profit petroleum to the county government and 5 per cent to the local community.

Constitutional Governance of Resource Revenues

A further dimension of constitutional governance concerns the management of the revenues generated by resource extraction. Because petroleum and mineral revenues are volatile, finite, and often very large relative to the rest of the economy, they create risks of macroeconomic instability, pro-cyclical spending, and the diversion of public funds. Constitutional and quasi-constitutional responses fall into four categories.

Fiscal rules limit the use of resource revenues in the annual budget. Norway's fiscal rule, adopted in 2001, provides that the structural non-oil deficit should over time correspond to the expected real return on the Government Pension Fund Global, now estimated at three per cent. Chile's structural balance rule adjusts fiscal targets for the long-term price of copper. Nigeria's Fiscal Responsibility Act 2007 introduced a budget framework based on a reference oil price, with revenues above that price saved in the Excess Crude Account, an arrangement whose legal basis was repeatedly contested between the federal and state governments.

Sovereign wealth funds set aside resource revenues for stabilisation, saving, or development purposes. Their governance is guided by the Santiago Principles. The Nigeria Sovereign Investment Authority Act 2011 established three funds — the Stabilisation Fund, the Future Generations Fund, and the Nigeria Infrastructure Fund — each with defined investment mandates. Ghana's Petroleum Revenue Management Act 2011 (Act 815) established the Ghana Stabilisation Fund and the Ghana Heritage Fund and created a Public Interest and Accountability Committee to monitor compliance, an innovative example of statutory civil society oversight.

Earmarking and distribution rules allocate revenues to particular purposes or populations. Some states have experimented with direct distribution of resource revenues to citizens, as in the Alaska Permanent Fund Dividend, paid annually to qualifying residents since 1982.

Transparency and audit obligations require the publication of resource revenues and contracts and their independent audit. The Nigeria Extractive Industries Transparency Initiative (NEITI) Act 2007 made Nigeria the first EITI implementing country to give the initiative a statutory basis, requiring periodic audits of the extractive sector and the publication of the results. Section 3 of the Act sets out NEITI's objectives, including ensuring due process and transparency in payments made by extractive industry companies to the federal government and its agencies.

These arrangements form part of the constitutional governance of resources in a broad sense, and their effectiveness depends on enforcement mechanisms — legislative oversight, independent audit institutions, judicial review, and civil

society monitoring. Doctoral research on the resource curse has increasingly focused on the design and performance of such arrangements.

Lessons from Comparison

Comparative analysis suggests several lessons. First, the formal ownership model matters less than the quality of the institutions that administer it; state ownership can be associated with excellent governance (Norway) or with chronic mismanagement. Second, revenue-sharing arrangements that leave producing regions and communities with a visible and meaningful share of benefits tend to reduce conflict, but must be designed to avoid creating new inequalities or local governance failures. Third, constitutional entrenchment of sustainability and stewardship duties can provide a basis for judicial review of resource decisions, although courts have generally been cautious about intervening in what they regard as policy matters. Fourth, the recognition of indigenous and community rights through constitutional, judicial, and treaty mechanisms has become a defining feature of resource governance, with profound implications for project development.

LEADING AUTHORITY

Government of the State of Kuwait v American Independent Oil Company (Aminoil), 1982

Facts. Aminoil held a sixty-year concession granted in 1948 by the Ruler of Kuwait over the Kuwaiti share of the Neutral Zone. The concession contained stabilisation clauses providing that the Ruler would not annul the agreement by general or special legislation or administrative measures, and that no alteration should be made except by mutual agreement. Over the years, the fiscal terms were revised several times. In 1977, Kuwait terminated the concession by decree and took over Aminoil's assets.

Holding. The tribunal (Professor Paul Reuter, President; Professor Hamed Sultan; Sir Gerald Fitzmaurice) held that the nationalisation was lawful. The stabilisation clauses did not prohibit nationalisation because such a far-reaching limitation on sovereignty would require express stipulation and would have to cover a relatively limited period; moreover, the contract had evolved through renegotiation. However, the stabilisation clauses gave rise to legitimate expectations relevant to the assessment of compensation, which was to be "appropriate" and was calculated by reference to the value of the undertaking and a reasonable rate of return. Kuwait was ordered to pay approximately USD 179 million, including interest.

Significance. Aminoil is the leading authority for the proposition that stabilisation clauses operate primarily to affect compensation rather than to invalidate sovereign acts. Its emphasis on the evolving nature of long-term contracts and on legitimate expectations anticipated modern debates over fair and equitable treatment.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Constitutional design for a new petroleum producer (hypothetical). The fictional Republic of Tamara, a federal state with five provinces, has recently discovered substantial onshore and offshore petroleum reserves. Its constitution is silent on resource ownership. A constitutional review commission has asked you to prepare a 3,000-word advisory paper addressing: (a) the choice between federal and provincial ownership of onshore and offshore resources; (b) the design of a revenue-sharing formula, including any derivation component and its treatment of offshore production; (c) the incorporation of stewardship and intergenerational equity duties; (d) the recognition of the rights of customary landholding communities; and (e) mechanisms to reduce the risk of resource-related conflict. Draw on comparative experience from at least four jurisdictions. *Tamara and its circumstances are fictitious.*

CHAPTER SUMMARY

- Resource sovereignty is an incident of territorial sovereignty, extended by UNCLOS to sovereign rights over the resources of the EEZ and continental shelf; modern instruments increasingly locate that sovereignty in the people, with the state as trustee.
- Subsurface ownership follows three broad models — regalian (state), accession (landowner), and claim-based systems — with major consequences for licensing, revenue distribution, and community attitudes.
- The public trust doctrine and related stewardship concepts impose fiduciary-type duties on states, enforceable in some jurisdictions by citizens and capable of affecting the security of investor rights.
- Federal systems allocate resource authority in widely differing ways; Nigeria's combination of federal ownership, a thirteen per cent derivation floor, the onshore/offshore dichotomy litigation, and recent electricity decentralisation illustrates the political sensitivity of the federal dilemma.
- Indigenous rights to land, consultation, consent, and benefit-sharing have been recognised through constitutional provisions, judicial decisions, and regional human rights jurisprudence, transforming the legal environment for resource projects.
- Resource nationalism is cyclical and takes forms ranging from fiscal adjustments to expropriation; it must be distinguished from good-faith regulatory reform such as Nigeria's Petroleum Industry Act 2021.
- Lawful expropriation requires public purpose, non-discrimination, due process, and compensation; the distinction between lawful and unlawful expropriation determines the measure and valuation date of compensation, with very large financial consequences in energy cases.

DOCTORAL RESEARCH QUESTIONS

1. Does the constitutional recognition of the people as bearers of resource sovereignty create enforceable obligations regarding the management of resource revenues, and how might such obligations be judicially enforced?
2. To what extent has the public trust doctrine been, or could it be, received into Nigerian law as a constraint on the grant of petroleum and mining rights?
3. Evaluate whether host community development trusts under the Petroleum Industry Act 2021 represent a meaningful response to demands for resource control, using comparative evidence from benefit-sharing regimes elsewhere.
4. Is the distinction between lawful and unlawful expropriation, with its consequences for valuation date, theoretically justified, or does it produce windfalls for claimants inconsistent with the compensatory purpose of reparation?
5. How should the valuation of undeveloped resource projects in investment arbitration account for the risk that climate policy will render reserves unburnable?

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4 CHAPTER

International Energy Institutions, Regulatory Systems, and Governance

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Describe the architecture of global energy governance and explain why it is fragmented and polycentric.
2. Analyse the legal foundations, mandates, powers, and limitations of the International Energy Agency, OPEC, the International Renewable Energy Agency, and the International Atomic Energy Agency.
3. Evaluate the role of regional energy regulatory institutions, including those of the European Union, the Energy Community, and African regional bodies.
4. Explain the rationale for independent regulatory agencies and the mechanisms by which their accountability is secured.
5. Assess the prospects for regulatory cooperation and institutional reform in global energy governance.

Introduction

Institutions matter. The rules of international energy law described in the previous chapters do not apply themselves; they are negotiated, interpreted, monitored, and enforced by institutions — international organisations, regional bodies, national regulators, courts, and tribunals. The design of these institutions shapes the content and effectiveness of the law. An institution with a narrow mandate, limited membership, and no binding powers will produce different outcomes from one with broad competence, universal membership, and the authority to adopt binding decisions.

Global energy governance is distinctive for the absence of a central institution. Unlike trade, which is governed by the WTO, or international civil aviation, which is governed by the International Civil Aviation Organization, energy has no single global organisation with comprehensive competence. Instead, governance

is distributed among a constellation of institutions with specialised mandates, overlapping memberships, and divergent interests. This chapter maps that constellation, examines the principal institutions in detail, and considers the role of national regulators and the prospects for reform.

4.1 Global Energy Governance Architecture

The Concept of Global Energy Governance

"Global energy governance" refers to the collective efforts of states and non-state actors to manage energy-related problems that transcend national borders through rules, institutions, and processes. The problems that global energy governance addresses include the security of supply for importing states and security of demand for exporting states, the stability of energy prices, the prevention of nuclear proliferation, the mitigation of climate change, the promotion of energy access in developing countries, the governance of transboundary infrastructure, and the protection of foreign investment.

Scholars have characterised the architecture of global energy governance as a "regime complex" — a set of partially overlapping and non-hierarchical institutions governing a particular issue area. The concept, developed by Kal Raustiala, David Victor, Robert Keohane, and others, captures the absence of a single overarching regime and the presence of multiple institutions whose rules and activities interact, sometimes reinforcing and sometimes conflicting with one another.

Why Is Energy Governance Fragmented?

Several factors explain the fragmentation of global energy governance.

Divergent interests. Producers and consumers have fundamentally opposed interests regarding prices: producers seek high and stable prices, consumers seek low and stable prices. These opposing interests have produced separate institutions — OPEC for producers, the IEA for consumers — rather than a single inclusive organisation.

Sovereignty sensitivities. Energy resources are closely tied to national sovereignty and security. States have been reluctant to cede authority over energy policy to international institutions, preferring arrangements that preserve national discretion.

Path dependence. Energy institutions were created in response to specific crises — the nuclear arms race (IAEA), the oil shocks of the 1970s (IEA), the collapse of the Soviet Union (Energy Charter), the climate crisis (UNFCCC, IRENA) — and each bears the imprint of the circumstances of its creation.

Technological diversity. Different energy sources raise different governance challenges: nuclear energy raises proliferation and safety concerns; oil raises supply security and price stability concerns; renewables raise technology diffusion and investment concerns. Specialised institutions have emerged for each.

Shifting centres of gravity. The growth of energy demand in emerging economies, particularly China and India, has shifted the centre of gravity of global energy markets away from the OECD states that dominated the institutions created in the twentieth century. Existing institutions have struggled to accommodate these new actors.

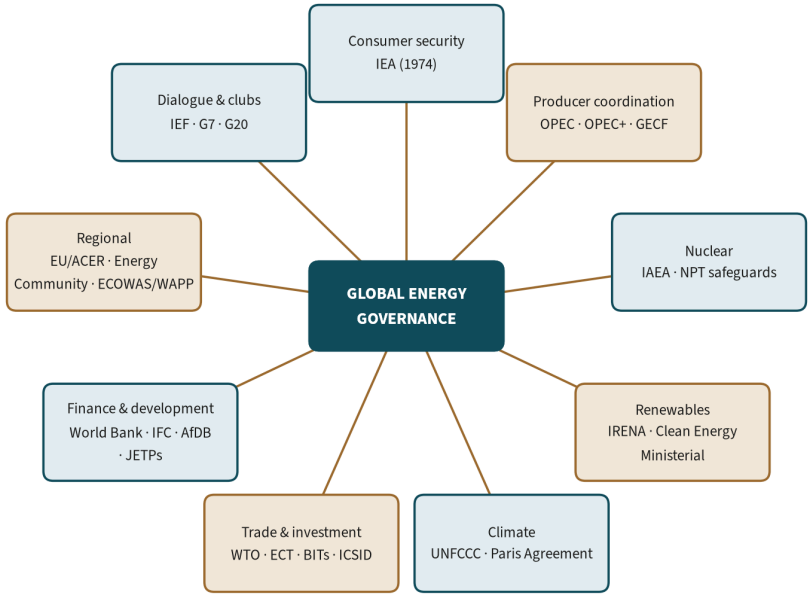


Figure 4.1 The polycentric architecture of global energy governance. *Source:* prepared for this edition.

Mapping the Architecture

The architecture of global energy governance may be mapped along two dimensions: the *functional* dimension, which identifies the governance function performed (security, markets, investment, technology, environment, development, nuclear non-proliferation), and the *membership* dimension, which distinguishes between universal, plurilateral, and regional institutions. Table 4.1 summarises the principal institutions.

Table 4.1 Principal international energy institutions. Membership figures are approximate and change over time; readers should consult the institutions' official sources for current data.

Institution	Founded	Legal basis	Member-ship	Core function	Binding pow-ers
International Energy Agency (IEA)	1974	Agreement on an International En-ergy Program	32 mem-ber coun-tries (OECD states) plus as-sociation countries	Oil supply secu-rity; energy data and analysis; clean energy tran-sitions	Emergency stockholding and response obligations
OPEC	1960	OPEC Statute	12 mem-ber coun-tries (2025)	Coordination of petroleum poli-cies; market sta-bilisation	Production decisions po-litically bind-ing; no exter-nal enforce-ment
International Renewable Energy Agency (IRENA)	2009	IRENA Statute	Approx. 170 members	Promotion of re-newable energy	None (advi-sory)
International Atomic En-ergy Agency (IAEA)	1957	IAEA Statute	Approx. 180 member states	Peaceful uses of nuclear energy; safeguards; safety standards	Safeguards agreements; standards binding when incorporated
Energy Char-ter Confer-ence	1994	Energy Charter Treaty	Contract-ing par-ties (con-tracted after EU with-drawal)	Investment pro-tection; transit; trade	Treaty obliga-tions; in-vestor–state arbitration
International Energy Forum (IEF)	1991	IEF Charter (char-ter 2011)	Approx. 70 coun-tries	Producer–con-sumer dialogue; JODI data	None
Gas Exporting Countries Fo-rum (GECF)	2008	GECF Statute	Gas-ex-porting states	Coordination of gas producer in-terests	None
UNFCCC / Paris Agree-ment	1992 / 2015	Treaties	Near-uni-versal	Climate mitiga-tion and adapta-tion	Procedural obligations; NDC cycle

Informal Governance: G7, G20, and Clubs

In addition to formal international organisations, informal groupings play a significant role in energy governance. The Group of Seven (G7) and the Group of Twenty (G20) have repeatedly addressed energy security, fossil fuel subsidies, and clean energy transitions. At the G20 summit in Pittsburgh in 2009, leaders committed to "rationalize and phase out over the medium term inefficient fossil

fuel subsidies that encourage wasteful consumption". The G7's Climate Club, launched in 2022–2023, seeks to accelerate the decarbonisation of industry. The Clean Energy Ministerial and Mission Innovation bring together energy ministers to promote clean energy technologies. Just Energy Transition Partnerships (JETPs), concluded between groups of donor states and South Africa (2021), Indonesia (2022), Vietnam (2022), and Senegal (2023), combine public and private finance to support the transition away from coal. Such informal arrangements lack legal personality and binding powers but can mobilise political commitment and finance.

4.2 The International Energy Agency and Its Institutional Role

Origins and Legal Foundation

The International Energy Agency was created in November 1974 in response to the oil embargo imposed by Arab members of OPEC in October 1973 following the Yom Kippur War and the associated quadrupling of oil prices. The embargo exposed the vulnerability of industrialised oil-importing states and the absence of any mechanism for coordinating their response. At the initiative of the United States, sixteen OECD states signed the Agreement on an International Energy Program (IEP Agreement) on 18 November 1974, establishing the IEA as an autonomous agency within the framework of the OECD.

The IEP Agreement is a treaty. Membership of the IEA is open only to members of the OECD that are willing and able to meet the requirements of the emergency response mechanism, including the holding of oil stocks. This requirement has been a significant barrier to the membership of major emerging economies such as China and India, which are not OECD members. To address this limitation, the IEA launched an "association" initiative in 2015, under which China, India, Indonesia, Brazil, South Africa, and other states have become association countries participating in IEA work without assuming the obligations of full membership.

Emergency Response Mechanism

The core legal obligation imposed by the IEP Agreement is the maintenance of emergency oil reserves. Article 2 requires each participating country to maintain emergency reserves sufficient to sustain consumption for at least sixty days with no net oil imports, a requirement subsequently increased to ninety days of net imports. The reserves may be held by governments (as in the United States Strategic Petroleum Reserve), by dedicated stockholding agencies, or by industry under obligation.

The IEP Agreement also established an emergency sharing system, triggered when the group or an individual participating country sustains or can reasonably be expected to sustain a reduction in the daily rate of its oil supplies of at least seven

per cent of the average daily rate of its final consumption during a base period. Upon activation, participating countries are required to implement demand restraint measures and to share available oil. In practice, the formal emergency sharing system has never been activated. Instead, the IEA has relied on "coordinated emergency response measures" (CERM), a more flexible mechanism under which members make stocks available in coordinated releases. Coordinated stock releases were undertaken in 1991 (Gulf War), 2005 (Hurricanes Katrina and Rita), 2011 (Libyan civil war), and twice in 2022 (following the Russian invasion of Ukraine), the latter being the largest in the Agency's history.

Evolution of the IEA's Mandate

Although established principally to address oil supply security, the IEA's mandate has broadened considerably. Its *World Energy Outlook*, published annually, is among the most influential sources of energy data and scenarios. In 2021, the IEA published *Net Zero by 2050: A Roadmap for the Global Energy Sector*, which concluded that achieving net zero emissions by 2050 would require no new oil and gas fields approved for development beyond those already committed as of 2021, and no new coal mines or mine extensions. The report was controversial among producers and some IEA members, illustrating the tensions between the Agency's historical mission of supply security and its newer role in promoting clean energy transitions. In 2022, IEA ministers mandated the Agency to help countries achieve net zero emissions, and the Agency has developed significant work programmes on critical minerals security, including the Critical Minerals Security Programme.

The IEA and Critical Minerals Security

The IEA's recent expansion into critical minerals illustrates how institutions created for one security problem adapt to another. The Agency's 2021 special report, *The Role of Critical Minerals in Clean Energy Transitions*, and its subsequent annual *Global Critical Minerals Outlook* documented the high geographical concentration of the extraction and, especially, the processing of minerals such as lithium, cobalt, nickel, graphite, and rare earth elements, and warned that supply disruptions could slow the energy transition and raise its cost. In 2024, IEA members launched the Critical Minerals Security Programme, which includes work on supply diversification, market transparency, and the feasibility of stockholding arrangements for critical minerals analogous in some respects to emergency oil stocks.

The analogy with oil is, however, imperfect. Oil is a fungible commodity traded on deep and liquid markets, whereas many critical minerals are traded in thin, opaque markets with product specifications tailored to individual buyers; stockholding is therefore more complex and potentially more expensive. More-

over, the security risks associated with critical minerals arise less from sudden supply interruptions than from structural concentration and the use of export controls as instruments of industrial and geopolitical strategy. Whether the IEA's legal toolkit, built around the IEP Agreement's oil-specific obligations, can be extended to minerals without treaty amendment is a significant institutional question. Chapter 8 examines the substantive law of critical minerals.

Legal Assessment

The IEA is a hybrid institution. Its emergency response obligations are legally binding treaty commitments, but its broader influence derives from its analytical authority, convening power, and technical expertise rather than from legal compulsion. Its limited membership constrains its legitimacy as a global institution, and its identification with consuming states limits its acceptance by producers. Nevertheless, it remains the most important institution for consumer-country cooperation and an indispensable source of the data on which energy policy and legal analysis depend.

4.3 OPEC and Petroleum Market Coordination

Origins and Legal Foundation

The Organization of the Petroleum Exporting Countries was founded at the Baghdad Conference of 10–14 September 1960 by Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela. Its Statute, adopted in 1961 and subsequently amended, defines its principal aims as "the co-ordination and unification of the petroleum policies of Member Countries and the determination of the best means for safeguarding their interests, individually and collectively" and the devising of "ways and means of ensuring the stabilization of prices in international oil markets with a view to eliminating harmful and unnecessary fluctuations" (Article 2). OPEC is headquartered in Vienna. Its membership has varied over time; Nigeria joined in 1971. Qatar left in 2019, Ecuador in 2020, and Angola in 2024.

The supreme authority of OPEC is the Conference, composed of delegations of the member countries, which meets at least twice a year and takes decisions by unanimity of full members (Article 11). The Board of Governors directs the management of the Organization, and the Secretariat, headed by the Secretary General, carries out executive functions.

Production Quotas and Market Management

OPEC's most visible function is the setting of production ceilings for its members. The system of production quotas was formally introduced in 1982 in response to falling prices and has been the principal instrument of OPEC market management since then. Quota decisions are adopted by the Conference and are politically binding on members, but the Statute provides no mechanism

for enforcing compliance, and over-production by members has been a recurring feature of OPEC history.

Since December 2016, OPEC members have coordinated production with ten non-OPEC producers, including the Russian Federation, Kazakhstan, Mexico, and Oman, under the Declaration of Cooperation, in an arrangement known as "OPEC+". In July 2019, the participants adopted the Charter of Cooperation, providing a framework for continued cooperation. OPEC+ has implemented significant production cuts, including an unprecedented cut of 9.7 million barrels per day agreed in April 2020 in response to the collapse in demand caused by the COVID-19 pandemic.

Quota Allocation and Baseline Disputes

The allocation of production quotas among members is among the most contentious aspects of OPEC's work, because the "baseline" from which each member's reductions are calculated determines its share of the collective market. Disputes over baselines have repeatedly strained the organisation. In 2023, OPEC+ participants agreed that the maximum sustainable production capacities of several members would be assessed by independent sources for the purpose of setting baselines for 2025, reflecting disagreement over whether some African producers, which had been persistently unable to produce up to their quotas, should have their baselines reduced. Angola announced its withdrawal from OPEC in December 2023, citing disagreement with its assigned production level.

For Nigeria, the quota system interacts with domestic production constraints. For extended periods in the early 2020s, Nigeria produced well below its OPEC allocation as a result of crude oil theft, pipeline vandalism, underinvestment, and the divestment of onshore assets by international oil companies. These constraints meant that the binding limit on Nigerian production was operational rather than legal, and they focused domestic policy attention on the security of pipeline infrastructure and the regulatory framework for onshore asset transfers under the Petroleum Industry Act 2021. The episode demonstrates that the legal significance of an international institution's decisions depends on the material circumstances of its members.

The Legal Character of OPEC Decisions

The legal character of OPEC Conference resolutions has never been authoritatively determined. They are adopted by the organ of an international organisation established by treaty, and in that sense they are acts of international law. Yet the Statute does not expressly confer on the Conference the power to adopt decisions binding on members regarding production, and members have consistently treated quota decisions as political commitments rather than legal obligations enforceable by sanctions. The OPEC+ Declaration of Cooperation and Charter

of Cooperation are, by their terms and the practice of the participants, non-binding instruments. The absence of legal enforcement is consistent with the general preference of states for flexibility in matters touching resource sovereignty, and it explains why OPEC's effectiveness has historically depended on the willingness of one or more members — usually Saudi Arabia — to act as "swing producer" by adjusting its own output to stabilise the market.

Legal Issues Surrounding OPEC

OPEC's activities raise several legal questions. The first concerns competition law. OPEC's coordination of production would, if carried out by private companies, constitute a cartel prohibited by the competition laws of most states. Private plaintiffs in the United States have sought to sue OPEC and its members under the Sherman Act. In *International Association of Machinists and Aerospace Workers v OPEC* (1981), the Ninth Circuit Court of Appeals affirmed the dismissal of such a suit on the basis of the act of state doctrine.[1] Legislation known as the "No Oil Producing and Exporting Cartels Act" (NOPEC), which would remove sovereign immunity and the act of state defence for such claims, has been introduced repeatedly in the United States Congress but has not been enacted.

The second concerns trade law. Production quotas are, in effect, restrictions on the production and therefore the export of oil, raising questions of compatibility with Article XI of the GATT, which prohibits quantitative restrictions on exports. Most OPEC members are members of the WTO, but no WTO challenge to OPEC production quotas has been brought. Arguments that quotas could be justified under Article XX(g) as measures relating to the conservation of exhaustible natural resources, or that production restrictions differ legally from export restrictions, have been advanced but not tested.

The third concerns the relationship between OPEC decisions and contractual obligations. Production quotas may require the restriction of production by joint ventures involving foreign companies, raising questions about whether quota compliance constitutes *force majeure* or a change in law under production sharing contracts and joint operating agreements. Many Nigerian production sharing contracts and joint venture arrangements contain provisions allocating the burden of OPEC-mandated production restrictions.

OPEC and the Energy Transition

OPEC faces a fundamental strategic challenge in the energy transition. If global oil demand peaks and declines, the value of members' reserves will fall, and competition for market share may intensify. OPEC has consistently argued that oil will remain a significant part of the global energy mix for decades and has warned

[1] *International Association of Machinists and Aerospace Workers v Organization of Petroleum Exporting Countries*, 649 F 2d 1354 (9th Cir 1981).

against underinvestment in oil supply. Within the UNFCCC process, OPEC members — particularly Saudi Arabia — have often resisted language calling for a phase-out of fossil fuels, and the formulation adopted at COP28, calling for "transitioning away from fossil fuels in energy systems", reflected a compromise in which producer-state resistance played a central role.

4.4 The International Renewable Energy Agency

Establishment and Legal Foundation

The International Renewable Energy Agency was established by the Statute of IRENA, opened for signature in Bonn on 26 January 2009 and entering into force on 8 July 2010. Its headquarters are in Masdar City, Abu Dhabi, in the United Arab Emirates — a symbolically significant choice, locating the principal international institution for renewable energy in a major oil-exporting state. Membership is open to all states members of the United Nations and to regional intergovernmental economic integration organisations; the European Union is a member. With approximately 170 members, IRENA has one of the broadest memberships of any energy institution.

Article II of the Statute defines IRENA's objective as promoting "the widespread and increased adoption and the sustainable use of all forms of renewable energy", taking into account national and domestic priorities and benefits derived from a combined approach of renewable energy and energy efficiency measures, and the contribution of renewable energy to environmental preservation, climate protection, economic growth and social cohesion, access to and security of energy supply, regional development, and intergenerational responsibility. Article III defines renewable energy to include bioenergy, geothermal energy, hydropower, ocean energy (including tidal, wave, and ocean thermal energy), solar energy, and wind energy.

Functions and Activities

IRENA's functions, set out in Article IV, are advisory and facilitative. They include analysing and systematising renewable energy practices, policy instruments, investment mechanisms, and best practices; providing policy advice and assistance to members on request; promoting knowledge and technology transfer; providing capacity building and education; and stimulating research and networking. IRENA publishes influential reports, including the *World Energy Transitions Outlook*, annual *Renewable Capacity Statistics*, and *Renewable Power Generation Costs*, the last of which has documented the dramatic decline in the costs of solar photovoltaic and wind power over the past decade.

IRENA has no power to adopt binding decisions. Its influence derives from its analytical work, its convening role, and its support for member states in designing renewable energy policies. In the context of COP28 in 2023, IRENA played a

central role in tracking progress towards the pledge, endorsed in the global stocktake decision, to triple global renewable energy capacity and double the global average annual rate of energy efficiency improvements by 2030.

Assessment

IRENA represents a distinctive model of international energy governance: a universal-membership institution with a promotional rather than regulatory mandate. Its strengths lie in its legitimacy, its expertise, and its capacity to support developing states. Its limitations lie in the absence of binding powers and of mechanisms for addressing the legal barriers to renewable energy deployment — such as trade disputes over local content requirements and investment disputes over the withdrawal of incentives — which are governed by other regimes.

4.5 The International Atomic Energy Agency

Establishment and Mandate

The International Atomic Energy Agency was established by its Statute, approved on 23 October 1956 and entering into force on 29 July 1957. Its creation followed President Eisenhower's "Atoms for Peace" address to the United Nations General Assembly in December 1953, which proposed an international agency to promote the peaceful uses of atomic energy while preventing the diversion of nuclear materials to military purposes. The IAEA is headquartered in Vienna and is an autonomous organisation within the United Nations system, reporting annually to the General Assembly and, where appropriate, to the Security Council.

Article II of the Statute defines the Agency's dual objective: to "seek to accelerate and enlarge the contribution of atomic energy to peace, health and prosperity throughout the world" and to "ensure, so far as it is able, that assistance provided by it or at its request or under its supervision or control is not used in such a way as to further any military purpose". This dual mandate — promotion and control — has shaped the Agency's work and generated tensions between its developmental and verification roles.

Safeguards

The IAEA's most significant legal function is the application of safeguards to verify that nuclear material is not diverted from peaceful uses to nuclear weapons or other nuclear explosive devices. Under Article III of the Treaty on the Non-Proliferation of Nuclear Weapons 1968 (NPT), each non-nuclear-weapon state party undertakes to accept safeguards, as set forth in an agreement to be negotiated and concluded with the IAEA, for the exclusive purpose of verification of the fulfilment of its obligations under the Treaty. These comprehensive safeguards agreements, based on a model document known as INFCIRC/153 (Corrected),

apply to all source or special fissionable material in all peaceful nuclear activities within the territory of the state.

Following the discovery in 1991 of Iraq's clandestine nuclear weapons programme, which comprehensive safeguards had failed to detect, the IAEA Board of Governors approved in 1997 the Model Additional Protocol (INFCIRC/540), which significantly expands the information that states must provide and the access that inspectors enjoy, including complementary access to sites. The Additional Protocol is voluntary but has been concluded by a large majority of states with comprehensive safeguards agreements.

Where the Agency finds that a state is in non-compliance with its safeguards agreement, the Board of Governors may report the non-compliance to all members and to the Security Council and General Assembly (Statute, Article XII.C). The Board found Iran in non-compliance in 2005, leading to referral to the Security Council and the adoption of sanctions resolutions, and later to the Joint Comprehensive Plan of Action of 2015, under which the IAEA was entrusted with extensive monitoring and verification responsibilities. The Board again found Iran in breach of its safeguards obligations in June 2025.

Safety Standards

Article III.A.6 of the Statute authorises the Agency "to establish or adopt ... standards of safety for protection of health and minimization of danger to life and property". The IAEA Safety Standards Series — comprising Safety Fundamentals, Safety Requirements, and Safety Guides — is not binding in itself, but the standards are binding on the Agency in relation to its own operations and on states in relation to operations assisted by the Agency, and they are widely incorporated into national regulation. The Convention on Nuclear Safety 1994 and the Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management 1997, both adopted under IAEA auspices, establish binding obligations reinforced by peer review mechanisms. These instruments are examined in Chapter 11.

Nuclear Newcomers and the Milestones Approach

A growing number of states without existing nuclear power programmes — including several in Africa — have expressed interest in developing nuclear energy. The IAEA supports such "newcomer" states through its Milestones Approach, which identifies nineteen infrastructure issues (including the national position, nuclear safety, management, funding and financing, the legal framework, safeguards, the regulatory framework, radiation protection, the electrical grid, human resource development, stakeholder involvement, site and supporting facilities, environmental protection, emergency planning, nuclear security, the nuclear fuel cycle, radioactive waste management, industrial involvement, and procurement)

that must be addressed across three phases of programme development, each concluding with a milestone. The Agency conducts Integrated Nuclear Infrastructure Review (INIR) missions to assess progress.

The legal framework is an explicit component of the Milestones Approach. A newcomer state is expected to adhere to the principal international nuclear instruments and to enact comprehensive national nuclear legislation establishing an independent regulatory body, licensing requirements, nuclear liability rules, and provisions on safeguards, security, and waste management. In Nigeria, the Nuclear Safety and Radiation Protection Act 1995 established the Nigerian Nuclear Regulatory Authority as the regulator for nuclear safety and radiological protection, while the Nigeria Atomic Energy Commission, established under the Nigeria Atomic Energy Commission Act 1976, is responsible for promoting the development of atomic energy. Ghana, Kenya, and Uganda are among the other African states that have engaged extensively with the IAEA on nuclear power infrastructure. The interaction between IAEA guidance and national legislation is a good illustration of how the soft law outputs of an international organisation shape domestic legal frameworks.

Assessment

The IAEA is the most legally powerful of the energy-specific international organisations, by virtue of its safeguards function and its role in the non-proliferation regime. Its experience demonstrates both the possibilities and the limits of international verification: safeguards have been strengthened in response to failures, but the Agency depends on the cooperation of states and on the Security Council for enforcement. Its promotional role has expanded with renewed interest in nuclear energy as a low-carbon source, including small modular reactors.

4.6 Regional Energy Regulatory Institutions

The European Union

The European Union possesses the most developed regional energy governance system in the world. Article 194 of the Treaty on the Functioning of the European Union, introduced by the Treaty of Lisbon in 2009, provides that Union policy on energy shall aim, in a spirit of solidarity between member states, to ensure the functioning of the energy market, ensure security of energy supply, promote energy efficiency and energy saving and the development of new and renewable forms of energy, and promote the interconnection of energy networks. Article 194(2) preserves "a Member State's right to determine the conditions for exploiting its energy resources, its choice between different energy sources and the general structure of its energy supply", a significant limitation on Union competence that reflects the sovereignty sensitivities discussed above.

EU energy law has developed through successive legislative packages. The Third Energy Package of 2009 required the unbundling of transmission system operators from generation and supply activities, strengthened the independence of national regulatory authorities, and established the Agency for the Cooperation of Energy Regulators (ACER) and the European Networks of Transmission System Operators for Electricity and Gas (ENTSO-E and ENTSOG). The Clean Energy for All Europeans Package of 2019 updated the framework for electricity markets, renewable energy, energy efficiency, and governance, including the Regulation on the Governance of the Energy Union and Climate Action (EU) 2018/1999, which requires member states to adopt integrated national energy and climate plans. The "Fit for 55" package, adopted between 2021 and 2023, aligned EU energy law with the target of reducing net greenhouse gas emissions by at least 55 per cent by 2030 relative to 1990 levels, established by the European Climate Law, Regulation (EU) 2021/1119, which also enshrines the objective of climate neutrality by 2050. The REPowerEU plan of 2022 sought to end dependence on Russian fossil fuels, and in 2024 the Union adopted reforms of electricity market design and a hydrogen and decarbonised gas market package.

The Court of Justice of the European Union has played a major role in shaping EU energy law, including through cases on unbundling, the independence of regulators, state aid for renewable energy (*PreussenElektra*, 2001), and the principle of energy solidarity. In *Germany v Poland* (2021), the Court held that the principle of energy solidarity in Article 194(1) is a legally binding principle that requires the EU institutions and member states to take into account the interests of other member states in energy security, annulling a Commission decision on the OPAL gas pipeline that had failed to examine the effects on Poland's security of supply.[2]

The Energy Community

The Energy Community, established by the Treaty Establishing the Energy Community signed in Athens on 25 October 2005, extends the EU's internal energy market rules to contracting parties in South-East Europe and the Black Sea region, currently comprising Albania, Bosnia and Herzegovina, Georgia, Kosovo (designation without prejudice to positions on status), Moldova, Montenegro, North Macedonia, Serbia, and Ukraine. Contracting parties undertake to implement the relevant EU *acquis* in energy, environment, competition, and renewables. The Energy Community has a Secretariat in Vienna and a dispute settlement procedure under which the Ministerial Council may declare that a contracting party has failed to comply with its obligations. The Energy Community has been

[2] Case C-848/19 P, *Federal Republic of Germany v Republic of Poland*, EU:C:2021:598 (CJEU, 15 July 2021).

a significant vehicle for the integration of Ukraine's energy system with that of the European Union, which culminated in the synchronisation of the Ukrainian and Moldovan electricity grids with the continental European grid in March 2022.

African Regional Institutions

Africa has developed a growing network of regional energy institutions, reflecting both the continent's vast energy resources and the imperative of expanding energy access.

The African Union. The AU's Agenda 2063 identifies energy as a priority, and the African Energy Commission (AFREC), established by a Convention adopted in 2001 and in force since 2008, is the AU's specialised agency for energy. The Programme for Infrastructure Development in Africa (PIDA) identifies priority regional energy projects. The African Single Electricity Market (AfSEM), launched in 2021, aims to create the world's largest electricity market by integrating Africa's regional power pools. The Agreement Establishing the African Continental Free Trade Area (AfCFTA), signed in 2018 and in force since 2019, has implications for trade in energy goods and services.

ECOWAS. In West Africa, the Economic Community of West African States has established the ECOWAS Energy Protocol (2003), modelled on the ECT, which provides for investment protection, trade, and transit in the energy sector; the ECOWAS Regional Electricity Regulatory Authority (ERERA), established in 2008 and based in Accra, which regulates cross-border electricity exchanges; the West African Power Pool (WAPP), a specialised institution of ECOWAS established in 1999 and given its current status in 2006, which coordinates the development of regional transmission infrastructure and the regional electricity market; and the ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), based in Praia. The West African Gas Pipeline, which transports Nigerian gas to Benin, Togo, and Ghana, is governed by a treaty concluded among the four states in 2003 and by an International Project Agreement with the pipeline company. The proposed Nigeria–Morocco Gas Pipeline and the Trans-Saharan Gas Pipeline would extend this infrastructure substantially.

Southern and Eastern Africa. The Southern African Power Pool (SAPP), established in 1995, is the most mature of the African power pools and operates a competitive regional electricity market. The Regional Electricity Regulators Association of Southern Africa (RERA) supports regulatory cooperation. The Eastern Africa Power Pool (EAPP) and the Central African Power Pool (CAPP) pursue similar objectives in their regions.

Development Finance Institutions as Energy Governors

Multilateral development banks and other development finance institutions are among the most influential actors in energy governance in developing countries,

although they are rarely described as energy institutions. Their influence operates through several channels: the conditions attached to policy-based lending, which have historically encouraged power sector unbundling, privatisation, and the establishment of independent regulators; the environmental and social standards attached to project finance, notably the World Bank's Environmental and Social Framework (2018) and the IFC Performance Standards; the provision of guarantees and political risk insurance, including through the Multilateral Investment Guarantee Agency; and the provision of technical assistance for the drafting of legislation and model contracts.

Development banks have also become central to the financing of energy access. Mission 300, launched by the World Bank Group and the African Development Bank in 2024, aims to connect 300 million people in Africa to electricity by 2030 through a combination of grid extension, mini-grids, and off-grid solutions. At the Mission 300 Africa Energy Summit held in Dar es Salaam in January 2025, participating governments presented national energy compacts setting out policy and regulatory reforms intended to attract investment. The African Development Bank's Desert to Power initiative seeks to develop solar generation across the Sahel. Such initiatives blend finance with policy conditionality, and their legal dimension — the reforms to tariff regulation, licensing, and procurement that they require — is a significant but understudied aspect of energy law in Africa.

The energy lending policies of development banks have also become a site of contestation over fossil fuel finance. The World Bank Group announced in 2017 that it would cease financing upstream oil and gas after 2019, save in exceptional circumstances, and the European Investment Bank's 2019 energy lending policy ended financing for most unabated fossil fuel projects. Developing-country governments, including Nigeria, have argued that such policies unfairly constrain the use of natural gas as a transition fuel in countries with low per capita emissions and large energy access deficits, an argument that connects institutional policy to the principle of common but differentiated responsibilities discussed in Chapter 18.

Other Regional Arrangements

Elsewhere, regional cooperation in energy takes a variety of forms. The Association of Southeast Asian Nations (ASEAN) has pursued the ASEAN Power Grid and the Trans-ASEAN Gas Pipeline projects through memoranda of understanding. In Latin America, the Central American Electrical Interconnection System (SIEPAC) is underpinned by the Framework Treaty for the Electricity Market of Central America (1996), which established a regional regulator, the Comisión Regional de Interconexión Eléctrica (CRIE). The Gulf Cooperation Council Interconnection Authority operates a regional transmission grid linking the six GCC states.

COMPARATIVE PERSPECTIVE

Regional integration models. The European Union represents a *supranational* model, in which binding legislation, an independent regional regulator (ACER), and a supreme court (the CJEU) drive deep market integration. The Energy Community represents an *acquis-export* model, extending EU rules to neighbouring states through treaty commitments. ECOWAS and SADC represent *intergovernmental* models, in which regional regulators and power pools coordinate national systems with more limited binding authority, and in which the pace of integration depends heavily on the construction of physical infrastructure and on the financial health of national utilities. The choice of model reflects not only political preferences regarding sovereignty but also the maturity of national markets and institutions.

4.7 Independent Regulators and Administrative Accountability

The Rise of the Independent Regulator

At the national level, the most significant institutional development in energy governance over the past four decades has been the spread of independent regulatory agencies. Before the liberalisation era, the electricity and gas sectors in most countries were dominated by vertically integrated, state-owned monopolies that were, in effect, self-regulating, subject to ministerial oversight. The petroleum sector was typically regulated by a ministry, sometimes acting through a national oil company that combined commercial and regulatory functions.

The liberalisation and privatisation of energy markets, beginning in the United Kingdom, Chile, and New Zealand in the 1980s and spreading globally in the 1990s, required new regulatory institutions. Where private companies own and operate natural monopoly networks, or where competition is introduced into generation and supply, an independent regulator is needed to set tariffs, ensure non-discriminatory network access, license market participants, monitor market conduct, and protect consumers. The model of the independent regulatory agency, long established in the United States in the form of state public utility commissions and the Federal Power Commission (later the Federal Energy Regulatory Commission), was transplanted across the world, frequently with the encouragement of the World Bank and other development institutions.

In Nigeria, the Electric Power Sector Reform Act 2005 established the Nigerian Electricity Regulatory Commission (NERC), now continued under the Electricity Act 2023. The Petroleum Industry Act 2021 established the NUPRC and the NMDPRA as regulators separate from the commercial NNPC Ltd, replacing the Department of Petroleum Resources, the Petroleum Products Pricing Regulatory Agency, and the Petroleum Equalisation Fund (Management) Board.

The Rationale for Independence

Several justifications have been advanced for the independence of energy regulators.

Credible commitment. Energy infrastructure is capital-intensive and long-lived, and investors require assurance that tariffs will allow them to recover their investment and earn a reasonable return. Political control of tariffs creates a risk that governments will hold down prices for electoral reasons, expropriating investors' sunk capital. Delegating tariff-setting to an independent regulator with a statutory mandate and secure tenure is a means by which governments can credibly commit not to behave opportunistically.

Expertise. Energy regulation is technically complex, requiring expertise in economics, engineering, accounting, and law. Specialised agencies can develop and retain such expertise more effectively than ministries.

Separation of roles. Where the state owns energy companies, independence separates the state's role as owner from its role as regulator, reducing conflicts of interest and promoting a level playing field between state-owned and private participants. The separation of the regulatory and commercial functions of national oil companies is a central theme of petroleum sector reform.

Insulation from capture. Independence is also intended to protect regulators from capture by the industries they regulate, although independence from government does not of itself guarantee independence from industry.

Elements of Independence

The independence of a regulator depends on several institutional features:

- **Legal foundation**: establishment by statute with a clearly defined mandate, rather than by executive decision;
- **Appointment and tenure**: appointment through a transparent process, fixed terms of office, and protection against removal except for defined cause;
- **Financial autonomy**: funding through levies on regulated entities or secure budgetary allocations, rather than discretionary appropriations;
- **Decision-making autonomy**: authority to take decisions without ministerial direction on individual matters, subject to general policy guidance;
- **Organisational autonomy**: control over staffing and internal organisation.

EU law imposes detailed independence requirements on national regulatory authorities. Under Article 57 of Directive (EU) 2019/944 on common rules for the internal market for electricity, member states must guarantee the independence of regulatory authorities and ensure that they exercise their powers impartially and transparently; regulatory staff and management must act independently from any market interest and must not seek or take direct instructions from any government or other public or private entity when carrying out regulatory tasks.

In *Commission v Germany* (2021), the Court of Justice held that Germany had failed to fulfil its obligations by conferring on the federal government powers to determine tariff methodologies that EU law reserved exclusively to the national regulatory authority.[3]

The Governance of National Oil Companies

National oil companies (NOCs) control the majority of the world's proven oil and gas reserves and are therefore among the most significant institutions in global energy governance, even though they are corporate entities rather than regulators. Their governance raises distinctive legal questions concerning the separation of commercial, policy, and regulatory functions; the relationship between the NOC and the state treasury; transparency and accountability; and the application of corporate law to entities whose shareholder is the state.

The "Norwegian model" is frequently cited as a benchmark. Under this model, policy functions are exercised by the Ministry of Energy, regulatory and resource management functions by a specialised agency (the Norwegian Offshore Directorate, as the Norwegian Petroleum Directorate was renamed in 2024), and commercial functions by a company operating on market terms (Equinor, formerly Statoil, partially listed since 2001), with the state's direct financial interests in petroleum licences managed separately by a wholly state-owned company, Petoro. Comparative research has questioned whether the Norwegian separation of functions is a necessary condition of good NOC performance, observing that it presupposes strong institutional capacity, and that some NOCs perform well under integrated models while some separated models perform poorly.

Nigeria's Petroleum Industry Act 2021 moved decisively towards a separation of functions. Section 53 provides for the incorporation of NNPC Limited under the Companies and Allied Matters Act, with ownership of its shares vested in the Federal Government and held by the Ministry of Finance Incorporated and the Ministry of Petroleum Resources on behalf of the Federation. NNPC Ltd is to operate as a commercial entity without recourse to government funding and to pay dividends to its shareholders, and the Act contemplates that its shares may in future be offered to the public. Regulatory functions previously exercised by NNPC divisions and the Department of Petroleum Resources were transferred to the NUPRC and the NMDPRA. Whether this formal separation will produce the commercial discipline and transparency its drafters intended depends on implementation, including the robustness of corporate governance at NNPC Ltd, the publication of audited accounts, and the treatment of legacy obligations such as fuel subsidy arrears.

[3] Case C-718/18, *European Commission v Federal Republic of Germany*, EU:C:2021:662 (CJEU, 2 September 2021).

Administrative Accountability

Independence must be balanced by accountability. An independent regulator exercises significant public power — it sets prices paid by millions of consumers, grants and revokes licences, and imposes penalties — and in a democratic system such power must be subject to mechanisms of accountability. These mechanisms include:

- **Legislative oversight**: reporting obligations to the legislature, appearances before parliamentary committees, and legislative approval of budgets;
- **Judicial review**: the right of affected persons to challenge regulatory decisions before courts or specialised tribunals on grounds of illegality, irrationality, and procedural impropriety, examined further in Chapter 27;
- **Procedural requirements**: obligations to consult on proposed decisions, to give reasons, to publish decisions, and to act transparently;
- **Appeals**: specialised appeal mechanisms, such as the Competition and Markets Authority's role in hearing appeals against certain decisions of Ofgem in the United Kingdom;
- **Performance review**: periodic review of regulatory performance by audit institutions or independent bodies.

The relationship between independence and accountability is a subject of ongoing scholarly debate. Some scholars emphasise the democratic deficit of independent agencies and argue for stronger political control; others emphasise the importance of independence for investment and argue that accountability is best secured through procedural transparency and judicial review. In the United States, the Supreme Court's decision in *Loper Bright Enterprises v Raimondo* (2024), overruling the *Chevron* doctrine of judicial deference to reasonable agency interpretations of ambiguous statutes, and its articulation of the "major questions doctrine" in *West Virginia v Environmental Protection Agency* (2022), have significantly altered the balance between regulatory agencies and courts, with major implications for energy and environmental regulation.[4]

Regulatory Independence in Practice: The Developing-Country Experience

The transplantation of the independent regulator model to developing countries has produced mixed results. Formal independence enshrined in legislation has often been undermined in practice by political interference in appointments, ministerial pressure on tariff decisions, inadequate funding, and limited technical capacity. Studies of electricity regulation in Africa have found that regulators frequently lack the authority to set cost-reflective tariffs in the face of political

[4] *Loper Bright Enterprises v Raimondo*, 603 US 369 (2024); *West Virginia v Environmental Protection Agency*, 597 US 697 (2022).

resistance, leading to utility financial deficits, underinvestment, and poor service quality. The Nigerian experience illustrates these challenges: NERC's efforts to implement cost-reflective tariffs under the Multi-Year Tariff Order framework have repeatedly been constrained by political considerations, and the accumulation of tariff shortfalls has required substantial government subsidies. The introduction of a "Band A" tariff structure in 2024, under which customers receiving at least twenty hours of supply daily pay higher, more cost-reflective tariffs, represents an attempt to align tariffs with service quality.

These experiences suggest that formal independence is a necessary but not sufficient condition for effective regulation. It must be supported by political commitment, adequate resources, technical capacity, and a broader governance environment that respects the rule of law. Scholars such as Brian Levy and Pablo Spiller have emphasised that regulatory design must be adapted to the institutional endowment of each country, and that in countries with weak checks and balances, contractual commitments and international guarantees may be needed to supplement regulatory independence.[5]

4.8 Regulatory Cooperation and Institutional Reform

Forms of Regulatory Cooperation

Regulatory cooperation in the energy sector takes several forms. At the bilateral level, regulators conclude memoranda of understanding to exchange information, coordinate decisions on cross-border infrastructure, and share expertise. At the regional level, regulators participate in associations such as the Council of European Energy Regulators (CEER), the Energy Regulators Regional Association (ERRA) for Central and Eastern Europe and beyond, the African Forum for Utility Regulators (AFUR), and RERA. At the global level, the International Confederation of Energy Regulators (ICER) provides a forum for cooperation among regional associations.

Regulatory cooperation is particularly important for cross-border infrastructure. The development of an interconnector or cross-border pipeline requires decisions by regulators in each jurisdiction on matters such as cost allocation, access rules, and tariffs. Within the European Union, Regulation (EU) 2022/869 on guidelines for trans-European energy infrastructure provides for the identification of projects of common interest and a framework for cross-border cost allocation decisions, with ACER empowered to decide where national regulators cannot agree.

[5] Brian Levy and Pablo T. Spiller, "The Institutional Foundations of Regulatory Commitment: A Comparative Analysis of Telecommunications Regulation" (1994) 10 *Journal of Law, Economics, and Organization* 201.

Proposals for Institutional Reform

The fragmentation of global energy governance has prompted numerous proposals for reform. These include:

Expanding the IEA. Proposals to open IEA membership to non-OECD states, particularly China and India, would require amendment of the IEP Agreement and the OECD framework. The association initiative represents an incremental step in this direction.

A World Energy Organization. Some scholars have proposed the creation of a comprehensive world energy organisation, analogous to the WTO, with universal membership and competence over energy trade, investment, and security. Such proposals face formidable political obstacles, given the divergence of producer and consumer interests and the reluctance of states to cede sovereignty over energy.

Strengthening the energy dimension of the climate regime. The UNFCCC and Paris Agreement are not energy institutions, but the decisions of the Conference of the Parties increasingly address energy directly, as in the COP28 global stocktake decision. Some scholars argue that the climate regime should become the primary forum for global energy governance; others caution that it lacks the expertise and instruments to address energy security and market issues.

Reforming the Energy Charter Treaty. The modernisation of the ECT, negotiated between 2020 and 2022 and adopted by the Energy Charter Conference in December 2024, sought to align the treaty with the Paris Agreement by permitting parties to exclude protection for fossil fuel investments, clarifying the right to regulate, and refining investment protection standards. The withdrawals of the European Union, Euratom, and several European states, and the continuing participation of other contracting parties, have left the treaty's future uncertain. These developments are examined in Chapter 14.

Networked governance. An alternative to institutional consolidation is to improve coordination among existing institutions through networked governance: joint initiatives, memoranda of understanding, shared data platforms, and the alignment of mandates. The joint work of the IEA, IRENA, the World Bank, the World Health Organization, and UN statistical bodies on tracking progress towards Sustainable Development Goal 7 illustrates this approach.

Data, Transparency, and Institutional Legitimacy

A final dimension of institutional reform concerns data and transparency. Energy governance depends on reliable data on production, consumption, stocks, prices, reserves, emissions, and investment. Much of that data is held by governments and companies that may have strategic or commercial reasons to withhold it, and the quality of data varies widely among states. The Joint Organisations Data Initiative, administered by the International Energy Forum with the participation

of APEC, Eurostat, GECF, IEA, OLADE, OPEC, and the United Nations Statistics Division, seeks to improve the transparency of oil and gas statistics. The EITI Standard requires implementing countries to disclose information along the extractive industry value chain, including contracts, licences, beneficial ownership, production, revenues, and the allocation of revenues. Satellite monitoring of methane emissions, gas flaring, and infrastructure is increasingly supplementing official data, as in the United Nations Environment Programme's International Methane Emissions Observatory.

Data has legal significance. Regulatory decisions, arbitral damages assessments, and climate litigation all depend on evidence, and the authority of institutions such as the IEA rests heavily on the credibility of their data and scenarios. Contests over data — over the reliability of emissions inventories, the assumptions of energy scenarios, or the reserves on which valuations depend — are therefore contests over institutional legitimacy. For doctoral researchers, the political economy of energy data is a promising and relatively unexplored field.

Institutional Reform and the Energy Transition

The energy transition poses fundamental challenges for energy institutions. Institutions designed for a world of centralised fossil fuel systems must adapt to a world of decentralised, digital, and decarbonised systems. Consumer-oriented institutions concerned with oil supply security must address new security challenges arising from critical mineral supply chains, cyber threats, and the variability of renewable generation. Producer-oriented institutions must address the prospect of declining demand. Regulators must design markets that accommodate variable renewable generation, storage, demand response, and distributed resources, and must balance the costs of transition against affordability for consumers. The capacity of institutions to adapt to these challenges will significantly influence the pace and fairness of the transition.

LEADING AUTHORITY

Federal Republic of Germany v Republic of Poland (OPAL), CJEU, 2021

Facts. In 2016, the European Commission approved a modification of the exemption from third-party access and tariff regulation previously granted to the OPAL pipeline, the German onshore extension of the Nord Stream pipeline, allowing Gazprom to use a greater share of OPAL's capacity. Poland challenged the decision before the General Court, arguing that it threatened Poland's security of gas supply by diverting transit flows from pipelines crossing Polish territory.

Holding. The General Court annulled the Commission's decision in 2019 for breach of the principle of energy solidarity. On appeal by Germany, the Court of Justice confirmed that energy solidarity under Article 194(1) TFEU is a general principle of EU law, which entails rights and obligations both for the EU and for the member

states and requires that the interests of all stakeholders liable to be affected be taken into account. The Commission had failed to examine the effects of its decision on the energy security of Poland and other member states.
Significance. The judgment transformed solidarity from a political aspiration into a justiciable legal principle that constrains EU energy decisions. It illustrates the role of courts in regional energy governance and the legal relevance of energy security considerations.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Designing an independent regulator (hypothetical). The fictional State of Morenia, a constituent state of a federal republic, has acquired legislative competence over its intrastate electricity market following a constitutional amendment. The Morenia State Assembly has asked you to prepare draft provisions for a Morenia Electricity Regulatory Commission Law addressing: (a) the Commission's statutory objectives, ranked in order of priority; (b) appointment, tenure, and removal of commissioners; (c) funding arrangements; (d) the relationship between the Commission and the State Commissioner for Energy, including the extent of permissible policy directions; (e) consultation, reason-giving, and publication obligations; (f) appeal and judicial review mechanisms; and (g) cooperation with the federal regulator during the transition of regulatory oversight. Accompany the draft provisions with a commentary of not more than 2,000 words explaining your choices by reference to comparative practice. *Morenia is fictitious.*

CHAPTER SUMMARY

- Global energy governance is a fragmented, polycentric regime complex, reflecting divergent producer and consumer interests, sovereignty sensitivities, path dependence, technological diversity, and shifting centres of economic gravity.
- The IEA, created in 1974, imposes binding emergency stockholding and response obligations on its OECD members and exercises broad analytical influence, but its limited membership constrains its global legitimacy.
- OPEC and the OPEC+ arrangements coordinate producer policies through politically binding but unenforceable production decisions, raising unresolved questions under competition and trade law.
- IRENA has near-universal membership and a promotional mandate without binding powers; the IAEA combines promotion with legally significant safeguards and safety functions.
- Regional institutions range from the supranational EU model to intergovernmental African power pools; the CJEU's recognition of energy solidarity as a binding principle illustrates the legal depth of EU integration.
- Independent regulators are justified by credible commitment, expertise, separation of roles, and insulation from capture, but must be accountable through

legislative oversight, judicial review, and procedural transparency; formal independence is often undermined in practice in developing countries.
- Reform proposals include expanding the IEA, creating a world energy organisation, strengthening the climate regime's energy role, modernising the ECT, and improving networked governance.

DOCTORAL RESEARCH QUESTIONS

1. Does the concept of a "regime complex" adequately explain the fragmentation of global energy governance, or does it underestimate the role of power asymmetries among states?
2. Assess the compatibility of OPEC+ production restrictions with Articles XI and XX(g) of the GATT. Why has no WTO member brought a challenge?
3. To what extent has the formal independence of energy regulators in sub-Saharan Africa translated into decisional independence in tariff-setting? Design a methodology to measure the gap.
4. How should the emergence of state-level electricity regulators in Nigeria be coordinated with federal regulation to avoid regulatory fragmentation in the national grid?
5. Could the energy solidarity principle articulated in *Germany v Poland* serve as a model for regional energy governance in ECOWAS?

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PART II

Petroleum, Gas, and Extractive Industries Law

The law governing the upstream petroleum industry, natural gas and LNG transactions, hard-rock mining, and the critical minerals on which the energy transition depends.

5 CHAPTER

International Petroleum Law and Upstream Resource Development

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal classification of petroleum resources and the stages of the upstream value chain.
2. Analyse the nature of exploration and production rights and the legal character of petroleum titles.
3. Compare licensing and concession systems with production sharing contracts and service contracts, and evaluate their respective allocation of risk and reward.
4. Analyse the structure and key provisions of joint operating agreements.
5. Evaluate the roles of national oil companies and state participation in upstream petroleum.
6. Explain the legal framework for unitisation and the development of transboundary reservoirs.
7. Analyse the legal regime governing decommissioning and abandonment liabilities.

Introduction

The upstream petroleum industry — the exploration for and production of crude oil and natural gas — is the historical core of international energy law. It was in the upstream sector that the foundational instruments of the field were developed: the concession, the licence, the production sharing contract, the joint operating agreement, and the unitisation agreement. It was disputes over upstream rights that generated the classic arbitral awards on stabilisation, internationalisation, and expropriation. And it is in the upstream sector that the tension between the sovereign interests of host states and the commercial interests of international oil companies is most acute.

The upstream sector has distinctive economic characteristics that shape its legal regime. Exploration is a high-risk activity: most exploration wells fail to discover commercial quantities of petroleum, and the costs of a single deep-water well can exceed one hundred million United States dollars. Development requires enormous capital investment, often billions of dollars, before any revenue is earned. Production continues for decades, during which prices, costs, technologies, and political conditions change unpredictably. And the resource is non-renewable and owned, in most jurisdictions, by the state. The legal instruments of the upstream sector are designed to allocate the risks and rewards of this activity between the state, as resource owner, and the investor, as provider of capital and technology.

This chapter examines the legal framework of upstream petroleum. It begins with the legal classification of petroleum resources, proceeds to the nature of exploration and production rights, and analyses the principal contractual and licensing systems. It then examines joint operating agreements, national oil companies, unitisation, and decommissioning. Throughout, Nigerian law — particularly the Petroleum Industry Act 2021 (PIA) — is used as a principal reference point, alongside the laws of the United Kingdom, Norway, and other leading jurisdictions.

5.1 Legal Classification of Petroleum Resources

What Is "Petroleum"?

The definition of "petroleum" in legislation and contracts determines the scope of rights granted and obligations imposed. Definitions vary, but most encompass naturally occurring hydrocarbons in liquid, gaseous, or solid form. Section 318 of the PIA defines "petroleum" as "crude oil, natural gas and condensates", and separately defines "crude oil" as "mineral oil, asphalt, ozokerite and all kinds of hydrocarbons and bitumen, whether in solid or in liquid form, in their natural state or obtained from oil shales, tar sands or any other mineral or rock, whether gaseous or not" and "natural gas" as "gas obtained from boreholes and wells and consisting primarily of hydrocarbons". Section 1(1) of the United Kingdom's Petroleum Act 1998 defines petroleum to include "any mineral oil or relative hydrocarbon and natural gas existing in its natural condition in strata", but expressly excludes coal and bituminous shales and other stratified deposits from which oil can be extracted by destructive distillation.

The precise definition can have significant legal consequences. Whether coal-bed methane, shale gas, oil shale, or helium falls within the definition of petroleum determines which licensing regime applies and who has the right to exploit the resource. The distinction between crude oil and natural gas matters because many

legal regimes treat them differently for fiscal purposes, ownership of associated and non-associated gas, and domestic supply obligations.

Associated and Non-Associated Gas

Natural gas may occur either in association with crude oil (associated gas), dissolved in the oil or as a gas cap above it, or in reservoirs that contain predominantly gas (non-associated gas). The distinction is legally important. Historically, many petroleum agreements granted rights to crude oil with only limited provision for associated gas, which was often flared because no market existed for it. Nigeria has long been among the world's largest gas flaring countries. The Associated Gas Re-Injection Act 1979 prohibited gas flaring without ministerial permission after 1 January 1984, but permission was routinely granted subject to payment of nominal fees. Section 104 of the PIA prohibits the flaring or venting of natural gas except in specified circumstances such as emergencies, or under an exemption granted by the NUPRC for facility start-up or for strategic operational reasons, and requires licensees and lessees to submit a natural gas flare elimination and monetisation plan. The Act further provides that payments of flare penalties shall not be cost-recoverable or tax-deductible. The Flare Gas (Prevention of Waste and Pollution) Regulations 2018 established the Nigerian Gas Flare Commercialisation Programme, under which flare gas is auctioned to third parties for commercial use.

Conventional and Unconventional Resources

A further classification distinguishes conventional resources — petroleum that can be produced from reservoirs by natural pressure, pumping, or conventional secondary recovery — from unconventional resources, such as shale oil and gas, tight gas, coal-bed methane, oil sands, and gas hydrates, which require specialised techniques such as horizontal drilling and hydraulic fracturing. The shale revolution in the United States, which transformed that country into the world's largest producer of both oil and natural gas, was made possible not only by technology but also by the American legal framework of private mineral ownership, which permitted rapid leasing and development without the need for state licensing. Elsewhere, unconventional resources have been subject to moratoria or bans on hydraulic fracturing, including in France (Law No 2011-835 of 13 July 2011), and, for a period, in the United Kingdom, where a moratorium imposed in 2019 was briefly lifted in 2022 and then reimposed.

Reserves Classification

The classification of petroleum reserves has legal and financial significance, since reserves underpin company valuations, borrowing capacity, and, in disputes, the assessment of damages. The Petroleum Resources Management System (PRMS), developed by the Society of Petroleum Engineers and other professional organ-

isations, classifies petroleum quantities according to their commerciality and the degree of uncertainty in recoverable volumes: reserves (proved, probable, and possible, commonly abbreviated as 1P, 2P, and 3P), contingent resources, and prospective resources. Securities regulators impose disclosure requirements based on reserves definitions; in the United States, the Securities and Exchange Commission's rules on oil and gas reserves disclosure apply to listed companies. The distinction between proved reserves and more speculative categories is frequently central to the valuation of expropriated petroleum assets in arbitration.

5.2 Exploration and Production Rights

The Nature of Petroleum Titles

In jurisdictions where petroleum is owned by the state, the right of an investor to explore for and produce petroleum derives from a grant by the state. The legal nature of that grant varies. It may be characterised as:

- an **administrative licence**, a permission granted by the state in the exercise of public power, which creates rights and obligations defined by statute and regulation and which may be varied or revoked in accordance with administrative law;
- a **contract**, a bilateral agreement between the state (or its agency) and the investor, creating reciprocal obligations enforceable in accordance with contract law;
- a **property right**, a real right in the petroleum or in the land, such as a lease or *profit à prendre*, enforceable against third parties; or
- a **hybrid** combining elements of each.

The characterisation has practical consequences. If the title is an administrative licence, the state may be able to vary its terms by legislation or regulation, subject to the protection afforded by administrative law and any applicable investment treaty. If it is a contract, unilateral variation will constitute a breach. If it is a property right, it may be mortgaged, assigned, and protected against third-party interference, and its taking may constitute expropriation.

In the United Kingdom, a petroleum licence granted under the Petroleum Act 1998 is a licence in the sense of a statutory permission, but it incorporates "model clauses" prescribed by regulation which take effect as terms of the licence and have a contractual character. The licence confers exclusive rights to "search and bore for, and get" petroleum within the licensed area. In *Shell UK Ltd v Enterprise Oil plc* and related litigation, English courts have treated licences as conferring valuable proprietary interests capable of assignment, subject to the consent of the regulator.

In Nigeria, petroleum titles under the PIA comprise the petroleum exploration licence (non-exclusive, for reconnaissance), the petroleum prospecting licence

(exclusive, for exploration and appraisal), and the petroleum mining lease (exclusive, for production). Under the PIA, a petroleum prospecting licence confers the exclusive right to drill exploration and appraisal wells and to carry out test production, while a petroleum mining lease confers the exclusive right to win, work, carry away, and dispose of petroleum. These titles are statutory instruments granted by the Minister on the recommendation of the NUPRC following a competitive bid round, and their terms are supplemented by the regulations made under the Act and by any associated contracts with NNPC Ltd.

Allocation of Rights: Competitive Bidding and Discretionary Award

States allocate petroleum rights through competitive bidding, discretionary award (negotiation), or first-come, first-served application systems. Competitive bidding, in which rights are awarded on the basis of bids comprising signature bonuses, work programme commitments, royalty or profit-share terms, or a combination of these, is now the dominant method in most producing states and is generally regarded as more transparent and efficient. The PIA requires that petroleum prospecting licences and petroleum mining leases be granted through a "fully transparent and competitive bid process" in accordance with regulations and guidelines issued by the NUPRC, subject to limited exceptions. Discretionary award was historically widespread in Nigeria and elsewhere and was associated with allegations of corruption, most notoriously in the award of Oil Prospecting Licence (OPL) 245, which generated criminal proceedings in Italy and Nigeria and civil litigation in England.

The design of bidding systems involves trade-offs. Signature bonuses generate immediate revenue for the state but may deter exploration in high-risk areas; work programme bidding encourages exploration activity but may lead to over-commitment; profit-share bidding defers state revenue but aligns state returns with project outcomes. Norway's licensing rounds, conducted through the Awards in Predefined Areas (APA) system for mature areas and numbered licensing rounds for frontier areas, award licences on the basis of discretionary criteria (technical competence, financial capacity, and the quality of work programmes) rather than monetary bids, reflecting a policy choice to capture rent through taxation rather than through bonuses.

Duration, Relinquishment, and Work Obligations

Petroleum titles are granted for defined periods. An exploration phase, typically lasting between three and ten years and often divided into sub-phases, is followed, upon discovery of a commercial field and approval of a development plan, by a production phase lasting twenty to thirty years or more. Under the PIA, a petroleum prospecting licence for onshore and shallow water acreage is granted for an initial exploration period of three years, which may be extended by a

further three years, with longer periods available for deep offshore and frontier acreage; a petroleum mining lease is granted for a term not exceeding twenty years, renewable in accordance with the Act.

Titles typically impose minimum work obligations — commitments to acquire seismic data and drill a specified number of wells within defined periods — backed by financial guarantees. Failure to perform the work programme may result in forfeiture of the guarantee and revocation of the title. Relinquishment provisions require the titleholder to surrender portions of the licensed area at specified intervals, ensuring that acreage that is not actively explored is returned to the state for re-allocation. The PIA provides for the relinquishment of fifty per cent of the area of a petroleum mining lease ten years after the grant, subject to specified conditions, a provision designed to address the problem of large, under-explored acreage held under legacy oil mining leases.

Assignment and Change of Control

Petroleum titles are valuable assets that are frequently transferred through farm-outs, sales, and corporate acquisitions. Because the identity, competence, and financial capacity of the titleholder are matters of public interest, most regimes require the consent of the state or regulator to any assignment of an interest in a title and, increasingly, to any change in control of the titleholder. The PIA prohibits the assignment, novation, or transfer of an interest in a licence or lease, or any change of control of the holder, without the prior consent of the Minister on the recommendation of the NUPRC. The divestment of onshore assets by international oil companies to Nigerian independents between 2021 and 2025 has generated significant regulatory scrutiny under these provisions, with the NUPRC requiring assurances concerning the purchasers' technical and financial capacity, their assumption of decommissioning and environmental liabilities, and the protection of host communities.

5.3 Petroleum Licensing and Concession Systems

Typology of Petroleum Arrangements

Petroleum arrangements between host states and investors are conventionally classified into two families: *concessionary* (or licensing) systems and *contractual* systems. Contractual systems are further divided into production sharing contracts and service contracts. The classification turns principally on the ownership of produced petroleum and the form of the state's financial return.

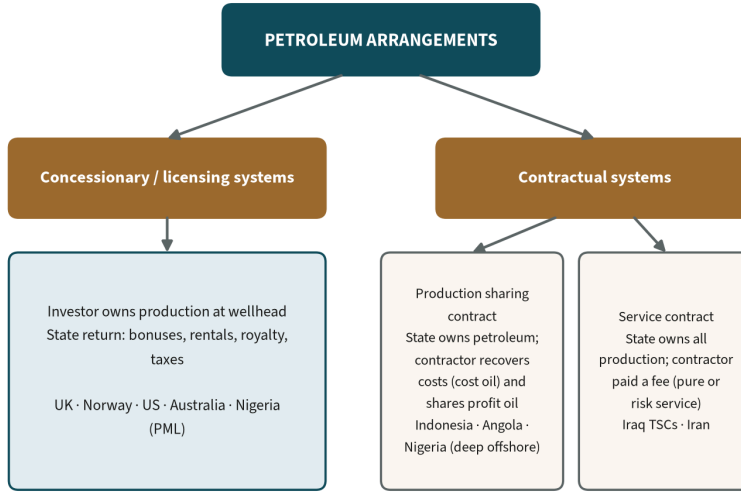


Figure 5.1 Principal families of petroleum arrangements, classified by ownership of production and the form of the state's return. *Source:* prepared for this edition.

Under a **concessionary or licensing system**, the investor acquires the exclusive right to explore for and produce petroleum within a defined area and owns the petroleum produced at the wellhead (or at another defined point), subject to the payment of royalties and taxes to the state. The state's return derives from fiscal instruments — bonuses, rentals, royalties, and taxes — rather than from a share of physical production. Modern licensing systems are used in the United Kingdom, Norway, the United States (for federal and state leases), Australia, Brazil (for its concession regime), and many other jurisdictions. Nigeria's historic oil mining lease system and the PIA's petroleum mining lease are licensing instruments, although in Nigeria they have been combined with joint ventures and production sharing contracts involving the national oil company.

Under a **production sharing contract (PSC)**, the state (usually acting through its national oil company) retains ownership of the petroleum, and the investor acts as a contractor that bears the exploration and development risk and costs. In return, the contractor is entitled to recover its costs from a portion of production ("cost oil") and to receive a share of the remaining production ("profit oil"), with the balance going to the state. Title to the contractor's share passes at a defined delivery point.

Under a **service contract**, the state retains ownership of all production, and the contractor is paid a fee for its services, either in cash or in kind. In a *pure* service contract, the fee is fixed and the state bears the risk; in a *risk* service contract, the contractor bears the exploration risk and is remunerated only if production

is achieved. Service contracts are used where constitutional or political considerations preclude any transfer of ownership of petroleum to foreign companies, notably in Iran (whose buyback contracts and the subsequent Iran Petroleum Contract reflect constitutional prohibitions on foreign ownership), Iraq (whose technical service contracts for major fields awarded from 2009 paid a fixed remuneration fee per barrel), and Mexico (prior to the 2013 reforms).

The Modern Licence

The modern petroleum licence differs fundamentally from the classical concession described in Chapter 1. Where the classical concession covered vast areas for very long periods with minimal state control, the modern licence covers a defined block for a limited exploration period, imposes work obligations and relinquishment requirements, subjects the licensee to comprehensive regulatory control over operations, and secures a substantial share of the resource rent for the state through a progressive fiscal regime. The modern licence is embedded in a statutory framework that regulates health and safety, environmental protection, resource management, decommissioning, and local content.

In the United Kingdom, for example, the North Sea Transition Authority (NSTA, formerly the Oil and Gas Authority) regulates licensing and resource management under the Petroleum Act 1998, the Energy Act 2016, and the model clauses; the Health and Safety Executive regulates safety under the Offshore Installations (Safety Case) Regulations 2015; the Offshore Petroleum Regulator for Environment and Decommissioning (OPRED), within the Department for Energy Security and Net Zero, regulates environmental matters and decommissioning; and HM Revenue and Customs administers a fiscal regime comprising ring-fence corporation tax, a supplementary charge, and the Energy Profits Levy introduced in 2022. Since 2021, the NSTA has been required to apply its "Strategy", which includes a central obligation on licensees and others to take the steps necessary to secure that the maximum value of economically recoverable petroleum is recovered from the strata beneath relevant UK waters, while also assisting the Secretary of State in meeting the net zero target.

Advantages and Disadvantages of Licensing

Licensing systems have several advantages. They are relatively simple to administer, since the state's return derives from fiscal instruments applied through the tax system. They provide investors with clear title to production, facilitating financing. They are compatible with general legislation applicable to all licensees, promoting transparency and equality of treatment. And they keep the state at arm's length from commercial operations, reducing the need for state technical capacity.

Their principal disadvantages, from the host state's perspective, are that the state has limited direct control over operations and limited access to information; that fiscal regimes based on royalties and taxes may fail to capture the full resource rent, especially when prices rise unexpectedly; and that the transfer of ownership of petroleum to foreign companies may be politically sensitive.

Local Content Obligations in Upstream Petroleum

Modern upstream regimes almost invariably impose local content obligations — requirements that licensees employ nationals, procure goods and services from domestic suppliers, use domestic facilities, and transfer technology and skills. Local content policies seek to convert petroleum extraction from an enclave activity into a driver of broader industrial development. They raise legal questions under trade law (where they discriminate against imported goods), investment law (where they impose performance requirements prohibited by some treaties), and domestic administrative law (where their enforcement is discretionary).

Nigeria's Nigerian Oil and Gas Industry Content Development Act 2010 (the NOGICD Act) is among the most comprehensive local content statutes in the world. It provides that Nigerian independent operators shall be given first consideration in the award of oil blocks, oil field licences, oil lifting licences, and all projects for which contracts are to be awarded in the Nigerian oil and gas industry, subject to the fulfilment of conditions specified by the Minister; that exclusive consideration shall be given to Nigerian indigenous service companies that demonstrate ownership of equipment, Nigerian personnel, and capacity to execute the work in bids for land and swamp operating areas; and that operators shall submit Nigerian content plans with every application for a licence, permit, or interest, demonstrating compliance with the minimum levels of Nigerian content set out in the Schedule to the Act for specified goods and services. The Act established the Nigerian Content Development and Monitoring Board to implement its provisions, and the Nigerian Content Development Fund, financed by a levy of one per cent of the value of every contract awarded in the upstream sector. Presidential directives issued in 2024 extended local content expectations to aspects of procurement and financing in the oil and gas industry.

Comparable statutes exist in Ghana (the Petroleum (Local Content and Local Participation) Regulations 2013, L.I. 2204), Angola, Brazil, Kazakhstan, and many other producer states. Empirical research suggests that local content policies are most effective when targets are realistic, linked to measurable capacity-building, and backed by investment in education, infrastructure, and access to finance; overly ambitious or poorly enforced targets may increase costs without delivering lasting capability. The design of local content provisions in licences and contracts — including the definitions of "local" goods, services, and companies,

the measurement methodology, the consequences of non-compliance, and the interaction with stabilisation clauses — is a significant area of legal practice.

5.4 Production Sharing Contracts

Origins

The production sharing contract was pioneered in Indonesia in the mid-1960s by the state oil company Permina (later Pertamina) under the leadership of Ibnu Sutowo, in response to political pressure to end the concession system and assert national control over petroleum. The Indonesian model, which provided for the state to retain ownership and management control while the foreign contractor bore the costs and risks and recovered them from production, was adopted by many other states in Asia, Africa, and Latin America. By the 1990s, the PSC had become the predominant form of petroleum arrangement in the developing world.

In Nigeria, PSCs were introduced in 1973 with Ashland Oil and became the primary vehicle for deep offshore development following the 1993 licensing round. The Deep Offshore and Inland Basin Production Sharing Contracts Act 1999 (as amended in 2019) established the fiscal framework for deep offshore PSCs, including royalty rates that decline with water depth. The 2019 amendment, which increased royalty rates and introduced a price-based royalty, was enacted after the federal government had lost significant revenue because the 1999 Act's provision for review of the royalty rates once oil prices exceeded USD 20 per barrel had never been implemented. The PIA now governs new upstream arrangements, while preserving the terms of existing PSCs subject to the conversion provisions discussed below.

Structure and Key Provisions

A typical PSC contains the following key provisions:

Parties and contract area. The PSC is concluded between the state or national oil company and the contractor (often a consortium of companies). It defines the contract area, usually by reference to geographical coordinates.

Term and phases. The PSC specifies an exploration period, divided into phases with associated work commitments, and a production period that commences upon a declaration of commerciality and approval of a development plan.

Work programme and budget. The contractor must submit annual work programmes and budgets for approval by a management committee on which the national oil company is represented.

Management committee. The management committee, composed of representatives of the national oil company and the contractor, supervises operations. The allocation of voting rights between the parties determines the degree of state control.

Cost recovery. The contractor is entitled to recover its exploration, development, and operating costs from a defined percentage of production — the "cost oil" or "cost recovery limit" — commonly between 50 and 80 per cent of production in any period. Costs that exceed the limit may usually be carried forward to future periods. Accounting procedures annexed to the PSC define which costs are recoverable and how they are to be allocated.

Profit oil split. Production remaining after royalty and cost oil — "profit oil" — is divided between the state and the contractor in accordance with an agreed formula. The split may be fixed or, more commonly, may vary according to a sliding scale based on cumulative production, daily production rates, the contractor's rate of return, or an "R-factor" (the ratio of the contractor's cumulative revenues to its cumulative costs).

Royalty and taxation. Many PSCs also require payment of a royalty and subject the contractor's share of profit oil to income tax. In some PSCs, the national oil company pays the contractor's income tax out of the state's share of profit oil (a "pay-on-behalf" arrangement), to enable the contractor to claim foreign tax credits in its home jurisdiction.

Title and lifting. Title to the contractor's share of production passes at the delivery point, and lifting arrangements govern the allocation of cargoes.

Ownership of assets. Title to equipment and facilities typically passes to the state either upon acquisition or upon full cost recovery, although the contractor retains the right to use them for operations.

Domestic supply obligations, local content, training, and stabilisation clauses are also common.

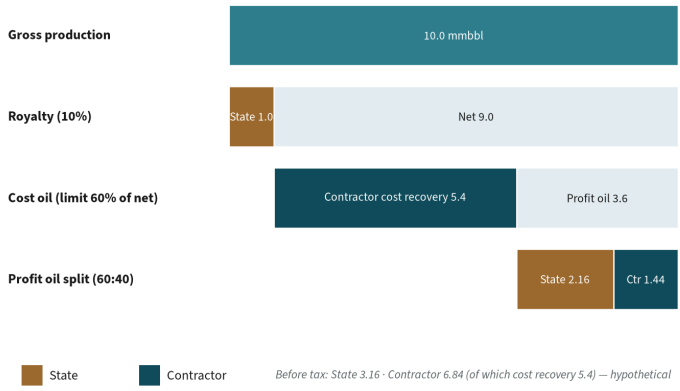


Figure 5.2 Illustrative production sharing waterfall. The percentages are hypothetical and are used solely to illustrate the mechanics of allocation. *Source:* prepared for this edition.

A Worked Illustration

The mechanics of a PSC may be illustrated with a simplified, hypothetical example. Suppose that in a given year a field produces 10 million barrels of oil, the PSC provides for a royalty of 10 per cent, a cost recovery limit of 60 per cent of production after royalty, and a profit oil split of 60:40 in favour of the state, and that the contractor has recoverable costs of 7 million barrels' worth for the year.

1. Royalty: 10 per cent of 10 million = 1 million barrels to the state, leaving 9 million barrels.
2. Cost oil: the cost recovery limit is 60 per cent of 9 million = 5.4 million barrels. The contractor recovers 5.4 million barrels; the unrecovered balance of 1.6 million barrels' worth is carried forward.
3. Profit oil: 9 million minus 5.4 million = 3.6 million barrels, of which the state receives 60 per cent (2.16 million) and the contractor 40 per cent (1.44 million).
4. Tax: if the contractor's profit oil is subject to income tax at, say, 30 per cent, a further portion of the contractor's share is payable to the state.

In this illustration, before tax the state receives 3.16 million barrels (31.6 per cent of production) and the contractor 6.84 million barrels, of which 5.4 million represent cost recovery. The "government take" — the share of the economic rent (revenues minus costs) accruing to the state — is a more meaningful measure than the share of gross production, and is the standard metric for comparing fiscal regimes. Chapter 16 examines fiscal design in more detail.

Advantages, Disadvantages, and Disputes

PSCs offer host states several advantages: formal ownership of petroleum is retained, satisfying political and constitutional requirements; the national oil company participates in management and gains access to information and expertise; and the state's share of production rises with profitability under progressive profit-sharing formulas. For investors, PSCs offer the right to recover costs from production and a share of production that can be booked as reserves under some accounting rules.

The disadvantages include administrative complexity, especially in auditing cost recovery; the incentive for contractors to "gold-plate" costs, since higher recoverable costs increase their share of production; and the potential for disputes over cost allocation, the interpretation of profit oil formulas, and the lifting of entitlements. The Nigerian experience includes significant disputes concerning the deep offshore PSCs. In the dispute concerning the Oil Mining Lease 133 (Erha) PSC, an arbitral tribunal in 2011 upheld claims by the contractors, Esso Exploration and Production Nigeria Ltd and Shell Nigeria Exploration and Production Co Ltd, against NNPC concerning the lifting of tax oil and royalty

oil. NNPC applied to the Federal High Court in Abuja, which set the award aside on the basis that the dispute concerned tax matters that were not arbitrable under Nigerian law, generating further appeals in Nigeria and parallel proceedings in the United States to enforce the award. The litigation illustrates the jurisdictional complexities that arise when fiscal provisions are embedded in contracts and the tension between the finality of arbitral awards and national courts' views on arbitrability.

The PIA and the Conversion of Existing Arrangements

The PIA introduced a framework for the voluntary conversion of existing oil prospecting licences and oil mining leases, including those underlying PSCs and joint ventures, to the new petroleum prospecting licences and petroleum mining leases. Conversion is incentivised by more favourable fiscal terms under the new regime, while holders that do not convert continue under their existing terms until expiry but cannot benefit from the new fiscal incentives and face restrictions on renewal. The Act also required the conversion of NNPC's joint venture interests into incorporated joint ventures within a specified period after its commencement. These transitional provisions illustrate a characteristic technique of petroleum law reform: the use of incentives rather than compulsion to migrate investors from legacy arrangements, thereby reducing the risk of investment treaty claims.

Stabilisation and Economic Equilibrium in PSCs

Because PSCs typically run for decades, contractors seek protection against changes in the legal and fiscal regime that would reduce the economic value of their entitlements. PSCs therefore frequently contain stabilisation provisions, which may take one of three principal forms. *Freezing clauses* provide that the law applicable to the contract is the law in force at the date of signature, and that subsequent legislation will not apply to the contractor to the extent that it is inconsistent with the contract. *Economic equilibrium clauses* permit the application of new laws but require the parties to restore the original economic balance of the contract, by adjusting fiscal terms or providing compensation, where a change in law materially affects the contractor's economic position. *Hybrid clauses* combine elements of both. Many modern PSCs prefer economic equilibrium clauses, because they preserve the state's legislative freedom while protecting the investor's economic expectations, and because freezing clauses have been criticised for insulating investors from environmental, labour, and human rights regulation.

The interaction between stabilisation and human rights has attracted particular attention. A study prepared for the United Nations Special Representative on Business and Human Rights in 2008 found that stabilisation clauses in many investment contracts in developing countries were drafted broadly enough to

exempt investors from new social and environmental legislation. The *Principles for Responsible Contracts*, endorsed by the Human Rights Council in 2011 as an addendum to the UN Guiding Principles, recommend that stabilisation clauses be drafted so as not to interfere with the state's good-faith efforts to implement laws, regulations, or policies in a non-discriminatory manner in order to meet its human rights obligations. Stabilisation clauses are examined further in Chapters 14 and 15.

Marginal Fields

Many mature producing provinces contain discovered but undeveloped fields that are too small to justify development by the original licensee. To promote their development by smaller companies, several jurisdictions have created special "marginal field" regimes. In Nigeria, the Petroleum (Amendment) Act 1996 introduced the marginal field programme, empowering the President to farm out marginal fields within an oil mining lease where they had been left unattended for not less than ten years from the date of first discovery. The programme facilitated the emergence of a cohort of Nigerian independent producers. The 2020–2021 marginal field bid round, conducted under the Marginal Field Operations (Fiscal Regime) Regulations and the Guidelines for Farm-out and Operation of Marginal Fields, awarded fifty-seven fields to Nigerian companies. The PIA continues to provide for the declaration and farm-out of fields that are not being developed, subject to the NUPRC's approval. Marginal field regimes illustrate how licensing law can be used as an instrument of industrial policy, but they also generate disputes with the original leaseholders concerning the identification of fields, access to infrastructure, and the terms of farm-out agreements.

Service Contracts in Practice

Service contracts, although less common than licences and PSCs, deserve attention because of their use in some of the world's largest oil provinces. The Iraqi technical service contracts awarded in 2009 and 2010 for super-giant fields such as Rumaila, West Qurna, and Majnoon paid contractors a fixed remuneration fee per barrel of incremental production above an agreed baseline, plus recovery of costs, with fees subject to an R-factor adjustment. Because contractors earned the same fee per barrel regardless of oil prices, the contracts transferred almost the entire price upside to the state, but they also left contractors exposed to production disruptions, export capacity constraints, and delays in cost recovery payments. Several contracts were subsequently renegotiated to lower production targets. The experience illustrates that the allocation of risk and reward in petroleum arrangements is determined not by the label attached to the contract but by the detailed economic and legal terms.

5.5 Joint Operating Agreements

Purpose of the Joint Operating Agreement

Petroleum titles are commonly held by several companies jointly, in order to spread risk, share capital costs, combine technical capabilities, and satisfy state participation requirements. The relationship among the co-venturers is governed by a joint operating agreement (JOA), a contract that establishes the rules for conducting joint operations, sharing costs and production, making decisions, and addressing default and withdrawal. The JOA does not usually create a partnership or a separate legal entity; most JOAs expressly provide that the parties' rights and obligations are several and not joint or collective, and that nothing in the agreement shall be construed as creating a partnership. The co-venturers hold undivided interests in the title as tenants in common (or the civil law equivalent), each entitled to take and separately dispose of its share of production.

The most widely used international model is the AIEN Model International Operating Agreement, first published in 1990 and most recently revised in 2012 (with subsequent updates). National models also exist, notably the UK Oil and Gas Authority (now NSTA) Master Deed and model JOAs developed under the industry's Standard Agreements initiative. In Nigeria, the historic NNPC joint ventures with international oil companies were governed by JOAs based in part on the AIEN model and in part on bespoke terms.

Key Provisions

Scope and participating interests. The JOA defines the joint property, the contract area, and each party's participating interest, which determines its share of costs, production, and voting power.

The operator. One party is designated as operator, responsible for conducting joint operations on behalf of all parties. The operator acts as an agent of the non-operators in some respects but is typically not entitled to profit from its role, recovering its costs through the joint account (including overhead charges) on a "no gain, no loss" basis. The JOA defines the operator's duties — commonly to conduct operations in a diligent, safe, and efficient manner in accordance with good oilfield practice — and limits its liability. A key feature of most international JOAs is that the operator is liable to the non-operators only for losses resulting from its "gross negligence or wilful misconduct", and in some forms only where such conduct is that of senior supervisory personnel. The interpretation of these terms has been considered in English cases, and their meaning depends heavily on the precise contractual wording.

Operating committee. Decisions on work programmes, budgets, and major operational matters are taken by an operating committee in which each party is represented. The JOA sets out voting thresholds, often requiring a majority

of participating interests together with a minimum number of parties ("pass marks"), and identifies matters requiring unanimity.

Work programmes, budgets, and authorisations for expenditure. The operator prepares annual work programmes and budgets for approval, and individual items of expenditure above specified thresholds require authorisations for expenditure (AFEs).

Sole risk and non-consent operations. Where a party wishes to undertake an operation that the operating committee has not approved, the JOA may permit it to proceed at its sole risk and cost. A party that elects not to participate in an approved operation ("non-consent") may be excluded from the benefits of that operation unless it pays a substantial premium — often several hundred per cent of the costs it avoided — to re-enter. These provisions balance the interests of parties wishing to proceed against those of parties wishing to limit risk.

Default. If a party fails to pay its share of costs ("cash calls"), the JOA provides remedies for the non-defaulting parties, which may include suspension of the defaulting party's rights to vote and to receive production, and ultimately forfeiture of its participating interest. Forfeiture provisions have been challenged as penalties. In *Cavendish Square Holding BV v Makdessi* and *ParkingEye Ltd v Beavis*, decided together in 2015, the United Kingdom Supreme Court reformulated the rule against penalties, holding that the test is whether the provision is a secondary obligation that imposes a detriment on the contract-breaker out of all proportion to any legitimate interest of the innocent party in the enforcement of the primary obligation.[1] In the oil and gas context, the legitimate interest of non-defaulting parties in ensuring the continuation of joint operations has generally been regarded as supporting the enforceability of forfeiture clauses, although each clause must be assessed on its terms.

Transfer and pre-emption. The JOA restricts transfers of participating interests, commonly requiring the consent of the other parties (not to be unreasonably withheld) and granting them pre-emption rights. Pre-emption provisions have been a frequent source of disputes, particularly regarding whether they are triggered by indirect transfers through changes of control of a party.

Withdrawal, decommissioning, and confidentiality. The JOA governs the right of a party to withdraw, the allocation of decommissioning costs (often addressed in a separate decommissioning security agreement), confidentiality, and dispute resolution.

[1] *Cavendish Square Holding BV v Makdessi; ParkingEye Ltd v Beavis* [2015] UKSC 67, [2016] AC 1172.

Joint Ventures with National Oil Companies

Where the state participates in upstream operations through its national oil company, the JOA governs the relationship between the national oil company and its private partners. In Nigeria, the joint venture arrangements between NNPC and international oil companies such as Shell, ExxonMobil, Chevron, TotalEnergies, and Eni historically required NNPC to fund its share of costs (typically 55 to 60 per cent) through cash calls. Chronic failure by NNPC to meet its cash call obligations, owing to budgetary constraints, led to underinvestment and the accumulation of arrears, prompting the development of alternative funding arrangements, including modified carry agreements, under which the private partners fund NNPC's share and recover it from production. The PIA's provision for the conversion of NNPC joint ventures into incorporated joint ventures, which could raise finance independently, was intended in part to address the cash call problem.

5.6 National Oil Companies and State Participation

Forms of State Participation

States participate in upstream petroleum in several ways:

1. **Full state operation**, in which the national oil company holds and operates all petroleum rights, as in Saudi Arabia (Saudi Aramco), Kuwait (Kuwait Petroleum Corporation), and Mexico (Pemex, historically).
2. **Majority or minority participation in joint ventures**, in which the national oil company holds an equity interest alongside private partners, either funding its share (paid participation) or being carried by the private partners through exploration (carried interest), as in Nigeria's joint ventures, Angola, and Ghana, where the Ghana National Petroleum Corporation holds a carried interest of not less than 15 per cent in petroleum agreements under the Petroleum (Exploration and Production) Act 2016 (Act 919).
3. **Contractual participation**, in which the national oil company is the contracting party under a PSC or service contract and participates in management through the management committee.
4. **Indirect participation through state-owned commercial companies or holding companies** that hold the state's direct financial interests, as in Norway's Petoro.

The Dual Role Problem

National oil companies frequently combine commercial and governmental functions: they may hold equity interests in joint ventures, act as the state's concessionaire or contracting party under PSCs, perform regulatory functions such as approving work programmes or monitoring compliance, market the state's share of production, and perform social or political functions such as fuel supply and

subsidy administration. This combination of roles creates conflicts of interest and impairs both commercial performance and regulatory effectiveness. A national oil company that is both a partner in a joint venture and the regulator of its partners cannot regulate impartially; one that is required to supply fuel at subsidised prices cannot operate on commercial principles.

The separation of the commercial and regulatory roles of national oil companies has been a central objective of petroleum sector reform, as discussed in Chapter 4 in the context of the Norwegian model and the PIA. The PIA's creation of NNPC Ltd as a commercial company without regulatory functions, and its vesting of regulatory functions in the NUPRC and NMDPRA, represents Nigeria's response to the dual role problem.

National Oil Companies Abroad

Many national oil companies have become international investors in their own right, operating outside their home states. Companies such as Saudi Aramco, Petronas, the China National Petroleum Corporation, the China National Offshore Oil Corporation, Equinor, and Petrobras hold significant upstream interests abroad. When national oil companies invest abroad, questions arise about their status as investors under investment treaties, their entitlement to sovereign immunity, and their susceptibility to the host state's competition and foreign investment screening laws. The ICSID Convention permits claims by "nationals of another Contracting State", and tribunals have held that state-owned companies may qualify as nationals unless they are acting as agents of the government or discharging an essentially governmental function — the test proposed by Aron Broches and applied in *Ceskoslovenska Obchodni Banka AS v Slovak Republic* (1999) and subsequent cases. In *Beijing Urban Construction Group Co Ltd v Republic of Yemen* (2017), the tribunal applied the Broches test to hold that a Chinese state-owned enterprise was a national for the purposes of the Convention.

5.7 Unitization and Cross-Border Reservoir Development

The Problem of Shared Reservoirs

Petroleum reservoirs do not respect licence boundaries or international borders. Where a reservoir straddles the boundary between two licence areas, or between two states, independent development by each owner would lead to the inefficiencies associated with the rule of capture: competitive drilling, wasteful duplication of infrastructure, loss of reservoir pressure, and reduced ultimate recovery. The solution is unitisation: the development of the reservoir as a single unit, with costs and production shared among the owners according to agreed "tract participation" percentages reflecting the proportion of the reservoir located within each area.

Domestic Unitisation

Within a single state, unitisation of a reservoir straddling licence boundaries may be achieved voluntarily or compelled by the regulator. Most petroleum regimes empower the regulator to require unitisation where it is necessary to secure the most effective recovery of petroleum. In the United Kingdom, the model clauses empower the Secretary of State (now acting through the NSTA) to require licensees to cooperate in the development of a field extending beyond the licensed area. In Nigeria, the PIA empowers the NUPRC to direct unitisation where a petroleum reservoir extends across licence boundaries and the holders cannot agree.

A unitisation and unit operating agreement (UUOA) establishes the unit, determines tract participations, appoints a unit operator, and provides for redeterminations of tract participations as geological knowledge improves. Redeterminations are a frequent source of disputes, because a change in tract participation can transfer very large values between the parties. UUOAs therefore contain detailed procedures for redetermination, often involving independent experts whose decisions are final and binding except for manifest error or fraud.

Transboundary Unitisation

Where a reservoir straddles an international boundary, unitisation requires cooperation between the states concerned. Customary international law does not clearly establish an obligation to unitise, but state practice, reflected in numerous bilateral agreements, demonstrates a strong preference for cooperation. Many maritime boundary agreements include a "mineral deposits clause", first appearing in the 1965 Agreement between the United Kingdom and Norway relating to the Delimitation of the Continental Shelf, which obliges the parties, where a single geological petroleum structure or field extends across the dividing line, to seek to reach agreement as to the manner in which the structure or field shall be most effectively exploited and the manner in which the proceeds shall be apportioned.

The Framework Agreement concerning Cross-Boundary Petroleum Co-operation between Norway and the United Kingdom of 2005 provides a comprehensive regime for the development of cross-boundary fields, including procedures for approval of unitisation agreements by both governments, apportionment of reserves, and the use of infrastructure on one side of the boundary to develop fields on the other. Similar arrangements exist in the Gulf of Mexico under the Agreement between the United States and Mexico concerning Transboundary Hydrocarbon Reservoirs in the Gulf of Mexico (2012).

Joint Development Zones

Where states cannot agree on a maritime boundary, they may establish a joint development zone (JDZ) in the disputed area, allowing petroleum activities to proceed without prejudice to their respective claims. The Nigeria–São Tomé and Príncipe Joint Development Zone, established by a treaty signed in Abuja in 2001, is administered by a Joint Ministerial Council and a Joint Development Authority, with revenues shared 60:40 in favour of Nigeria. Other examples include the Malaysia–Thailand Joint Development Area (1979) and the arrangements in the Timor Sea between Australia and Timor-Leste, which were superseded by the maritime boundary treaty of 2018 concluded following the first-ever compulsory conciliation under Annex V of UNCLOS. Joint development is examined further in Chapter 20.

5.8 Decommissioning and Abandonment Liabilities

The Decommissioning Challenge

At the end of their productive lives, petroleum installations must be decommissioned: wells must be plugged and abandoned, platforms and pipelines removed or made safe, and sites remediated. Decommissioning is technically complex and extremely expensive. Estimates of the total cost of decommissioning offshore infrastructure on the United Kingdom Continental Shelf alone run to tens of billions of pounds over the coming decades, and globally the liability runs to hundreds of billions of dollars. The energy transition is likely to accelerate decommissioning as fields become uneconomic earlier than anticipated.

Decommissioning raises four principal legal questions: what standard of decommissioning is required; who is liable to carry it out and pay for it; how the state can ensure that funds will be available when decommissioning becomes due; and how liability is affected by transfers of interests during the life of a field.

International Standards

UNCLOS Article 60(3) provides that "any installations or structures which are abandoned or disused shall be removed to ensure safety of navigation, taking into account any generally accepted international standards established in this regard by the competent international organization", and that "such removal shall also have due regard to fishing, the protection of the marine environment and the rights and duties of other States". The competent organisation is the International Maritime Organization, whose *Guidelines and Standards for the Removal of Offshore Installations and Structures on the Continental Shelf and in the Exclusive Economic Zone* (1989) require the removal of installations standing in less than 75 metres of water and weighing less than 4,000 tonnes in air (excluding the deck and superstructure), and of all installations placed after 1 January 1998 in waters of

less than 100 metres depth, while permitting partial removal of larger installations subject to specified conditions.

In the North-East Atlantic, OSPAR Decision 98/3 on the Disposal of Disused Offshore Installations, adopted in 1998 following the controversy over Shell's proposal to dispose of the Brent Spar storage buoy in deep water, prohibits the dumping and leaving wholly or partly in place of disused offshore installations, subject to derogations for the footings of large steel jackets weighing more than 10,000 tonnes and for concrete gravity-based installations, where the competent authority is satisfied that there are significant reasons why an alternative disposal option is preferable to reuse, recycling, or final disposal on land. The Brent Spar episode illustrates the role of public opinion and civil society in shaping decommissioning law.

The London Convention 1972 and its 1996 Protocol on the prevention of marine pollution by dumping also regulate the disposal of platforms at sea.

Domestic Regimes

In the United Kingdom, Part IV of the Petroleum Act 1998 establishes a detailed regime under which the Secretary of State may require persons responsible for offshore installations and pipelines to submit decommissioning programmes for approval. Section 29 notices may be served on a wide class of persons, including current and former licensees, owners of installations, and companies associated with them. The Energy Act 2008 extended the power to require security for decommissioning liabilities. Crucially, liability under Part IV can extend to former licensees: a company that has sold its interest may be called upon to carry out decommissioning if the purchaser defaults. This "residual liability" regime gives the state a powerful protection against insolvency of current operators but complicates transactions, since sellers must manage the risk of being called back.

In Nigeria, the PIA requires the holder of a petroleum mining lease to submit a decommissioning and abandonment plan and to establish a decommissioning and abandonment fund, to be maintained with a financial institution and managed in accordance with regulations issued by the NUPRC. Contributions to the fund are to be made over the life of the field, ensuring that adequate resources will be available. The PIA also provides that the obligations relating to decommissioning and abandonment are not extinguished by the transfer of an interest without the express agreement of the regulator, and the NUPRC's regulations on decommissioning and abandonment set out detailed requirements. The adequacy of decommissioning provisioning has been a central issue in the review of onshore divestments by international oil companies, given the age of much of the onshore infrastructure in the Niger Delta and the legacy of environmental contamination.

Liability Allocation in Contracts

Contracts allocate decommissioning liabilities among co-venturers and between buyers and sellers. JOAs typically provide that decommissioning costs are shared in proportion to participating interests at the time decommissioning is carried out, but require parties to provide security in advance — often through decommissioning security agreements (DSAs) — to protect against the default of any party. In asset sales, the seller typically seeks an indemnity from the buyer for decommissioning liabilities, supported by security such as letters of credit or parent company guarantees, because the seller may remain exposed to residual statutory liability. Disputes over the adequacy of security and the calculation of decommissioning cost estimates are frequent.

Decommissioning and the Energy Transition

The energy transition creates new opportunities and challenges for decommissioning. Infrastructure that would otherwise be removed may be repurposed for carbon dioxide transport and storage, hydrogen production or transport, or offshore wind support. Such repurposing requires legal frameworks that address the transfer of liability for legacy infrastructure, the long-term liability for stored carbon dioxide, and the interaction between petroleum, carbon storage, and renewable energy licensing regimes. The United Kingdom's Energy Act 2023 and the European Union's CCS Directive 2009/31/EC address some of these questions, examined in Chapter 12. At the same time, the risk that fields will be abandoned prematurely, by operators lacking the resources to decommission them properly, raises concerns about stranded liabilities that may ultimately fall on the state and taxpayers.

Onshore Abandonment and Legacy Pollution

Onshore, decommissioning intersects with the remediation of legacy pollution. In the Niger Delta, decades of oil spills, gas flaring, and inadequate decommissioning of wells and pipelines have left extensive environmental damage. The United Nations Environment Programme's *Environmental Assessment of Ogoniland* (2011) found widespread contamination of land, groundwater, and surface water and recommended a clean-up programme that it estimated would take up to thirty years. The Hydrocarbon Pollution Remediation Project (HYPREP) was established by the Federal Government in 2016 to implement the recommendations. The PIA's environmental remediation fund provisions, discussed in Chapter 17, are designed to ensure that operators provide for the remediation of environmental damage arising from their operations. The legal liability of parent companies for pollution caused by their subsidiaries in Nigeria has been addressed by English and Dutch courts, notably in *Okpabi v Royal Dutch Shell plc* (2021) and *Four Nigerian Farmers v Shell* (2021), considered in Chapter 21.

LEADING AUTHORITY

Ghana/Côte d'Ivoire Maritime Delimitation (ITLOS Special Chamber, 2017) — petroleum activities in disputed areas

Facts. Ghana had for decades licensed petroleum activities up to a line it regarded as the customary equidistance boundary with Côte d'Ivoire, and significant discoveries had been made, including the TEN fields. Côte d'Ivoire contested the boundary and claimed that Ghana's unilateral petroleum activities in the disputed area violated Côte d'Ivoire's sovereign rights and Article 83(3) of UNCLOS, which requires states, pending agreement on delimitation, to "make every effort to enter into provisional arrangements of a practical nature" and "not to jeopardize or hamper the reaching of the final agreement". In 2015 the Special Chamber had prescribed provisional measures requiring Ghana not to undertake new drilling in the disputed area.

Holding. The Special Chamber delimited the boundary using the equidistance/relevant circumstances methodology, producing a line close to that claimed by Ghana. It rejected Ghana's argument that a tacit agreement on the boundary had arisen from the parties' oil practice, but also held that Ghana had not violated Côte d'Ivoire's sovereign rights, because the activities had been carried out in an area that was ultimately attributed to Ghana, and that Ghana had not violated Article 83(3).

Significance. The decision is important for upstream petroleum practice in disputed maritime areas. It confirms that oil practice alone rarely establishes a tacit maritime boundary, and it illustrates the risks — including provisional measures halting new drilling — that investors face when operating in contested waters.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Allocating risk under a production sharing contract (hypothetical). The fictional Republic of Kandara is preparing a model PSC for its first deep-water licensing round. You are counsel to the Kandaran Ministry of Petroleum. Prepare (a) an annotated term sheet of the key fiscal provisions (royalty, cost recovery limit, profit oil split based on an R-factor, taxation, and state participation), explaining the policy rationale for each choice; (b) a short model calculation applying your terms to a hypothetical field producing 50,000 barrels per day at an assumed price of USD 70 per barrel, with stated assumptions; and (c) a note identifying the three provisions most likely to generate disputes, with drafting recommendations to reduce that risk. *Kandara, its institutions, and all figures are hypothetical and for teaching purposes only.*

CHAPTER SUMMARY

- Petroleum is defined by statute and contract; the distinctions between crude oil and gas, associated and non-associated gas, and conventional and unconven-

tional resources have significant legal and fiscal consequences, as illustrated by Nigeria's gas flaring provisions.

- Petroleum titles may be characterised as administrative licences, contracts, property rights, or hybrids; the characterisation affects the state's power to vary terms and the investor's protection.
- Competitive and transparent allocation of rights is increasingly required by law, as under the PIA, in reaction to the abuses associated with discretionary award.
- Petroleum arrangements fall into concessionary/licensing systems, production sharing contracts, and service contracts, distinguished by ownership of production and the form of the state's return; the economic substance matters more than the label.
- Joint operating agreements govern relations among co-venturers, including the operator's role and liability, operating committee decisions, sole risk, default, and transfers; NNPC cash call problems illustrate the difficulties of state participation.
- National oil companies' dual commercial and regulatory roles create conflicts of interest that reforms such as the PIA seek to resolve.
- Shared reservoirs require unitisation; transboundary fields are governed by mineral deposits clauses, framework agreements, and joint development zones.
- Decommissioning is governed by UNCLOS, IMO guidelines, regional instruments such as OSPAR Decision 98/3, and domestic regimes that increasingly require advance funding and impose residual liability.

DOCTORAL RESEARCH QUESTIONS

1. Is the conventional classification of petroleum arrangements into licences, PSCs, and service contracts analytically useful, given that fiscal terms can make any form economically equivalent to another?
2. Evaluate the PIA's incentive-based conversion regime as a technique for migrating legacy arrangements without triggering investment treaty claims.
3. Has state practice in maritime boundary agreements given rise to a customary obligation to cooperate in, or to unitise, the development of transboundary petroleum deposits?
4. How should residual decommissioning liability be designed to protect the state without deterring the transfer of mature assets to new operators better placed to extend field life?
5. Should international law impose obligations on states to ensure that fields are not abandoned without adequate remediation, and what would be the source of such obligations?

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6 CHAPTER

Natural Gas Law, LNG Transactions, and International Gas Markets

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the physical, economic, and commercial characteristics of natural gas that distinguish its legal regime from that of crude oil.
2. Analyse the regulation of gas exploration and production, including gas-specific licensing, domestic supply obligations, and flaring rules.
3. Evaluate the legal models for gas transportation and pipeline access, including negotiated and regulated third-party access and unbundling.
4. Compare the principal structures used to develop liquefied natural gas (LNG) projects and identify the legal risks associated with each.
5. Analyse the key provisions of long-term gas sales and purchase agreements, including take-or-pay, delivery, quality, and destination provisions.
6. Explain the law and practice of price review and the mechanisms by which gas contracts accommodate changing market conditions.
7. Evaluate the international legal framework governing cross-border gas pipelines and transit.
8. Assess the causes and resolution of gas supply disputes and the legal limits on regulatory intervention in gas markets.

Introduction

Natural gas occupies a distinctive and increasingly contested position in international energy law. For much of the twentieth century it was treated as an unwanted by-product of oil production, flared or reinjected because there was no economical way to bring it to market. Over the past five decades it has become a global commodity in its own right, supplying almost a quarter of world primary energy, fuelling a large share of electricity generation, providing heat for homes and industry, and serving as a feedstock for fertilisers and petrochemicals. Its lower carbon intensity relative to coal and oil has led many governments to describe

it as a "bridge" or "transition" fuel, while its methane emissions and long-lived infrastructure have led critics to warn that new gas investment risks locking in emissions incompatible with the goals of the Paris Agreement. The legal regime of natural gas must therefore accommodate both its role in energy security and its contested position in the energy transition.

The legal regime of natural gas differs from that of oil in ways that flow directly from the physical properties of the commodity. Oil is a dense liquid that can be stored easily and transported cheaply by tanker, rail, or pipeline to almost any market; this fungibility gave rise, from the 1980s onwards, to a global oil market with transparent spot and futures prices. Gas, by contrast, has a very low energy density at atmospheric pressure. It must be transported either through high-pressure pipelines, which are fixed, capital-intensive, and tie producer and consumer together for decades, or as liquefied natural gas, which requires cooling the gas to approximately minus 162 degrees Celsius, shipping it in specialised vessels, and regasifying it at the destination. Both modes of transport require very large, specific, and long-lived investments. As a consequence, gas markets historically developed as regional rather than global markets, organised around long-term contracts that allocated the risks of these investments between sellers and buyers, and subject to extensive regulation of the natural monopoly elements of the supply chain.

Three transformations over the past four decades have reshaped the legal regime of gas. The first is liberalisation: the introduction of competition in gas supply in North America from the 1980s, in the United Kingdom from the 1990s, and in the European Union from the late 1990s onwards, requiring the unbundling of transmission from supply and the creation of rights of third-party access to pipelines. The second is the growth of the LNG trade, which has expanded from a handful of point-to-point projects linking specific producers with Japanese and European buyers into a global market involving dozens of exporting and importing countries, with an increasing share of spot and short-term trade and greater price convergence among regions. The third is the politicisation of gas supply, illustrated most dramatically by the Russian Federation's curtailment of gas deliveries to Europe in 2021 and 2022, which precipitated an energy crisis, extraordinary regulatory interventions, and a fundamental restructuring of European gas supply.

This chapter examines the legal regime of natural gas from wellhead to burner tip. It begins with the legal and commercial characteristics of gas (section 6.1) and the regulation of gas exploration and production (section 6.2). It then examines gas transportation and pipeline access (section 6.3), the structures used to develop LNG projects (section 6.4), long-term gas sales and purchase agreements (section 6.5), and the mechanisms of price review, take-or-pay, and contractual

flexibility (section 6.6). It concludes with the legal framework for cross-border gas infrastructure and transit (section 6.7) and the law of gas supply disputes and regulatory intervention (section 6.8). The treatment of submarine pipelines under the law of the sea is reserved for Chapter 20, and the geopolitics of pipeline dependence for Chapter 25; this chapter focuses on the commercial and regulatory law of gas.

6.1 Legal and Commercial Characteristics of Natural Gas

Physical Properties and Their Legal Consequences

Natural gas is a mixture of hydrocarbon gases composed predominantly of methane, together with varying proportions of heavier hydrocarbons (ethane, propane, butane, and pentanes), and non-hydrocarbon components such as carbon dioxide, nitrogen, hydrogen sulphide, and water vapour. Before gas can be delivered to a transmission system, it must usually be processed to remove impurities and to extract natural gas liquids, which are valuable products in their own right. The composition of gas varies from field to field, and its energy content — its calorific or heating value — varies accordingly. Gas is therefore sold not by volume alone but by energy content, commonly measured in British thermal units (Btu), therms, gigajoules, or kilowatt hours, and gas contracts and network codes specify detailed quality specifications, including ranges for calorific value, the Wobbe index (a measure of the interchangeability of gases in burners), and maximum levels of contaminants.

These physical characteristics have several legal consequences. First, the specification of quality and the allocation of responsibility for off-specification gas are central issues in gas contracts and network codes. A seller that delivers gas outside the contractual specification may be in breach, and the buyer may be entitled to reject it, while a transmission system operator may refuse to accept gas that would compromise the safety or integrity of the network. Second, the measurement of gas — volume, pressure, temperature, and composition — requires agreed metering standards and procedures for resolving measurement disputes. Third, because gas from different sources is commingled in pipelines and storage, title to gas is usually a matter of contractual allocation rather than physical identification; the gas a buyer receives is not physically the same gas that its seller delivered. Gas law therefore relies heavily on concepts such as nominations, allocations, and balancing, which translate physical flows into contractual entitlements.

Natural Monopoly and Network Characteristics

High-pressure transmission pipelines and low-pressure distribution networks exhibit strong economies of scale and are commonly regarded as natural monopolies: it is usually more efficient for a single network to serve a given area than for competing networks to be duplicated. Natural monopoly provides the classic

justification for economic regulation: in the absence of competition, the owner of a network could charge monopoly prices or refuse access to competitors in upstream production or downstream supply. Gas law therefore distinguishes between the potentially competitive segments of the supply chain — production, wholesale trading, and retail supply — and the monopoly segments — transmission, distribution, and, in some cases, storage and LNG terminals — which are subject to regulation of tariffs, access, and investment.

Asset Specificity and the Long-Term Contract

The economics of gas infrastructure are characterised by high asset specificity. A pipeline linking a particular field to a particular market, or an LNG liquefaction plant built to process gas from a particular reservoir, has little value in any alternative use. Once the investment is made, its owner is exposed to opportunistic behaviour by its counterparties — the "hold-up" problem analysed in transaction cost economics by Oliver Williamson and others. A producer that has built a pipeline to a single buyer may find the buyer demanding lower prices once the pipeline is complete; a buyer that has built power stations relying on a single supplier may find the supplier demanding higher prices.

The long-term gas contract emerged as the principal legal device for managing this risk. By committing the buyer to purchase, or pay for, specified quantities of gas over periods of fifteen to twenty-five years, and the seller to deliver them, the long-term contract provides the revenue certainty necessary to finance the infrastructure, while price indexation and price review mechanisms allow the price to adjust to changing market conditions without reopening the fundamental bargain. The long-term contract is thus a form of private governance structure for a relationship in which neither party can easily exit. Its central provisions — take-or-pay, destination restrictions, price indexation, and price review — are examined in sections 6.5 and 6.6.

Regional Markets and Pricing Systems

Partly because of the costs of transport, gas markets developed historically as regional markets with distinct pricing systems. Four broad systems may be identified.

Gas-on-gas competition prevails in North America, the United Kingdom, and much of continental Europe, where liberalised markets with liquid trading hubs determine prices through the interaction of supply and demand. The principal hubs are Henry Hub in Louisiana, which serves as the delivery point for the New York Mercantile Exchange natural gas futures contract; the National Balancing Point (NBP) in Great Britain, a virtual trading point within the National Gas transmission system; and the Title Transfer Facility (TTF) in the Netherlands, which has become the benchmark for continental European gas prices.

Oil indexation was the traditional pricing method in continental Europe and remains significant in Asia. Under oil-indexed contracts, the gas price is linked by formula to the prices of competing fuels, typically crude oil or oil products. In Asian LNG contracts, the price has commonly been expressed as a percentage ("slope") of the Japan Customs-Cleared Crude price (the "JCC" or "Japanese Crude Cocktail") or, increasingly, Brent crude, sometimes with an "S-curve" that moderates the effect of very high or very low oil prices. The rationale for oil indexation, which originated in the pricing of Groningen gas in the Netherlands in the 1960s, was the "market value" or "replacement value" principle: gas should be priced at a level that allowed it to compete with the alternative fuel the buyer would otherwise use.

Regulated pricing prevails in many producing countries and some importing countries, where governments set gas prices administratively, often below the cost of supply, to support domestic industry and power generation or to protect consumers. Regulated domestic gas prices have been a significant obstacle to gas development in many countries, including Nigeria, where the historically low regulated domestic price discouraged investment in gas supply to the domestic market.

Spot and short-term LNG pricing has grown with the expansion of the LNG trade, and Asian spot LNG is now commonly priced by reference to the Japan Korea Marker (JKM), an assessed price published by a price reporting agency, while increasing quantities of LNG from the United States are sold on the basis of Henry Hub prices plus a fixed liquefaction fee.

The International Gas Union's periodic wholesale gas price surveys have documented a long-term shift from oil indexation towards gas-on-gas competition, particularly in Europe, although the extreme price volatility of hub prices during the 2021–2023 crisis led some buyers to reconsider the value of oil-linked contracts as a hedge.

Gas in the Energy Transition

The role of gas in the energy transition is legally significant because it shapes the policy frameworks within which gas investment takes place. The European Union's Taxonomy Regulation, through a Complementary Climate Delegated Act adopted in 2022, classifies certain gas-fired electricity generation and heat activities as transitional activities that may qualify as environmentally sustainable, subject to stringent emissions thresholds and conditions, including a commitment to switch to renewable or low-carbon gases by 2035. The inclusion of gas (and nuclear) in the taxonomy was controversial and was challenged before the General Court of the European Union by Austria and by environmental organisations; in September 2025 the General Court dismissed Austria's action. The

EU's Regulation on the reduction of methane emissions in the energy sector, adopted in 2024, imposes measurement, reporting, and verification obligations on oil, gas, and coal operators, restricts routine venting and flaring, and will progressively apply equivalent monitoring requirements to imported fossil fuels — a measure with significant implications for exporting states, including Nigeria. At the international level, the outcome of the first global stocktake under the Paris Agreement, adopted at COP28 in Dubai in 2023, called on parties to contribute to "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner", while recognising that "transitional fuels can play a role in facilitating the energy transition while ensuring energy security".

For gas-producing developing countries, the transition raises profound questions of equity. African governments, including Nigeria, have argued that gas is essential to expanding energy access, industrialisation, and clean cooking, and that it would be unjust for countries that bear little historical responsibility for climate change to be denied financing for gas development. Nigeria declared 2021–2030 the "Decade of Gas", reflecting a strategy of using its large gas reserves — among the largest in Africa — to drive domestic industrialisation and power generation. The tension between this position and the climate policies of international lenders, which have restricted financing for upstream gas, is examined further in Chapters 18 and 23.

6.2 Gas Exploration and Production Regulation

The Historical Neglect of Gas in Petroleum Legislation

Petroleum legislation was designed historically around oil. Early concessions and petroleum statutes often made little distinction between oil and gas, or treated gas as an incidental by-product. As a result, petroleum regimes frequently failed to provide adequate incentives for gas development. Gas discovered in the course of oil exploration might be reinjected, flared, or left undeveloped because the fiscal terms designed for oil made gas development uneconomic, because there was no domestic market or infrastructure, or because the rights to gas were unclear. The development of gas-specific legal provisions — governing rights to non-associated gas, the fiscal treatment of gas, the obligations of producers to supply gas to the domestic market, and the restriction of flaring — has been a significant feature of petroleum law reform over the past three decades.

Rights to Gas Under Petroleum Titles

Petroleum titles usually confer rights to both oil and gas within the licence area, but they may differ in their treatment of the two. Some regimes provide for separate gas licences or for distinct terms applicable to non-associated gas discoveries, recognising that gas development requires longer periods for market development and a different fiscal structure. A common device is the extended "gas holding" or

"gas retention" period, which allows a licensee that has discovered non-associated gas but cannot yet develop it commercially to retain the discovery area for a defined period while it seeks a market, without being required to relinquish it. Without such provisions, licensees may be forced to relinquish gas discoveries before markets have developed.

Under Nigeria's Petroleum Industry Act 2021, the general framework for petroleum prospecting licences and petroleum mining leases applies to both oil and gas, but the Act contains specific provisions recognising the distinct characteristics of gas. Holders of petroleum prospecting licences that discover non-associated gas may be granted a period to establish its commerciality before conversion to a lease, and the Act's fiscal provisions treat gas production more favourably than oil, notably by subjecting gas to companies income tax rather than the hydrocarbon tax applicable to crude oil, and by providing reduced royalty rates for gas, with lower rates for gas delivered to the domestic market. These provisions reflect the policy of the Act to promote the development of gas for domestic use.

Domestic Gas Supply Obligations

Many gas-producing states impose domestic supply obligations requiring producers to make a proportion of their gas available for the domestic market, often at regulated prices. Such obligations reflect the policy of ensuring that national resources contribute to domestic energy supply and industrial development, rather than being exported exclusively. They may, however, deter investment if the regulated price is below the cost of supply or the export price.

In Nigeria, the Nigerian Gas Master Plan of 2008 and the National Domestic Gas Supply and Pricing Regulations 2008 introduced a domestic gas delivery obligation requiring producers to supply specified volumes of gas to the domestic market. The PIA continues this approach, empowering the Nigerian Midstream and Downstream Petroleum Regulatory Authority (NMDPRA) to determine the domestic gas delivery obligation and to set a domestic base price for gas supplied to the strategic sectors (notably power generation) until a willing-buyer, willing-seller market develops. In Western Australia, the Domestic Gas Reservation Policy, introduced in 2006, requires LNG export projects to make gas equivalent to fifteen per cent of their LNG production available to the domestic market, and the federal Australian Domestic Gas Security Mechanism, introduced in 2017, empowers the government to restrict LNG exports where there is a domestic shortfall. In 2022 and 2023, Australia introduced emergency price caps on wholesale gas and a mandatory code of conduct for the east coast gas market in response to the price surge triggered by the European energy crisis. Similar obligations exist in Indonesia, Egypt, Israel, and other producing countries.

Domestic supply obligations raise legal questions in international trade law, since they may function as export restrictions, and in investment law, since their introduction or tightening after an investment is made may affect the investor's legitimate expectations. They are discussed further in Chapter 13 in the context of export restrictions.

Gas Flaring Regulation

Gas flaring — the burning of associated gas at the wellsite — wastes a valuable resource, emits carbon dioxide and black carbon, and exposes nearby communities to air pollution, noise, heat, and light. The World Bank's Global Gas Flaring Tracker reports that over 140 billion cubic metres of gas are flared globally each year, with Russia, Iraq, Iran, the United States, Venezuela, Algeria, Libya, and Nigeria among the largest flaring countries. Venting — the direct release of unburned gas, predominantly methane — is even more damaging to the climate, since methane is a far more potent greenhouse gas than carbon dioxide over short time horizons.

Legal approaches to flaring include outright prohibitions with exceptions for safety and emergencies, flare fees or penalties, requirements for gas utilisation plans as a condition of development approval, and mechanisms to allocate flare gas to third parties that can commercialise it. Nigeria's experience with flaring regulation, examined in Chapter 5, illustrates both the difficulties of enforcement and the evolution of policy from prohibition to commercialisation through the Flare Gas (Prevention of Waste and Pollution) Regulations 2018 and the Nigerian Gas Flare Commercialisation Programme. The *Gbemre* case, also discussed in Chapter 5 and Chapter 21, established the human rights dimension of flaring in Nigerian law.

Internationally, the World Bank's Zero Routine Flaring by 2030 Initiative, launched in 2015, has been endorsed by governments, oil companies, and development institutions, which commit not to engage in routine flaring in new oil field developments and to end legacy routine flaring as soon as possible and no later than 2030. The Global Methane Pledge, launched at COP26 in 2021, commits participating countries to collective action to reduce global anthropogenic methane emissions by at least thirty per cent from 2020 levels by 2030. The Oil and Gas Methane Partnership 2.0 (OGMP 2.0), administered by the United Nations Environment Programme, has established a reporting framework for company methane emissions that has been incorporated by reference into the EU Methane Regulation. These instruments are largely voluntary, but they are increasingly being translated into binding domestic and regional law.

Gas Processing and Natural Gas Liquids

Gas processing — the separation of natural gas liquids, removal of impurities, and conditioning of gas to pipeline specification — occupies a regulatory borderland between upstream and midstream activity. In some regimes, processing facilities associated with a specific field are treated as part of upstream operations under the petroleum licence; in others, they require separate licences. The PIA treats gas processing as a midstream activity requiring a licence from the NMDPRA, distinct from the upstream licence regulated by the Nigerian Upstream Petroleum Regulatory Commission (NUPRC). The classification affects the applicable fiscal regime, the identity of the regulator, and the rights of third parties to access the facilities. Access to upstream infrastructure, including processing facilities, by third parties is examined in section 6.3.

Fiscal Treatment of Gas

Because gas development is typically less profitable than oil development, owing to higher transport costs, lower prices per unit of energy, and the need for market development, many petroleum fiscal regimes provide more favourable terms for gas. Common incentives include lower royalty rates, exemption from or reduced rates of resource rent taxes, accelerated depreciation for gas infrastructure, and tax holidays for gas utilisation projects. Nigeria's Associated Gas Framework Agreement of 1992 and the incentives for gas utilisation in the Companies Income Tax Act, now carried forward in modified form in the PIA, are examples. The design of gas fiscal incentives must balance the need to make gas development commercially viable against the risk of conferring windfall gains on projects that would have proceeded without them, and against the need to avoid distortions between oil and gas production that might encourage artificial classification of production. Fiscal regimes generally are examined in Chapter 16.

6.3 Gas Transportation and Pipeline Access

The Gas Supply Chain and Its Regulation

The physical supply chain for pipeline gas comprises upstream gathering pipelines, which collect gas from wells and convey it to processing facilities; high-pressure transmission pipelines, which carry processed gas over long distances; storage facilities, including depleted reservoirs, salt caverns, and aquifers, which balance seasonal and short-term fluctuations in demand; and low-pressure distribution networks, which deliver gas to end-users. LNG terminals, considered in section 6.4, perform an analogous function at the interface between maritime and pipeline transport. Each element raises distinct regulatory issues, but the central questions are the same: who may own and operate the infrastructure; on what terms must the owner grant access to third parties; how are tariffs determined; and how is new investment encouraged.

Models of Industry Structure

Historically, gas industries in most countries were organised as vertically integrated monopolies, whether state-owned (as in France, Italy, and much of the developing world) or privately owned but regulated (as in the United States). A single company, or a small number of regional companies, purchased gas from producers, transported it, and sold it to distributors or end-users. Liberalisation sought to introduce competition in the potentially competitive segments of the chain by separating them from the monopoly networks and requiring network owners to provide access to competing suppliers on non-discriminatory terms.

The United States led this process. Under the Natural Gas Act of 1938, interstate pipelines were regulated by the Federal Power Commission (now the Federal Energy Regulatory Commission, FERC) as merchants that bought and resold gas. Following the partial decontrol of wellhead prices under the Natural Gas Policy Act of 1978, FERC Order No 436 (1985) encouraged pipelines to offer open-access transportation services, and Order No 636 (1992) required interstate pipelines to unbundle their sales and transportation services, transforming them into pure transporters and creating a competitive wholesale gas market. The United Kingdom followed with the privatisation of British Gas in 1986, the Gas Act 1995, and the separation of transportation from supply, producing one of the most liquid gas markets in the world. The European Union adopted a sequence of liberalisation directives, culminating in the Third Energy Package of 2009, comprising Directive 2009/73/EC concerning common rules for the internal market in natural gas and Regulation (EC) No 715/2009 on conditions for access to the natural gas transmission networks. In 2024, the EU adopted the Hydrogen and Decarbonised Gas Market Package — Directive (EU) 2024/1788 and Regulation (EU) 2024/1789 — which recasts the gas market rules and extends them to the emerging hydrogen sector.

Third-Party Access

Third-party access (TPA) is the right of a person other than the owner of a pipeline or other facility to use its capacity to transport gas. TPA may be negotiated or regulated.

Under **negotiated TPA**, the owner and the access seeker negotiate the terms of access commercially, subject to a general obligation to negotiate in good faith and not to discriminate, and to the oversight of a regulator or competition authority in the event of a dispute. Negotiated TPA is common for upstream pipelines and processing facilities, where the circumstances of each access request are highly specific. In the United Kingdom, for example, the Energy Act 2011 and the Infrastructure Code of Practice govern access to upstream oil and gas infra-

structure on the UK Continental Shelf, and the North Sea Transition Authority may determine the terms of access in default of agreement.

Under **regulated TPA**, the terms of access, including tariffs, are published in advance and approved or set by an independent regulator. Regulated TPA is the norm for transmission and distribution networks in the European Union, where Directive 2009/73/EC (now the recast Directive) requires member states to ensure a system of third-party access to transmission and distribution systems and LNG facilities based on published tariffs, applicable to all eligible customers and applied objectively and without discrimination. Access to storage may be either negotiated or regulated, at the option of the member state.

Access may be refused on limited grounds. The EU directive permits a natural gas undertaking to refuse access on the basis of lack of capacity, or where access would prevent it from carrying out public service obligations, or on the basis of serious economic and financial difficulties with take-or-pay contracts. Refusals must be duly substantiated.

Unbundling

Third-party access is unlikely to be effective if the network owner is also a supplier of gas in competition with the access seekers, because it will have both the incentive and the means to discriminate against them — for example, by underinvesting in capacity, manipulating information, or delaying connections. Liberalisation has therefore been accompanied by requirements for the *unbundling* of network activities from supply and production activities. Unbundling may take several forms, of increasing stringency:

1. **Accounting unbundling**, requiring separate accounts for network and supply activities, to prevent cross-subsidisation.
2. **Functional or legal unbundling**, requiring network activities to be carried out by a separate legal entity with independent management, although it may remain within a vertically integrated group.
3. **Independent system operator (ISO)**, under which the integrated group retains ownership of the network but its operation is entrusted to an independent operator.
4. **Independent transmission operator (ITO)**, under which the network remains within the integrated group but is subject to detailed rules ensuring its independence in management and decision-making.
5. **Ownership unbundling**, requiring the separation of ownership of transmission networks from production and supply interests.

The Third Energy Package required member states to choose among ownership unbundling, the ISO model, and the ITO model for transmission system operators, and the choice was the subject of intense political controversy, with France

and Germany defending the ITO model to protect their integrated incumbents. The "third country clause" in Article 11 of the Directive required certification of transmission system operators controlled by persons from third countries, permitting refusal of certification where the grant would put at risk the security of energy supply of the member state or the Union — a provision directed in large part at Gazprom.

Tariff Regulation

The regulation of transportation tariffs seeks to allow network owners to recover their efficiently incurred costs, including a reasonable return on capital, while preventing monopoly pricing. The principal methodologies are *cost-of-service* or *rate-of-return* regulation, which sets tariffs to recover a defined revenue requirement based on operating costs, depreciation, and a return on the regulated asset base; and *incentive regulation*, such as price caps or revenue caps of the "RPI–X" type pioneered in the United Kingdom, under which tariffs are set for multi-year periods and the network owner retains the benefits of efficiency improvements. Tariff structures may be *distance-related* (point-to-point), *postage stamp* (uniform irrespective of distance), or *entry–exit*, under which users book capacity separately at entry and exit points and gas can flow freely within the system between them. The EU Network Code on Harmonised Transmission Tariff Structures for Gas (Regulation (EU) 2017/460) requires entry–exit systems and sets out detailed rules for reference price methodologies.

Pipeline Regulation in Nigeria

Nigeria's gas transportation infrastructure has historically been owned and operated by the Nigerian Gas Company, a subsidiary of NNPC, now succeeded within NNPC Ltd's corporate structure. The PIA establishes a framework for midstream gas regulation by the NMDPRA, including licensing of gas transportation pipelines, gas distribution systems, gas processing facilities, and gas storage; the establishment of a Gas Transportation Network Code governing access to the transportation network; open access to pipelines and facilities on a non-discriminatory basis, subject to available capacity; and the regulation of tariffs. The Act requires the separation of gas transportation network operation from gas wholesale and supply activities, reflecting the European unbundling model. Major infrastructure projects, including the Ajaokuta–Kaduna–Kano (AKK) pipeline and the Obiafu–Obrikom–Oben (OB3) pipeline, are intended to connect gas resources in the south with demand centres in the north, but have faced delays associated with financing, security, and construction challenges.

6.4 Liquefied Natural Gas Project Structures

The LNG Value Chain

The LNG value chain comprises upstream production of feed gas; liquefaction, in which the gas is treated and cooled to approximately minus 162 degrees Celsius, reducing its volume by a factor of around six hundred; shipping in specialised LNG carriers; and regasification at import terminals, either onshore or on floating storage and regasification units (FSRUs). Each stage requires very large capital investments: a large liquefaction plant may cost tens of billions of United States dollars, a modern LNG carrier more than two hundred million dollars, and a regasification terminal several hundred million dollars or more. The capital intensity of the chain, together with its specificity and long life, means that LNG projects have historically required long-term contracts covering most of their output before financing can be secured.

Physical chain (top) and principal legal instruments (bottom)

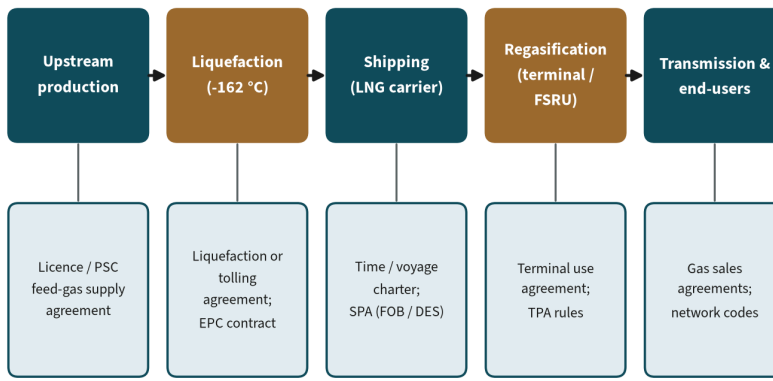


Figure 6.1 The LNG value chain and the principal contracts that link its stages. *Source:* prepared for this edition.

Principal Project Structures

LNG export projects have been organised according to several structures, distinguished by the relationship between ownership of the upstream gas, the liquefaction facilities, and the LNG product.

The integrated structure. Under the integrated structure, the same parties (in the same proportions) own the upstream gas reserves and the liquefaction plant, and sell LNG directly to buyers. The upstream and liquefaction activities are treated as a single project for fiscal and commercial purposes. The structure aligns the interests of upstream and midstream investors and avoids the need for

a transfer price between them, but it requires all investors to participate in both segments. Many projects in Qatar, Australia, and elsewhere have used integrated structures.

The merchant structure. Under the merchant structure, the liquefaction company is owned separately from the upstream producers (or by them in different proportions). It purchases feed gas from the producers under gas supply agreements, liquefies it, and sells LNG to buyers on its own account, taking the margin between its purchase and sales prices. Nigeria LNG Limited (NLNG), owned by NNPC (49 per cent), Shell (25.6 per cent), TotalEnergies (15 per cent), and Eni (10.4 per cent), operates on a broadly merchant basis, purchasing feed gas from upstream joint ventures and selling LNG under long-term contracts to buyers in Europe and elsewhere. The merchant structure requires a negotiated feed gas price, which allocates value between upstream and midstream, and it may raise fiscal issues where the liquefaction company is subject to a different tax regime from the upstream producers.

The tolling structure. Under the tolling structure, the liquefaction company does not take title to the gas. It provides liquefaction services to the gas owners (or to LNG offtakers) in return for a tolling fee, which is typically fixed and payable regardless of whether capacity is used, providing a utility-like return to the plant owners. The gas owners retain title through the liquefaction process and sell the LNG themselves. The tolling model was used for later trains of Atlantic LNG in Trinidad and Tobago and for several of the export projects developed in the United States from the mid-2010s, such as Freeport LNG, whose capacity is contracted to offtakers under liquefaction tolling agreements. Other United States projects, such as Cheniere's Sabine Pass, sell LNG under sale and purchase agreements priced at a percentage of Henry Hub plus a fixed fee, which replicate the economics of tolling. Under the United States model, buyers enter into long-term "liquefaction services" or sale and purchase agreements under which they pay a fixed liquefaction fee for capacity and purchase feed gas at Henry Hub prices plus a percentage, giving them flexibility to lift cargoes or cancel them (paying the fixed fee) and to resell LNG anywhere in the world.

Hybrid structures. Many projects combine elements of these structures, and the choice depends on the fiscal regime, the identity and objectives of the investors, the availability of financing, and the requirements of the host state.

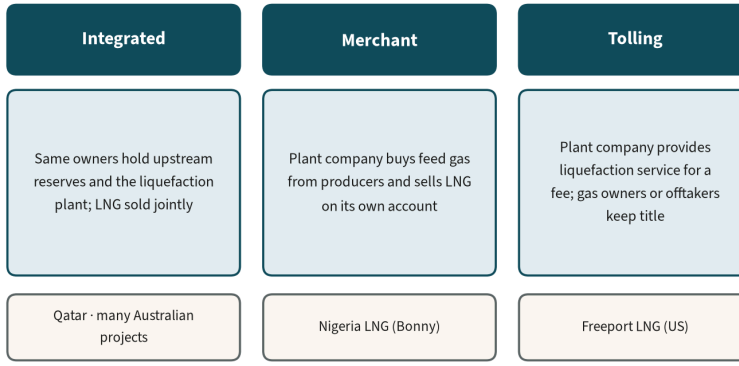


Figure 6.2 Integrated, merchant, and tolling structures for LNG export projects. *Source:* prepared for this edition.

The Host State Framework

LNG projects typically require a dedicated legal framework, often in the form of a project-specific statute or host government agreement, providing fiscal terms, guarantees against expropriation, stability of the legal regime, rights to export LNG, and rights to retain export proceeds offshore. The Nigeria LNG (Fiscal Incentives, Guarantees and Assurances) Act 1990 is an example: it conferred on NLNG a package of fiscal incentives, guarantees, and assurances, including exemption from certain taxes for an initial period, a guarantee against nationalisation, the right to maintain offshore accounts, and a commitment that the Act could not be amended without the consent of the company's shareholders. The NLNG Act illustrates both the attractions and the controversies of project-specific legislation. When the Nigerian Maritime Administration and Safety Agency (NIMASA) sought to impose levies on NLNG's shipping activities, and when the Federal Government sought to subject NLNG to contributions to the Niger Delta Development Commission, the company invoked the protections of the Act, generating litigation concerning the extent to which a statute can bind future legislatures. The question whether a legislature may validly restrict its own future legislative competence by statute raises fundamental constitutional issues; in common law systems influenced by the doctrine of parliamentary sovereignty, such entrenchment is generally ineffective as a matter of domestic law, although it may still give rise to liability under contract or investment treaties.

Shipping

LNG shipping arrangements depend on the delivery terms of the sales contract. Under a *free on board* (FOB) contract, the buyer takes delivery at the loading port and is responsible for shipping; under a *delivered ex-ship* (DES) or *cost, insurance and freight* (CIF) contract, the seller is responsible for shipping and delivers the LNG at the discharge port. LNG carriers are chartered under long-term time charters, voyage charters, or, increasingly, short-term charters for spot cargoes. The legal issues include the compatibility of vessels with terminals (ship–shore compatibility), the allocation of boil-off gas (LNG that evaporates during the voyage), demurrage, and the application of maritime conventions.

Regasification and FSRUs

The legal structure of regasification terminals varies. Some terminals are owned by the buyers that use them; others are owned by independent operators that offer capacity to multiple users under terminal use agreements, subject to regulated or negotiated third-party access. In the European Union, LNG terminals are subject to regulated third-party access, but major new infrastructure may obtain an exemption under Article 36 of Directive 2009/73/EC (now carried forward in the recast) where the investment enhances competition and security of supply and would not take place without the exemption. Many European terminals, including the Adriatic LNG terminal in Italy and the Gate terminal in Rotterdam, have obtained exemptions.

Floating storage and regasification units have dramatically reduced the time and cost required to establish LNG import capacity, enabling countries such as Egypt, Pakistan, Bangladesh, and, following the 2022 crisis, Germany to begin importing LNG within months rather than years. Germany's LNG Acceleration Act (LNG-Beschleunigungsgesetz) of May 2022 expedited permitting for FSRUs and onshore terminals by, among other things, temporarily exempting projects from certain environmental impact assessment requirements, illustrating the use of emergency legislation to accelerate energy infrastructure.

LNG Export Authorisation

Some states require specific authorisation for the export of LNG. In the United States, section 3 of the Natural Gas Act requires authorisation by the Department of Energy for exports of natural gas. Exports to countries with which the United States has a free trade agreement requiring national treatment for trade in natural gas are deemed to be consistent with the public interest and must be granted without modification or delay; exports to non-FTA countries must be authorised unless the Department finds that they would not be consistent with the public interest. In January 2024, the Department announced a temporary pause on pending decisions on non-FTA exports while it updated the economic and envi-

ronmental analyses underlying its public interest determinations; the pause was lifted in January 2025 by the incoming administration. The episode illustrates the regulatory risk facing LNG projects, whose commercial viability depends on export authorisation, and the growing incorporation of climate considerations into export licensing.

6.5 Long-Term Gas Sales and Purchase Agreements

Function and Forms

The gas sales and purchase agreement (GSPA, or SPA in the LNG context) is the central commercial instrument of the gas industry. It allocates between seller and buyer the volume risk (the risk that the buyer will not need, or the seller will not be able to supply, the contracted quantities), the price risk (the risk that market prices will move against one party), and the operational risks of delivery. Long-term GSPAs of fifteen to twenty-five years' duration underpin the financing of most major pipeline and LNG export projects. Medium-term contracts (two to ten years), short-term contracts (up to two years), and spot transactions have grown in importance with the development of liquid hubs and the LNG spot market, and are commonly concluded under master agreements that set out general terms, with individual transactions confirmed by short confirmation notices.

Several standard forms are used in the industry. For trading on European hubs, the European Federation of Energy Traders (EFET) General Agreement Concerning the Delivery and Acceptance of Natural Gas is widely used; in North America, the North American Energy Standards Board (NAESB) Base Contract for Sale and Purchase of Natural Gas performs a similar function. For LNG, the Association of International Energy Negotiators (AIEN) has published a Model Master LNG Sale and Purchase Agreement, and the International Group of Liquefied Natural Gas Importers (GIIGNL) has published a Master LNG Sale and Purchase Agreement for spot and short-term trades. Long-term SPAs, by contrast, are typically bespoke, although they follow well-established structures. Appendix D contains a model international gas sales agreement with annotations.

Contract Quantity Provisions

The quantity provisions of a long-term GSPA typically define:

- the **Annual Contract Quantity (ACQ)**, the quantity of gas the seller undertakes to make available and the buyer undertakes to take or pay for in each contract year;
- the **Daily Contract Quantity (DCQ)**, usually the ACQ divided by the number of days in the year, and the **Maximum Daily Quantity**, often 110 or 120 per cent of the DCQ, which defines the buyer's daily flexibility;

- **downward quantity tolerance** or **downward flexibility**, permitting the buyer to reduce its annual nominations by a specified percentage without incurring take-or-pay liability;
- adjustments to the ACQ for gas not taken because of the seller's failure to deliver, force majeure, or the buyer's rejection of off-specification gas; and
- the **build-up period** at the start of the contract and the **plateau period** during which the ACQ remains constant.

LNG SPAs specify quantities in terms of the Annual Contract Quantity expressed in tonnes or million British thermal units, and contain an **Annual Delivery Programme** procedure under which the parties agree before each contract year on the schedule of cargoes, subject to adjustments for operational reasons. The seller may have rights to reduce quantities for scheduled maintenance of the liquefaction facilities, and the buyer may have rights to divert or cancel cargoes, depending on the terms of the contract.

Delivery Point and Transfer of Title and Risk

The GSPA defines the delivery point at which title and risk pass from seller to buyer. In pipeline contracts, the delivery point may be at the outlet of the seller's processing facilities, at a border crossing, at a trading hub, or at the buyer's facilities. In LNG contracts, the delivery point depends on whether the contract is FOB (title and risk pass at the loading port, usually as the LNG passes the flange connecting the loading arms to the vessel's manifold) or DES (title and risk pass at the discharge port). Under a CIF contract, risk passes at the loading port but the seller arranges and pays for shipping and insurance. The choice affects which party controls the vessel and, consequently, the ability to divert cargoes.

Quality and Measurement

GSPAs specify the quality of gas to be delivered and the consequences of off-specification deliveries. The buyer may be entitled to reject off-specification gas, or to accept it subject to a price reduction or compensation for costs incurred in treating it. The allocation of risk for off-specification gas is particularly important where the gas is to be used in sensitive equipment, such as gas turbines, or injected into networks with strict specifications. Measurement provisions define the methods, standards, and equipment to be used for measuring quantity and quality, the procedures for calibration and testing, and the resolution of measurement disputes, often by reference to an independent surveyor in the LNG context.

Destination Clauses and Diversion Rights

Traditional long-term gas contracts frequently contained **destination clauses** (or territorial restriction clauses) prohibiting the buyer from reselling gas outside a defined territory, and **profit-sharing mechanisms** requiring the buyer to share with the seller any profit from resale to a different destination. Sellers defended

these provisions as necessary to prevent arbitrage between markets in which they sold at different prices, and to protect their pricing strategies. Buyers and competition authorities criticised them as restrictions on competition and on the development of integrated markets.

In the European Union, destination clauses in contracts with EU buyers have been treated as restrictions of competition contrary to Article 101 of the Treaty on the Functioning of the European Union (TFEU). From the early 2000s, the European Commission pursued investigations concerning the contracts of suppliers including Gazprom, Sonatrach, and Nigeria LNG, resulting in the removal of destination clauses from their contracts with European buyers. In its 2018 decision accepting commitments from Gazprom in Case AT.39816 (*Upstream Gas Supplies in Central and Eastern Europe*), the Commission addressed concerns that Gazprom had pursued an overall strategy of partitioning Central and Eastern European gas markets, including through territorial restrictions, and Gazprom committed to remove contractual barriers to the cross-border resale of gas and to facilitate gas flows to isolated markets. In 2018 the Commission also opened an investigation into whether Qatar Petroleum's long-term LNG supply agreements with European importers contained territorial restrictions.

In Asia, the Japan Fair Trade Commission published a report in 2017 concluding that destination restrictions in FOB LNG contracts were likely to violate Japan's Antimonopoly Act, and that profit-sharing mechanisms and certain restrictions in DES contracts were also problematic. The report has encouraged Japanese buyers to negotiate greater flexibility in new contracts, contributing, together with the growth of United States LNG supplied on destination-free FOB terms, to the increasing liquidity and flexibility of the global LNG market.

Force Majeure and Seller's Failure to Deliver

GSPAs contain detailed force majeure provisions excusing performance where a party is prevented from performing by events beyond its reasonable control. In the gas context, force majeure clauses often enumerate specific events, such as failures of facilities, governmental action, and natural disasters, and may treat certain events — such as failures of the seller's upstream supply or of the buyer's downstream market — as either within or outside the clause depending on the parties' negotiated allocation of risk. Force majeure in energy contracts is examined in Chapter 15; here it is sufficient to note that the treatment of upstream and transportation failures has been a recurring source of disputes in gas contracts, and that the 2022 crisis generated significant litigation concerning Gazprom's declaration of force majeure in relation to deliveries to European buyers.

GSPAs also provide remedies for the seller's failure to deliver contracted quantities otherwise than as a result of force majeure. These commonly include

a price discount on shortfall quantities, sometimes expressed as a percentage of the contract price, and may be the buyer's exclusive remedy for shortfalls up to a specified level. The adequacy of such remedies was tested during the 2022 crisis, when replacement gas cost many times the contract price; buyers argued that the contractual shortfall remedies did not apply to deliberate non-performance, and sought full damages in arbitration.

6.6 Price Review, Take-or-Pay, and Contractual Flexibility

Take-or-Pay

The **take-or-pay** clause is the defining feature of the long-term gas contract. It obliges the buyer to pay for a minimum quantity of gas in each contract year (the "take-or-pay quantity", often 80 to 90 per cent of the ACQ, after allowing for downward flexibility) whether or not it actually takes delivery of that quantity. If the buyer takes less than the minimum, it must pay for the shortfall, but it typically acquires a right to take the paid-for but untaken gas ("make-up gas") in subsequent years, usually after it has taken the ACQ for that year, and sometimes at the price applicable at the time of payment. Conversely, if the buyer takes more than the ACQ in a year, it may accumulate "carry-forward" credits that reduce its minimum take obligation in future years.

The economic function of take-or-pay is to allocate volume risk to the buyer, providing the seller with predictable revenues sufficient to service the debt raised to finance the production and transportation facilities. In return, the buyer obtains security of supply over the long term. Lenders to gas projects typically require a substantial proportion of the project's output to be committed under take-or-pay contracts with creditworthy buyers before they will provide financing.

The enforceability of take-or-pay clauses has been challenged on the ground that they are penalties. Under English law, the rule against penalties applies only to clauses that operate upon a breach of contract; a clause that provides for payment as a primary obligation in the event of a contractual contingency, rather than a breach, is not subject to the rule. In *M&J Polymers Ltd v Imerys Minerals Ltd*, the High Court considered a take-or-pay clause in a supply contract and held that, although the rule against penalties was potentially applicable on the particular construction of that clause, the clause was commercially justifiable and not a penalty.[1] Following the restatement of the penalty rule in *Cavendish Square Holding BV v Makdessi* (discussed in Chapter 5), the predominant view is that a properly drafted take-or-pay clause, which creates a primary obligation to pay for gas made available, does not engage the penalty rule at all, and that even if it did, it would usually be justified by the seller's legitimate interest in recovering its

[1] *M&J Polymers Ltd v Imerys Minerals Ltd* [2008] EWHC 344 (Comm), [2008] 1 Lloyd's Rep 541.

investment. In the United States, courts have generally upheld take-or-pay clauses, treating them as alternative performance obligations, although extensive litigation followed the collapse of gas prices in the 1980s, when pipeline companies that had contracted to take gas at high prices sought to escape their take-or-pay obligations.

Price Indexation

The price provisions of a long-term GSPA typically combine a base price with an indexation formula that adjusts the price periodically to reflect changes in the value of gas. In oil-indexed contracts, the formula links the gas price to the prices of crude oil, gas oil, or fuel oil, often averaged over a preceding period (for example, six or nine months) and applied with a time lag, smoothing the effect of short-term volatility. In hub-indexed contracts, the price is linked to a hub price, such as TTF or NBP, perhaps with a fixed discount or premium. Hybrid formulas combine oil and hub indexation.

Price Review

Because long-term contracts run for decades, and because the indexation formula may cease to reflect the market value of gas as markets evolve, most long-term GSPAs contain a **price review** (or price reopener) clause permitting either party to request a revision of the price formula at specified intervals (commonly every three or five years) and, in some cases, at any time upon the occurrence of exceptional changes in market circumstances ("wildcard" reviews). Price review clauses typically provide that if, since the last price review, there has been a significant change in the economic circumstances in the buyer's market beyond the control of the parties, and the contract price no longer reflects the value of gas in that market, the parties shall negotiate in good faith a revision of the price formula. If they fail to agree within a specified period, either party may refer the matter to arbitration or to an expert.

Price review clauses raise distinctive legal issues. The *trigger* question is whether the conditions for a revision are satisfied — whether there has been a sufficiently significant change in circumstances of the specified kind. The *revision* question is what the revised price should be — what price reflects the value of gas in the buyer's market, often measured by reference to the prices paid by comparable buyers under comparable contracts or by reference to the "netback" value of gas from the price of competing fuels. The *scope* question is what the tribunal is empowered to change — whether it may alter only the base price, or also the indexation, the weighting of indices, or the introduction of hub indexation.

The wave of price review arbitrations that followed the decoupling of European hub prices from oil-indexed contract prices after 2009 — when the development of hub trading and the arrival of LNG intended for the United States market led to hub prices falling well below oil-indexed prices — trans-

formed the European gas market. European buyers, committed to purchasing large quantities at oil-indexed prices while their customers could buy cheaper gas at the hubs, invoked price review clauses against their suppliers, including Gazprom, Statoil (now Equinor), and Sonatrach. Many disputes were settled through renegotiation, often introducing a significant element of hub indexation; others were decided by arbitral tribunals, which in a number of reported cases accepted that the emergence of hub prices was a relevant change in circumstances and introduced hub indexation into the contract formula. Because most price review awards are confidential, the jurisprudence is known primarily through practitioners' commentaries and the limited number of awards that have entered the public domain through court proceedings.

One such case is *Atlantic LNG Co of Trinidad and Tobago v Gas Natural Aprovisionamientos SDG SA*, in which a price review award in favour of Gas Natural concerning LNG supplied from Trinidad was challenged before the United States District Court for the Southern District of New York. The court confirmed the award in 2008, rejecting the argument that the tribunal had exceeded its powers in determining the revised price.[2] The case illustrates the deference shown by courts in leading arbitral seats to tribunals' interpretation of price review clauses.

Hardship and Price Review Compared

Price review clauses perform a function similar to hardship clauses in general commercial contracts and to doctrines such as *imprévision* in French administrative law, *Wegfall der Geschäftsgrundlage* (now codified in § 313 of the German Civil Code), and Article 6.2.2 of the UNIDROIT Principles of International Commercial Contracts. They differ, however, in that they are designed to operate routinely and periodically, to keep the price aligned with the market, rather than to address exceptional events that fundamentally alter the equilibrium of the contract. A price review tribunal is not concerned with whether performance has become excessively onerous for one party, but with whether the contract price reflects the market value of gas. Chapter 15 examines hardship clauses in energy contracts generally.

Contractual Flexibility in a Changing Market

The trend in gas contracting over the past two decades has been towards greater flexibility: shorter contract durations, smaller volumes, greater downward quantity tolerance, removal of destination restrictions, hub indexation, and rights to divert and cancel cargoes. In the LNG market, portfolio players — international oil companies and trading houses that aggregate supply from multiple projects and sell to multiple buyers — have intermediated between sellers seeking long-term commitments and buyers seeking flexibility.

[2] *Atlantic LNG Co of Trinidad and Tobago v Gas Natural Aprovisionamientos SDG SA*, (SDNY 2008).

The 2021–2023 crisis complicated this trend. The extreme volatility of hub prices and the scarcity of uncommitted LNG led some buyers, particularly in Asia and in Europe, to sign long-term contracts again, including twenty-year contracts with Qatari and United States suppliers, while European buyers sought to balance supply security against the risk that gas demand would fall over the life of the contract as a result of decarbonisation policies. The resulting contracts frequently include provisions addressing the energy transition, such as rights to reduce volumes in response to changes in climate policy, commitments to reduce methane emissions across the supply chain, and provisions for the future supply of low-carbon gases. The capacity of contract law to accommodate the energy transition is examined further in Chapter 15.

6.7 Cross-Border Gas Infrastructure and Transit

The Distinctive Legal Problem of Cross-Border Pipelines

A cross-border gas pipeline links a producing state, a consuming state, and frequently one or more transit states through a single, fixed, and very expensive physical asset. Once built, the pipeline cannot be relocated, and each state through which it passes acquires a degree of leverage over the others: the producing state can withhold supply, the consuming state can refuse to purchase or delay payment, and the transit state can interrupt flows, demand higher tariffs, or divert gas for its own use. Unlike LNG, which can be redirected to alternative markets, pipeline gas offers no easy exit. The legal framework for cross-border pipelines must therefore address not only the ordinary questions of project finance and regulation but also the political risks that arise from the interdependence of sovereign states.

The legal architecture of cross-border pipelines typically operates at three levels: an **intergovernmental agreement** (IGA) among the states concerned, establishing the principles governing the project, the rights and obligations of the states, and the framework for dispute resolution; **host government agreements** (HGAs) between each state and the project company, setting out the detailed legal and fiscal regime applicable in each territory; and the commercial contracts among the project participants, shippers, and buyers. This structure was developed and refined in the Baku–Tbilisi–Ceyhan (BTC) oil pipeline project, completed in 2006, and has been applied to gas pipelines including the South Caucasus Pipeline, the Trans-Anatolian Natural Gas Pipeline (TANAP), and the Trans Adriatic Pipeline (TAP), which together form the Southern Gas Corridor bringing gas from Azerbaijan to Europe.

International Law of Transit

General international law does not confer a comprehensive right of transit for energy across the territory of another state. Several treaty regimes, however, address transit.

Article V of the General Agreement on Tariffs and Trade 1994 provides that there shall be "freedom of transit through the territory of each contracting party, via the routes most convenient for international transit, for traffic in transit to or from the territory of other contracting parties", and that such traffic shall not be subject to unnecessary delays or restrictions and shall be exempt from customs duties and transit duties, other than charges commensurate with the administrative expenses entailed by transit or with the cost of services rendered. Whether Article V applies to fixed infrastructure such as pipelines, as distinct from goods moving by conventional means of transport, has been debated. The WTO panel in *Russia — Measures Concerning Traffic in Transit* (2019), which concerned Russian restrictions on road and rail transit from Ukraine, is significant chiefly for its interpretation of the national security exception in Article XXI, but it confirmed the justiciability of Article XXI defences, a matter of considerable importance for energy trade (discussed in Chapters 13 and 25).

Article 7 of the Energy Charter Treaty provides that each contracting party shall take the necessary measures to facilitate the transit of energy materials and products consistent with the principle of freedom of transit and without distinction as to the origin, destination, or ownership of such energy materials and products or discrimination as to pricing on the basis of such distinctions, and without imposing any unreasonable delays, restrictions, or charges. Article 7(6) provides that a contracting party through whose area energy materials and products transit shall not, in the event of a dispute over any matter arising from that transit, interrupt or reduce the existing flow prior to the conclusion of the conciliation procedure set out in Article 7(7), except where specifically provided in a contract or other agreement or permitted in accordance with the conclusion of the conciliator. Article 7 does not, however, oblige a contracting party to permit the construction or modification of energy transport facilities or to allow new or additional transit through existing facilities that it demonstrates would endanger the security or efficiency of its energy systems.

Negotiations on a more detailed **Energy Charter Transit Protocol** began in 2000 but were suspended after Russia, which had signed but never ratified the ECT and which terminated its provisional application in 2009, objected to key provisions, including the treatment of existing long-term transit contracts and the "right of first refusal" for holders of expiring transit capacity. The failure of the Transit Protocol, and Russia's withdrawal from provisional application of the ECT, left European gas transit governed largely by bilateral arrangements

and by EU law. The modernisation of the ECT adopted in December 2024, and the withdrawal of the EU and many of its member states from the Treaty, are discussed in Chapters 4 and 14.

Article 125 of the United Nations Convention on the Law of the Sea confers on land-locked states a right of access to and from the sea and freedom of transit through the territory of transit states by all means of transport, the terms and modalities of which are to be agreed through bilateral, subregional, or regional agreements. Submarine pipelines on the continental shelf and in the exclusive economic zone are governed by Articles 58 and 79 of UNCLOS, examined in Chapter 20.

Transit Tariffs and Transit Disputes

Transit states are compensated for the use of their territory through transit fees or tariffs, which may be set by reference to the costs of providing the service, the value of the gas transported, or the bargaining power of the parties. Transit tariffs have been a recurrent source of disputes. The gas disputes between Russia and Ukraine in January 2006 and January 2009 — the latter resulting in a complete cessation of gas flows through Ukraine to south-eastern Europe for almost two weeks in mid-winter — arose from linked disputes over the price of gas supplied to Ukraine, the tariff for transit of Russian gas to Europe, and Ukraine's accumulated debts. They exposed the vulnerability of European consumers to disputes between producer and transit states and prompted the European Union to adopt measures to strengthen security of supply, including the requirement of reverse-flow capacity on interconnectors, and to support the diversification of supply routes. The 2009 crisis was resolved by a set of ten-year contracts between Gazprom and Naftogaz, the disputes under which are examined in the casebox at the end of this chapter. Gas transit through Ukraine to the EU ended on 1 January 2025, when the transit agreement concluded in 2019 expired and was not renewed.

Regional Gas Pipeline Frameworks in Africa

Africa offers several examples of regional and transnational gas pipeline frameworks. The **West African Gas Pipeline** (WAGP), which transports Nigerian gas to Benin, Togo, and Ghana, is governed by the Treaty on the West African Gas Pipeline Project signed by the four states in 2003 and by an International Project Agreement between the states and the West African Gas Pipeline Company. The Treaty established the West African Gas Pipeline Authority as a regional regulator, with powers over tariffs and access, and provided for a harmonised legal and fiscal regime for the pipeline across the four states. The WAGP has suffered from supply interruptions, attributable in part to supply shortfalls in Nigeria and to damage to the pipeline, and from payment arrears by downstream buyers, illustrating the

challenges of regional gas integration where domestic supply and payment discipline are weak.

Other initiatives include the proposed **Nigeria–Morocco Gas Pipeline** (also referred to as the African Atlantic Gas Pipeline), intended to transport Nigerian gas along the West African coast to Morocco and potentially to Europe, for which intergovernmental memoranda have been signed with numerous ECOWAS states and Mauritania; and the **Trans-Saharan Gas Pipeline**, intended to link Nigeria with Algeria via Niger, for which the three governments signed a memorandum of understanding in 2022. These projects face formidable challenges of financing, security, and coordination, but they illustrate the aspiration to use gas infrastructure as an instrument of regional integration. The ECOWAS Energy Protocol of 2003, modelled on the Energy Charter Treaty, contains provisions on investment protection and transit, and the African Continental Free Trade Area (AfCFTA) may in time provide a broader framework for cross-border energy trade (examined in Chapter 13).

Cross-Border LNG: The Greater Tortue Ahmeyim Project

Cross-border cooperation is not confined to pipelines. The Greater Tortue Ahmeyim (GTA) gas field straddles the maritime boundary between Mauritania and Senegal. In 2018, the two governments concluded an Inter-Governmental Cooperation Agreement providing for the joint development of the field as a single unit, with resources and revenues shared equally in the initial phase, subject to future redeterminations, and for the liquefaction of the gas on a floating LNG vessel located at a hub near the maritime boundary. The project, operated by BP with Kosmos Energy and the national oil companies of both states, illustrates the application of the unitisation principles examined in Chapter 5 to a transboundary LNG development, and the use of a bilateral treaty framework to provide the legal stability required for financing in states with no prior history of petroleum production. First LNG cargoes from the project were exported in 2025.

Elsewhere in Africa, the Mozambique LNG project in the Rovuma Basin illustrates the vulnerability of LNG projects to security risks: TotalEnergies declared force majeure in April 2021 following insurgent attacks on the town of Palma near the project site, suspending construction for an extended period, while the Coral South floating LNG project, operated by Eni, began production in 2022. The force majeure declaration, and the subsequent review of the human rights situation around the project commissioned by the operator, raise issues examined in Chapters 15 and 21.

Host Government Agreements and Human Rights

Host government agreements for cross-border pipelines have attracted criticism for conferring extensive stabilisation protections on project companies, poten-

tially restricting the ability of host states to adopt new environmental, labour, and human rights legislation. In response to such criticism concerning the BTC pipeline, the project participants executed a Human Rights Undertaking in 2003 clarifying that the stabilisation provisions would not be invoked to challenge host state regulation in the fields of health, safety, environment, and human rights that was consistent with international standards. The episode contributed to the development of the *Principles for Responsible Contracts* discussed in Chapter 5 and has influenced the drafting of later agreements.

6.8 Gas Supply Disputes and Regulatory Intervention

Typology of Gas Disputes

Gas disputes may be classified into several categories: price review disputes between sellers and buyers under long-term contracts (examined in section 6.6); disputes over volumes, including take-or-pay claims and claims for failure to deliver; disputes over quality and measurement; transit disputes; disputes between investors and host states arising from regulatory measures, such as changes in fiscal terms, domestic supply obligations, or export restrictions; and disputes arising from emergency interventions in gas markets. Many of these disputes are resolved through international commercial arbitration under the rules of the ICC, the LCIA, or the Arbitration Institute of the Stockholm Chamber of Commerce (SCC), or through investment treaty arbitration.

Supply Failure and Diversion: The Egyptian Disputes

The supply failures in Egypt following the political upheavals of 2011 generated a series of significant awards. Egypt had committed to supply gas for export, both by pipeline to Israel and Jordan through the Arab Gas Pipeline and the East Mediterranean Gas (EMG) pipeline, and to LNG export plants at Damietta and Idku. As domestic demand outstripped production and the pipeline to Israel was repeatedly attacked in the Sinai, Egypt diverted gas to the domestic market and terminated the EMG supply contract in 2012.

The resulting claims included an ICC arbitration brought by the Israel Electric Corporation against the Egyptian General Petroleum Corporation (EGPC) and the Egyptian Natural Gas Holding Company (EGAS), in which the tribunal in 2015 awarded the Israel Electric Corporation approximately USD 1.76 billion; and the ICSID case *Ampal-American Israel Corp v Egypt*, in which the tribunal found that Egypt had breached the fair and equitable treatment standard by failing to protect the pipeline and by the manner in which the supply contract was terminated. In *Unión Fenosa Gas SA v Arab Republic of Egypt* (2018), an ICSID tribunal held that Egypt had breached the fair and equitable treatment standard of the Spain–Egypt bilateral investment treaty by failing to supply gas to the Damietta LNG plant, in which the claimant held an interest, and awarded damages of

approximately USD 2 billion. The tribunal rejected Egypt's defence of necessity, holding that Egypt had contributed to the situation of necessity through its domestic gas pricing policies, which had encouraged domestic consumption. These disputes were eventually resolved through settlements, including arrangements that facilitated the resumption of LNG exports from Damietta and the reversal of the EMG pipeline to carry Israeli gas to Egypt.

The Egyptian disputes illustrate the tension between export commitments and domestic supply obligations, the limits of the necessity defence in investment law (examined in Chapter 14), and the way in which regional gas flows can reverse dramatically as production patterns change.

The European Gas Crisis of 2021–2023

The most significant gas supply crisis of the modern era began in 2021, when Gazprom reduced deliveries to European buyers to the minimum contractual levels and ceased selling additional volumes on the spot market, while European storage levels fell. Following the Russian invasion of Ukraine in February 2022, Russia required European buyers to pay for gas in roubles through a mechanism established by Presidential Decree No 172 of 31 March 2022, and suspended supplies to buyers that refused to comply, including buyers in Poland, Bulgaria, Finland, Denmark, and the Netherlands. Gazprom progressively reduced flows through the Nord Stream pipeline, citing technical problems with turbines, before halting them entirely in August 2022. In September 2022, explosions damaged both Nord Stream pipelines. European hub prices reached unprecedented levels, peaking in August 2022 at more than ten times their historical average.

The crisis generated extensive commercial arbitration between European buyers and Gazprom. In 2024, the German importer Uniper announced that an arbitral tribunal had awarded it damages of more than EUR 13 billion for Gazprom's failure to deliver contracted gas, and that it had been granted the right to terminate its long-term contracts. Other European buyers obtained similar awards. Gazprom responded by obtaining anti-arbitration and anti-suit injunctions from Russian courts under Articles 248.1 and 248.2 of the Russian Arbitrazh Procedure Code, introduced in 2020, which assert exclusive jurisdiction of Russian courts over disputes involving sanctioned Russian persons. The enforcement of awards against Gazprom's assets outside Russia raises complex questions examined in Chapter 26.

Regulatory Intervention: The European Emergency Measures

The European Union responded to the crisis with an unprecedented series of emergency measures, many adopted under Article 122 TFEU, which permits the Council to adopt measures appropriate to the economic situation "in a spirit of solidarity between Member States", in particular if severe difficulties arise in the

supply of certain products, notably in the area of energy. The principal measures included:

- **Regulation (EU) 2022/1032** on gas storage, amending the Security of Gas Supply Regulation (EU) 2017/1938, which required member states to fill their underground storage facilities to 80 per cent of capacity by 1 November 2022 and 90 per cent by 1 November in subsequent years;
- **Council Regulation (EU) 2022/1369** on coordinated demand-reduction measures, which established a voluntary target to reduce gas consumption by 15 per cent compared with the average of the previous five years, with the possibility of a mandatory target in a "Union alert";
- **Council Regulation (EU) 2022/2576** on enhancing solidarity, which established the AggregateEU mechanism for joint purchasing of gas and default solidarity rules;
- **Council Regulation (EU) 2022/2578** establishing a market correction mechanism, which capped prices for TTF front-month derivatives in defined circumstances; and
- the **REPowerEU** plan of May 2022, which set the objective of ending EU dependence on Russian fossil fuels, and subsequent legislative proposals in 2025 to phase out remaining imports of Russian gas.

At national level, member states nationalised or recapitalised gas importers facing insolvency as a result of the need to purchase replacement gas at spot prices to meet their supply obligations — notably Germany's stabilisation of Uniper, in which the federal government acquired a stake of approximately 99 per cent — placed Russian-owned subsidiaries under trusteeship, as Germany did with Gazprom Germania (later renamed SEFE), and introduced price caps, windfall taxes, and subsidies for consumers.

The Principle of Energy Solidarity

The crisis gave new prominence to the principle of energy solidarity in Article 194(1) TFEU, which provides that EU energy policy shall aim "in a spirit of solidarity between Member States" to ensure the functioning of the energy market, ensure security of energy supply, promote energy efficiency and renewable energy, and promote the interconnection of networks. In *Germany v Poland* (Case C-848/19 P), decided in July 2021, the Court of Justice confirmed the judgment of the General Court in the OPAL case (examined as a casebox in Chapter 4), holding that the principle of energy solidarity is a legally binding principle that entails rights and obligations for both the EU and its member states, and that EU institutions must assess, in adopting measures in the energy field, whether there is a risk to the energy interests of member states. The principle has been invoked in support of the emergency measures adopted in 2022 and 2023.

Limits on Regulatory Intervention

Regulatory interventions in gas markets, however justified by emergency, raise legal questions concerning their compatibility with property rights, contract rights, investment treaties, and the rule of law. Windfall taxes and price caps have been challenged before national courts and the Court of Justice of the European Union, and some interventions have given rise to investment treaty claims. The legal limits of emergency intervention, including the doctrine of necessity in international law and the treatment of emergency measures under investment treaties, are examined in Chapters 14 and 25. The general lesson of the crisis for gas law is that the commercial framework of long-term contracts and liberalised markets operates within a political and security environment that can override it, and that the law must provide mechanisms both for decisive intervention in emergencies and for the protection of legitimate expectations and the restoration of normal market functioning.

Gas Supply Security Regulation Beyond Europe

The European experience is the most fully developed, but security of gas supply regulation exists in many jurisdictions. In Japan and Korea, which depend almost entirely on imported LNG, governments have encouraged buyers to maintain diversified portfolios of long-term contracts and have established strategic buffer LNG arrangements; Japan introduced a "strategic buffer LNG" mechanism in 2023 under which the state-backed Japan Organization for Metals and Energy Security (JOGMEC) may procure LNG for use in emergencies. China has imposed gas storage obligations on suppliers and local governments following winter shortages in 2017 and 2018, when an aggressive coal-to-gas switching policy outpaced supply. In the United States, the Federal Energy Regulatory Commission and the North American Electric Reliability Corporation have examined the interdependence of gas and electricity systems following Winter Storm Uri in Texas in February 2021, during which freezing of gas production and processing facilities contributed to the failure of the Texas electricity system and the loss of life. These experiences demonstrate that gas security is a function not only of supply sources but also of the physical resilience of infrastructure and the coordination of gas and power markets.

In Nigeria, the principal security of supply challenges are domestic: inadequate gas supply to power stations, attributable to pipeline vandalism, insufficient processing and transportation capacity, payment arrears in the electricity value chain, and the historic preference of producers for export markets that pay in foreign currency. The PIA's domestic gas delivery obligation, the Gas Transportation Network Code, and the initiatives of the Decade of Gas are intended to address these challenges, but their success depends on resolving the liquidity crisis in the

electricity sector examined in Chapter 9, on the security of pipeline infrastructure, and on the gradual movement of domestic gas prices towards cost-reflective levels.

LEADING AUTHORITY

The Gazprom–Naftogaz Arbitrations (SCC, 2017–2018)

Background. Following the January 2009 gas crisis, Gazprom and Naftogaz of Ukraine concluded two ten-year contracts: a supply contract, under which Gazprom sold gas to Naftogaz for consumption in Ukraine, and a transit contract, under which Naftogaz transported Russian gas across Ukraine to European buyers. The supply contract contained a take-or-pay obligation for large annual quantities at an oil-indexed price that, following the fall in oil prices and the deterioration of relations after 2014, Naftogaz regarded as far above the market value of gas. In 2014 the parties commenced arbitration under the Rules of the Arbitration Institute of the Stockholm Chamber of Commerce, with claims by Gazprom for unpaid gas and take-or-pay amounts exceeding USD 30 billion, and claims by Naftogaz for a revision of the supply contract price, the removal of the take-or-pay obligation and the prohibition on re-export, and compensation for Gazprom's failure to deliver contracted transit volumes.

Awards. In the supply arbitration, the tribunal's final award of December 2017 substantially reduced Gazprom's take-or-pay claim, revised the contract price to reflect the market in continental Europe, and invalidated the prohibition on re-export, while ordering Naftogaz to pay for gas delivered but unpaid. In the transit arbitration, the tribunal's final award of February 2018 held that Gazprom had failed to deliver the agreed transit volumes and awarded Naftogaz damages of approximately USD 4.6 billion. After netting off the amounts awarded in the two proceedings, Gazprom owed Naftogaz approximately USD 2.56 billion. Gazprom's challenges before the Swedish courts were unsuccessful, and Naftogaz commenced enforcement proceedings against Gazprom's assets in several European jurisdictions. In December 2019, alongside the conclusion of a new five-year transit arrangement, Gazprom paid Naftogaz approximately USD 2.9 billion in settlement of the awards, and the parties withdrew pending claims.

Significance. The arbitrations are among the largest commercial arbitrations in history. They demonstrate the willingness of arbitral tribunals to revise price and volume provisions in long-term gas contracts where market circumstances have fundamentally changed and the contract so permits, and to sever export restrictions incompatible with competition law. They also illustrate the interaction between commercial arbitration and geopolitics: the enforcement of the awards and their eventual settlement were intertwined with intergovernmental negotiations over the future of gas transit through Ukraine, mediated by the European Commission.

PRACTICE NOTE

Drafting checklist: key protections for a buyer in a long-term LNG SPA

1. Downward quantity tolerance sufficient to absorb foreseeable demand fluctuations, including those resulting from decarbonisation policies.
2. Destination flexibility (FOB delivery or DES with diversion rights) and the absence of profit-sharing on resale, subject to applicable competition law.
3. A price review clause with a clear trigger, a defined reference market, express authority for the tribunal to change the indexation, and a time limit for determination.
4. Shortfall remedies that are not exclusive in cases of wilful or deliberate non-delivery, and a cap on the seller's ability to invoke upstream supply failures as force majeure.
5. Quality specifications compatible with the buyer's terminal and downstream network, with clear allocation of the costs of treating off-specification LNG.
6. Methane intensity reporting and reduction commitments across the supply chain, aligned with applicable import regulations such as the EU Methane Regulation.
7. Termination rights for prolonged force majeure and for sanctions that render performance unlawful, and a neutral seat of arbitration in a New York Convention state.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Simulated price review arbitration (hypothetical). *All parties, facts, and figures in this exercise are fictitious and created solely for teaching purposes.* Coastal Gas Exporters Ltd, a fictional liquefaction company in the fictional Republic of Ardenia, supplies 2 million tonnes per annum of LNG to Northern Utilities AG, a fictional utility in the fictional state of Valdora, under a twenty-year DES SPA concluded in 2015. The price is 14 per cent of the Brent crude oil price (three-month average) plus USD 0.50 per MMBtu. Clause 12 permits either party, every four years, to request a price review if "there has been a significant change in circumstances in the Buyer's Market, beyond the control of the Parties, as a result of which the Contract Price no longer reflects the value of LNG in the Buyer's Market", and provides that in default of agreement the tribunal "shall determine a revised Contract Price that reflects the value of LNG in the Buyer's Market". In 2025, Northern Utilities requests a price review, arguing that the Valdoran market is now priced predominantly by reference to TTF and that hub prices have been consistently below the contract price. Coastal Gas Exporters argues that the shift to hub pricing predated the 2021 review and that the 2021–2023 crisis showed that oil-indexed prices provide valuable security.

Divide into teams representing the claimant, the respondent, and the tribunal. (a) Prepare written submissions of not more than 2,000 words on the trigger question and the revision question, identifying the evidence each party would need (for example, comparable contract data and expert evidence on market value). (b) Conduct a thirty-minute oral hearing. (c) As the tribunal, draft the dispositive section of a

reasoned award, addressing whether the tribunal has authority to introduce hub indexation into the price formula.

CHAPTER SUMMARY

- The physical properties of natural gas — low energy density, the need for fixed transport infrastructure, and variable composition — explain its distinctive legal regime: long-term contracts, regulation of network monopolies, and detailed provisions on quality, measurement, and balancing.
- Gas markets have developed as regional markets with distinct pricing systems — gas-on-gas competition at hubs such as Henry Hub, NBP, and TTF; oil indexation; regulated pricing; and spot LNG — although LNG trade has promoted convergence.
- Gas-specific upstream regulation addresses rights to non-associated gas, domestic supply obligations, fiscal incentives, and the restriction of flaring and venting, increasingly through binding methane regulation.
- Liberalisation has required third-party access to pipelines and LNG terminals, unbundling of network and supply activities, and regulation of tariffs, models that Nigeria's PIA has adopted for the midstream gas sector.
- LNG projects are structured as integrated, merchant, tolling, or hybrid projects, within host-state frameworks that frequently include project-specific statutes or agreements.
- Long-term gas sales agreements allocate volume risk through take-or-pay clauses and price risk through indexation and price review clauses; competition law has curtailed destination restrictions.
- Cross-border pipelines are governed by intergovernmental and host government agreements and by transit regimes in GATT Article V and ECT Article 7; African initiatives such as the WAGP illustrate the promise and difficulties of regional integration.
- Gas supply disputes — from the Egyptian supply failures to the European crisis of 2021–2023 — reveal the interaction between commercial contracts, investment protection, emergency regulation, and geopolitics.

DOCTORAL RESEARCH QUESTIONS

1. Has the take-or-pay clause outlived its economic rationale in liberalised gas markets, or does the experience of the 2021–2023 crisis demonstrate its continuing value for security of supply?
2. To what extent do price review tribunals act as market regulators, and what are the implications of the confidentiality of their awards for the development of a coherent body of gas law?

3. Evaluate the adequacy of GATT Article V and ECT Article 7 as frameworks for energy transit in the light of the Russia–Ukraine transit experience and the failure of the Energy Charter Transit Protocol.
4. How should the law reconcile domestic gas supply obligations in producing developing countries with export commitments and investment protection, in the light of the Egyptian experience?
5. Does the principle of energy solidarity in Article 194 TFEU, as interpreted in *Germany v Poland*, offer a model for regional energy integration in West Africa?
6. Can the extraterritorial reach of import-side methane regulation, such as the EU Methane Regulation, be reconciled with the principle of common but differentiated responsibilities and with WTO law?

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7 CHAPTER

Mining Law and the Governance of Mineral Resources

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal foundations of mineral rights and the principal systems for allocating mining titles.
2. Analyse the structure of exploration permits, mining licences, and the cadastral systems that administer them.
3. Evaluate the role of mining concessions and investment agreements and the controversies surrounding their stability provisions.
4. Identify the environmental and social obligations imposed on mining operations by domestic and international law.
5. Assess legal mechanisms for community development and benefit sharing, including community development agreements.
6. Analyse the regulation of mine closure, rehabilitation, and financial assurance.
7. Evaluate approaches to the regulation and formalisation of artisanal and small-scale mining.
8. Critically assess international standards for responsible mining and their interaction with domestic law.

Introduction

Mining is one of humanity's oldest economic activities and one of its most legally complex. Minerals extracted from the earth supply the metals of modern infrastructure and technology, the fertilisers of agriculture, the materials of construction, and, increasingly, the components of the energy transition: copper for electrical networks, lithium, cobalt, nickel, and graphite for batteries, rare earth elements for permanent magnets in wind turbines and electric motors, and silver and silicon for solar panels. At the same time, mining transforms landscapes, consumes vast quantities of water and energy, generates enormous volumes of waste rock and tailings, displaces communities, and has been associated in many

countries with conflict, corruption, and human rights abuses. The law of mining must reconcile the interests of states seeking revenue and development, investors seeking secure and profitable titles, communities seeking protection and a fair share of benefits, and the broader public interest in environmental protection.

Mining law shares many features with petroleum law, examined in Chapter 5: both regulate the extraction of non-renewable subsurface resources, both are founded on the state's ownership or control of those resources (examined in Chapter 3), and both rely on systems of titles, fiscal regimes, and contracts to allocate rights and revenues between states and investors. But mining differs from petroleum in important respects. Mineral deposits are fixed in place and do not migrate across boundaries in the manner of petroleum reservoirs, so the rule of capture and unitisation are rarely relevant. The diversity of minerals — from bulk commodities such as iron ore and coal, to base metals such as copper and zinc, to precious metals and gemstones, to industrial minerals and construction materials — and the diversity of operators — from multinational corporations operating giant open-pit mines to millions of individual artisanal miners using hand tools — produce a far more heterogeneous regulatory landscape. Mining also has a much larger physical footprint than most petroleum operations and a longer environmental legacy, since tailings facilities and waste dumps must be managed in perpetuity after mining ends.

The modern history of mining law in developing countries has been characterised by successive waves of reform. In the 1960s and 1970s, many newly independent states nationalised their mining industries, as Zambia did with its copper mines in 1969–1970 and Zaire (now the Democratic Republic of the Congo) did with the Union Minière du Haut-Katanga in 1967. The poor performance of many state mining enterprises, combined with the debt crises of the 1980s, led in the 1990s to a wave of liberalising reforms, often supported by the World Bank, which privatised state enterprises and adopted investor-friendly mining codes offering secure titles, low royalties, and generous fiscal incentives. The commodity price boom of the 2000s, driven by Chinese demand, led many states to conclude that these codes had transferred too great a share of mineral rents to investors, prompting a third wave of reforms — revised mining codes in the Democratic Republic of the Congo (2018), Tanzania (2017), Zambia, Guinea, and elsewhere — that increased royalties and state participation, curtailed stability guarantees, and strengthened local content and community development requirements. The Africa Mining Vision, adopted by African Union heads of state in 2009, articulates the aspiration for "transparent, equitable and optimal exploitation of mineral resources to underpin broad-based sustainable growth and socio-economic development".

This chapter examines the core institutions of mining law. It begins with mineral rights and mining titles (section 7.1), exploration permits and mining licences (section 7.2), and mining concessions and investment agreements (section 7.3). It then examines environmental and social obligations (section 7.4), community development and benefit sharing (section 7.5), and mine closure and rehabilitation (section 7.6). It concludes with the regulation of artisanal and small-scale mining (section 7.7) and international standards for responsible mining (section 7.8). Chapter 8 addresses the distinctive legal issues raised by critical minerals, and Chapter 20 considers deep-seabed mining under the law of the sea.

7.1 Mineral Rights and Mining Titles

Ownership of Minerals

The starting point of any mining regime is the ownership of minerals in their natural state. As explained in Chapter 3, legal systems allocate subsurface mineral ownership according to two broad models: the *accession* model, under which minerals belong to the owner of the surface land (the position at common law, retained for most minerals in the United States in respect of private land); and the *domanial* or *regalian* model, under which minerals are owned or controlled by the state irrespective of surface ownership (the position in most civil law jurisdictions and in almost all developing countries). Even where the accession principle applies, certain minerals are often reserved to the Crown or state: in the United Kingdom, gold and silver in their natural state belong to the Crown under the *Case of Mines* (1568), and coal was nationalised under the Coal Industry Nationalisation Act 1946 and is now vested in the Coal Authority.

In Nigeria, section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 vests "the entire property in and control of all minerals, mineral oils and natural gas in, under or upon any land in Nigeria or in, under or upon the territorial waters and the Exclusive Economic Zone of Nigeria" in the Government of the Federation, and "mines and minerals, including oil fields, oil mining, geological surveys and natural gas" appear in the Exclusive Legislative List. Section 1 of the Nigerian Minerals and Mining Act 2007 reiterates that the property in and control of all mineral resources in, under, or upon any land in Nigeria is vested in the Government of the Federation for and on behalf of the people of Nigeria. The exclusive federal competence over minerals has been controversial in a federation where states and communities bear the environmental and social costs of mining, and it has complicated the regulation of artisanal mining, which is often more effectively supervised at local level. Proposals to devolve control of solid minerals to the states have featured in constitutional reform debates.

In South Africa, the Mineral and Petroleum Resources Development Act 2002 (MPRDA) made a fundamental change. Before the Act, mineral rights in

South Africa could be privately owned and severed from surface ownership, and were concentrated in the hands of a small number of predominantly white-owned mining houses. Section 3(1) of the MPRDA declares that mineral and petroleum resources are "the common heritage of all the people of South Africa" and that the state is "the custodian thereof for the benefit of all South Africans". The Act abolished private ownership of unexploited mineral rights and required holders of "old order rights" to convert them into new order rights granted by the state within transitional periods. In *Agri South Africa v Minister for Minerals and Energy* (2013), the Constitutional Court held by a majority that the MPRDA did not effect an expropriation of the old order rights requiring compensation, because the state had not acquired the rights for itself but had become their custodian.[1] The decision is significant for the constitutional theory of resource ownership and for the distinction between deprivation and expropriation, examined in Chapter 3.

The Nature of Mining Titles

Where minerals belong to the state, a person may lawfully explore for or extract them only by virtue of a title granted by the state. As with petroleum titles, the legal nature of mining titles varies among legal systems. In civil law systems influenced by French law, mining titles have traditionally been characterised as concessions — administrative acts conferring real rights in immovable property that may be mortgaged, transferred with consent, and protected against arbitrary withdrawal. In common law systems, mining leases and licences are statutory rights, the nature of which depends on the construction of the statute, but which are commonly treated as property for the purposes of constitutional protection and as capable of being mortgaged and transferred. In many African mining codes, mining titles are expressly declared to be real rights or immovable property, distinct from the ownership of the surface land, in order to facilitate their use as security for project finance.

Mining titles usually confer the exclusive right to carry out the specified activity within the title area for the specified minerals, together with ancillary rights to use land, water, and timber, to construct infrastructure, and to dispose of the minerals produced. They are subject to obligations: to carry out work programmes, to pay fees, royalties, and taxes, to comply with environmental, health, safety, and labour requirements, to report to the regulator, and to rehabilitate the land at the end of operations.

Systems for Allocating Mining Titles

Mining regimes employ three principal methods for allocating titles: first-come, first-served; competitive tender; and administrative discretion.

[1] *Agri South Africa v Minister for Minerals and Energy* [2013] ZACC 9, 2013 (4) SA 1 (CC).

Under the **first-come, first-served** system, the first qualified applicant to apply for an available area is entitled to the title, provided that it meets the statutory requirements. The system has its origins in the "free mining" tradition of the American West, under which prospectors could stake claims on public land by physically marking the boundaries and recording the claim; the United States General Mining Law of 1872 still permits citizens to locate claims to "valuable mineral deposits" on open federal lands. In modern regimes, the first-come, first-served system is typically administered through an electronic mining cadastre that records applications in order of receipt and grants titles to the first applicant that satisfies objective criteria. Its advantages are simplicity, transparency, and the reduction of administrative discretion, which limits opportunities for corruption. It is well suited to exploration licences over areas where geological information is limited and the value of the area is uncertain.

Under the **competitive tender** system, titles over areas of known prospectivity — for example, areas where the state has carried out geological surveys or where previous exploration has identified a deposit — are allocated through a competitive bidding process, with bids evaluated on the basis of work programmes, financial terms, technical capacity, or a combination of these. Competitive tender allows the state to capture a greater share of the value of known deposits, but requires greater administrative capacity and robust procedures to prevent manipulation.

Under **administrative discretion**, the minister or another authority grants titles on the basis of a discretionary assessment of the merits of applications. Discretionary systems have been associated with corruption and the allocation of rights to politically connected persons, and the trend in mining law reform has been to reduce discretion and to rely on objective criteria administered by an independent cadastre.

Many regimes combine these methods: first-come, first-served for exploration licences in open areas, competitive tender for areas of known prospectivity, and some residual discretion for strategic deposits.

The Mining Cadastre

The mining cadastre — the official register of mining titles and applications — is the central administrative institution of a modern mining regime. A well-functioning cadastre records the location, holder, type, status, and obligations of every title; prevents overlapping grants; processes applications according to transparent rules; monitors compliance with title obligations, such as payment of annual fees and fulfilment of work programmes; and makes information publicly available. Electronic cadastres, often implemented with support from the World Bank and other development partners, have been introduced in many African

states, including Nigeria, Ghana, Tanzania, Zambia, the Democratic Republic of the Congo, and Mozambique, many offering online portals that enable investors and the public to view title information.

In Nigeria, the Minerals and Mining Act 2007 established the Mining Cadastre Office (MCO) as the agency responsible for the administration of mineral titles and the maintenance of the cadastral register. The Act provides for applications to be processed on a first-come, first-served basis, and the MCO operates an electronic mining cadastre system (eMC+), which enables online applications and public access to the cadastral map. The Act provides for the revocation of titles for failure to pay annual service fees or to comply with work obligations, and the MCO has periodically revoked large numbers of dormant titles held for speculative purposes.

Surface Rights and Land Access

Because mineral ownership is usually separated from surface ownership, mining law must regulate the relationship between title holders and surface owners or occupiers. Mining codes typically confer on title holders a right of access to the land, subject to obligations to obtain the consent of, or give notice to, surface owners and occupiers; to pay compensation for disturbance of surface rights, damage to crops, and loss of use of land; and in some cases to acquire the land through compulsory acquisition by the state. The adequacy of compensation, the treatment of customary land rights, and the relationship between mining law and land law are recurrent sources of conflict, particularly in Africa, where much land is held under customary tenure without formal registration.

In Nigeria, the Land Use Act 1978 vests all land in each state in the governor, to be held in trust for the use and common benefit of all Nigerians, and the Minerals and Mining Act 2007 provides that land required for mining purposes shall have priority over other uses, subject to the payment of compensation to the owners or occupiers for disturbance of surface rights, damage to crops, and other loss. In South Africa, the Constitutional Court held in *Maledu v Itereleng Bakgatla Mineral Resources (Pty) Ltd* (2018) that the holder of a mining right could not evict a community holding informal land rights protected by the Interim Protection of Informal Land Rights Act 1996 without complying with that Act, which requires the consent of the community or the expropriation of the rights with compensation.[2] In *Baleni v Minister of Mineral Resources* (2018), concerning the proposed Xolobeni titanium mine on the Wild Coast, the High Court held that the Minister could not grant a mining right over land occupied by the Umgungundlovu community under informal rights without obtaining the community's

[2] *Maledu v Itereleng Bakgatla Mineral Resources (Pty) Ltd* [2018] ZACC 41, 2019 (2) SA 1 (CC).

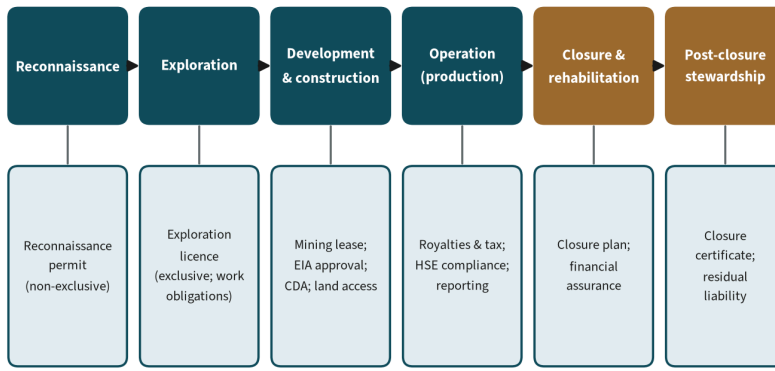
full and informed consent.[3] These decisions are significant for the emergence of community consent requirements in domestic mining law, examined further in Chapter 21.

7.2 Exploration Permits and Mining Licences

The Sequence of Mining Titles

Most modern mining codes establish a sequence of titles corresponding to the stages of the mining life cycle: reconnaissance, exploration, development and production, and closure. Each stage involves different activities, risks, and levels of investment, and the titles granted for each stage differ in their duration, area, rights, and obligations. The holder of an exploration title that discovers a commercially viable deposit typically has a right — subject to conditions — to obtain a mining title over the deposit. This "security of tenure", which links the exploration investment to the right to exploit any resulting discovery, is regarded by investors as the most important feature of a mining regime: no rational company will invest millions of dollars in exploration if the state may refuse a mining licence and award the deposit to someone else.

Stages of the mining life cycle (top) and the corresponding legal instruments (bottom)



Security of tenure links exploration to the right to mine; closure obligations are planned and secured from the outset

Figure 7.1 The mining life cycle and the corresponding sequence of legal titles and approvals. *Source:* prepared for this edition.

Reconnaissance Permits

Reconnaissance is the preliminary search for mineral occurrences by geological, geophysical, and geochemical surveys, remote sensing, and surface sampling,

[3] *Baleni v Minister of Mineral Resources* [2018] ZAGPPHC 829, 2019 (2) SA 453 (GP).

without significant ground disturbance. Reconnaissance permits are typically non-exclusive, cover large areas, and are granted for short periods (one or two years). Under the Nigerian Minerals and Mining Act 2007, a reconnaissance permit confers a non-exclusive right to search for minerals by surface geological and geophysical surveys, and does not entitle the holder to engage in drilling, excavation, or other subsurface activities.

Exploration Licences

Exploration involves detailed investigation of a defined area to locate and evaluate mineral deposits, including drilling, trenching, bulk sampling, and feasibility studies. Exploration licences are exclusive, cover defined areas (often subject to a maximum size), and are granted for initial periods of three to five years, renewable for further periods subject to the fulfilment of work obligations and the relinquishment of a proportion of the area on renewal. Work obligations may be expressed as minimum expenditure commitments or as minimum work programmes. Relinquishment requirements prevent the holding of large areas for speculative purposes and return unexplored land to the state for allocation to others.

Under the Nigerian Act, an exploration licence confers the exclusive right to conduct exploration for the minerals specified in the licence within the licence area. It is granted for an initial period of three years, renewable twice for periods of two years each, subject to the holder having complied with its obligations, and the licence area is subject to maximum size limits. The holder is required to commence exploration within a specified period, to expend minimum amounts on exploration, to submit reports, and to pay annual service fees. Upon discovery of a commercially viable deposit, the holder may apply for a mining lease or a small-scale mining lease.

Exploration licensing raises recurrent legal issues: the transferability of licences (which is usually subject to ministerial consent), the treatment of licences held by "juniors" — small exploration companies, often listed on the Toronto, Australian, or London AIM stock exchanges, that finance exploration through equity markets and seek to sell discoveries to larger companies; the extent of the holder's right to a mining licence; and the confidentiality of exploration data, which must usually be submitted to the state geological survey and may be released after a period.

Mining Licences and Leases

A mining licence or lease confers the exclusive right to develop and operate a mine within a defined area and to sell the minerals produced. Mining licences are granted for long periods — commonly twenty to thirty years, or the estimated life of the mine — renewable for further periods. The application for a mining licence

must usually be supported by a feasibility study demonstrating the technical and economic viability of the deposit, a mining plan, an environmental impact assessment and environmental management plan, a community development plan, and evidence of financial and technical capacity.

Under the Nigerian Act, a mining lease is granted for a period of up to twenty-five years, renewable every twenty-four years, over an area not exceeding a specified maximum. The holder must commence development within a specified period of grant, comply with an approved Environmental Impact Assessment and an Environmental Protection and Rehabilitation Programme, enter into a community development agreement with the host community (discussed in section 7.5), pay royalties and other fees, and comply with the Act's provisions on health, safety, and employment. The Act also provides for small-scale mining leases, quarry leases (for the extraction of construction materials such as granite, marble, and laterite), and water use permits.

The Right to Proceed from Exploration to Mining

The extent to which an exploration licence holder is entitled to a mining licence is critical. Some regimes confer an automatic right to a mining licence upon satisfaction of objective criteria; others confer a right to apply, leaving the state a discretion to refuse on grounds such as environmental considerations, the public interest, or inadequate financial terms. A discretion to refuse undermines security of tenure but preserves the state's ability to protect the public interest.

The tension between security of tenure and regulatory discretion has generated major investment disputes. In *Crystallex International Corp v Bolivarian Republic of Venezuela* (2016), an ICSID tribunal held that Venezuela had breached the fair and equitable treatment standard and expropriated the claimant's investment in the Las Cristinas gold deposit by refusing an environmental permit required for exploitation and subsequently rescinding the mining contract, and awarded damages of approximately USD 1.2 billion plus interest.[4] In *Tethyan Copper Co Pty Ltd v Islamic Republic of Pakistan* (2019), concerning the Reko Diq copper-gold deposit in Balochistan, an ICSID tribunal held that the refusal of a mining lease to the claimant, after it had carried out extensive exploration and a feasibility study, breached the fair and equitable treatment standard of the Australia–Pakistan bilateral investment treaty and constituted an expropriation, and awarded approximately USD 5.9 billion in damages and interest — one of the largest awards in ICSID history.[5] The dispute was resolved in 2022 through a settlement

[4] *Crystallex International Corp v Bolivarian Republic of Venezuela*, ICSID Case No ARB(AF)/11/2, Award (4 April 2016).

[5] *Tethyan Copper Co Pty Ltd v Islamic Republic of Pakistan*, ICSID Case No ARB/12/1, Award (12 July 2019).

under which Barrick Gold, one of the claimant's shareholders, agreed to develop the project in partnership with the federal and provincial governments.

These awards are frequently cited by investors as demonstrating the importance of security of tenure, and by critics of investment arbitration as demonstrating the risk that awards of speculative future profits impose enormous liabilities on developing countries. The methodology of damages in such cases, including the use of discounted cash flow valuation for projects that have never produced, is examined in Chapter 26.

Transfer and Change of Control

Mining titles are valuable assets that are frequently bought and sold, both directly and through transactions involving the shares of title-holding companies. Most mining codes require ministerial consent to the transfer of titles, and increasingly to changes of control of title holders, in order to ensure that transferees meet the qualification requirements and to enable the state to capture a share of the gains through capital gains taxation. Indirect transfers — the sale of shares in an offshore holding company that indirectly owns a mining title — have been a particular concern, since they may allow substantial gains to escape taxation in the host state. Several states, including Uganda, Tanzania, Mozambique, and India, have introduced legislation to tax indirect transfers, and the issue is examined in Chapter 16. The Platform for Collaboration on Tax (comprising the IMF, OECD, United Nations, and World Bank) published a toolkit on the taxation of offshore indirect transfers in 2020.

7.3 Mining Concessions and Investment Agreements

The Role of Mining Agreements

Many large mining projects, particularly in developing countries, are governed not only by the general mining code but also by a project-specific agreement between the state and the investor — variously called a mining agreement, mineral development agreement, mining convention, or investment agreement. Such agreements may supplement the code by providing details specific to the project, or derogate from it by granting special fiscal terms, stability guarantees, or other incentives. In some states, mining agreements must be ratified by the legislature, giving them the force of law, as in Liberia, Sierra Leone, and Ghana (where the Constitution requires parliamentary ratification of agreements relating to the grant of rights to minerals).

Mining agreements typically address: the project area and title; development obligations and timetable; fiscal terms, including royalty rates, income tax rates, depreciation, and import duty exemptions; state participation; foreign exchange and the right to maintain offshore accounts; employment and local procurement; infrastructure, including access to rail, ports, and power; community develop-

ment; environmental obligations; stabilisation; and dispute resolution, usually by international arbitration.

Stabilisation in Mining Agreements

Stabilisation clauses in mining agreements are designed to protect investors from adverse changes in the legal and fiscal regime over the long life of the project. Mining codes themselves may offer stability guarantees: the 2002 Mining Code of the Democratic Republic of the Congo, for example, guaranteed holders of mining rights that the fiscal and customs regime in force at the time of grant would remain unchanged for ten years. Ghana's Minerals and Mining Act 2006 (Act 703) permits the Minister to enter into stability agreements guaranteeing that the holder will not be adversely affected by changes in law for a period not exceeding fifteen years, and development agreements for investments exceeding a specified threshold.

The effectiveness of stabilisation in mining has been sorely tested by the "resource nationalism" of the past two decades. When commodity prices rose dramatically in the 2000s, many states concluded that stabilised fiscal terms negotiated in a period of low prices were inequitable, and sought to renegotiate or override them. The DRC's 2018 Mining Code, which increased the royalty rate for cobalt and other "strategic substances" to ten per cent, introduced a super-profits tax, and reduced the stability period, applied the new provisions to existing title holders notwithstanding the stability guarantee in the 2002 Code, provoking protests from major mining companies. Tanzania's Natural Wealth and Resources (Permanent Sovereignty) Act 2017 and the Natural Wealth and Resources Contracts (Review and Re-negotiation of Unconscionable Terms) Act 2017 required the review of agreements containing "unconscionable" terms, prohibited the submission of disputes relating to natural resources to international arbitration, and asserted the permanent sovereignty of the people of Tanzania over natural resources. In the same year, the Tanzanian government imposed an export ban on gold and copper concentrates and issued tax assessments of approximately USD 190 billion against Acacia Mining, a subsidiary of Barrick Gold. The dispute was resolved through a framework agreement in 2019 under which the government acquired a sixteen per cent free-carried interest in Barrick's Tanzanian mines and a fifty per cent share of economic benefits, and Barrick paid USD 300 million in settlement of the tax claims.

The Renegotiation of Mining Agreements

The renegotiation of mining agreements raises fundamental questions about the balance between *pacta sunt servanda* and the state's sovereign right to regulate in the public interest. Investors argue that stability is essential to attract long-term investment and that states should be held to their bargains; states argue that

agreements concluded under conditions of information asymmetry, weak negotiating capacity, or corruption may lack legitimacy, and that circumstances may change so fundamentally that the original bargain becomes inequitable. Several states have conducted systematic reviews of mining agreements: Liberia reviewed concessions concluded during its civil war; Guinea conducted a review under its 2011 Mining Code, leading to the revocation of the rights of BSG Resources over the Simandou iron ore deposit on the ground that they had been obtained through corruption; and the DRC reviewed sixty-one mining contracts in 2007–2008.

The Guinea case illustrates the relevance of corruption to the validity of mining titles. BSGR commenced ICSID arbitration against Guinea following the revocation of its rights, and Vale, which had paid BSGR USD 500 million for a fifty-one per cent interest in the project, commenced LCIA arbitration against BSGR. In 2019 an LCIA tribunal found that BSGR had obtained its rights through bribery and fraudulently misrepresented the position to Vale, and ordered BSGR to pay Vale approximately USD 2 billion. The English courts subsequently enforced the award. The ICSID proceedings were settled in 2019 through an agreement under which BSGR relinquished its claims to Simandou. The Simandou project, divided into blocks held by Rio Tinto and a consortium including Chinese interests, began exporting iron ore in late 2025, illustrating both the immense value of major mineral deposits and the decades-long delays that legal and political disputes can cause.

Model Mining Development Agreement

In response to concerns about the imbalance of negotiating capacity in mining agreements, the International Bar Association's Mining Law Committee published a Model Mine Development Agreement (MMDA) in 2011, designed as a template that reflects a balance of the interests of states, investors, and communities. The MMDA includes provisions on community development, environmental and social impact assessment, transparency, and dispute resolution, and is accompanied by commentary explaining the policy choices involved. Other resources include the Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development (IGF) Mining Policy Framework of 2013, the African Minerals Development Centre's Country Mining Vision guidebook, and the negotiation support initiatives of the African Legal Support Facility, established by the African Development Bank to provide legal advice to African governments in complex commercial negotiations, including mining agreements.

Mining Investment Disputes and Social Licence

A significant body of investment arbitration concerns the interaction between mining investments and community opposition. In *Bear Creek Mining Corp v*

Republic of Peru (2017), the claimant had obtained rights to the Santa Ana silver project near the border with Bolivia, but the government revoked the decree authorising the project following violent protests by local Aymara communities. The tribunal held that the revocation constituted an indirect expropriation under the Canada–Peru free trade agreement, rejecting Peru's argument that the claimant's failure to obtain a "social licence to operate" should defeat or reduce its claim, although the tribunal limited damages to the amounts actually invested.[6] A partially dissenting arbitrator, Professor Philippe Sands, would have reduced damages on the ground that the investor had contributed to the social unrest by failing to engage adequately with the affected communities. In *Copper Mesa Mining Corp v Republic of Ecuador* (2016), the tribunal reduced the damages awarded to the investor by thirty per cent on account of contributory fault, because the investor's contractors had used armed security personnel against protesters. In *Eco Oro Minerals Corp v Republic of Colombia* (2021), the tribunal held that Colombia's prohibition of mining in the Santurbán páramo, a high-altitude wetland ecosystem, breached the minimum standard of treatment under the Canada–Colombia FTA, but declined to find that the general environmental exception in the treaty precluded liability, and later awarded only limited damages. In *Gabriel Resources Ltd v Romania* (2024), concerning the Roșia Montană gold project, the tribunal dismissed the investor's claims, finding that the investor had not secured the social licence necessary for the project and that the state's conduct did not breach the treaty.

These cases illustrate the growing significance of community acceptance in mining investment law and the extent to which arbitral tribunals are prepared to consider the investor's own conduct in assessing liability and damages. They are examined further in Chapters 14 and 21.

7.4 Environmental and Social Obligations

The Environmental Footprint of Mining

Mining has profound environmental impacts across its life cycle. Exploration may involve the clearing of vegetation, the construction of access roads, and drilling. Development and operation involve the removal of overburden, the excavation of open pits or underground workings, the disposal of waste rock, the processing of ore using chemicals such as cyanide (in gold processing) and sulphuric acid (in copper heap leaching), the consumption of large volumes of water and energy, and the storage of tailings — the finely ground residue remaining after the extraction of valuable minerals — in tailings storage facilities that may be among the largest engineered structures on earth. Acid mine drainage, which occurs when sulphide

[6] *Bear Creek Mining Corp v Republic of Peru*, ICSID Case No ARB/14/21, Award (30 November 2017).

minerals exposed by mining react with air and water to produce sulphuric acid that mobilises heavy metals, can contaminate watercourses for centuries. The catastrophic failure of tailings dams — at Mount Polley in British Columbia in 2014, at Fundão (Mariana) in Brazil in 2015, and at Córrego do Feijão (Brumadinho) in Brazil in 2019, the last of which killed 270 people — has demonstrated the scale of the risks.

Environmental Impact Assessment and Permitting

Virtually all modern mining codes and environmental statutes require an environmental impact assessment (EIA), or an environmental and social impact assessment (ESIA), before a mining licence is granted or operations commence. The EIA process, examined generally in Chapter 17, requires the proponent to identify and evaluate the likely environmental and social impacts of the project, to consider alternatives, to propose measures to avoid, minimise, mitigate, and compensate for adverse impacts, and to consult the public and affected communities. The outcome is typically an environmental management plan or programme, the implementation of which becomes a condition of the licence.

In Nigeria, the Environmental Impact Assessment Act 1992 (now the Environmental Impact Assessment Act, Cap E12, Laws of the Federation of Nigeria 2004) requires an EIA for mining projects, and the Minerals and Mining Act 2007 requires the holder of a mining lease, small-scale mining lease, or quarry lease to submit an Environmental Impact Assessment statement and an Environmental Protection and Rehabilitation Programme (EPRP) to the Mines Environmental Compliance Department of the Ministry of Solid Minerals Development. The Federal Ministry of Environment administers the EIA Act, and the National Environmental Standards and Regulations Enforcement Agency (NESREA) has issued regulations on mining and processing of coal, ores, and industrial minerals.

Water, Biodiversity, and Protected Areas

Mining's demands on water are a growing source of conflict, particularly in arid regions such as the Atacama Desert in Chile, where copper and lithium mining compete with agriculture and indigenous communities for scarce water. Mining laws increasingly require water use permits, impose limits on extraction, and require the treatment of discharges. In Chile, the 2022 reform of the Water Code (Law No 21.435) recognised access to water for human consumption and sanitation as a human right and limited the duration of new water rights, with significant implications for mining.

Many jurisdictions prohibit or restrict mining in protected areas, such as national parks, wetlands, and areas of high biodiversity value. Colombia's prohibition of mining in páramo ecosystems, upheld by its Constitutional Court in a 2016 decision and challenged in the *Eco Oro* arbitration discussed in section

7.3, is an example. International instruments such as the Ramsar Convention on Wetlands (1971), the World Heritage Convention (1972), and the Convention on Biological Diversity (1992) inform these restrictions, and the Kunming-Montreal Global Biodiversity Framework, adopted in December 2022, sets the target of effectively conserving thirty per cent of the world's terrestrial and marine areas by 2030 (the "30 by 30" target), which may constrain the availability of land for mining. The International Council on Mining and Metals (ICMM) committed its members in 2003 not to explore or mine in World Heritage properties, a commitment that has been influential in the industry.

Tailings Governance

The tailings dam disasters of the past decade have transformed the governance of tailings facilities. Following the Brumadinho disaster, Brazil enacted Law No 14.066 of 2020, amending its National Dam Safety Policy, which prohibits the construction of "upstream" tailings dams — a design, in which successive embankments are built on top of previously deposited tailings, that was used at both Fundão and Brumadinho — and requires the decommissioning of existing upstream dams. Chile and Peru had prohibited upstream dams in seismically active areas some years earlier.

At the international level, the **Global Industry Standard on Tailings Management** (GISTM), published in August 2020 following an independent review co-convened by the ICMM, the United Nations Environment Programme, and the Principles for Responsible Investment, establishes requirements for the safe management of tailings facilities, with the goal of zero harm to people and the environment. It requires operators to take a precautionary approach, to design and operate facilities to minimise risk, to establish accountability at board level, to engage meaningfully with affected communities, to prepare for emergencies, and to disclose information about their facilities. ICMM members committed to implementing the standard at their highest-consequence facilities by August 2023 and at all other facilities by August 2025. Although the GISTM is a voluntary industry standard, it is increasingly referenced by regulators, investors, and insurers, and its requirements may inform the standard of care in negligence litigation.

Litigation Arising from Mining Disasters

The legal consequences of the Fundão disaster illustrate the multiple layers of liability arising from mining catastrophes. The collapse of the dam, operated by Samarco Mineração SA, a joint venture between Vale and BHP, released about forty million cubic metres of tailings into the Doce River basin, killing nineteen people and causing extensive environmental and economic damage along hundreds of kilometres of river to the Atlantic coast. In Brazil, the companies entered into a settlement in 2016 establishing the Renova Foundation to carry out reme-

diation and compensation programmes, and in October 2024 the companies and Brazilian public authorities signed a revised settlement valued at approximately BRL 170 billion. In England, more than six hundred thousand claimants, including municipalities, businesses, and individuals, brought proceedings against BHP's English and Australian parent companies. The Court of Appeal in *Municipio de Mariana v BHP Group plc* (2022) reversed the High Court's decision striking out the claims as an abuse of process, allowing them to proceed,[7] and a liability trial took place in 2024 and 2025. The case is one of the largest group actions ever brought in the English courts and is examined further, alongside parent company liability generally, in Chapters 21 and 27.

Social Obligations

Mining codes and licences impose a range of social obligations on title holders, including obligations to employ and train nationals, to procure goods and services locally, to respect labour and occupational health and safety standards, to protect cultural heritage, and to contribute to community development. The destruction by Rio Tinto in May 2020 of the 46,000-year-old rock shelters at Juukan Gorge in the Pilbara region of Western Australia, sacred to the Puutu Kunti Kurrama and Pinikura peoples, although lawful under a ministerial consent granted under the Aboriginal Heritage Act 1972 (WA), led to the resignation of Rio Tinto's chief executive, a parliamentary inquiry, and the enactment of the Aboriginal Cultural Heritage Act 2021 (WA). That Act was itself repealed in 2023 after objections from landholders that its requirements were unworkable, and the 1972 Act was restored with amendments. The episode illustrates the gap between legal compliance and social acceptability, and the speed with which reputational and investor pressures can transform corporate practice.

Occupational health and safety in mining is governed by domestic legislation and by international standards, notably the ILO Safety and Health in Mines Convention 1995 (No 176), which requires ratifying states to adopt and enforce laws ensuring safe working conditions in mines, to establish inspection systems, and to confer rights on workers, including the right to remove themselves from danger. Labour rights in extractive industries are examined in Chapter 21.

7.5 Community Development and Benefit Sharing

The Rationale for Benefit Sharing

Communities located near mining operations bear a disproportionate share of the costs of mining — loss of land and livelihoods, environmental degradation, social disruption, and health risks — while the principal benefits, in the form of royalties and taxes, accrue to the central government. This mismatch fuels

[7] *Municipio de Mariana v BHP Group (UK) Ltd* [2022] EWCA Civ 951.

conflict between companies, communities, and governments, and has led to the development of legal mechanisms for sharing the benefits of mining with affected communities. Benefit sharing may take the form of fiscal transfers from central to local government, direct payments or equity participation for communities, local employment and procurement commitments, and investments in community infrastructure and services. These mechanisms aim not only to compensate communities for the costs they bear but also to secure their support for mining operations — the "social licence to operate".

Community Development Agreements

The community development agreement (CDA) — also called an impact and benefit agreement (IBA) in Canada, an indigenous land use agreement in Australia, or a social and labour plan in South Africa — is a contract between a mining company and a host community that sets out the company's obligations to the community and the community's support for the project. CDAs may be voluntary or mandated by law.

Section 116 of Nigeria's Minerals and Mining Act 2007 requires the holder of a mining lease, small-scale mining lease, or quarry lease to conclude a community development agreement with the host community before commencing development. The CDA must contain undertakings with respect to the social and economic contribution that the project will make to the sustainability of the community, and may address educational scholarships, apprenticeships, technical training, and employment opportunities for indigenes of the community; financial or other forms of contributory support for infrastructural development and maintenance, such as education, health, or other community services, roads, water, and power; assistance with the creation, development, and support of small-scale and micro-enterprises; agricultural product marketing; and methods and procedures of environmental and socio-economic management and local governance enhancement. The agreement must be reviewed every five years, and the Mines Inspectorate Department of the Ministry monitors compliance. The Nigerian Minerals and Mining Regulations 2011 elaborate the procedures for negotiating CDAs.

The CDA model has been adopted in many other African mining codes, including those of Sierra Leone (Mines and Minerals Act 2009), Guinea (Mining Code 2011, as amended in 2013), Kenya (Mining Act 2016), and Mozambique. The Kenyan Mining Act 2016 also provides that a share of royalties shall be distributed to county governments and communities: seventy per cent to the national government, twenty per cent to the county government, and ten per cent to the community where the mining operations occur.

Challenges in Negotiating and Implementing CDAs

The effectiveness of CDAs depends on several factors that are frequently absent in practice. The first is the identification of the community: mining impacts may extend across multiple villages, ethnic groups, and administrative units, and the determination of which communities are parties to the agreement, and who represents them, is contentious. Traditional leaders may lack democratic legitimacy or may capture benefits for themselves or their families, and women, youth, and marginalised groups may be excluded. The second is capacity: communities typically lack the technical, legal, and financial expertise to negotiate effectively with mining companies, and the law rarely provides for independent advice to be funded. The third is enforceability: CDAs are often vaguely worded, lacking clear obligations, timeframes, and dispute resolution mechanisms, and communities may lack the means to enforce them through the courts. The fourth is sustainability: community development projects may be poorly designed, may create dependency, or may cease when the mine closes.

Good practice guidance, such as the World Bank's *Mining Community Development Agreements: Source Book* (2012), emphasises the importance of inclusive representation, independent advice for communities, clear and measurable commitments, transparent governance of community funds, monitoring and evaluation, and accessible grievance mechanisms.

Fiscal Decentralisation and Revenue Sharing

In addition to direct agreements, benefit sharing occurs through the allocation of mineral revenues to subnational governments. Nigeria's federal revenue allocation formula provides for thirteen per cent of revenues derived from natural resources to be allocated to the states from which they derive (the "derivation principle" in the proviso to section 162(2) of the Constitution), although in practice the derivation principle has been applied principally to oil revenues. In Ghana, the Minerals Development Fund Act 2016 (Act 912) provides for a share of mineral royalties to be allocated to mining communities and local authorities. In Peru, the *canon minero* allocates fifty per cent of the income tax paid by mining companies to regional and local governments in the producing areas. The design of revenue sharing must balance the claims of producing regions against those of the nation as a whole, and must address the capacity of subnational governments to manage revenues effectively. Resource revenue distribution is examined further in Chapter 23.

Equity Participation by Communities

Some regimes provide for community equity participation in mining projects. South Africa's Mining Charter, issued under the MPRDA, has required black economic empowerment ownership of mining companies, including a share for

host communities and employees. The 2018 Mining Charter required new mining rights to have a minimum of thirty per cent black economic empowerment shareholding, including a carried interest of five per cent for host communities. In *Minerals Council of South Africa v Minister of Mineral Resources and Energy* (2021), the High Court held that the Mining Charter was a policy instrument rather than binding legislation and set aside several of its provisions, including those purporting to require the continuation of empowerment ownership after historical transactions had been completed.[8] The case illustrates the limits of policy instruments as a basis for legally binding obligations.

7.6 Mine Closure and Rehabilitation

The Closure Challenge

Every mine eventually closes, whether because the ore body is exhausted, because commodity prices fall below the cost of production, or because of accidents, disputes, or regulatory action. Closure is not the end of a mine's legal life. Open pits, underground workings, waste rock dumps, and tailings facilities remain, and may require monitoring, maintenance, and treatment of water for decades or even in perpetuity. Historically, many mines were simply abandoned at the end of their lives, leaving a legacy of contaminated land and water, unsafe structures, and stranded communities. Thousands of abandoned mines in the United States, Canada, Australia, South Africa, and Europe, and many more in developing countries, impose ongoing costs on governments and communities. In Nigeria, the legacy of colonial and post-colonial tin and columbite mining on the Jos Plateau has left extensive areas of mine ponds and spoil heaps that have affected agriculture and public health. The modern law of mine closure seeks to ensure that closure is planned from the outset, that the costs are borne by the operator rather than the public, and that land is rehabilitated to a safe, stable, and productive condition.

Closure Planning

Modern mining regimes require mining operators to prepare closure plans as part of their applications for mining licences, and to update them periodically during the life of the mine. A closure plan typically identifies the post-mining land use objectives, agreed where possible with affected communities and authorities; describes the measures for decommissioning infrastructure, stabilising landforms, revegetating disturbed land, managing water, and securing tailings facilities; estimates the cost of closure; and establishes criteria by which successful completion of closure will be assessed. Progressive rehabilitation — the rehabilitation of

[8] *Minerals Council of South Africa v Minister of Mineral Resources and Energy* (High Court, Gauteng Division, Pretoria, 21 September 2021).

disturbed areas during the life of the mine, rather than at the end — is increasingly required, since it reduces the eventual closure liability and demonstrates the operator's commitment.

Closure plans must also address the social dimensions of closure: the loss of employment, the loss of revenue for local governments and suppliers, and the loss of services, such as health clinics, schools, and water supply, that the mine may have provided to the community. Social closure planning seeks to reduce the dependence of communities on the mine during its operating life and to support economic diversification. The ICMM's *Integrated Mine Closure: Good Practice Guide* (2nd edn, 2019) addresses both environmental and social dimensions.

Financial Assurance

The central legal problem of mine closure is ensuring that funds are available to carry out closure when it is required. Operators may become insolvent, particularly at the end of a mine's life when revenues decline; parent companies may dispose of subsidiaries holding closure liabilities; and assets may be sold to under-capitalised purchasers. Mining laws therefore require operators to provide financial assurance — security for the estimated cost of closure — which the state may call upon if the operator fails to carry out its obligations. Forms of financial assurance include cash deposits, trust funds, bank guarantees and letters of credit, insurance bonds, and, in some jurisdictions, corporate guarantees or self-bonding (reliance on the financial strength of the operator itself). Self-bonding has been criticised as inadequate, following the bankruptcies of several major United States coal companies in 2015 and 2016, which left states facing large unfunded reclamation liabilities.

In Nigeria, the Minerals and Mining Act 2007 requires the holder of a mineral title to contribute to an Environmental Protection and Rehabilitation Fund, established to guarantee the environmental obligations of holders, in amounts determined by reference to the approved EPRP. South Africa's National Environmental Management Act 1998, as amended, and the Financial Provisioning Regulations made under it require holders of mining rights to make financial provision for the rehabilitation and remediation of environmental impacts, including latent and residual impacts, and to assess and adjust the provision annually. Western Australia's Mining Rehabilitation Fund Act 2012 established a pooled fund to which most tenement holders contribute an annual levy, used to rehabilitate abandoned mine sites, replacing the previous system of unconditional performance bonds for most tenements.

Relinquishment and Completion Criteria

A critical question in mine closure law is when the operator's obligations end. Most regimes require the operator to demonstrate that closure objectives have

been met and to obtain a certificate of closure or relinquishment from the regulator, after which the financial assurance is released and responsibility for the site passes to the state. Regulators are often reluctant to accept relinquishment, because of the risk of latent impacts emerging later, while operators seek certainty. Some regimes provide for residual liability funds or for long-term stewardship arrangements to manage sites that require perpetual care, such as those with acid mine drainage requiring permanent water treatment. The development of clear, measurable, and agreed completion criteria is essential to the effective operation of the closure regime.

Closure and the Energy Transition

The energy transition creates both new closure challenges and new opportunities. The phase-out of coal mining in many countries — in Germany, the Coal Phase-out Act (Kohleausstiegsgesetz) of 2020 provides for the end of coal-fired power generation by 2038 at the latest, and in the United Kingdom the last coal-fired power station closed in 2024 — requires the closure and rehabilitation of large mining areas and support for affected communities, issues addressed by just transition policies examined in Chapters 18 and 23. Former mine sites may be repurposed for renewable energy generation, pumped hydro storage, or other uses, and mine waste may contain recoverable critical minerals. The reprocessing of tailings to recover minerals and the regulation of "mining waste as a resource" are emerging areas of law, examined in Chapter 8.

7.7 Artisanal and Small-Scale Mining Regulation

The Scale and Significance of ASM

Artisanal and small-scale mining (ASM) — mining by individuals, groups, families, or cooperatives with minimal mechanisation, often in the informal sector — is a major economic activity in many developing countries. The World Bank has estimated that around forty-five million people are directly engaged in ASM worldwide, and many more depend on it indirectly. ASM produces a significant share of the world's gold, tin, tantalum, tungsten, coloured gemstones, and diamonds, and a substantial share of the world's cobalt is produced by artisanal miners in the Democratic Republic of the Congo. For many rural households, ASM is a vital source of income and a buffer against poverty and agricultural failure.

At the same time, ASM is associated with serious problems: dangerous working conditions and frequent fatalities; child labour; environmental degradation, including deforestation, river siltation, and mercury pollution; conflict and the financing of armed groups; tax evasion and smuggling; and conflict with large-scale mining companies over overlapping rights. The use of mercury to amalgamate gold in artisanal and small-scale gold mining (ASGM) is the largest source of an-

thropogenic mercury emissions globally. In Nigeria, the lead poisoning epidemic in Zamfara State in 2010, caused by the artisanal processing of lead-rich gold ore in villages, killed several hundred children and was one of the worst lead poisoning outbreaks in modern history, prompting a major remediation effort.

Legal Approaches to ASM

Historically, many mining laws ignored ASM or treated it as illegal, focusing exclusively on large-scale industrial mining. Criminalisation has generally failed: it drives ASM underground, exposes miners to extortion and violence, deprives the state of revenue, and makes it impossible to address environmental and safety problems. The prevailing approach in contemporary mining law is formalisation — bringing ASM within the legal framework through simplified licensing, technical and financial support, and integration into formal supply chains.

Formalisation strategies typically include: the creation of a distinct category of ASM licences or permits, with simplified application procedures, lower fees, and reduced technical requirements; the designation of areas reserved for ASM; the encouragement of cooperatives or associations to facilitate licensing, training, and access to finance; the provision of geological information, extension services, and equipment; the establishment of official buying centres to purchase ASM production, particularly of gold, at fair prices; and the integration of environmental, health, and safety requirements adapted to the scale of operations.

Nigeria's Minerals and Mining Act 2007 provides for the grant of small-scale mining leases to individuals, cooperatives, and companies, and requires artisanal miners to form cooperatives in order to obtain titles. The Act established the Artisanal and Small-Scale Mining Department within the Ministry to provide extension services, training, and support. The Presidential Artisanal Gold Development Initiative (PAGDI), launched in 2019, sought to formalise artisanal gold mining by establishing a framework under which the Central Bank of Nigeria would purchase gold from artisanal miners through licensed buying centres for inclusion in the country's external reserves. In 2024 the federal government established a Mining Marshals unit within the Nigeria Security and Civil Defence Corps to combat illegal mining, reflecting concern over the links between illegal mining, insecurity, and the financing of armed groups in the north-west.

In Ghana, illegal artisanal mining, known as *galamsey*, has become a major political issue because of its devastating impact on rivers and forests. The Minerals and Mining (Amendment) Act 2019 (Act 995) increased penalties for illegal mining, and the government has periodically imposed bans on small-scale mining and deployed military task forces, with limited long-term success. The experience demonstrates the difficulty of reconciling the livelihood needs of artisanal miners with environmental protection where governance is weak.

The Minamata Convention

The **Minamata Convention on Mercury**, adopted in 2013 and in force since 2017, addresses mercury use in ASGM. Article 7 requires each party with ASGM within its territory that is more than insignificant to take steps to reduce, and where feasible eliminate, the use of mercury and mercury compounds in, and the emissions and releases to the environment of mercury from, such mining and processing. Such parties must develop and implement a national action plan in accordance with Annex C, which must include national objectives and reduction targets, actions to eliminate the worst practices (whole ore amalgamation, open burning of amalgam, burning of amalgam in residential areas, and cyanide leaching in sediment, ore, or tailings to which mercury has been added without first removing the mercury), steps to facilitate formalisation, and strategies for promoting the reduction of mercury use. Nigeria is a party to the Convention and has developed a national action plan for ASGM.

ASM and Large-Scale Mining

Conflicts between ASM and large-scale mining (LSM) are common, particularly where large-scale companies obtain titles over areas historically worked by artisanal miners. Such conflicts have led to violence, including forced evictions and abuses by security forces, examined in Chapter 21. Approaches to managing ASM–LSM coexistence include designating portions of large-scale concessions for ASM, establishing partnerships in which large-scale companies purchase ASM production or provide technical support, and resolving disputes through dialogue mechanisms. The Voluntary Principles on Security and Human Rights, examined in section 7.8, address the conduct of security forces in such situations.

7.8 International Standards for Responsible Mining

The Proliferation of Standards

Mining is governed not only by domestic law and international treaties but by a dense and growing body of international standards developed by international financial institutions, industry associations, multi-stakeholder initiatives, and civil society. These standards address environmental and social performance, human rights, transparency, supply chain due diligence, and governance. Although most are not legally binding in themselves, they acquire legal significance in several ways: through incorporation into loan agreements and investment conditions; through incorporation into domestic law and licence conditions; through reference in contracts between companies and their customers; through their use by courts and tribunals as evidence of the standard of care or of good industry practice; and through the reputational and market pressures that lead companies to commit to them publicly. This section surveys the principal standards; supply chain due diligence standards specific to minerals, notably the OECD Due

Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas, are examined in Chapter 8.

International Finance Corporation Performance Standards and the Equator Principles

The **IFC Performance Standards on Environmental and Social Sustainability**, adopted by the International Finance Corporation (the private sector arm of the World Bank Group) and most recently revised in 2012, are the most influential environmental and social standards for project finance in emerging markets. They comprise eight standards: PS1 (Assessment and Management of Environmental and Social Risks and Impacts); PS2 (Labor and Working Conditions); PS3 (Resource Efficiency and Pollution Prevention); PS4 (Community Health, Safety, and Security); PS5 (Land Acquisition and Involuntary Resettlement); PS6 (Biodiversity Conservation and Sustainable Management of Living Natural Resources); PS7 (Indigenous Peoples); and PS8 (Cultural Heritage). The Performance Standards are supported by Guidance Notes and by the World Bank Group's Environmental, Health, and Safety (EHS) Guidelines, including industry-specific guidelines for mining. Compliance with the Performance Standards is a condition of IFC financing, and the IFC's Compliance Advisor Ombudsman (CAO) receives complaints from communities affected by IFC-financed projects.

The **Equator Principles**, first adopted in 2003 by a group of commercial banks and now in their fourth iteration (EP4, effective from October 2020), commit participating financial institutions — over one hundred banks in more than thirty countries — to apply the IFC Performance Standards (and, in designated countries with robust governance, host country law) to project finance transactions above a specified threshold. EP4 strengthened requirements in relation to human rights, climate change risk assessment, and free, prior, and informed consent of indigenous peoples. Through the Equator Principles, the IFC Performance Standards have become a de facto global benchmark for the environmental and social performance of major mining projects, regardless of whether the IFC itself is involved.

The decision of the United States Supreme Court in *Jam v International Finance Corp* (2019), holding that international organisations such as the IFC enjoy only the same restrictive immunity from suit as foreign states under the Foreign Sovereign Immunities Act, opened the possibility of litigation against the IFC in United States courts by communities harmed by projects it financed, although the subsequent proceedings in that case, concerning a coal-fired power plant in India, were dismissed on other grounds.[9]

[9] *Jam v International Finance Corp*, 586 US 199 (2019).

Industry Standards

The **International Council on Mining and Metals** (ICMM), whose members include most of the world's largest mining companies, has adopted Mining Principles comprising ten principles and associated performance expectations covering ethical business, decision-making, human rights, risk management, health and safety, environmental performance, conservation of biodiversity, responsible production, social performance, and stakeholder engagement. Members must implement the principles, report on their performance, and obtain independent assurance. ICMM has also adopted position statements on indigenous peoples and mining (committing members to work to obtain the consent of indigenous peoples for new projects), on climate change, on water stewardship, and on tailings governance, and co-convened the development of the Global Industry Standard on Tailings Management discussed in section 7.4.

National industry initiatives include the Mining Association of Canada's **Towards Sustainable Mining** (TSM) programme, launched in 2004 and adopted by mining associations in several other countries, which requires member companies to assess and report their performance against protocols on tailings management, indigenous and community relationships, biodiversity, climate change, safety and health, crisis management, water stewardship, and preventing child and forced labour, with external verification.

Multi-Stakeholder Standards

The **Initiative for Responsible Mining Assurance** (IRMA), established in 2006 by a coalition of non-governmental organisations, affected communities, labour unions, mining companies, and purchasers of minerals, has developed the IRMA Standard for Responsible Mining, which provides for independent third-party audits of mine sites against comprehensive requirements covering business integrity, planning for positive legacies, social responsibility, and environmental responsibility. IRMA is distinctive in its equal governance by six stakeholder houses and in its requirement for site-level rather than corporate-level assessment. Purchasers of minerals, including automotive and electronics manufacturers, increasingly require their suppliers to undergo IRMA audits.

The **Extractive Industries Transparency Initiative** (EITI), launched in 2003, requires implementing countries to publish information on the extractive sector value chain, including the award of contracts and licences, beneficial ownership, production, revenue collection, revenue allocation, and social and economic spending, and to reconcile company payments with government receipts. Nigeria implements the EITI through the Nigeria Extractive Industries Transparency Initiative (NEITI) Act 2007, which makes Nigeria one of the few countries to have given EITI a statutory basis, and NEITI publishes regular audits

of the oil, gas, and solid minerals sectors. The EITI is examined further in Chapter 16 in the context of revenue transparency.

The **Kimberley Process Certification Scheme**, established in 2003 following United Nations General Assembly resolutions on "conflict diamonds" that financed civil wars in Angola, Sierra Leone, Liberia, and the Democratic Republic of the Congo, requires participating states to ensure that shipments of rough diamonds are accompanied by government-validated certificates attesting that they are conflict-free, and to trade rough diamonds only with other participants. The Kimberley Process has been criticised for its narrow definition of conflict diamonds, which is limited to diamonds used by rebel movements to finance wars against legitimate governments and does not extend to human rights abuses by governments or security forces, and for its consensus-based decision-making, which has blocked proposals for reform.

The **Voluntary Principles on Security and Human Rights**, adopted in 2000 by the governments of the United States and the United Kingdom, extractive companies, and non-governmental organisations, provide guidance to extractive companies on maintaining the safety and security of their operations within a framework that ensures respect for human rights. They address risk assessment, relations with public security forces (police and military), and relations with private security providers, and they have been incorporated into some host government agreements, security arrangements, and financing conditions.

Regional Frameworks

At the regional level, the **Africa Mining Vision** (2009) and its Action Plan provide a policy framework for African mining that emphasises linkages between mining and the broader economy, beneficiation and value addition, artisanal mining, human and institutional capacity, and the management of mineral revenues. The African Minerals Development Centre, established to support implementation, has developed guidance on Country Mining Visions. In West Africa, the **ECOWAS Directive on the Harmonisation of Guiding Principles and Policies in the Mining Sector** (C/DIR.3/05/09), adopted in 2009, requires member states to harmonise their mining laws in accordance with common principles concerning human rights, transparency, environmental protection, community development, and the rights of local communities, including requirements for free, prior, and informed consent of local communities before exploration and prior to each subsequent phase of mining operations. The ECOWAS Directive is notable as one of the first binding regional instruments to require community consent in mining, although its implementation in member states has been uneven. The **ECOWAS Mining and Minerals Development Policy** of 2011 elaborates these principles.

Evaluating the Standards Landscape

The proliferation of standards raises questions of legitimacy, coherence, and effectiveness. Standards developed by industry associations may be criticised as self-serving or lacking independent verification; multi-stakeholder standards may be more credible but are costly to implement; and the coexistence of multiple overlapping standards imposes compliance burdens and may confuse stakeholders. Standards are, moreover, most effective in relation to large companies with reputational exposure and access to international capital markets, and least effective in relation to state-owned enterprises, companies from jurisdictions with weaker accountability mechanisms, and the informal sector. Their legal status is ambiguous: they are not law in the traditional sense, yet they shape behaviour, inform legal standards, and may become binding through contract and regulation. The relationship between such transnational private regulation and formal law, analysed in Chapter 2, is one of the most important areas for doctoral research in natural resources law.

LEADING AUTHORITY

Reko Diq: From the Supreme Court of Pakistan to ICSID (2013–2022)

Facts. In 1993, the Balochistan Development Authority concluded the Chagai Hills Exploration Joint Venture Agreement (CHEJVA) with BHP Minerals for exploration of the Reko Diq area. BHP's interest was later acquired by Tethyan Copper Company, owned jointly by Barrick Gold and Antofagasta. Tethyan discovered a world-class copper-gold deposit and, after completing a feasibility study, applied in 2011 for a mining lease, which the Balochistan authorities refused.

Domestic proceedings. In January 2013, the Supreme Court of Pakistan, in proceedings brought by public interest petitioners, declared the CHEJVA void *ab initio* on the ground that it had been concluded in violation of the applicable mining rules and in excess of the authority of the officials who signed it.

International proceedings. Tethyan commenced ICSID arbitration under the Australia–Pakistan bilateral investment treaty. In its 2017 decision on jurisdiction and liability, the tribunal held that the refusal of the mining lease was motivated by the provincial government's intention to develop the project itself and breached the fair and equitable treatment standard and the prohibition of unlawful expropriation; it rejected Pakistan's allegations that the investment had been procured through corruption, finding them unproven. In July 2019, the tribunal awarded damages of approximately USD 4.1 billion plus interest, bringing the total to approximately USD 5.9 billion. Pakistan's application for annulment was pending when, in 2022, the parties reached a settlement under which Barrick would develop the project with the federal and Balochistan governments and Pakistani state-owned enterprises, and the award would be discharged. The Supreme Court of Pakistan, in an advisory opinion in December 2022, held that the new arrangements were not contrary to law or to its 2013 judgment.

Significance. The case illustrates the interaction between domestic and international legal orders in mining disputes: a domestic judgment invalidating a mining agreement did not prevent an international tribunal from finding the state liable for breaching treaty protections. It raises questions about the valuation of undeveloped mining projects, the role of corruption allegations in investment arbitration, and the use of settlements to convert awards into renegotiated partnerships.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Regulatory compliance review of a mining lease application (hypothetical). *The company, community, and facts in this exercise are fictitious.* Ogbo Minerals Ltd, a fictional Nigerian company with a foreign parent, holds an exploration licence over 40 square kilometres in a fictional local government area in Nigeria's north-central zone and has identified a commercially viable lithium-bearing pegmatite deposit. Part of the deposit is worked by an estimated 1,500 artisanal miners organised loosely into three informal groups, and a seasonal stream used by two villages for drinking water crosses the area. The company has prepared a feasibility study and wishes to apply for a mining lease.

Acting as counsel to the company, prepare a compliance memorandum (not more than 3,000 words) that (a) identifies the approvals, documents, and agreements required under Nigerian law before development may commence, with reference to the Minerals and Mining Act 2007, the Nigerian Minerals and Mining Regulations 2011, and the Environmental Impact Assessment Act; (b) advises on the negotiation of a community development agreement, including identification of the host community and representation of women and youth; (c) proposes a strategy for engagement with the artisanal miners consistent with Nigerian law and the IFC Performance Standards; and (d) identifies the international standards that prospective purchasers of the company's lithium concentrate are likely to require it to meet.

CHAPTER SUMMARY

- Mineral ownership is vested in the state in most jurisdictions; in Nigeria, section 44(3) of the Constitution and section 1 of the Minerals and Mining Act 2007 vest minerals in the Federal Government, while South Africa's MPRDA treats minerals as the common heritage of the people under state custodianship.
- Mining titles follow the mining life cycle — reconnaissance, exploration, mining — and are allocated by first-come, first-served, competitive tender, or discretion, ideally through a transparent electronic cadastre.
- Security of tenure, linking exploration to the right to mine, is central to investment; its denial has generated major awards, including *Crystallex* and *Tethyan Copper*.

- Mining agreements and stability clauses have been challenged by waves of resource nationalism, as in the DRC's 2018 Code and Tanzania's 2017 legislation; corruption can vitiate titles, as in the Simandou dispute.
- Environmental obligations include EIA, water and biodiversity protection, and tailings governance, transformed by the Brumadinho and Mariana disasters and the GISTM; social obligations include cultural heritage protection, highlighted by Juukan Gorge.
- Community development agreements, mandated by section 116 of the Nigerian Act, and revenue sharing mechanisms seek to share mining benefits with affected communities, but face challenges of representation, capacity, and enforceability.
- Mine closure law requires closure planning, financial assurance, and completion criteria to prevent public liability for abandoned mines.
- Artisanal and small-scale mining is best addressed through formalisation; the Minamata Convention requires action plans to reduce mercury use in ASGM.
- International standards — IFC Performance Standards, Equator Principles, ICMM, IRMA, EITI, Kimberley Process, Voluntary Principles, and the ECOWAS Mining Directive — increasingly shape mining practice and law.

DOCTORAL RESEARCH QUESTIONS

1. Is security of tenure compatible with the state's continuing duty to regulate mining in the public interest? How should mining codes structure the transition from exploration to mining to balance these interests?
2. Evaluate the effectiveness of mandatory community development agreements under section 116 of Nigeria's Minerals and Mining Act 2007, using empirical evidence from at least two mining regions.
3. To what extent should investment tribunals take account of an investor's failure to obtain a "social licence to operate" in determining liability and damages? Analyse *Bear Creek*, *Copper Mesa*, and *Gabriel Resources*.
4. Has the Global Industry Standard on Tailings Management altered the legal standard of care owed by mining companies and their parent companies to communities downstream of tailings facilities?
5. Assess the ECOWAS Mining Directive's requirement of community consent as a model for regional harmonisation, and explain the gap between its terms and member state implementation.
6. Should the formalisation of artisanal mining be pursued through mining law, labour law, environmental law, or a dedicated legal framework?

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8 CHAPTER

Critical Minerals, Strategic Resources, and Supply Chain Security

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain how states and international organisations define and classify "critical" and "strategic" minerals, and the legal consequences of such classification.
2. Analyse the distinctive legal and policy issues associated with lithium, cobalt, nickel, copper, and rare earth elements.
3. Evaluate the legal regimes governing state ownership of strategic minerals and export controls on minerals and processing technologies, including their compatibility with international trade law.
4. Assess the use of industrial policy to secure mineral supply chains, including the EU Critical Raw Materials Act and United States legislation.
5. Analyse mandatory and voluntary frameworks for mineral supply chain due diligence.
6. Identify the human rights and labour risks associated with critical mineral production and the legal mechanisms for addressing them.
7. Evaluate the regulation of mineral processing, recycling, and the circular economy.
8. Critically assess the emerging architecture of critical mineral agreements and the geopolitical competition for mineral resources.

Introduction

The energy transition is, among other things, a transition from a fuel-intensive energy system to a material-intensive one. A conventional car requires little more than steel, aluminium, and copper; an electric car requires, in addition, around two hundred kilograms of minerals for its battery and motor, including lithium, nickel, cobalt, manganese, graphite, and rare earth elements. An onshore wind plant requires around nine times more mineral resources than a gas-fired power plant of the same capacity. Solar photovoltaic panels require silicon, silver, copper,

and, in some technologies, tellurium, indium, and gallium. Electricity networks, which must expand dramatically to accommodate renewable generation and electrification, require vast quantities of copper and aluminium. The International Energy Agency has projected that, in scenarios consistent with the goals of the Paris Agreement, demand for minerals for clean energy technologies could increase several-fold by 2040, with demand for lithium potentially growing by a factor of more than ten.

The minerals on which the transition depends are characterised by a combination of features that generate distinctive legal and political challenges. Their production is geographically concentrated: the Democratic Republic of the Congo produces around seventy per cent of the world's cobalt; Australia, Chile, and China dominate lithium production; Indonesia has become the world's largest producer of nickel; and China accounts for around sixty to seventy per cent of rare earth mining. Processing is even more concentrated: China refines the majority of the world's lithium, cobalt, graphite, and rare earths, and controls most of the world's capacity to manufacture battery cells and permanent magnets. Supply is relatively inelastic, because new mines typically take ten to fifteen years or more from discovery to production. Many deposits are located in countries with weak governance, high levels of poverty, and histories of conflict, and in or near territories of indigenous peoples. Production is associated with serious environmental impacts and, in some contexts, with child labour, forced labour, and other human rights abuses.

These features have transformed critical minerals from a niche concern of defence planners and materials scientists into a central preoccupation of energy, trade, industrial, and foreign policy. Governments in importing states have adopted critical mineral strategies, legislation, and international partnerships aimed at diversifying supply and reducing dependence on China. Governments in producing states, particularly in Africa and Latin America, have sought to capture a greater share of the value of their mineral resources by requiring domestic processing, restricting exports of unprocessed ores, increasing state participation, and renegotiating agreements. China, for its part, has used export controls on minerals and processing technologies as instruments of economic statecraft. The legal regime of critical minerals sits at the intersection of mining law (examined in Chapter 7), trade law (Chapter 13), investment law (Chapter 14), human rights law (Chapter 21), and the law of geopolitics and national security (Chapter 25), and it is evolving with extraordinary speed.

This chapter examines the legal classification of critical and strategic minerals (section 8.1); the distinctive issues associated with lithium, cobalt, nickel, copper, and rare earth elements (section 8.2); strategic mineral ownership and export controls (section 8.3); resource security and industrial policy (section 8.4); supply

chain due diligence (section 8.5); human rights and labour risks in mineral production (section 8.6); processing, recycling, and circular economy regulation (section 8.7); and geopolitical competition and critical mineral agreements (section 8.8). The institutional dimension of critical minerals governance, including the role of the International Energy Agency, was introduced in Chapter 4.

8.1 Legal Classification of Critical and Strategic Minerals

The Concept of Criticality

There is no universally accepted definition of a "critical mineral". The concept of criticality, as developed in the policy literature, generally combines two dimensions: the *economic importance* of a mineral — the extent to which it is essential to economic activity, technologies, or national security, and the difficulty of substituting it — and its *supply risk* — the likelihood of disruption to its supply, arising from geological scarcity, geographic concentration of production and processing, political instability or poor governance in producing countries, export restrictions, low rates of recycling, and similar factors. A mineral is "critical" if it scores highly on both dimensions. The related concept of a "strategic" mineral, which has its origins in military planning during and after the Second World War, emphasises importance for national defence or for strategic sectors of the economy, such as the energy transition, digital technologies, and aerospace.

Criticality assessments are contingent and dynamic. They depend on the perspective of the assessing state (a mineral critical to an importing state may be of little concern to a producing state), on technological developments (new battery chemistries may reduce or increase demand for particular minerals), and on market and geopolitical conditions. Most jurisdictions therefore review their lists periodically.

Classification in Major Jurisdictions

The European Union. The European Commission first published a list of critical raw materials in 2011 and has updated it every three years since. The Critical Raw Materials Act — Regulation (EU) 2024/1252, which entered into force in May 2024 — placed the classification on a legislative footing. Annex II lists thirty-four critical raw materials, assessed on the basis of economic importance and supply risk according to a methodology set out in the Regulation, and Annex I lists seventeen strategic raw materials, selected from among the critical raw materials on the basis of their importance for technologies supporting the green and digital transitions and defence and aerospace applications, the projected growth in demand, and the difficulty of increasing production. The strategic list includes lithium, nickel (battery grade), cobalt, natural graphite, manganese (battery grade), copper, rare earth elements for magnets, gallium, germanium, silicon metal, tungsten, and platinum group metals. The classification has legal

consequences: strategic raw materials are the subject of the Act's benchmarks for domestic capacity and diversification and are eligible for strategic project status, examined in section 8.4.

The United States. In the United States, the Energy Act of 2020 defines a "critical mineral" as a non-fuel mineral or mineral material essential to the economic or national security of the United States, the supply chain of which is vulnerable to disruption, and that serves an essential function in the manufacturing of a product, the absence of which would have significant consequences for the economic or national security of the United States. The Act requires the Secretary of the Interior, acting through the United States Geological Survey (USGS), to publish and periodically review a list of critical minerals. The 2022 list identified fifty critical minerals, and the USGS's subsequent review in 2025 expanded the list, adding minerals including copper. Separately, the Department of Energy maintains a list of "critical materials" for energy, and the Department of Defense manages the National Defense Stockpile of strategic and critical materials under the Strategic and Critical Materials Stock Piling Act. The classification has legal consequences in, among other things, the eligibility requirements for the clean vehicle tax credit under section 30D of the Internal Revenue Code (as amended by the Inflation Reduction Act of 2022), streamlined permitting, and federal financing.

Other jurisdictions. Australia, Canada, Japan, India, the United Kingdom, South Korea, and China have also adopted lists of critical or strategic minerals. China's National Mineral Resources Plan (2016–2020) identified twenty-four strategic minerals, including oil, gas, coal, rare earths, lithium, cobalt, and graphite. Producing states have increasingly adopted their own classifications, often for different purposes: whereas importing states classify minerals as critical to identify supply risks, producing states classify them as strategic to justify enhanced state control, higher royalties, and local processing requirements. The Democratic Republic of the Congo's Mining Code of 2018 permits the government to declare minerals "strategic substances" subject to a ten per cent royalty, and in 2018 the government so classified cobalt, germanium, and columbite-tantalite (coltan).

The African Perspective

African institutions have increasingly articulated their own understanding of critical minerals, emphasising their potential to drive industrialisation rather than merely their role in supplying importing economies. The African Union's Africa Mining Vision (2009), discussed in Chapter 7, already emphasised beneficiation and linkages, and the African Union has since developed an African Green Minerals Strategy aimed at leveraging the continent's mineral endowment for green industrialisation, regional value chains, and economic transformation. The Africa

Finance Corporation, the African Development Bank, and the United Nations Economic Commission for Africa have supported initiatives to develop regional battery and electric vehicle value chains, such as the cooperation agreement between the Democratic Republic of the Congo and Zambia, signed in 2022, to develop a battery precursor value chain linking Congolese cobalt and Zambian copper and manganese.

In Nigeria, the Ministry of Solid Minerals Development has identified a number of minerals, including lithium, as priority minerals for development, and the federal government has adopted a policy of promoting local processing of lithium and other minerals rather than the export of raw ore. The first lithium processing plants in Nigeria, owned by Chinese investors and located in Kaduna and Nasarawa States, began operations in 2024.

Legal Consequences of Classification

Classification as critical or strategic may have a range of legal consequences, depending on the jurisdiction:

- **Permitting**: projects involving critical minerals may benefit from expedited permitting procedures, as under the EU Critical Raw Materials Act's strategic project regime and various United States initiatives.
- **Financing**: projects may be eligible for public financing, loan guarantees, or grants, as under Title III of the United States Defense Production Act and the programmes of export credit agencies.
- **Trade**: classification may support export restrictions in producing states or tariff and procurement preferences in importing states.
- **Investment screening**: foreign investment in critical mineral assets may be subject to enhanced scrutiny on national security grounds, as under the Canadian government's November 2022 policy on investments by state-owned enterprises in critical minerals, which led to orders requiring three Chinese companies to divest their interests in Canadian lithium companies.
- **Fiscal**: producing states may impose higher royalties on strategic minerals, as in the DRC.
- **Stockpiling**: governments may maintain strategic stockpiles of critical minerals, analogous to strategic petroleum reserves (examined in Chapter 25).

Because the label "critical" or "strategic" may trigger significant legal consequences, the methodology and transparency of classification are themselves legally significant, and the classification decisions of administrative agencies may be subject to judicial review.

8.2 Lithium, Cobalt, Nickel, Copper, and Rare Earth Elements

Each of the principal energy transition minerals has a distinctive geology, market structure, geography of production, and set of legal issues. This section examines five of the most significant.

Lithium

Geology and production. Lithium is produced from two principal types of deposit: hard-rock deposits, principally spodumene-bearing pegmatites mined in Australia (the world's largest producer), Zimbabwe, Nigeria, Brazil, and China; and brine deposits in the salt flats (*salares*) of the "lithium triangle" of Argentina, Bolivia, and Chile, from which lithium-rich brine is pumped into evaporation ponds and concentrated over many months. Direct lithium extraction technologies, which extract lithium from brine without evaporation ponds, are being developed and may reduce water consumption and environmental impacts. Lithium is processed into lithium carbonate or lithium hydroxide for use in battery cathodes, and the majority of the world's lithium chemical processing capacity is located in China.



Figure 8.1 Lithium brine evaporation ponds in a high-altitude salt flat. Illustrative image.

Legal regimes. Lithium has been subject to distinctive legal regimes in several Latin American states, reflecting its classification as a strategic resource. In Chile, lithium was declared a strategic mineral and "non-concessible" (not subject to ordinary mining concessions) by Decree-Law No 2.886 of 1979, on account of its potential use in nuclear fusion; lithium can be exploited only by the state, by state enterprises, or under special administrative contracts (*contratos especiales de operación de litio*, CEOLs) or leases granted by the state. The leases held by the private companies SQM and Albemarle over the Salar de Atacama, granted through the state development agency CORFO, have been the subject of repeated

renegotiation and litigation. In April 2023, the Chilean government announced a National Lithium Strategy providing for state control of strategic salt flats through public-private partnerships in which the state holds a majority, and in 2024 the state copper company Codelco concluded an agreement with SQM to form a joint venture, majority-owned by Codelco from 2031, for the exploitation of the Salar de Atacama.

In Bolivia, which holds some of the world's largest lithium resources in the Salar de Uyuni, the Constitution of 2009 and the state's model of resource sovereignty have reserved lithium exploitation to the state company Yacimientos de Litio Bolivianos (YLB), which has sought partnerships with foreign companies for direct lithium extraction technologies. In Mexico, a 2022 reform of the Mining Law declared lithium to be of public utility, reserved its exploration, exploitation, and use to the state, and created a state lithium company (LitioMx). In Argentina, by contrast, a federal country in which mineral resources belong to the provinces under Article 124 of the National Constitution, lithium is subject to ordinary mining concessions, and a combination of provincial ownership, a liberal investment regime, and the incentive regime for large investments (RIGI) introduced in 2024 has attracted substantial foreign investment.

Africa. Africa's lithium resources, principally in Zimbabwe, Namibia, the Democratic Republic of the Congo, Mali, Ghana, and Nigeria, have attracted significant investment, much of it from Chinese companies. In December 2022 Zimbabwe banned the export of unprocessed lithium ore, permitting exports only of concentrates, and announced its intention to ban the export of concentrates from 2027 to require domestic refining. Namibia similarly banned the export of unprocessed lithium ore in 2023. Ghana's first lithium mining lease, granted in 2023 to Barari DV Ghana (a subsidiary of the Australian company Atlantic Lithium) for the Ewoyaa project, provided for a royalty of ten per cent, a thirteen per cent free-carried interest for the state, and a further six per cent equity stake for the Minerals Income Investment Fund, and its ratification by Parliament was delayed amid debates over whether the terms secured sufficient value for Ghana. The DRC's Manono deposit, among the largest hard-rock lithium deposits in the world, has been the subject of a dispute between the Australian company AVZ Minerals and the Congolese state and state mining company Cominière, including ICC and ICSID proceedings.

Cobalt

Production and market. Cobalt is produced predominantly as a by-product of copper mining (and, to a lesser extent, nickel mining), which means that its supply responds primarily to copper and nickel prices rather than to cobalt demand. The Democratic Republic of the Congo, principally in the Copperbelt provinces of Lualaba and Haut-Katanga, accounts for around seventy per cent of global

production, and a substantial share of that production comes from mines owned wholly or partly by Chinese companies, notably CMOC Group's Tenke Fungurume and Kisanfu mines. A significant share of Congolese cobalt is produced by artisanal miners, often working in hazardous conditions, including in some cases children.

Legal issues. The legal issues associated with cobalt include the fiscal terms of the DRC's 2018 Mining Code (including its classification of cobalt as a strategic substance); the renegotiation of the "Sicomines" infrastructure-for-minerals agreement concluded in 2008 between the DRC and a consortium of Chinese companies, under which the Chinese partners were to finance infrastructure in exchange for mining rights, renegotiated in 2024 to increase the DRC's share of benefits; the formalisation of artisanal cobalt mining, including through the creation in 2019 of the state-owned Entreprise Générale du Cobalt (EGC), which holds a monopoly on the purchase, processing, and export of artisanal cobalt; and the due diligence obligations of companies sourcing cobalt, examined in section 8.5. In February 2025, in response to a collapse in cobalt prices caused by oversupply, the DRC government suspended cobalt exports for four months, subsequently extending the suspension and then replacing it with an export quota system — an example of a producing state using export restrictions as an instrument of market management, analogous to production quotas in the oil market.

Nickel

Production and market. Nickel is used in stainless steel and, increasingly, in high-nickel battery cathodes. Indonesia has become the dominant producer, accounting for around half of global mined supply following a dramatic expansion of production since 2020, predominantly from laterite deposits processed through rotary kiln-electric furnace and high-pressure acid leaching (HPAL) technologies, in industrial parks developed largely with Chinese investment. Other significant producers include the Philippines, Russia, New Caledonia, Canada, and Australia. The expansion of Indonesian supply has caused prices to fall sharply, forcing the closure of mines in Australia and New Caledonia.

Legal issues. Indonesia's nickel policy is the most prominent example of a resource-rich state using export restrictions to drive downstream industrialisation. Indonesia's Mining Law No 4 of 2009 required domestic processing and refining of minerals; after a partial relaxation, the government banned exports of nickel ore with effect from January 2020. The ban attracted substantial investment in nickel smelting and refining in Indonesia, increasing the value of Indonesian nickel exports dramatically, but it was challenged by the European Union before the WTO in *Indonesia — Measures Relating to Raw Materials* (DS592). In November 2022, the panel found that the export ban was inconsistent with Article XI:1 of

the GATT 1994 and could not be justified under Article XI:2(a), as a measure temporarily applied to prevent or relieve a critical shortage, or under Article XX(d), as a measure necessary to secure compliance with laws not inconsistent with the GATT. Indonesia appealed the report to the non-functioning Appellate Body, preventing its adoption — an "appeal into the void" — and has maintained the ban. The dispute is examined further in section 8.3 and in Chapter 13.

Indonesia's nickel boom has also raised serious environmental and social concerns, including deforestation, coastal and marine pollution from mine runoff and tailings, the reliance of smelters on captive coal-fired power plants, labour conditions and fatal accidents at smelters, and impacts on indigenous communities. These concerns have led some purchasers to scrutinise the environmental and social performance of Indonesian nickel suppliers, and the question whether Indonesian nickel processed with Chinese investment qualifies for incentives under the United States Inflation Reduction Act has been politically contested.

Copper

Production and market. Copper is the archetypal transition metal, essential for electricity generation, transmission, and distribution, electric vehicles, and renewable energy installations. Chile and Peru are the largest producers, followed by the Democratic Republic of the Congo, China, the United States, Australia, Russia, Zambia, and Mexico. Copper demand is projected to grow strongly, and many analysts anticipate a structural deficit by the end of the 2020s, as ore grades in existing mines decline and the development of new mines faces permitting delays, community opposition, water constraints, and rising costs.

Legal issues. The legal issues associated with copper reflect the challenges of developing large new mines. The closure of the Cobre Panamá mine illustrates the risks. In 2023, the government of Panama and First Quantum Minerals concluded a revised concession contract for the mine, one of the largest new copper mines in the world, following a 2017 Supreme Court ruling that the original concession law was unconstitutional. The new contract, approved by the National Assembly as Law 406, provoked mass protests over environmental concerns and perceived inadequate benefits for Panama. In November 2023, the Supreme Court of Panama unanimously declared Law 406 unconstitutional, and the government ordered the closure of the mine. First Quantum commenced arbitration proceedings under the contract and under the Canada–Panama free trade agreement, although it subsequently suspended the proceedings to facilitate negotiations with the government. The case illustrates the vulnerability of mining projects to constitutional challenges and social opposition, and the interaction between domestic constitutional law and international investment protection.

In Chile, the Mining Royalty Law (Law No 21.591) of 2023 introduced a new royalty on large copper producers, combining an *ad valorem* component with a component based on operating margins, with an effective tax burden capped at a specified percentage of operating margin. In Zambia, frequent changes in mining fiscal terms have been a source of tension with investors, and in 2022 the government introduced changes to make mineral royalties deductible for income tax purposes in order to attract investment.

Rare Earth Elements

Production and market. The rare earth elements comprise the fifteen lanthanides, together with scandium and yttrium. They are not, in fact, geologically rare, but economically viable concentrations are uncommon, and their separation from one another is chemically complex, environmentally hazardous (generating radioactive and acidic waste), and technically demanding. Neodymium, praseodymium, dysprosium, and terbium are used in high-performance permanent magnets for wind turbines, electric vehicle motors, electronics, and defence systems. China dominates every stage of the rare earth supply chain: it accounts for around sixty to seventy per cent of mining, around ninety per cent of separation and refining, and a similar proportion of magnet manufacturing. Other producers include the United States (Mountain Pass, California), Australia (Mount Weld, with processing by Lynas in Malaysia and Australia), Myanmar (where unregulated mining in Kachin State has caused severe environmental damage), and, increasingly, Brazil, Vietnam, and several African countries.

Legal issues. Rare earths have been at the centre of trade disputes and export control controversies. In *China — Measures Related to the Exportation of Rare Earths, Tungsten, and Molybdenum* (2014), the WTO Appellate Body upheld findings that China's export duties and export quotas on rare earths were inconsistent with its WTO obligations and not justified on conservation grounds, examined in section 8.3. China's subsequent export controls on rare earth processing technologies (2023) and on rare earth elements and magnets (2025) have prompted urgent efforts by importing states to develop alternative supply chains, including through public financing, offtake agreements, and price floors. In 2025, the United States Department of Defense agreed to acquire an equity stake in MP Materials, the owner of the Mountain Pass mine, and to guarantee a minimum price for its neodymium-praseodymium output — an unprecedented intervention illustrating the shift towards state capitalism in critical mineral supply chains.

8.3 Strategic Mineral Ownership and Export Controls

State Ownership and Control of Strategic Minerals

As examined in Chapter 3, almost all states assert ownership or control over subsurface minerals. In relation to minerals classified as strategic, however, many

states go further, reserving exploitation to the state or to state enterprises, requiring majority state participation, or subjecting the minerals to special regimes. The Chilean, Bolivian, and Mexican regimes for lithium, examined in section 8.2, are examples, as are the DRC's regime for strategic substances and the reservation of radioactive minerals (uranium and thorium) to the state in many jurisdictions. Such reservations are an exercise of permanent sovereignty over natural resources and are, in principle, consistent with international law, provided that they are applied prospectively or, where they affect existing rights, in compliance with the requirements for lawful expropriation, including compensation (examined in Chapters 3 and 14).

State ownership of strategic minerals is often justified on grounds of national security, the capture of resource rents, and the promotion of downstream industrialisation. Its effectiveness depends on the capacity and governance of state enterprises, which have frequently struggled to attract the capital and technology necessary to develop complex projects. Bolivia's experience with lithium, where state control has not yet produced significant output despite immense resources, illustrates the difficulties; Chile's model of public-private partnerships under state control represents an attempt to combine state ownership with private capital and expertise.

Export Restrictions by Producing States

Producing states have increasingly used export restrictions — bans, quotas, export taxes, and licensing requirements — to promote domestic processing, conserve resources, manage prices, or raise revenue. The Organisation for Economic Co-operation and Development (OECD), which maintains an inventory of export restrictions on industrial raw materials, has documented a more than five-fold increase in such restrictions between 2009 and 2020, with the largest increases in critical minerals. Examples include Indonesia's nickel ore export ban, Zimbabwe's and Namibia's restrictions on unprocessed lithium, the DRC's cobalt export suspension and quotas, and the restrictions on the export of unprocessed bauxite in Guinea and other states.

International Trade Law Disciplines

Export restrictions are subject to the disciplines of international trade law, examined generally in Chapter 13. The central provision is **Article XI:1 of the GATT 1994**, which prohibits "prohibitions or restrictions other than duties, taxes or other charges, whether made effective through quotas, import or export licences or other measures" on the exportation of any product destined for the territory of another WTO member. Export *duties*, by contrast, are not prohibited by the GATT, although some members, notably China, accepted commitments to eliminate export duties in their accession protocols.

Export restrictions may be justified under several exceptions:

- **Article XI:2(a)** permits export prohibitions or restrictions "temporarily applied to prevent or relieve critical shortages of foodstuffs or other products essential to the exporting contracting party".
- **Article XX(g)** permits measures "relating to the conservation of exhaustible natural resources if such measures are made effective in conjunction with restrictions on domestic production or consumption", subject to the requirements of the chapeau of Article XX that measures not be applied in a manner that would constitute arbitrary or unjustifiable discrimination or a disguised restriction on international trade.
- **Article XX(i)** permits measures "involving restrictions on exports of domestic materials necessary to ensure essential quantities of such materials to a domestic processing industry during periods when the domestic price of such materials is held below the world price as part of a governmental stabilization plan", subject to conditions.
- **Article XXI(b)** permits measures that a member "considers necessary for the protection of its essential security interests", including those "taken in time of war or other emergency in international relations".

The WTO dispute settlement bodies have interpreted these exceptions narrowly in the context of mineral export restrictions. In *China — Measures Related to the Exportation of Various Raw Materials* (2012), concerning Chinese export duties and quotas on bauxite, coke, fluorspar, magnesium, manganese, silicon carbide, silicon metal, yellow phosphorus, and zinc, the Appellate Body held that China could not invoke Article XX to justify export duties inconsistent with paragraph 11.3 of its Accession Protocol, because that paragraph contained no reference to the GATT exceptions. It also held that a measure is "temporarily applied" under Article XI:2(a) only if it is applied for a limited time to bridge a passing need, and that a "critical shortage" refers to deficiencies in quantity that are crucial or that reach a vitally important or decisive stage. In *China — Rare Earths* (2014), the Appellate Body upheld the panel's findings that China's export quotas on rare earths, tungsten, and molybdenum were not justified under Article XX(g), because China had not demonstrated that the quotas were made effective in conjunction with restrictions on domestic production or consumption in an even-handed manner; on the contrary, the quotas appeared designed to secure supply for domestic users at the expense of foreign users. China removed the export duties and quotas in 2015.

In *Indonesia — Raw Materials* (DS592), discussed in section 8.2, the panel held that Indonesia's nickel ore export ban was a prohibition contrary to Article XI:1, that it was not "temporarily applied" to relieve a "critical shortage" within

Article XI:2(a), since it was intended to continue indefinitely, and that it was not justified under Article XX(d). The panel's report has not been adopted owing to Indonesia's appeal to the non-functioning Appellate Body.

These decisions establish that WTO law significantly constrains the use of export restrictions to promote downstream processing. Critics argue that the rules are asymmetric: they constrain the industrial policy tools available to resource-rich developing countries, which historically were used by today's industrialised countries during their own development, while importing states are relatively free to use subsidies and other measures to build processing capacity. The legitimacy of the WTO disciplines on export restrictions, and the scope for reform, are significant themes for doctoral research, explored in Chapter 13.

Export Controls on Minerals and Technologies by China

China's use of export controls on critical minerals and processing technologies has transformed the landscape. The Export Control Law of the People's Republic of China (2020) established a comprehensive framework for controlling exports of dual-use items, military products, nuclear materials, and other items related to national security and interests, and authorised countermeasures against countries that abuse export controls to endanger China's national security and interests. Under this framework and related regulations, including the Regulations on the Export Control of Dual-Use Items (2024), China has imposed licensing requirements on exports of:

- **gallium and germanium** and their compounds, from August 2023;
- **graphite**, including certain high-purity natural and synthetic graphite products, from December 2023;
- **rare earth extraction, separation, and magnet manufacturing technologies**, added to the Catalogue of Technologies Prohibited or Restricted from Export in December 2023;
- **antimony** and superhard materials, from September 2024;
- **seven medium and heavy rare earth elements** (samarium, gadolinium, terbium, dysprosium, lutetium, scandium, and yttrium) and related magnets, from April 2025.

In December 2024, China prohibited exports of gallium, germanium, antimony, and superhard materials to the United States in response to United States export controls on semiconductors. In October 2025, China announced a substantial expansion of its rare earth export controls, including provisions asserting jurisdiction over foreign-made products containing Chinese-origin rare earths or produced with Chinese technology; following a meeting between the presidents of China and the United States at the end of October 2025, China announced that it would suspend the implementation of the expanded measures for one year.

These measures, and their partial suspension as part of a broader trade negotiation, illustrate the use of critical minerals as instruments of economic statecraft and the vulnerability of importing states to concentrated supply chains. China has justified the measures as necessary for the protection of national security and the fulfilment of non-proliferation obligations; importing states have characterised them as economic coercion. Their consistency with WTO law would turn largely on the interpretation of Article XXI, examined in Chapters 13 and 25.

Export Controls by Importing States and Investment Screening

Importing states also use export controls and investment screening in relation to critical minerals. The United States, the European Union, Japan, and other states control exports of technologies related to semiconductor manufacturing, advanced batteries, and other strategic sectors, and the United States' foreign direct product rules extend these controls extraterritorially. Investment screening regimes — such as the Committee on Foreign Investment in the United States (CFIUS) under the Foreign Investment Risk Review Modernization Act of 2018, the EU Foreign Direct Investment Screening Regulation (EU) 2019/452 (the revision of which was under negotiation in 2025), the United Kingdom's National Security and Investment Act 2021, and the Australian Foreign Acquisitions and Takeovers Act 1975 — have been used to block or condition acquisitions of critical mineral assets by foreign investors, particularly Chinese state-linked companies. Australia has blocked several Chinese investments in rare earth and lithium companies on national interest grounds. The use of investment screening in the mineral sector raises questions about its consistency with investment treaties and with the principle of non-discrimination, and about the risk that screening becomes an instrument of geopolitical rivalry rather than a narrowly targeted security measure.

8.4 Resource Security and Industrial Policy

From Market Reliance to Industrial Policy

For several decades, the dominant approach of industrialised states to mineral supply was reliance on global markets: minerals were commodities to be purchased wherever they were cheapest, and the state's role was limited to maintaining open trade and, in some cases, strategic stockpiles for defence purposes. The concentration of supply chains in China, China's demonstrated willingness to use export restrictions, the supply chain disruptions of the COVID-19 pandemic, and the Russian invasion of Ukraine have led to a fundamental shift. Industrialised states now pursue active industrial policies aimed at building domestic and allied capacity across the mineral value chain — mining, processing, component manufacturing, and recycling — using legislation, subsidies, public financing, permitting reforms, and international partnerships.

The EU Critical Raw Materials Act

The EU Critical Raw Materials Act (Regulation (EU) 2024/1252) is the most comprehensive legislative framework for critical mineral security. Its principal elements are:

Benchmarks. Article 5 establishes benchmarks for strengthening the different stages of the strategic raw materials value chain, providing that by 2030 the Union's capacity should be sufficient for extraction to produce at least ten per cent, processing to produce at least forty per cent, and recycling to produce at least twenty-five per cent of the Union's annual consumption of strategic raw materials, and that the Union should diversify its imports so that no more than sixty-five per cent of its annual consumption of each strategic raw material at any relevant stage of processing comes from a single third country. The benchmarks are not legally binding targets for member states, but they guide the implementation of the Act.

Strategic projects. The Act establishes a procedure for the recognition of strategic projects — projects that contribute meaningfully to the security of the Union's supply of strategic raw materials, are technically feasible, are implemented sustainably, and, for projects in third countries, are mutually beneficial for the Union and the third country concerned. Strategic projects benefit from priority status in permitting, with maximum permit-granting periods (twenty-seven months for extraction projects and fifteen months for processing and recycling projects within the Union), a presumption that they are in the public interest for the purposes of certain environmental assessments, and support in accessing finance. In March 2025, the Commission selected the first forty-seven strategic projects within the Union, and in June 2025 it selected thirteen strategic projects in third countries, including projects in Africa and Latin America.

Monitoring and risk preparedness. The Act requires the Commission and member states to monitor supply chains, to conduct stress tests, and to coordinate strategic stocks, and requires large companies manufacturing strategic technologies to carry out risk assessments of their supply chains.

Circularity and sustainability. The Act requires member states to adopt national programmes to increase the collection and recycling of critical raw materials, and requires certain products to disclose information on the recyclability and recycled content of permanent magnets.

International partnerships. The Act provides a framework for strategic partnerships with third countries, building on the memoranda of understanding concluded by the EU with Canada, Ukraine, Kazakhstan, Namibia, Argentina, Chile, Zambia, the Democratic Republic of the Congo, Greenland, Rwanda, Norway, Uzbekistan, Australia, and others.

United States Industrial Policy

The United States has adopted a range of measures to build domestic critical mineral supply chains. The **Inflation Reduction Act of 2022** introduced, among other incentives, a clean vehicle tax credit under section 30D of the Internal Revenue Code, part of which was conditional on a rising percentage of the value of critical minerals in the vehicle's battery being extracted or processed in the United States or in a country with which the United States has a free trade agreement, or recycled in North America, and excluded vehicles whose batteries contained critical minerals extracted, processed, or recycled by a "foreign entity of concern". The FTA requirement led the United States to conclude "critical minerals agreements" with Japan (March 2023) and to negotiate similar agreements with the European Union and the United Kingdom, raising questions about whether such limited agreements qualify as "free trade agreements". The clean vehicle tax credit was terminated for vehicles acquired after 30 September 2025 by the One Big Beautiful Bill Act of July 2025, illustrating the vulnerability of industrial policy to political change.

The **Defense Production Act** of 1950, Title III of which authorises the President to provide financial assistance to expand domestic productive capacity for materials essential to national defence, has been invoked to support critical mineral projects, and in 2022 President Biden authorised its use for battery minerals. In March 2025, President Trump issued an Executive Order on Immediate Measures to Increase American Mineral Production, directing agencies to expedite permitting for priority mineral projects and to expand the use of Defense Production Act authorities and federal financing. The Department of Energy's Loan Programs Office, the Development Finance Corporation, and the Export-Import Bank of the United States have provided financing for domestic and overseas critical mineral projects, and in 2025 the federal government took equity stakes in several critical mineral companies, including MP Materials and the lithium developer Lithium Americas.

Other Jurisdictions

Japan has pursued mineral security for decades through the Japan Organization for Metals and Energy Security (JOGMEC), which provides equity financing, loan guarantees, and exploration support for overseas mineral projects and maintains national stockpiles. Following the Chinese restrictions on rare earth exports to Japan in 2010, during a territorial dispute, JOGMEC and Japanese companies invested in the Australian rare earth producer Lynas, helping to establish one of the few significant sources of rare earths outside China. Australia's Critical Minerals Strategy 2023–2030 and its Critical Minerals Facility, administered by Export Finance Australia, provide financing for critical mineral projects, and the Future Made in Australia Act 2024 introduced production tax incentives for

critical mineral processing. In October 2025, Australia and the United States concluded a framework agreement on critical minerals providing for joint investment in projects. Canada's Critical Minerals Strategy (2022) and India's National Critical Mineral Mission (2025) pursue similar objectives.

Industrial Policy in Producing States

Producing states pursue their own industrial policies, aimed at capturing a greater share of the value chain. These include requirements for domestic processing (export bans on unprocessed ores, as in Indonesia, Zimbabwe, and Namibia), local content and beneficiation requirements, state participation, and incentives for investment in processing and manufacturing. The success of Indonesia's nickel strategy in attracting downstream investment has inspired other producing states, but its replicability is uncertain: Indonesia combined its export ban with a large domestic market, abundant coal-fired power, and massive Chinese investment, and its strategy has generated significant environmental and social costs. Producing states must also navigate the constraints of WTO law and investment treaties, and the risk that export restrictions will simply divert investment to other sources.

The Tension Between Security and Sustainability

Industrial policies aimed at accelerating mineral supply create tensions with environmental protection, indigenous rights, and community consent. Streamlined permitting, presumptions of public interest, and political pressure to approve projects quickly may compromise the rigour of environmental assessment and the quality of consultation. The EU Critical Raw Materials Act seeks to address this tension by requiring that strategic projects be implemented sustainably, including in compliance with environmental and social standards, but civil society organisations have criticised the selection of strategic projects for insufficient transparency and inadequate consideration of community opposition. The reconciliation of mineral security with sustainability and human rights is one of the defining legal challenges of the energy transition.

8.5 Supply Chain Due Diligence

The Concept of Supply Chain Due Diligence

Mineral supply chains are long, complex, and opaque. A cobalt atom in a smartphone battery may have been mined by an artisanal miner in the Congolese Copperbelt, sold to a trader at a local depot, processed at a refinery in the DRC or China, converted into cathode material in China, South Korea, or Japan, incorporated into a battery cell, and assembled into a device sold by a brand headquartered in the United States or Europe. At each stage, minerals from different sources are mixed, and the connection between the final product and the condi-

tions of extraction is obscured. Supply chain due diligence is the process by which companies identify, assess, prevent, mitigate, and account for the adverse impacts associated with their supply chains, including impacts caused by their suppliers and their suppliers' suppliers.

The concept derives from the UN Guiding Principles on Business and Human Rights (2011), examined in Chapter 21, which provide that business enterprises should carry out human rights due diligence to identify, prevent, mitigate, and account for how they address their adverse human rights impacts, including impacts directly linked to their operations, products, or services by their business relationships. In the mineral sector, due diligence was first developed in response to the financing of armed conflict in the eastern Democratic Republic of the Congo by the trade in tin, tantalum, tungsten, and gold (the "3TG" or "conflict minerals").

The OECD Due Diligence Guidance

The **OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas**, first adopted in 2010 and approved as an OECD Council Recommendation in 2011 (now in its third edition, 2016), is the leading international standard for mineral supply chain due diligence. It establishes a five-step framework:

1. **Establish strong company management systems**, including a supply chain policy consistent with the model policy in Annex II of the Guidance, internal structures to support due diligence, a system of controls and transparency over the supply chain (including chain of custody or traceability systems), and a grievance mechanism.
2. **Identify and assess risks in the supply chain**, including by mapping the supply chain and assessing the conditions of extraction, trade, handling, and export.
3. **Design and implement a strategy to respond to identified risks**, which may involve continuing trade while pursuing measurable risk mitigation, temporarily suspending trade, or disengaging from a supplier.
4. **Carry out independent third-party audits** of due diligence practices at identified points in the supply chain, notably smelters and refiners.
5. **Report annually** on supply chain due diligence.

The Guidance distinguishes between "upstream" companies (from the mine to smelters and refiners) and "downstream" companies (from smelters and refiners to retailers), recognising that smelters and refiners constitute a "choke point" in the supply chain through which minerals from many sources pass and at which audits can be most effectively focused. Annex II identifies the most serious risks — including torture and cruel, inhuman, or degrading treatment, forced or com-

pulsory labour, the worst forms of child labour, gross human rights violations, war crimes, direct or indirect support to non-state armed groups, and bribery and money laundering — and the expected responses to them. Although originally developed for 3TG, the Guidance applies to all minerals, and supplements have addressed specific issues. The OECD has also developed guidance for responsible business conduct in other sectors, including the OECD Due Diligence Guidance for Responsible Business Conduct (2018).

Mandatory Due Diligence: Dodd-Frank Section 1502

The first mandatory mineral due diligence regime was **section 1502 of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010** in the United States, which required companies listed on United States exchanges to determine whether their products contained 3TG originating in the DRC or adjoining countries and, if so, to exercise due diligence on the source and chain of custody of those minerals and to file a report with the Securities and Exchange Commission (SEC). The SEC's implementing rule, adopted in 2012, required companies to describe products that had not been found to be "DRC conflict free". In *National Association of Manufacturers v SEC* (2014, reaffirmed 2015), the United States Court of Appeals for the District of Columbia Circuit held that the requirement to describe products as not "DRC conflict free" violated the First Amendment by compelling speech, while upholding other aspects of the rule.[1] Section 1502 was praised for raising awareness and driving industry due diligence, but was also criticised for causing a *de facto* embargo on minerals from the eastern DRC, which harmed the livelihoods of artisanal miners without proportionately reducing conflict financing.

The EU Conflict Minerals Regulation

The **EU Conflict Minerals Regulation** — Regulation (EU) 2017/821, which became fully applicable on 1 January 2021 — requires Union importers of 3TG ores, concentrates, and metals above specified volume thresholds to comply with supply chain due diligence obligations aligned with the OECD Guidance, including management systems, risk management, third-party audits, and disclosure. Unlike Dodd-Frank, the Regulation applies to minerals from any conflict-affected and high-risk area, not only the DRC region, and focuses on upstream importers rather than downstream manufacturers. Member state authorities are responsible for enforcement, and the Commission has published an indicative list of conflict-affected and high-risk areas.

[1] *National Association of Manufacturers v Securities and Exchange Commission*, 748 F.3d 359 (DC Cir 2014), reaffirmed on rehearing, 800 F.3d 518 (DC Cir 2015).

The EU Batteries Regulation

The **EU Batteries Regulation** — Regulation (EU) 2023/1542 — extends mandatory due diligence to the principal battery minerals. It requires economic operators placing batteries on the Union market, subject to a turnover threshold, to adopt and implement battery due diligence policies in respect of cobalt, natural graphite, lithium, and nickel, and chemical compounds based on them, covering risks related to human rights, labour rights, the environment, and community life, in line with internationally recognised due diligence instruments including the UN Guiding Principles and the OECD Guidance. Operators must have their due diligence policies verified by a notified body and must make information publicly available. The due diligence obligations were originally due to apply from August 2025, but their application was postponed to 2027 by an amendment adopted in 2025 as part of the Commission's simplification agenda. The Regulation also introduces a "battery passport" — an electronic record containing information on each industrial and electric vehicle battery, including its composition, carbon footprint, and recycled content — and requirements on carbon footprint declarations, recycled content, and recycling efficiency, examined in section 8.7.

General Mandatory Human Rights Due Diligence

Beyond mineral-specific regimes, general mandatory human rights and environmental due diligence laws apply to companies in mineral supply chains. France's Duty of Vigilance Law (*Loi relative au devoir de vigilance*) of 2017 requires large French companies to establish, publish, and implement a vigilance plan identifying and preventing serious violations of human rights, health and safety, and the environment resulting from their activities and those of their subsidiaries, subcontractors, and suppliers. Germany's Supply Chain Due Diligence Act (*Lieferkettensorgfaltspflichtengesetz*), in force since 2023, imposes similar obligations on large companies operating in Germany. The EU **Corporate Sustainability Due Diligence Directive** — Directive (EU) 2024/1760 — requires large companies to identify, prevent, mitigate, and remediate adverse human rights and environmental impacts in their own operations, those of their subsidiaries, and those of their business partners in their "chain of activities", and to adopt climate transition plans. The Directive's transposition and application dates were postponed, and its scope and requirements were reduced, through the "Omnibus" simplification package negotiated in 2025. These general regimes are examined in Chapters 21 and 22.

Import Bans Based on Forced Labour

A further category of supply chain regulation consists of import bans on goods produced with forced labour. In the United States, section 307 of the Tariff Act of 1930 prohibits the importation of goods mined, produced, or manufactured

wholly or in part by forced labour, and the **Uyghur Forced Labor Prevention Act** of 2021 establishes a rebuttable presumption that goods mined, produced, or manufactured wholly or in part in the Xinjiang Uyghur Autonomous Region of China, or by entities on a designated list, are made with forced labour and are prohibited from importation. The Act has significant implications for supply chains of polysilicon (used in solar panels), aluminium, and battery materials. The EU **Forced Labour Regulation** — Regulation (EU) 2024/3015 — prohibits products made with forced labour from being placed on the Union market or exported from it, with application from December 2027.

Industry and Multi-Stakeholder Initiatives

Industry initiatives play a central role in implementing due diligence. The **Responsible Minerals Initiative** (RMI) operates the Responsible Minerals Assurance Process, which audits smelters and refiners against standards aligned with the OECD Guidance, and publishes lists of conformant facilities; it has developed standards for cobalt, mica, and other minerals in addition to 3TG. The **London Bullion Market Association** Responsible Gold Guidance requires refiners accredited to the London Good Delivery list to undergo annual audits. The **International Conference on the Great Lakes Region** (ICGLR), an intergovernmental organisation of twelve African states, operates a Regional Certification Mechanism for 3TG under its 2006 Protocol against the Illegal Exploitation of Natural Resources. Initiatives addressing artisanal cobalt include the Fair Cobalt Alliance and the cobalt standard of the DRC's Entreprise Générale du Cobalt.

Evaluating Due Diligence

Mineral supply chain due diligence has become a central feature of the legal regime of critical minerals, but its effectiveness is debated. Proponents argue that it has improved transparency, reduced the financing of armed groups in some areas, and established expectations of corporate responsibility. Critics argue that due diligence has too often become a box-ticking exercise focused on audits and certifications rather than on actual improvements in conditions at mine sites; that it may lead companies to disengage from high-risk areas, harming the communities that depend on mining rather than improving conditions; that traceability systems are vulnerable to fraud, as illustrated by allegations concerning the laundering of minerals from conflict areas through neighbouring countries; and that the costs fall disproportionately on artisanal and small-scale producers. The design of due diligence regimes that promote responsible engagement rather than disengagement is a key challenge for regulators.

8.6 Human Rights and Labour Risks in Mineral Production

The Spectrum of Risks

The production of critical minerals is associated with a wide spectrum of human rights and labour risks. These include the worst forms of child labour, particularly in artisanal mining; forced labour, including in state-sponsored labour transfer programmes; unsafe working conditions, resulting in deaths and injuries from collapses, accidents, and exposure to toxic substances; violations of the rights of indigenous peoples, including the right to free, prior, and informed consent; forced evictions and inadequate resettlement; violence by public and private security forces; environmental harms affecting the rights to health, water, and an adequate standard of living; and the financing of armed conflict. These risks are examined generally in Chapter 21; this section highlights their particular manifestations in critical mineral supply chains.

Child Labour and Artisanal Cobalt

In 2016, Amnesty International and African Resources Watch published *"This is what we die for": Human rights abuses in the Democratic Republic of the Congo power the global trade in cobalt*, documenting the use of child labour and hazardous working conditions in artisanal cobalt mining in the DRC and tracing cobalt from artisanal sources through processing companies to battery manufacturers and consumer brands. The report contributed to the development of industry due diligence for cobalt and to the inclusion of cobalt in the EU Batteries Regulation. In *Doe v Apple Inc*, a group of former child miners and the families of deceased child miners from the DRC sued Apple, Alphabet, Dell, Microsoft, and Tesla in the United States, alleging that the companies had knowingly benefited from forced child labour in their cobalt supply chains in violation of the Trafficking Victims Protection Reauthorization Act. The District Court dismissed the claims in 2021, and in March 2024 the Court of Appeals for the District of Columbia Circuit affirmed, holding that purchasing cobalt in the global market did not amount to participation in a "venture" with the suppliers that used forced labour within the meaning of the statute.[2] The case illustrates the difficulty of establishing the legal liability of downstream companies for abuses in their supply chains under existing law, and the significance of mandatory due diligence legislation in filling that gap.

Forced Labour and State-Sponsored Labour Programmes

Reports by researchers, journalists, and civil society organisations have documented the involvement of state-sponsored labour transfer programmes in the Xinjiang Uyghur Autonomous Region in the production of polysilicon, alu-

[2] *Doe v Apple Inc*, 96 F.4th 403 (DC Cir 2024).

minium, and other materials used in energy transition technologies. The United Nations Office of the High Commissioner for Human Rights, in its 2022 assessment of human rights concerns in Xinjiang, concluded that serious human rights violations had been committed and that the extent of arbitrary and discriminatory detention may constitute international crimes. Such programmes raise the risk that solar panels, batteries, and other clean energy products incorporate materials produced with forced labour. The Uyghur Forced Labor Prevention Act and the EU Forced Labour Regulation, examined in section 8.5, are the principal legal responses.

Indigenous Peoples and Lithium

A large share of known lithium, nickel, copper, and other transition mineral resources is located on or near the lands of indigenous peoples. A study published in *Nature Sustainability* in 2022 estimated that more than half of energy transition mineral projects are located on or near indigenous peoples' lands. In the lithium triangle, indigenous Atacameño (Lickanantay), Colla, and Kolla communities have raised concerns about the impact of brine extraction on scarce water resources, wetlands, and traditional livelihoods, and have challenged lithium projects before national courts and the Inter-American human rights system. In Argentina, communities in the Salinas Grandes and Laguna de Guayatayoc basin have invoked their right to free, prior, and informed consent under ILO Convention No 169 and the Argentine Constitution. In the United States, the Thacker Pass lithium project in Nevada, located near a site of a historic massacre of Paiute people, was challenged by indigenous groups, ranchers, and environmental organisations; the Court of Appeals for the Ninth Circuit in 2023 rejected the challenges to the project's federal approvals.[3] These disputes illustrate the tension between the urgency of the energy transition and the rights of indigenous peoples, which some commentators have described as a risk of "green colonialism" or "green extractivism".

Labour Conditions in Processing and Manufacturing

Labour risks are not confined to mining. Nickel smelters in Indonesia's industrial parks have experienced fatal accidents, including an explosion at a smelter in the Indonesia Morowali Industrial Park in December 2023 that killed around twenty workers, and reports have documented long working hours, inadequate safety measures, and restrictions on freedom of association. Battery manufacturing and recycling involve exposure to hazardous substances. The ILO's core conventions on freedom of association, forced labour, child labour, discrimination, and occupational safety and health, and the ILO Declaration on Fundamental Principles

[3] *Bartell Ranch LLC v McCullough* (9th Cir, July 2023) (memorandum disposition), affirming the District Court of Nevada.

and Rights at Work (as amended in 2022 to include a safe and healthy working environment), provide the international framework.

The UN Secretary-General's Panel

In recognition of the risks associated with the expansion of critical mineral production, the United Nations Secretary-General established in 2024 the **Panel on Critical Energy Transition Minerals**, comprising governments, intergovernmental organisations, industry, and civil society. The Panel's report, published in September 2024, set out seven guiding principles: that human rights must be at the core of all mineral value chains; that the integrity of the planet, its environment, and biodiversity must be safeguarded; that justice and equity must underpin mineral value chains; that development should be fostered through benefit sharing, value addition, and economic diversification; that investment, finance, and trade should be responsible and fair; that transparency, accountability, and anti-corruption measures are necessary to ensure good governance; and that multilateral and international cooperation should underpin global action. The Panel recommended, among other things, the establishment of a high-level expert group to advance the principles, a global traceability, transparency, and accountability framework, and a global mining legacies fund to address abandoned mines. Although not binding, the principles represent the first United Nations-endorsed framework specifically addressed to energy transition minerals.

8.7 Mineral Processing, Recycling, and Circular Economy Regulation

The Processing Bottleneck

The concentration of mineral processing — the conversion of ores and concentrates into refined metals and chemicals suitable for industrial use — is the most acute vulnerability in critical mineral supply chains. Even where mining is geographically diverse, processing may be concentrated in a single country: lithium mined in Australia is shipped as spodumene concentrate to China for conversion to lithium hydroxide; cobalt mined in the DRC is refined predominantly in China; and almost all rare earth separation and magnet manufacturing takes place in China. Processing is capital-intensive, technically demanding, energy-intensive, and environmentally hazardous, and the economics of new processing capacity outside China are challenged by the scale, experience, and low costs of Chinese producers, as well as by price volatility that Chinese oversupply can intensify.

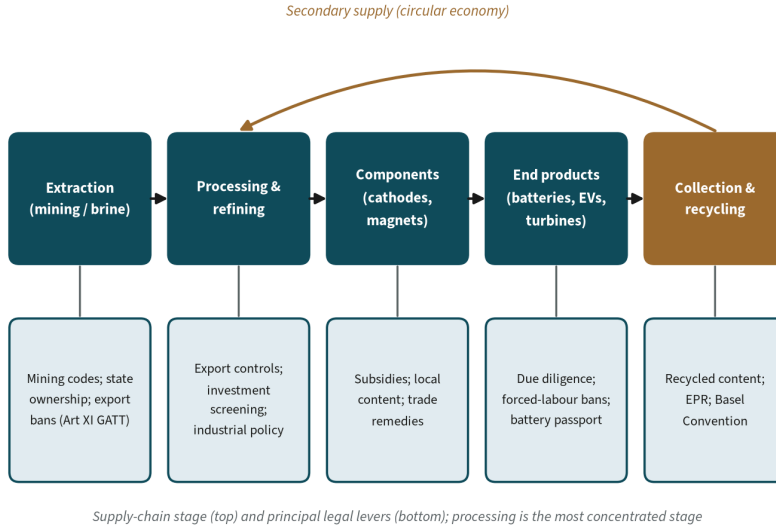


Figure 8.2 The critical mineral supply chain and the principal legal levers applied at each stage. *Source:* prepared for this edition.

The law relating to processing has several dimensions. Environmental regulation of processing facilities — governing emissions to air and water, waste management (including radioactive waste from rare earth processing), and energy use — is critical, and the strength of environmental regulation affects both the location of processing and its environmental footprint. The controversy over the Lynas rare earth processing plant in Kuantan, Malaysia, which faced protests and litigation over the management of low-level radioactive waste and was required by the Malaysian government to relocate its cracking and leaching operations to Australia, illustrates the issues. Trade and industrial policy measures — export restrictions on unprocessed ores in producing states and subsidies for processing in importing states — shape the location of processing, as examined in sections 8.3 and 8.4. Investment law protects processing investments, and investment screening may restrict foreign ownership of processing facilities.

The Circular Economy

The circular economy — an economic model in which materials are kept in use for as long as possible through reuse, repair, remanufacturing, and recycling, and waste is minimised — offers a partial solution to the supply challenges of critical minerals. Recycling of end-of-life batteries, electronics, magnets, and other products can provide a secondary source of critical minerals that is located in consuming countries, reduces the environmental impacts of primary mining, and reduces dependence on concentrated supply chains. However, recycling

cannot meet growing demand in the near term, because the stock of end-of-life products is small relative to projected demand, particularly for batteries, most of which are still in use. The IEA has projected that recycling could reduce primary supply requirements for lithium, cobalt, and nickel by a meaningful share by 2040 in scenarios with ambitious recycling policies, but primary mining will remain essential for decades.

The EU Batteries Regulation: Circularity Requirements

The EU Batteries Regulation (Regulation (EU) 2023/1542), introduced in section 8.5, is the most comprehensive regulatory framework for battery circularity. Its principal circularity provisions include:

- **Recycled content**: from 2031, industrial batteries with a capacity above 2 kWh, electric vehicle batteries, and certain other batteries must contain minimum shares of recovered cobalt (16 per cent), lead (85 per cent), lithium (6 per cent), and nickel (6 per cent), rising in 2036 to 26 per cent for cobalt, 12 per cent for lithium, and 15 per cent for nickel;
- **Recycling efficiency and material recovery**: recycling processes must achieve minimum recycling efficiencies (for lithium-based batteries, 65 per cent by the end of 2025 and 70 per cent by the end of 2030), and minimum material recovery rates (for lithium, 50 per cent by the end of 2027 and 80 per cent by the end of 2031; for cobalt, copper, lead, and nickel, 90 per cent by the end of 2027 and 95 per cent by the end of 2031);
- **Extended producer responsibility**: producers must organise and finance the collection and treatment of waste batteries, with rising collection targets for portable batteries;
- **Carbon footprint**: electric vehicle batteries and certain other batteries must be accompanied by a carbon footprint declaration and will become subject to maximum carbon footprint thresholds;
- **Battery passport**: from February 2027, electric vehicle batteries, industrial batteries above 2 kWh, and light means of transport batteries must have a digital battery passport.

The Regulation also addresses removability and replaceability of portable batteries and labelling requirements. Its extraterritorial effect is significant: any battery placed on the EU market, wherever manufactured, must comply, and the recycled content and carbon footprint requirements create incentives for battery manufacturers worldwide to adopt EU standards — an example of the "Brussels effect" in energy transition regulation.

Waste and the Transboundary Movement of E-Waste

The recycling of critical minerals depends on the collection and transport of waste products, which are subject to the law on hazardous waste. The **Basel Conven-**

tion on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (1989) requires the prior informed consent of importing and transit states for transboundary movements of hazardous wastes, and prohibits exports to states that have prohibited imports. The Basel Ban Amendment, which entered into force in 2019, prohibits exports of hazardous waste from OECD and EU member states and Liechtenstein to other states for any purpose. Amendments to the Convention's Annexes adopted in 2022, which entered into force on 1 January 2025, extend the prior informed consent procedure to all electronic and electrical waste, whether or not hazardous. The **Bamako Convention** (1991) of the African Union goes further, prohibiting the import of hazardous wastes into Africa from non-parties. These regimes are designed to prevent the dumping of hazardous waste in developing countries, a practice exemplified by the 1988 Koko incident in Nigeria, in which several thousand tonnes of toxic waste from Italy were dumped in the port town of Koko in present-day Delta State, prompting Nigeria's Harmful Waste (Special Criminal Provisions, etc.) Act 1988. The legal challenge is to distinguish legitimate trade in recyclable materials, which can support the circular economy, from the export of waste for disposal or substandard recycling in countries lacking adequate facilities, which harms workers and communities, as in the informal e-waste processing sites of Agbogbloshie in Accra, Ghana, and parts of Lagos.

Mine Waste as a Resource

Mine wastes and tailings may contain recoverable quantities of critical minerals that were not economically extractable when the waste was produced. The reprocessing of tailings to recover cobalt, copper, rare earths, and other minerals is attracting investment, and the EU Critical Raw Materials Act requires member states to compile databases of closed extractive waste facilities with potential for recovery of critical raw materials. Reprocessing raises legal questions concerning ownership of tailings (whether they belong to the state as unworked minerals, to the operator as property, or to the owner of the land), the allocation of liability for historical contamination, and the applicability of mining, waste, and environmental law to reprocessing operations.

8.8 Geopolitical Competition and Critical Mineral Agreements

The Geopolitics of Critical Minerals

Critical minerals have become an arena of intense geopolitical competition. The principal axis of competition is between China, which has built dominance across critical mineral supply chains through decades of state-supported investment in domestic and overseas mining and processing, and the United States and its allies, which seek to reduce their dependence on Chinese supply. Producing states in Africa, Latin America, and Asia find themselves courted by competing powers,

which offers opportunities to negotiate better terms but also risks entanglement in great power rivalry. The geopolitics of critical minerals is examined further in Chapter 25 in the context of energy security; this section focuses on the legal instruments through which states cooperate and compete.

Types of Critical Mineral Agreements

A diverse and rapidly evolving array of international instruments addresses critical minerals:

Plurilateral partnerships among consumer states. The **Minerals Security Partnership** (MSP), launched in June 2022 by the United States and a group of partners including Australia, Canada, the European Union, Japan, South Korea, and the United Kingdom, aims to catalyse public and private investment in responsible critical mineral supply chains, including through a Minerals Security Partnership Finance Network. The **G7 Critical Minerals Action Plan**, adopted at the Kananaskis summit in June 2025, commits G7 members to cooperation on standards-based markets, investment, recycling, and innovation. The IEA's **Critical Minerals Security Programme**, launched in 2024, supports its members in assessing supply risks and coordinating responses.

Bilateral critical minerals agreements. The Agreement between the Government of the United States of America and the Government of Japan on Strengthening Critical Minerals Supply Chains (March 2023) commits the parties not to impose export duties on critical minerals traded between them, to promote labour rights and environmental protection in critical mineral supply chains, and to cooperate on investment screening and recycling. Its principal practical function was to enable Japan to be treated as an FTA partner for the purposes of the IRA clean vehicle credit.

Strategic partnerships between consumers and producers. The European Union has concluded memoranda of understanding on strategic partnerships on raw materials with numerous producing states, typically covering integration of value chains, cooperation on research and innovation, application of environmental, social, and governance standards, infrastructure development, and capacity building. The United States has supported the **Lobito Corridor**, a rail and port infrastructure project linking the copper and cobalt mining regions of the DRC and Zambia to the Angolan port of Lobito on the Atlantic coast, in partnership with the European Union, the African Development Bank, and the Africa Finance Corporation, as an alternative to routes to the Indian Ocean.

Minerals-for-security and minerals-for-reconstruction arrangements. In April 2025, the United States and Ukraine concluded an agreement establishing a United States–Ukraine Reconstruction Investment Fund, under which Ukraine contributes revenues from new licences for natural resources, including

critical minerals, oil, and gas, to a jointly governed fund that invests in Ukrainian projects. In June 2025, the DRC and Rwanda signed a peace agreement facilitated by the United States, accompanied by a regional economic integration framework addressing mineral supply chains in the Great Lakes region. These arrangements, which link access to minerals with security guarantees or peace processes, raise questions about the permanent sovereignty of states over their natural resources, the legitimacy of agreements concluded in conditions of conflict and dependence, and the risk of reproducing historical patterns of extraction.

Resource-for-infrastructure agreements. Chinese state-backed companies have concluded numerous agreements under which infrastructure is financed and constructed in exchange for mining rights or mineral offtake, notably the Sicomines agreement in the DRC (examined in section 8.2). Such agreements have been criticised for opacity, undervaluation of mineral rights, and inadequate benefits for host states, and several have been renegotiated.

The Legal Character of Critical Mineral Agreements

Many critical mineral instruments are memoranda of understanding or political declarations rather than legally binding treaties. Their non-binding character provides flexibility and enables rapid conclusion, but it limits their enforceability and their capacity to provide the legal certainty required for long-term investment. Where they are binding, as in the case of the United States–Japan agreement, they are often narrow in scope. The emerging architecture of critical mineral agreements is thus fragmented and largely soft, contrasting with the comprehensive multilateral frameworks that govern trade generally under the WTO. Some commentators have proposed a multilateral critical minerals agreement, or a critical minerals plurilateral within the WTO, to establish common rules on export restrictions, subsidies, standards, and investment; others have proposed producer-consumer dialogues analogous to the International Energy Forum in the oil sector.

Producer Coordination

Producing states have explored coordination to strengthen their bargaining position. Proposals have been made for a lithium producers' organisation among Argentina, Bolivia, and Chile, analogous to OPEC, and Indonesia has floated the idea of a nickel producers' organisation. African states have pursued coordination through the African Union and regional economic communities, including the development of regional value chains. Such coordination faces the obstacles that undermined earlier producer associations for copper, bauxite, and other minerals in the 1970s — divergent interests, the availability of substitutes, and the entry of new producers — but it reflects the determination of producing states to avoid a

repetition of the historical pattern in which resource wealth flowed outward while host countries remained dependent on raw material exports.

Assessment

The legal regime of critical minerals is in the midst of rapid and contested formation. It is shaped by the imperatives of the energy transition, the pursuit of national security, the aspirations of producing states for development and industrialisation, and the claims of communities, workers, and indigenous peoples for respect for their rights. The instruments that constitute the regime — national legislation, trade rules, export controls, industrial policy, due diligence obligations, and bilateral and plurilateral agreements — are fragmented and sometimes contradictory. A central question for international energy and natural resources law is whether this regime can be developed in a manner that secures mineral supply for the energy transition while respecting the permanent sovereignty of producing states, ensuring that they and their populations share equitably in the benefits, and protecting human rights and the environment.

LEADING AUTHORITY

Indonesia — Measures Relating to Raw Materials (WTO, DS592, Panel Report 2022)

Measures. Indonesia banned the export of nickel ore with effect from 1 January 2020, and required domestic processing of nickel ore before export. The European Union, whose stainless steel industry relied on imported nickel ore, challenged the measures.

Findings. The panel found that the export ban was a prohibition on exportation within the meaning of Article XI:1 of the GATT 1994. It rejected Indonesia's defence under Article XI:2(a), holding that the ban was not "temporarily applied", since Indonesia intended it to remain in place indefinitely, and that Indonesia had not demonstrated a "critical shortage" of nickel ore, given its very large reserves. It also rejected Indonesia's defence under Article XX(d), holding that the ban was not "necessary" to secure compliance with Indonesia's mining laws, since a requirement in domestic law could not itself justify a measure inconsistent with the GATT simply because the measure implemented it. Having rejected the defences, the panel did not need to address the domestic processing requirement in detail.

Aftermath. Indonesia appealed in December 2022 to the Appellate Body, which has been unable to hear appeals since December 2019 because of the blocking of appointments by the United States; the report therefore has not been adopted and is not binding. Indonesia has maintained the ban. The European Union's Enforcement Regulation (Regulation (EU) No 654/2014), as amended in 2021 by Regulation (EU) 2021/167, permits it to adopt countermeasures where a dispute is blocked by an appeal into the void, although it has not done so in this dispute.

Significance. The case is the leading recent authority on the application of GATT disciplines to export restrictions designed to promote downstream processing. It

confirms the narrow scope of Article XI:2(a) and Article XX(d), and illustrates the practical limits of WTO enforcement in the absence of a functioning Appellate Body. It has been widely discussed in debates about the policy space available to resource-rich developing countries.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Treaty interpretation and negotiation simulation: a critical minerals partnership (hypothetical). *The states, companies, and facts in this exercise are fictitious.* The fictional Republic of Kavuma, a West African state with large lithium and graphite deposits, is negotiating a "Strategic Partnership on Sustainable Raw Materials" with the fictional Northern Economic Union, a consumer bloc of industrialised states. Kavuma's negotiating objectives are to secure investment in domestic lithium refining, to retain the right to impose an export ban on unprocessed spodumene from 2028, and to secure financing for a rail corridor. The Union's objectives are to secure long-term supply of lithium concentrate, to ensure that Kavuma does not impose export duties or restrictions on partners, and to require compliance with environmental and human rights standards consistent with its due diligence legislation.

Working in two negotiating teams, (a) prepare a negotiating brief identifying each side's legal constraints, including under WTO law (assuming both are WTO members), any applicable investment treaties, and regional obligations such as the ECOWAS Mining Directive and the AfCFTA; (b) negotiate and draft the key provisions of a partnership instrument (no more than 1,500 words), deciding whether it should be legally binding; and (c) write an individual reflective memorandum (1,000 words) assessing whether the agreed text respects Kavuma's permanent sovereignty over natural resources.

CHAPTER SUMMARY

- "Critical" and "strategic" minerals are classified by reference to economic importance and supply risk; classifications such as those in the EU Critical Raw Materials Act and the United States Energy Act of 2020 carry significant legal consequences.
- Lithium, cobalt, nickel, copper, and rare earth elements each present distinctive legal issues, from Chile's state-controlled lithium regime to the DRC's cobalt export controls, Indonesia's nickel ore ban, Panama's Cobre Panamá litigation, and China's rare earth dominance.
- Export restrictions on minerals are constrained by GATT Article XI and narrowly interpreted exceptions, as in *China — Raw Materials*, *China — Rare Earths*, and *Indonesia — Raw Materials*; China's export controls on minerals and technologies have become instruments of economic statecraft.

- Importing states pursue industrial policies combining benchmarks, accelerated permitting, public finance, and partnerships, notably under the EU Critical Raw Materials Act and United States legislation.
- Supply chain due diligence, grounded in the OECD Guidance, has become mandatory under Dodd-Frank section 1502, the EU Conflict Minerals Regulation, the EU Batteries Regulation, and general due diligence laws, supplemented by forced labour import bans.
- Critical mineral production is associated with child labour, forced labour, indigenous rights violations, and unsafe conditions; liability of downstream companies remains difficult to establish, as *Doe v Apple* shows.
- Circular economy regulation, especially the EU Batteries Regulation, and the law on transboundary waste shape the recycling of critical minerals.
- Critical mineral agreements form a fragmented, largely soft-law architecture of partnerships, bilateral agreements, and minerals-for-security arrangements, raising questions about permanent sovereignty and equity.

DOCTORAL RESEARCH QUESTIONS

1. Should WTO law be reformed to provide greater policy space for resource-rich developing countries to use export restrictions to promote downstream processing? Evaluate the arguments in the light of *Indonesia — Raw Materials*.
2. Are China's export controls on critical minerals and technologies justifiable under GATT Article XXI, and what are the implications of the answer for the security exception generally?
3. Does mandatory supply chain due diligence improve conditions at mine sites, or does it primarily shift compliance costs and encourage disengagement from high-risk areas? Design an empirical study to test the question.
4. Analyse the legal character and legitimacy of minerals-for-security agreements, such as the United States–Ukraine Reconstruction Investment Fund, by reference to the principle of permanent sovereignty over natural resources.
5. To what extent does the EU Batteries Regulation exercise extraterritorial regulatory power, and is that power consistent with international law and with the principle of common but differentiated responsibilities?
6. Is there a case for a multilateral critical minerals agreement, and what institutional form should it take?

FURTHER READING

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- International Energy Agency, *The Role of Critical Minerals in Clean Energy Transitions* (IEA 2021) and *Global Critical Minerals Outlook* (annual, from 2023).
- OECD, *OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas* (3rd edn, OECD Publishing 2016).
- Owen, John R. et al, "Energy Transition Minerals and Their Intersection with Land-Connected Peoples" (2023) 6 *Nature Sustainability* 203.
- Sovacool, Benjamin K. et al, "Sustainable Minerals and Metals for a Low-Carbon Future" (2020) 367 *Science* 30.
- United Nations Secretary-General's Panel on Critical Energy Transition Minerals, *Resourcing the Energy Transition: Principles to Guide Critical Energy Transition Minerals towards Equity and Justice* (United Nations 2024).



PART III

Electricity, Renewable Energy, and Emerging Technologies

The law of electricity systems and power markets, renewable energy development, nuclear energy and its international regime, and the emerging technologies — hydrogen, storage, carbon capture, offshore renewables, and digital grids — that are reshaping energy law.

9 CHAPTER

International Electricity Law and Power Market Regulation

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the technical and economic characteristics of electricity that shape the legal structure of electricity industries.
2. Analyse the regulation of generation, transmission, and distribution, including licensing, tariff regulation, and the role of system operators.
3. Evaluate the models and legal instruments of electricity market liberalisation, with particular reference to the European Union, the United States, the United Kingdom, and Nigeria.
4. Assess the law of grid access and connection, including third-party access rights and the allocation of network capacity.
5. Analyse the structure and key provisions of power purchase agreements, including those underpinning independent power projects in developing countries.
6. Evaluate the legal frameworks for cross-border electricity trade, including the West African Power Pool and the EU internal electricity market.
7. Identify the legal mechanisms for consumer protection and energy affordability in electricity markets.
8. Analyse the regulation of market abuse and competition in electricity markets and the enforcement powers of regulators.

Introduction

Electricity is the energy carrier of modern life and, increasingly, of the energy transition. It powers homes, hospitals, schools, factories, and digital infrastructure; its reliable supply is a precondition of economic development and of the realisation of numerous human rights. The decarbonisation of energy systems depends overwhelmingly on electricity: on the replacement of fossil-fuelled generation with renewable and other low-carbon sources, and on the electrification of transport,

heating, and industrial processes that currently rely on direct combustion of fossil fuels. The International Energy Agency has described the current period as the beginning of an "age of electricity", in which global electricity demand is growing faster than overall energy demand, driven by industrial growth, air conditioning, electric vehicles, and data centres. At the same time, around seven hundred million people, the large majority of them in sub-Saharan Africa, still lack access to electricity, and hundreds of millions more have access only to unreliable supply.

The law of electricity has historically been predominantly domestic. Electricity systems were built as national or regional monopolies, regulated by domestic statutes and agencies, and international law played a limited role. That position has changed. Cross-border electricity trade is expanding through regional power pools and interconnected markets; foreign investment in generation and networks is protected by investment treaties and has given rise to numerous arbitrations; international development institutions have shaped the reform of electricity sectors across the developing world; the European Union has built a supranational electricity market governed by detailed legislation; and international climate law is driving the transformation of generation. International electricity law, as treated in this chapter, therefore comprises both the comparative law of domestic electricity regulation — the models and techniques that states have adopted and that circulate among them — and the international and regional legal frameworks that govern cross-border trade, investment, and cooperation.

This chapter begins with the legal structure of electricity industries (section 9.1) and the regulation of generation, transmission, and distribution (section 9.2). It then examines electricity market liberalisation (section 9.3), grid access and third-party rights (section 9.4), and power purchase agreements (section 9.5). It concludes with cross-border electricity trading (section 9.6), consumer protection and energy affordability (section 9.7), and market abuse, competition, and regulatory enforcement (section 9.8). Throughout, particular attention is paid to Nigeria's Electricity Act 2023, which fundamentally restructured the legal framework of the Nigerian electricity supply industry, and to the regional framework of the Economic Community of West African States (ECOWAS). Renewable energy law is examined in Chapter 10, and the technologies of the future grid — storage, smart grids, and digital infrastructure — in Chapter 12.

9.1 Legal Structure of Electricity Industries

The Physical Characteristics of Electricity

The legal structure of electricity industries is shaped by the physical characteristics of electricity, which distinguish it from every other commodity. Four characteristics are particularly important.

First, **electricity cannot be economically stored in large quantities**, at least with technologies widely deployed until recently. Supply and demand must therefore be balanced continuously, second by second, across the entire interconnected system. If generation falls short of demand, the frequency of the system falls, and if the imbalance is not corrected quickly, protective devices disconnect generators and loads, potentially causing a cascading blackout. The need for continuous balancing requires a central coordinator — the system operator — with authority to dispatch generation, procure reserves, and take emergency action. The development of battery storage at scale, examined in Chapter 12, is beginning to change this characteristic, but the need for real-time balancing remains fundamental.

Second, **electricity flows according to the laws of physics, not the terms of contracts**. Electricity injected into a meshed network flows along all available paths in inverse proportion to their impedance (Kirchhoff's laws), so that a contract for the sale of electricity from a generator in one location to a consumer in another does not determine the physical path that the electricity takes. This "loop flow" phenomenon means that transactions between two parties affect the network capacity available to others, including in neighbouring countries, and complicates the allocation of transmission rights and costs.

Third, **demand for electricity is highly variable and, in the short term, relatively insensitive to price**. Demand varies by time of day, day of the week, and season, and most consumers historically had no means of observing or responding to real-time prices. Generation capacity must therefore be sufficient to meet peak demand plus a reserve margin, and much capacity is used only for a few hours a year.

Fourth, **the networks through which electricity is transmitted and distributed are natural monopolies**, as in the gas sector (examined in Chapter 6). It would be inefficient to duplicate transmission lines or distribution networks, and network owners therefore enjoy market power that must be regulated.

The Electricity Value Chain

The electricity supply industry comprises four principal functions:

1. **Generation** — the production of electricity in power stations using fossil fuels, nuclear energy, hydropower, wind, solar, geothermal, biomass, or other sources.
2. **Transmission** — the transport of electricity at high voltage over long distances from generators to load centres, and the operation of the interconnected system.
3. **Distribution** — the transport of electricity at lower voltages from the transmission system to end-users.

4. **Supply** (or retail) — the sale of electricity to end-users, including metering, billing, and customer service.

To these should be added **system operation** — the real-time balancing of supply and demand and the dispatch of generation, which may be performed by the transmission owner or by an independent entity — and **market operation** — the administration of wholesale markets, settlement, and related functions.

Models of Industry Structure

The legal structure of electricity industries can be classified according to the degree of vertical integration and competition. A widely used typology, derived from the work of Sally Hunt and Graham Shuttleworth, distinguishes four models.

Model 1: Vertically integrated monopoly. A single utility — state-owned or privately owned but regulated — owns and operates generation, transmission, distribution, and supply within its territory, with no competition at any level. This was the dominant model worldwide until the 1980s and remains common in many developing countries and in parts of the United States.

Model 2: Single buyer. The integrated utility, or a separate entity, acts as the sole purchaser of electricity from generators, including independent power producers (IPPs) that sell under long-term power purchase agreements, and resells it to distributors or consumers. Competition exists only for the right to build new generation, typically through competitive procurement of PPAs. The single buyer model was widely adopted in Asia and Africa in the 1990s as a means of attracting private investment in generation without full liberalisation.

Model 3: Wholesale competition. Generators compete to sell electricity to distribution companies and large consumers through a wholesale market, which may be organised as a mandatory pool, a power exchange, or bilateral contracts, with non-discriminatory access to the transmission network. Distribution companies retain monopoly supply to small consumers.

Model 4: Retail competition. All consumers, including households, may choose their supplier, and distribution networks are open to competing retailers. This is the model required by EU law and adopted in the United Kingdom, Australia, parts of the United States, and other jurisdictions.

	Competition in generation	Wholesale buyer choice	Retail choice	Typical examples
1. Vertically integrated monopoly	No	No	No	Many developing states; SE United States
2. Single buyer	Yes — for new PPAs	No	No	1990s Asian & African IPP models
3. Wholesale competition	Yes	Yes	No	Transitional systems; Nigeria (in law)
4. Retail competition	Yes	Yes	Yes	EU; Great Britain; Texas; Australia

Figure 9.1 Models of electricity industry structure, from vertically integrated monopoly to retail competition. *Source:* prepared for this edition.

These models are ideal types; real electricity industries frequently combine elements of several. Many developing countries, including Nigeria, have adopted a transitional model in which a liberalised framework exists in law but the market operates in practice through regulated contracts and administered prices, pending the development of the conditions for effective competition.

Ownership

Electricity industries may be state-owned, privately owned, or mixed. State ownership was justified historically by the natural monopoly characteristics of networks, the strategic importance of electricity, the need for very large investments, and the goal of universal access. Privatisation, beginning in Chile in the 1980s and in England and Wales with the Electricity Act 1989, was justified by the expected efficiency gains of private ownership, the need to mobilise private capital, and the fiscal pressures facing governments. The results of privatisation have been mixed, and in recent years some jurisdictions have taken steps towards renationalisation or increased state ownership, as in France, where the state acquired full ownership of Électricité de France (EDF) in 2023, and in Germany, where municipalities have repurchased distribution networks.

The Nigerian Electricity Supply Industry: Structural Evolution

The Nigerian electricity supply industry illustrates the evolution from vertically integrated monopoly to an unbundled, privatised, and increasingly decentralised structure. From 1972, electricity was supplied by the National Electric Power Authority (NEPA), a vertically integrated state monopoly that became a byword

for unreliable supply. The National Electric Power Policy of 2001 and the Electric Power Sector Reform Act 2005 (EPSRA) provided for the unbundling of NEPA into the Power Holding Company of Nigeria (PHCN), and subsequently into eighteen successor companies — six generation companies (GenCos), eleven distribution companies (DisCos), and the Transmission Company of Nigeria (TCN) — and established the Nigerian Electricity Regulatory Commission (NERC) as an independent regulator. In November 2013, the federal government completed the privatisation of the GenCos and DisCos, while retaining ownership of TCN, whose management was contracted for a period to a private operator. The Nigerian Bulk Electricity Trading Plc (NBET) was established as a transitional bulk trader, purchasing power from GenCos and IPPs under PPAs and reselling it to DisCos under vesting contracts.

The privatised industry has faced severe challenges: insufficient gas supply to thermal power stations; transmission constraints; high technical and commercial losses; tariffs below cost-recovery levels, resulting in large subsidy obligations and a liquidity crisis in which DisCos failed to remit full payment for energy received; widespread estimated billing and inadequate metering; and the inability of many DisCos to attract investment. Several DisCos were taken over by their creditors, including the Asset Management Corporation of Nigeria (AMCON), following defaults on acquisition financing.

The Electricity Act 2023 and Constitutional Decentralisation

Two legal developments in 2023 transformed the structure of the Nigerian electricity industry. First, the Constitution of the Federal Republic of Nigeria (Fifth Alteration) Acts of 2023 amended the Exclusive and Concurrent Legislative Lists to empower state Houses of Assembly to make laws with respect to the generation, transmission, and distribution of electricity in areas covered by the national grid within the state, extending the existing state competence over areas not covered by the national grid. Secondly, the Electricity Act 2023, signed into law in June 2023, repealed EPSRA and established a new framework that recognises the competence of states to regulate electricity markets within their territories. Under the Act, a state that establishes its own electricity market and regulatory body assumes regulatory oversight of the market within the state, and NERC must transfer regulatory functions over that market to the state regulator. By 2025, a number of states — beginning with Enugu, followed by Ekiti, Ondo, Imo, Oyo, Edo, Kogi, Ogun, Niger, Plateau, and others — had enacted electricity laws and established state regulators, and NERC had transferred regulatory oversight to them.

The Electricity Act 2023 also provides for the establishment of the Nigerian Independent System Operator (NISO), separating system operation from the transmission business of TCN; for the promotion of renewable energy and

a requirement that generation companies include renewable sources in their generation mix; for the development of a competitive wholesale market; for the protection of consumers; and for integrated resource planning. The constitutional and statutory decentralisation of electricity regulation in Nigeria is one of the most significant recent developments in African electricity law, and raises novel questions concerning the relationship between federal and state regulation, the regulation of interstate transmission and trading, and the coherence of the national market. These questions are examined in section 9.2 and in Chapter 24.

9.2 Generation, Transmission, and Distribution Regulation

Licensing

Most electricity regimes require persons carrying out generation, transmission, distribution, supply, or trading activities to hold a licence (or, in some jurisdictions, an authorisation, permit, or concession) issued by the regulator or the ministry. Licensing serves several functions: it enables the state to verify that licensees have the technical, financial, and managerial capacity to perform their activities safely and reliably; it provides the legal instrument through which conditions — technical standards, tariff rules, consumer protection obligations, reporting requirements, and compliance with codes — are imposed; and it gives the regulator enforcement powers, including the ultimate sanction of revocation.

Licence conditions are typically of two kinds: standard conditions applicable to all licensees of a given class, and special conditions applicable to particular licensees. In Great Britain, for example, the Electricity Act 1989 provides for licences for generation, transmission, distribution, supply, interconnection, and the operation of the electricity system, each subject to standard conditions modifiable by the Gas and Electricity Markets Authority (acting through Ofgem) under statutory procedures. Licence conditions require licensees to comply with industry codes, such as the Grid Code, the Distribution Code, the Balancing and Settlement Code, and the Connection and Use of System Code, which set out detailed technical and commercial rules and are themselves subject to regulated modification procedures.

Under Nigeria's Electricity Act 2023, no person may construct, own, or operate an undertaking for generation (above a specified threshold capacity), transmission, system operation, distribution, supply, or trading of electricity without a licence issued by NERC or, within a state that has established its own market, by the state regulator, subject to the Act's provisions on interstate and national functions. Small generation facilities below the threshold capacity are exempt from licensing but may be required to obtain a permit. The Act empowers the regulator to impose licence conditions, to amend licences, and to suspend or cancel licences for breach.

Generation Regulation

In liberalised systems, generation is in principle a competitive activity, and regulation focuses on safety, environmental performance, and compliance with technical standards and market rules rather than on prices. Entry into generation is subject to licensing, planning, and environmental permitting, and may be subject to resource adequacy policies (examined in section 9.3). The environmental regulation of generation — emissions limits under instruments such as the EU Industrial Emissions Directive (Directive 2010/75/EU), carbon pricing under emissions trading systems, and restrictions on coal-fired generation — has become a principal driver of the generation mix.

In single-buyer systems, generation investment is determined by the procurement of PPAs, often through competitive tenders conducted by the single buyer or a procurement agency, and the regulation of generation is largely contractual. In vertically integrated systems, generation investment is determined by the utility's integrated resource planning, subject to regulatory approval.

Transmission Regulation

Transmission is a natural monopoly and is regulated in all systems. Transmission regulation addresses five principal matters.

Ownership and unbundling. As in the gas sector, liberalised electricity regimes require the separation of transmission from generation and supply, to prevent discrimination against competing generators. EU law, under Directive (EU) 2019/944, requires member states to apply ownership unbundling, the independent system operator (ISO) model, or the independent transmission operator (ITO) model to transmission system operators (TSOs), and the third country certification provisions apply to electricity as to gas.

Tariffs. Transmission tariffs must allow the transmission owner to recover efficient costs and earn a reasonable return, while sending efficient signals for the location of generation and load. Tariff methodologies are subject to regulatory approval, and the allocation of costs between generators and consumers and between locations is often contentious.

Investment and planning. The expansion of transmission is essential for the integration of renewable generation, which is often located far from load centres, and for the electrification of the economy. Transmission planning involves the identification of needs, the selection of projects, the allocation of costs, and the obtaining of permits. In the United States, FERC Order No 1000 (2011) required transmission providers to participate in regional transmission planning processes and to adopt cost allocation methods for regional projects, and Order No 1920 (2024) required long-term, forward-looking regional transmission planning over a twenty-year horizon that takes account of factors including changes in the

resource mix and policy requirements. In the European Union, the Ten-Year Network Development Plan prepared by the European Network of Transmission System Operators for Electricity (ENTSO-E) and the Trans-European Networks for Energy (TEN-E) Regulation (EU) 2022/869 identify projects of common interest that benefit from accelerated permitting and eligibility for EU financing.

System operation. The system operator is responsible for real-time balancing, dispatch, procurement of ancillary services (such as frequency response and reserves), management of congestion, and emergency actions. System operation may be performed by the TSO, by an ISO, or, in the United States, by regional transmission organisations (RTOs) and independent system operators that operate across the networks of multiple transmission owners. The independence of the system operator from market participants is essential to non-discriminatory dispatch. The establishment of the Nigerian Independent System Operator under the Electricity Act 2023 reflects this principle.

Reliability standards. System operators and transmission owners must comply with reliability standards governing the security of the system. In North America, the North American Electric Reliability Corporation (NERC), certified by FERC as the Electric Reliability Organization under the Energy Policy Act of 2005, develops and enforces mandatory reliability standards, which became legally binding following the Northeast blackout of August 2003. In the European Union, network codes adopted under Regulation (EU) 2019/943, such as the System Operation Guideline (Regulation (EU) 2017/1485), establish harmonised operational security requirements.

Distribution Regulation

Distribution networks are natural monopolies regulated by tariff controls, quality of service standards, and connection obligations. The role of distribution networks is changing fundamentally with the growth of distributed energy resources — rooftop solar, small-scale storage, electric vehicles, heat pumps, and demand response — which transform distribution networks from passive conduits delivering electricity in one direction into active systems managing bidirectional flows. Distribution system operators (DSOs) are increasingly required to procure flexibility services, to facilitate the connection of distributed resources, and to coordinate with TSOs. EU law, under Directive (EU) 2019/944, requires DSOs above a specified size to be legally and functionally unbundled from supply activities and empowers them to procure flexibility services through market-based procedures.

Tariff Regulation

Tariff regulation is the core of economic regulation of electricity networks and, in non-competitive systems, of end-user prices. The two principal approaches,

introduced in Chapter 6 in the context of gas, are cost-of-service (rate-of-return) regulation and incentive regulation. In Great Britain, Ofgem's RIIO framework ("Revenue = Incentives + Innovation + Outputs") sets revenue allowances for network companies over five-year price control periods, with incentives linked to outputs such as reliability, connections, customer satisfaction, and environmental performance. In the United States, state public utility commissions regulate retail rates and distribution, while FERC regulates wholesale rates and interstate transmission under the Federal Power Act of 1935, which requires that rates be "just and reasonable" and not unduly discriminatory.

In Nigeria, NERC has regulated end-user tariffs through the Multi-Year Tariff Order (MYTO) methodology, which sets tariffs for each DisCo on the basis of projected costs, including generation, transmission, and distribution costs, losses, and a return on investment. Because full cost-reflective tariffs would have been politically unacceptable, the federal government historically subsidised the gap between cost-reflective and allowed tariffs, accumulating large subsidy obligations. In April 2024, NERC introduced a significant increase in tariffs for customers in "Band A" — those receiving a minimum of twenty hours of supply per day — while maintaining lower tariffs for customers receiving less reliable supply. The service-based tariff structure links tariffs to the quality of supply, and DisCos that fail to deliver the specified hours of supply to Band A feeders must reclassify them, reducing the applicable tariff. The tariff reform illustrates the use of regulatory design to link price increases to service improvements and to address the political economy of tariff reform.

Federal and State Regulation

In federal systems, the allocation of regulatory competence between federal and state authorities is a fundamental feature of electricity law. In the United States, the Federal Power Act divides jurisdiction: FERC regulates interstate transmission and wholesale sales of electricity in interstate commerce, while states regulate retail sales, distribution, and generation facilities. The boundary has been the subject of frequent litigation. In *FERC v Electric Power Supply Association* (2016), the Supreme Court upheld FERC's jurisdiction to regulate compensation for demand response in wholesale markets, holding that demand response directly affects wholesale rates.[1] In *Hughes v Talen Energy Marketing LLC* (2016), decided the same year, the Court held that a Maryland programme guaranteeing a new generator a rate different from the wholesale market clearing price was pre-empted, because it set an interstate wholesale rate in contravention of FERC's exclusive authority.[2]

[1] *FERC v Electric Power Supply Association*, 577 US 260 (2016).
[2] *Hughes v Talen Energy Marketing LLC*, 578 US 150 (2016).

Nigeria's constitutional and statutory reforms of 2023 introduced a comparable division. Under the Electricity Act 2023, NERC retains regulatory oversight of interstate and international electricity matters, including the transmission network operated by TCN, system operation by NISO, and trading across state boundaries, while state regulators oversee markets within their states, including distribution and retail supply. The transition raises questions about the treatment of existing DisCo licences, many of which cover multiple states; the determination of tariffs where wholesale costs are regulated federally and retail tariffs by states; and the coherence of technical codes and standards. Several DisCos have established subsidiaries to hold licences in individual states, and the relationship between federal and state regulators will be shaped by the courts and by intergovernmental cooperation.

9.3 Electricity Market Liberalization

The Rationale for Liberalisation

Electricity market liberalisation — the introduction of competition in generation and supply, combined with the regulation of networks and the unbundling of competitive and monopoly activities — was driven by a combination of economic theory, technological change, and political ideology. Economists argued that the natural monopoly characteristics of electricity were confined to networks, and that competition in generation and supply would promote efficiency, innovation, and lower prices. The development of combined-cycle gas turbines, which could be built at smaller scale and lower cost than traditional coal and nuclear plants, reduced economies of scale in generation and lowered barriers to entry. The broader privatisation and deregulation movement of the 1980s and 1990s, and the influence of international financial institutions, spread the model across the world.

Pioneers: Chile, England and Wales, and Norway

Chile was the first country to liberalise its electricity industry, under the General Electricity Services Law of 1982, which introduced a wholesale market based on marginal cost dispatch and the separation of generation, transmission, and distribution. England and Wales followed with the Electricity Act 1989, which privatised the Central Electricity Generating Board and the area boards and established a mandatory wholesale pool, later replaced in 2001 by the New Electricity Trading Arrangements (NETA), based on bilateral contracts and a balancing mechanism, and extended to Great Britain as a whole in 2005 as the British Electricity Trading and Transmission Arrangements (BETTA). Norway's Energy Act of 1990 introduced competition while retaining substantial public ownership, and the Norwegian market became the nucleus of the Nordic power exchange, Nord Pool, which has expanded to cover the Nordic and Baltic countries.

The European Union Internal Electricity Market

The European Union has built a supranational internal electricity market through four successive legislative packages: the First Electricity Directive (96/92/EC), the Second Directive (2003/54/EC), the Third Energy Package (2009), and the Clean Energy for All Europeans package (2019), comprising Directive (EU) 2019/944 on common rules for the internal market for electricity and Regulation (EU) 2019/943 on the internal market for electricity, together with Regulation (EU) 2019/942 establishing the European Union Agency for the Cooperation of Energy Regulators (ACER). The legislation requires retail competition for all consumers, unbundling of networks, regulated third-party access, independent national regulatory authorities, and the integration of national markets through common rules for cross-border trading, examined in section 9.6.

In response to the energy price crisis of 2021–2023, the European Union adopted a reform of electricity market design in 2024, through Regulation (EU) 2024/1747 and Directive (EU) 2024/1711. The reform retained the marginal pricing model of the wholesale market but sought to shield consumers from short-term price volatility by promoting long-term contracts: it requires that direct price support schemes for new investments in renewable and nuclear generation take the form of two-way contracts for difference (under which the generator receives a top-up payment when the market price falls below a strike price and repays the difference when it rises above it), facilitates power purchase agreements, strengthens consumer rights to fixed-price contracts, and empowers the Council to declare a regional or Union-wide electricity price crisis permitting member states to set regulated prices below cost for households and small businesses.

The United States

The United States has a patchwork of market structures. FERC Order No 888 (1996) required public utilities owning interstate transmission to provide open-access transmission service on a non-discriminatory basis under an open access transmission tariff, and encouraged the formation of independent system operators. Order No 2000 (1999) encouraged the formation of regional transmission organisations. Today, RTOs and ISOs — including PJM Interconnection, the Midcontinent ISO, the California ISO, the Electric Reliability Council of Texas (ERCOT), the Southwest Power Pool, ISO New England, and the New York ISO — operate organised wholesale markets covering around two-thirds of United States electricity demand, while the South-east and much of the West retain vertically integrated utilities. Retail competition was introduced in some states, including Texas, Pennsylvania, Illinois, and much of the North-east, but abandoned or never adopted in others, particularly following the California electricity crisis of 2000–2001, in which flawed market design, market manipulation

by traders including Enron, and supply shortages caused soaring prices, rolling blackouts, and the bankruptcy of Pacific Gas and Electric.

Wholesale Market Design

Wholesale electricity markets may be organised in various ways. In a **mandatory pool**, all generation is sold into a central pool, which dispatches generation in order of bid price and pays all dispatched generators the marginal (clearing) price. In **bilateral markets**, generators and buyers contract directly, with a balancing mechanism operated by the system operator to resolve imbalances. In **power exchanges**, standardised products are traded anonymously on an exchange. Most markets combine these elements, with forward bilateral contracting, day-ahead and intraday auctions on power exchanges, and real-time balancing by the system operator.

A central feature of most wholesale markets is **marginal pricing**, under which all generators dispatched in a given period receive the price bid by the most expensive generator needed to meet demand. Marginal pricing provides efficient short-term dispatch signals, but it means that when gas-fired generators set the marginal price, all generators, including those with low costs such as renewable and nuclear plants, receive the high gas-based price. During the 2021–2023 crisis, this produced very high electricity prices and windfall profits for inframarginal generators, prompting the EU to adopt Council Regulation (EU) 2022/1854, which imposed a temporary cap on market revenues of inframarginal generators and a temporary solidarity contribution on fossil fuel companies, and prompting the electricity market design reform described above.

Markets also differ in their treatment of location. **Nodal pricing** (locational marginal pricing), used in most United States RTOs, sets different prices at each node of the network, reflecting transmission constraints and losses. **Zonal pricing**, used in Europe, sets a single price across large bidding zones, often corresponding to national borders, with congestion between zones managed through cross-border capacity allocation. The configuration of bidding zones is politically sensitive; Great Britain considered and, in 2025, rejected the introduction of zonal or nodal pricing in its Review of Electricity Market Arrangements.

Resource Adequacy and Capacity Mechanisms

A persistent concern in liberalised markets is whether energy-only markets — in which generators are paid only for the electricity they produce — provide sufficient incentives for investment in capacity needed to ensure security of supply at times of peak demand. Many jurisdictions have introduced **capacity mechanisms** that pay generators (and, increasingly, storage and demand response providers) for being available. Capacity mechanisms include capacity markets, strategic reserves, and capacity payments. In the European Union, capacity mech-

anisms are subject to the State aid rules and to conditions in Regulation (EU) 2019/943, including emission limits that exclude new coal-fired plants. In *Tempus Energy Ltd v European Commission* (2018), the General Court annulled the Commission's decision approving the British capacity market under the State aid rules without opening a formal investigation, on the ground that the Commission should have had doubts about the treatment of demand-side response; the British capacity market was suspended for around a year until the Commission re-approved it following a formal investigation, and in 2021 the Court of Justice set aside the General Court's judgment on appeal.

Liberalisation in Developing Countries

Electricity sector reform in developing countries, promoted from the 1990s by the World Bank and other development institutions, typically followed a standard sequence: corporatisation of the state utility, establishment of an independent regulator, introduction of IPPs, unbundling, privatisation of distribution and generation, and the eventual creation of a competitive wholesale market. Experience has shown that the standard model often fails to deliver its expected benefits in contexts characterised by tariffs below cost-recovery levels, weak payment discipline, high losses, limited regulatory capacity, and political interference. Scholars and practitioners have increasingly argued for "hybrid" models adapted to local conditions, which may retain significant state roles while introducing private participation and competition selectively. The Nigerian experience since privatisation in 2013 illustrates both the difficulty of implementing a liberalised model in the absence of financial viability and the potential of decentralisation to create new dynamics of reform.

9.4 Grid Access and Third-Party Rights

The Importance of Access

Access to the grid is the lifeblood of competition and of the energy transition. A generator that cannot connect to the network, or that cannot obtain the transmission capacity necessary to deliver its output to buyers, cannot compete; a consumer that cannot use the network to buy from a competing supplier cannot exercise choice. The law of grid access determines who may connect to and use the network, on what terms, at what cost, and with what priority when capacity is scarce. It has become one of the most important and contested areas of electricity law, as the queue of renewable and storage projects seeking connection has grown far faster than the capacity of networks to accommodate them.

Third-Party Access to Networks

Liberalised electricity regimes, like gas regimes, require network owners to grant third-party access to their networks on non-discriminatory terms. EU law, under

Article 6 of Directive (EU) 2019/944, requires member states to ensure a system of third-party access to transmission and distribution systems based on published tariffs, applicable to all customers and applied objectively and without discrimination between system users. Access may be refused only where the operator lacks the necessary capacity, and refusals must be duly substantiated. The Court of Justice has emphasised that the right of access is a fundamental element of the internal market. In *VEMW* (Case C-17/03, 2005), the Court held that the grant of priority access to cross-border transmission capacity to a former monopoly under pre-liberalisation contracts was discriminatory and contrary to the Electricity Directive, notwithstanding the age of the contracts.

In the United States, FERC Order No 888 required public utilities to provide open access transmission service under an open access transmission tariff, distinguishing between network integration service and point-to-point service, and requiring utilities to take transmission service for their own wholesale sales under the same tariff ("functional unbundling").

Connection Rights and Procedures

Connection to the network is distinct from use of the network. Connection procedures determine how a generator, storage facility, or large consumer obtains a physical connection to the grid, including the studies required to assess the impact of the connection on the network, the reinforcements needed, the allocation of connection costs, and the timeline. In many jurisdictions, connection queues have become a major bottleneck for the energy transition. In the United States, the capacity of generation and storage projects seeking interconnection grew to more than two thousand gigawatts by the mid-2020s — far exceeding existing installed capacity — and the median time from interconnection request to commercial operation lengthened to several years. FERC Order No 2023 (2023) reformed interconnection procedures by replacing the "first-come, first-served" serial study process with a "first-ready, first-served" cluster study process, imposing stricter financial commitments and site control requirements on applicants, and imposing penalties on transmission providers that fail to complete studies on time.

Great Britain faced a similar challenge, with a connection queue exceeding seven hundred gigawatts by 2024, many of the projects in it speculative or unlikely to proceed. Ofgem approved reforms in 2025 replacing the first-come, first-served approach with a process that prioritises projects that are ready to proceed and needed to meet the government's Clean Power 2030 objectives. The EU's Action Plan for Grids (2023) and the electricity market design reform of 2024 require transparency of available grid capacity and encourage flexible connection agreements, under which projects may connect more quickly in exchange for accepting curtailment in certain conditions.

Allocation of Connection Costs

The allocation of the costs of connecting new generation — between the connecting party and network users generally — affects the location and economics of generation investment. Under a **deep connection charging** approach, the connecting party pays the full cost of the connection assets and any wider network reinforcement required; under a **shallow** approach, it pays only for the assets connecting it to the nearest point of the network, with wider reinforcement costs socialised through network tariffs; and under **shallowish** approaches, it pays for some reinforcement. Deep charging sends strong locational signals but may deter investment, particularly by first movers in areas that require reinforcement; shallow charging encourages connection but may lead to inefficient location decisions.

Priority Dispatch and Curtailment

Renewable energy has historically benefited from **priority access** to the grid and **priority dispatch**, under which system operators must dispatch renewable generation ahead of conventional generation, subject to the security of the system. The EU Renewable Energy Directive 2009/28/EC required member states to provide priority or guaranteed access and priority dispatch for renewable electricity. The 2019 Clean Energy package narrowed priority dispatch to small installations and demonstration projects, reflecting the view that renewable generation had matured and should be integrated into markets, while requiring compensation for curtailment of renewable generation in specified circumstances under Article 13 of Regulation (EU) 2019/943. As the share of variable renewable generation grows, curtailment — the reduction of output from renewable generators because of network constraints or oversupply — has increased, and the allocation of the costs and risks of curtailment has become an important issue in grid access law and in PPAs.

Grid Access in Nigeria and Developing Countries

In Nigeria, the transmission network's limited capacity — wheeling capacity has frequently been below the available generation capacity — has constrained the evacuation of power from generation plants, resulting in "stranded" generation and payments to GenCos and IPPs for capacity that cannot be used. The Grid Code for the Nigeria Electricity Transmission System, issued by NERC, sets out the connection and operational requirements for users of the transmission system, and NERC's regulations on embedded generation, mini-grids, and eligible customers define access rights for distributed generators and large consumers. The **Eligible Customer Regulations**, first issued by NERC in 2017 and since revised, permit large consumers meeting specified thresholds to purchase electricity directly from generators or traders rather than from their local DisCo,

subject to payment of network charges — a limited form of retail competition. The **Mini-Grid Regulations** (2016, revised 2023) establish a framework for the development of isolated and interconnected mini-grids, including provisions for the treatment of mini-grids when the main grid arrives, and have supported significant investment in rural electrification, examined in Chapter 23.

Access for Distributed Energy Resources

Distributed energy resources (DERs) raise new questions of access. Consumers who install rooftop solar may wish to export surplus electricity to the grid ("prosumers"), and aggregators may wish to combine many small resources to participate in wholesale markets. EU law confers rights on "active customers" to generate, store, sell, and participate in flexibility and energy efficiency schemes, and on "citizen energy communities" and "renewable energy communities" to engage in generation, supply, aggregation, and storage. FERC Order No 2222 (2020) requires RTOs and ISOs to allow aggregations of distributed energy resources to participate in wholesale markets. The compensation of prosumers for exported electricity — through net metering, feed-in tariffs, or market-based arrangements — and the allocation of network costs between prosumers and other consumers have been contentious, as illustrated by debates over the revision of net metering in California in 2022 and 2023, which significantly reduced the compensation paid to new rooftop solar customers for exported electricity.

9.5 Power Purchase Agreements

Function and Types

A power purchase agreement (PPA) is a contract for the sale of electricity (and, often, associated capacity, ancillary services, and environmental attributes) by a generator to a buyer over a defined period. PPAs serve different functions in different market structures.

In **single-buyer systems** and developing countries, the PPA between an independent power producer and a state-owned utility or bulk purchaser is the foundation of private investment in generation. The long-term PPA — typically twenty to twenty-five years — provides the revenue certainty that lenders require to finance the project, and allocates the principal risks between the IPP and the offtaker, often with support from the government (examined in Chapter 16).

In **liberalised markets**, PPAs are used by generators and buyers to hedge price risk, and by corporate buyers to procure renewable electricity. **Corporate PPAs** — contracts between generators, usually of renewable energy, and corporate consumers such as technology companies, manufacturers, and retailers — have grown rapidly, driven by corporate sustainability commitments. Corporate PPAs may be **physical** (involving the delivery of electricity through the grid, often via a licensed supplier in a "sleeved" arrangement) or **virtual** or **financial** (a contract

for difference under which the parties settle the difference between an agreed strike price and the market price, without physical delivery, while the generator sells its output into the market and the buyer purchases its electricity from its supplier).

Appendix E contains a sample renewable energy PPA with annotations. The following discussion focuses on the long-term IPP PPA in developing countries, which raises the most significant legal issues.

Key Provisions of an IPP PPA

Capacity and energy payments. The tariff structure of an IPP PPA for a dispatchable plant typically comprises a **capacity payment** (or availability payment), payable for making the plant's capacity available to the offtaker regardless of whether it is dispatched, designed to cover the IPP's fixed costs, debt service, and return on equity; and an **energy payment**, payable for electricity actually delivered, designed to cover variable costs, principally fuel. The capacity payment is the economic equivalent of take-or-pay in gas contracts, allocating demand risk to the offtaker. For variable renewable generation, PPAs typically provide for payment per unit of energy delivered, with "deemed energy" payments where the plant is available but cannot deliver because of grid constraints or curtailment by the offtaker.

Dispatch. The PPA provides for the offtaker (or system operator) to dispatch the plant within specified technical parameters, and allocates the risk of non-dispatch.

Fuel supply. For thermal plants, the PPA allocates fuel supply risk. The IPP may procure fuel itself under a fuel supply agreement, with fuel costs passed through in the energy payment, or the offtaker may supply fuel under a tolling arrangement. In Nigeria, the unreliability of gas supply to power plants has been a major source of risk, and PPAs have addressed whether the IPP is entitled to capacity payments when it cannot generate because of gas shortfalls.

Currency and indexation. Because IPPs in developing countries typically finance projects in foreign currency, PPAs commonly denominate tariffs in, or index them to, the United States dollar, transferring currency risk to the offtaker. Large devaluations, such as those experienced by Nigeria in 2016 and 2023–2024, Egypt in 2016, and Pakistan, can render dollar-indexed tariffs unaffordable for the offtaker and consumers, creating pressure for renegotiation.

Change in law. PPAs typically protect the IPP against changes in law that increase its costs or reduce its revenues, through tariff adjustments or compensation.

Force majeure. PPAs distinguish between **natural force majeure** (events outside the control of either party, such as natural disasters), and **political force majeure** (events within the control of the government, such as changes in law,

expropriation, and political violence), and allocate the consequences differently: political force majeure events commonly entitle the IPP to continued capacity payments and, if prolonged, to terminate and receive a termination payment.

Termination and termination payments. The PPA specifies events of default by each party and the termination payments payable on termination. Upon termination for offtaker default or prolonged political force majeure, the termination payment typically covers outstanding debt and a return on equity; upon termination for IPP default, it may be limited to outstanding debt or less. Termination payments are often backed by government guarantees or put-and-call option agreements under which the government is obliged to purchase the plant.

Credit support. Because state-owned offtakers in many developing countries are not creditworthy, PPAs are supported by credit enhancement mechanisms: letters of credit for short-term payment obligations, escrow accounts, government guarantees or support agreements, and partial risk guarantees from multilateral development banks such as the World Bank's International Development Association (IDA) and the Multilateral Investment Guarantee Agency (MIGA).

Dispute resolution. PPAs typically provide for expert determination of technical disputes and international arbitration of other disputes.

The Azura-Edo IPP

The Azura-Edo IPP, a 461 MW open-cycle gas-fired power plant in Edo State that achieved commercial operation in 2018, was the first project-financed IPP in Nigeria to reach financial close under the post-privatisation framework. Its PPA with NBET was supported by a put-call option agreement with the federal government and by a partial risk guarantee from the World Bank backing NBET's payment obligations under a letter of credit, together with political risk insurance from MIGA. The project's financing structure became a template for subsequent projects, but its experience also illustrated the risks: as the liquidity crisis in the Nigerian electricity market deepened, the World Bank's guarantee was reportedly called upon to cover arrears in NBET's payments. The project illustrates the extent to which long-term PPAs in developing countries depend on sovereign and multilateral credit support to compensate for the weakness of the offtaker.

PPA Disputes and Renegotiation

Long-term PPAs have generated significant disputes, particularly where economic crises, currency devaluations, or changes in government have made tariffs appear excessive. In the aftermath of the Asian financial crisis of 1997–1998, the Indonesian state electricity company PLN sought to renegotiate or cancel numerous IPP PPAs concluded under the Suharto government. In *Himpurna California Energy Ltd v PT (Persero) Perusahaan Listrik Negara* (1999), an UNCITRAL tribunal

awarded the claimant more than USD 390 million for PLN's breach of an energy sales contract for a geothermal project, while reducing the damages claimed to take account of the economic crisis; when the award was not paid, the claimants recovered under political risk insurance from the Overseas Private Investment Corporation. In India, the Dabhol power project, developed by a consortium led by Enron, became the subject of disputes with the Maharashtra State Electricity Board in 2001 after the board refused to pay tariffs it regarded as excessive, leading to arbitrations, litigation in India and elsewhere, and eventually a settlement under which the plant was transferred to Indian state ownership.

More recently, Pakistan's government undertook in 2024 a programme of renegotiating or terminating PPAs with IPPs, citing the burden of capacity payments on consumers and the public finances, and several IPPs agreed to early termination or conversion of their contracts to "take-and-pay" terms under which payment is made only for energy delivered. Ghana renegotiated several IPP PPAs from 2020 onwards to reduce excess capacity charges. These episodes illustrate the tension between the sanctity of contracts necessary to attract investment and the political and economic pressures that arise when long-term PPAs impose burdens perceived as unsustainable. They also raise questions about the role of investment treaty protection and the extent to which renegotiation under pressure is consistent with the fair and equitable treatment standard, examined in Chapter 14.

Corporate PPAs and the Energy Transition

Corporate PPAs have become a major driver of renewable energy investment in Europe and North America and are growing in emerging markets. Their legal issues include the allocation of "shape" and "volume" risk (the mismatch between the generator's variable output and the buyer's demand profile); the treatment of curtailment and negative prices; the transfer of guarantees of origin or renewable energy certificates (examined in Chapter 10); change in law; credit support; and accounting treatment, including whether a virtual PPA must be accounted for as a derivative. The EU electricity market design reform of 2024 requires member states to remove unjustified barriers to PPAs and to ensure the availability of instruments, such as guarantee schemes, to reduce the financial risks of PPAs for buyers. In Nigeria, the eligible customer framework and the bilateral trading opportunities created by the Electricity Act 2023 have facilitated the emergence of corporate PPAs between generators and industrial consumers, including captive and embedded renewable projects.

9.6 Cross-Border Electricity Trading

Benefits and Challenges

Interconnecting national electricity systems offers substantial benefits. Interconnection enables countries to share generation reserves, reducing the capacity each

must maintain to ensure security of supply; to exploit differences in generation costs, resource endowments, and demand patterns, so that low-cost hydropower in one country can displace expensive thermal generation in another; to integrate variable renewable generation over larger areas, smoothing fluctuations in wind and solar output; and to provide mutual assistance in emergencies. For small developing countries, regional integration can make possible the development of large, low-cost generation projects that would be too large for any single national market.

Cross-border electricity trade nonetheless faces significant obstacles: the physical characteristics of electricity, which require close coordination of system operation across borders; differences in market structures, regulatory frameworks, and technical standards; concerns about dependence on foreign supply and the loss of national control; the creditworthiness of buyers; and the need to agree on the allocation of interconnector costs and capacity. The legal frameworks for cross-border trade range from bilateral agreements between utilities, through regional power pools with common rules, to fully integrated supranational markets.

The European Union Internal Electricity Market

The European Union has developed the world's most advanced framework for cross-border electricity trade. Regulation (EU) 2019/943 and the network codes and guidelines adopted under it establish common rules for cross-border trading, capacity allocation, and congestion management. The **Capacity Allocation and Congestion Management Guideline** (Regulation (EU) 2015/1222) established **single day-ahead coupling** (SDAC) and **single intraday coupling** (SIDC), under which cross-border transmission capacity is allocated implicitly together with energy through a common price coupling algorithm (EUPHEMIA), maximising the use of interconnectors and producing price convergence across bidding zones when capacity is not congested. Regulation (EU) 2019/943 requires TSOs to make available for cross-border trade at least seventy per cent of the transmission capacity of interconnectors and internal critical network elements, subject to transitional arrangements. ACER oversees the implementation of these rules, decides on certain cross-border matters where national regulators cannot agree, and monitors the market. The EU has also set an interconnection target of fifteen per cent of installed generation capacity for each member state by 2030.

Following the United Kingdom's withdrawal from the European Union, Great Britain left the EU's internal electricity market and market coupling arrangements. The Trade and Cooperation Agreement of 2020 contains an energy title that provides for cooperation on electricity trading over interconnectors and commits the parties to develop new arrangements for the efficient use of interconnectors, although the more efficient implicit allocation of capacity

has not yet been restored. The episode illustrates both the depth of integration achieved within the EU and the costs of exit.

The West African Power Pool

The **West African Power Pool** (WAPP) was established by Decision A/DEC.5/12/99 of the ECOWAS Authority of Heads of State and Government in 1999, and its Articles of Agreement were adopted in 2006, establishing it as a specialised institution of ECOWAS. The WAPP's mission is to integrate the national power systems of the fourteen continental ECOWAS member states into a unified regional electricity market, with the objective of providing reliable and affordable electricity. Its members are the national utilities, transmission companies, and generation companies of member states. The WAPP has developed a master plan of priority regional generation and transmission projects, including the **CLSG interconnection** linking Côte d'Ivoire, Liberia, Sierra Leone, and Guinea, the **OMVG interconnection** linking the Gambia River Basin Development Organisation states (Gambia, Guinea, Guinea-Bissau, and Senegal), and the **North Core** project linking Nigeria, Niger, Benin, Togo, and Burkina Faso.

The **ECOWAS Regional Electricity Regulatory Authority** (ERERA), established in 2008 by a Supplementary Act of the ECOWAS Authority, is the regional regulator, responsible for regulating cross-border electricity exchanges among member states, approving regional market rules, setting tariffs for regional transmission, and resolving disputes. The **ECOWAS Regional Electricity Market** was launched in phases, with the first phase commencing in 2018 on the basis of bilateral contracts and regional market rules approved by ERERA, and later phases envisaging a day-ahead market. The legal framework is supported by the ECOWAS Energy Protocol of 2003, examined in Chapter 6, and by the ECOWAS directive on the organisation of the regional electricity market.

Nigeria has historically been a significant exporter of electricity to neighbouring states, supplying Benin, Togo, and Niger under bilateral contracts. In August 2023, following the military coup in Niger and the imposition of sanctions by ECOWAS, Nigeria suspended electricity supply to Niger, illustrating the vulnerability of cross-border electricity trade to political crises and the use of energy interdependence as an instrument of regional pressure. Supply was restored in 2024 after ECOWAS lifted most sanctions. Nigeria's own domestic supply shortages have periodically generated political pressure to reduce exports, and the payment performance of foreign buyers has been a recurrent concern.

Other Regional Power Pools

The **Southern African Power Pool** (SAPP), established in 1995 under an intergovernmental memorandum of understanding among member states of the Southern African Development Community (SADC), was the first regional

power pool in Africa. It operates competitive day-ahead, intraday, and forward markets alongside bilateral contracts, and the **Regional Electricity Regulators Association of Southern Africa** (RERA) facilitates regulatory harmonisation. The **Eastern Africa Power Pool** (EAPP), established in 2005, and the **Central African Power Pool** (CAPP), established in 2003, are at earlier stages of development, while the **Comité Maghrébin de l'Électricité** coordinates North African interconnection.

At the continental level, the **African Single Electricity Market** (AfSEM), launched by the African Union in 2021 with the support of the European Union, aims to integrate the regional power pools into a single continental market by 2040, through a continental transmission master plan, harmonised regulatory frameworks, and continental market rules. The **Continental Power System Master Plan**, adopted in 2023, identifies priority interconnections. The AfSEM is linked to the African Continental Free Trade Area, which may provide a legal framework for trade in electricity services.

Elsewhere, the **ASEAN Power Grid**, established under ASEAN agreements and memoranda of understanding, has developed through bilateral interconnections and the Lao PDR–Thailand–Malaysia–Singapore Power Integration Project, the first multilateral electricity trade in ASEAN. The **Gulf Cooperation Council Interconnection Authority** (GCCIA) operates an interconnection linking the six GCC states, used principally for mutual support in emergencies. In South Asia, India's cross-border electricity trade with Bhutan, Nepal, and Bangladesh is governed by bilateral agreements and Indian regulations on cross-border trade.

International Law and Cross-Border Electricity

The international law governing cross-border electricity trade is relatively undeveloped. Electricity is generally treated as a good for the purposes of GATT 1994, although the classification of electricity transmission and distribution services under the General Agreement on Trade in Services is uncertain. The Energy Charter Treaty's transit provisions, examined in Chapter 6, apply to electricity as well as to gas. Regional frameworks such as the ECOWAS Energy Protocol and the EU legislation provide the most developed rules. The legal issues raised by long-distance interconnections — such as the proposed Xlinks project to transmit solar and wind power from Morocco to Great Britain through submarine cables, and the proposed interconnections between North Africa and Europe — include the regulatory treatment of interconnectors, the allocation of costs and capacity, security of supply, and the law of submarine cables, examined in Chapter 20.

9.7 Consumer Protection and Energy Affordability

Electricity as an Essential Service

Electricity is an essential service, without which households cannot light their homes, cook, heat or cool their dwellings, store food and medicines, communicate, study, or work. Access to affordable and reliable electricity is closely connected to the realisation of human rights, including the rights to an adequate standard of living, to housing, to health, and to education, as examined in Chapter 23. Sustainable Development Goal 7 calls for universal access to affordable, reliable, sustainable, and modern energy by 2030. Electricity consumers are, moreover, in a structurally weak position relative to suppliers: they depend on a continuous supply that they cannot easily replace, they face complex contracts and tariffs, and in many markets they have little or no choice of supplier. Electricity law therefore contains extensive provisions for consumer protection, which have become more important as liberalisation has exposed consumers to market risks and as the energy price crisis of 2021–2023 created severe affordability pressures.

Consumer Rights in EU Law

Directive (EU) 2019/944 establishes a comprehensive set of consumer rights, including: the right to a contract with specified information; the right to switch supplier within three weeks (and, from 2026, within twenty-four hours) free of charge; the right to clear billing information, at least once a year based on actual consumption; the right to a dynamic electricity price contract for consumers with smart meters; the right to request a smart meter; access to out-of-court dispute settlement mechanisms; and protection for vulnerable customers and against disconnection. Member states must define the concept of vulnerable customers, which may refer to energy poverty, and ensure that the rights and obligations linked to vulnerable customers are applied, including measures to prohibit disconnection in critical times. The 2024 market design reform strengthened these rights, including by giving consumers the right to fixed-price, fixed-term contracts and requiring suppliers to have appropriate hedging strategies to limit the risk that supplier failures harm consumers, and by establishing a framework for suppliers of last resort.

Price Regulation for Households

Many jurisdictions regulate retail electricity prices for households, either through comprehensive tariff regulation (in non-liberalised systems) or through price caps in liberalised markets. In Great Britain, the Domestic Gas and Electricity (Tariff Cap) Act 2018 required Ofgem to impose a cap on standard variable and default tariffs, which has been adjusted periodically by reference to wholesale and other costs. During the 2022 crisis, the cap rose dramatically, and the government

introduced the **Energy Price Guarantee**, which limited the unit rates paid by households below the level of the cap, with the government compensating suppliers for the difference. In the European Union, the Court of Justice held in *Federutility* (Case C-265/08, 2010) that public intervention in the price of supply of natural gas to final consumers is compatible with the internal market provided it pursues a general economic interest, is proportionate, and complies with conditions of transparency, non-discrimination, and limited duration; the reasoning has been applied to electricity, and the 2019 Directive codified the conditions under which member states may regulate prices for households and microenterprises, including energy poor and vulnerable households.

Consumer Protection in Nigeria

In Nigeria, consumer protection in the electricity sector has centred on metering, billing, quality of supply, and complaint handling. A large proportion of electricity consumers have historically been unmetered and subjected to **estimated billing**, which consumers regard as arbitrary and excessive. NERC has adopted a series of measures to address the metering gap, including the Meter Asset Provider Regulations (2018, revised), which permit independent meter asset providers to finance and install meters; the National Mass Metering Programme, launched in 2020; and orders capping estimated bills for unmetered customers by reference to the consumption of metered customers on the same feeder. The NERC Customer Service Standards of Performance for Distribution Companies, and the Customer Complaints Handling Standards and Procedures, establish timelines for addressing complaints, connecting new customers, and restoring supply, with compensation for failures. Consumers may escalate complaints to NERC Forum Offices, which operate as a quasi-judicial dispute resolution mechanism. The Electricity Act 2023 contains provisions on consumer protection and empowers the regulator to establish rules on customer rights and obligations, and the Federal Competition and Consumer Protection Act 2018 applies to the electricity sector.

The service-based tariff structure introduced by NERC, in which tariffs vary according to the hours of supply received by each feeder band, links consumer payments to service quality, and NERC has imposed sanctions on DisCos that failed to deliver the minimum hours of supply to Band A feeders. The effectiveness of these measures depends on the regulator's capacity to monitor service delivery and enforce compliance.

Disconnection and the Right to Electricity

The disconnection of electricity supply for non-payment raises acute questions of consumer protection and human rights. Many jurisdictions restrict disconnection: prohibiting it during winter months or extreme heat, for households with vulnerable members (such as those dependent on medical equipment), or

without adequate notice and opportunity to pay. In South Africa, the courts have developed a body of jurisprudence on the procedural fairness of electricity disconnections by municipalities. In *Joseph v City of Johannesburg* (2009), the Constitutional Court held that tenants who were not parties to the supply contract between their landlord and the municipality were nonetheless entitled to procedural fairness, including adequate notice, before their supply was disconnected, because the municipality's obligation to provide basic services gave rise to a public law relationship with the users of those services.[3] The Court's reasoning rests on the constitutional duty of local government to ensure the provision of services to communities in a sustainable manner.

The prolonged "load shedding" — scheduled rotational power cuts — imposed by South Africa's state utility Eskom because of generation shortfalls also gave rise to litigation. In *United Democratic Movement v Eskom Holdings SOC Ltd* (2023), the High Court in Pretoria held that the state's failure to address the impact of load shedding on hospitals, clinics, schools, and police stations infringed constitutional rights, and ordered the Minister of Public Enterprises to take steps to ensure sufficient supply to such facilities. The case illustrates the potential of constitutional litigation to translate the human rights dimensions of electricity supply into enforceable obligations, a theme examined in Chapters 23 and 27.

Energy Poverty and Social Tariffs

Energy poverty — the inability of households to secure adequate energy services at affordable cost — affects households in both developed and developing countries. Legal and policy responses include social or "lifeline" tariffs, under which low-income or low-consumption households pay reduced prices for an initial block of consumption; direct income support; energy efficiency programmes targeting low-income households; and protections against disconnection. Nigeria's tariff structure has included a lifeline tariff for residential customers with very low consumption. The EU's Social Climate Fund, established by Regulation (EU) 2023/955 to address the social impacts of the extension of carbon pricing to buildings and road transport, will support vulnerable households from 2026. The design of social tariffs must balance the goal of protecting vulnerable consumers against the risks of subsidising consumption by better-off households and undermining the financial viability of utilities. Energy poverty and universal access are examined further in Chapter 23.

[3] *Joseph v City of Johannesburg* [2009] ZACC 30, 2010 (4) SA 55 (CC).

9.8 Market Abuse, Competition, and Regulatory Enforcement

Vulnerability of Electricity Markets to Abuse

Electricity markets are unusually vulnerable to the exercise of market power and to manipulation. Because electricity cannot be stored economically at scale and demand is inelastic in the short term, a generator that withholds capacity at times of tight supply can raise prices dramatically; because the marginal generator sets the price for all, even a small generator may have significant market power at peak times; and because markets are governed by complex rules, traders may exploit flaws in market design to profit at the expense of other participants and consumers. The California electricity crisis of 2000–2001, examined in the casebox below, demonstrated the scale of harm that market abuse can inflict, and led to the development of specialised regimes for the prevention and sanctioning of market manipulation in energy markets.

Market Manipulation and Insider Trading: REMIT and United States Law

In the European Union, the **Regulation on Wholesale Energy Market Integrity and Transparency** (REMIT) — Regulation (EU) No 1227/2011, amended by Regulation (EU) 2024/1106 — prohibits insider trading and market manipulation in wholesale energy markets, requires market participants to publish inside information in a timely manner, and requires the reporting of transactions and orders to ACER. Market manipulation includes transactions or orders that give false or misleading signals as to supply, demand, or price, that secure the price at an artificial level, or that employ fictitious devices or any other form of deception. ACER monitors trading activity, and national regulatory authorities investigate and enforce the prohibitions, imposing penalties under national law. The 2024 amendments strengthened ACER's investigatory powers in cross-border cases, extended REMIT to algorithmic trading, and harmonised penalties. Enforcement decisions have included substantial fines for wash trades, layering, and the withholding of capacity, such as the fine imposed by the French Energy Regulatory Commission's dispute settlement and sanctions committee (CoRDiS) on VITOL in 2018 for manipulation of gas markets, and decisions by Ofgem against generators for charging excessive prices in the balancing mechanism.

In the United States, the **Energy Policy Act of 2005** amended the Federal Power Act and the Natural Gas Act to prohibit the use of any manipulative or deceptive device or contrivance in connection with the purchase or sale of electricity or natural gas subject to FERC's jurisdiction, and gave FERC authority to impose civil penalties of up to USD 1 million per day per violation (adjusted for inflation). FERC's Anti-Manipulation Rule, modelled on Rule 10b-5 under

the Securities Exchange Act of 1934, has been the basis of numerous enforcement actions. In *FERC v Barclays Bank plc*, FERC assessed a civil penalty of USD 435 million against Barclays and four of its traders for manipulating electricity prices in western United States markets between 2006 and 2008 by trading physical products to benefit financial swap positions; the matter was settled in 2017 for USD 105 million. The Commodity Futures Trading Commission has parallel jurisdiction over manipulation in energy futures and swaps.

Competition Law in Electricity Markets

General competition law applies to electricity markets alongside sector-specific regulation. In the European Union, Article 102 TFEU, which prohibits the abuse of a dominant position, has been applied to incumbent electricity companies in a series of cases. In 2008, the European Commission accepted commitments from E.ON to divest generation capacity and its extra-high-voltage transmission network in Germany, following an investigation into suspected strategic withholding of generation capacity to raise prices and discrimination in the procurement of balancing services. In 2010, the Commission accepted commitments from EDF to release capacity tied up in long-term supply contracts with large industrial customers in France, which allegedly foreclosed the market to competing suppliers. In *DEI v Commission*, concerning the Greek incumbent's exclusive access to lignite deposits, the Commission found that Greece had infringed Article 106 in conjunction with Article 102 by granting the state-owned Public Power Corporation privileged access to lignite, and the Court of Justice in 2014 upheld the Commission's approach, holding that it was not necessary to prove actual abuse where state measures created a situation of inequality of opportunity between economic operators.[4] The **State aid rules** of Articles 107 and 108 TFEU are also of central importance, governing the compatibility of renewable support schemes, capacity mechanisms, and emergency interventions, as examined in Chapter 10.

In the United Kingdom, the Competition and Markets Authority's **Energy Market Investigation** (2014–2016) found that a combination of features of the retail market, including weak customer engagement, gave suppliers a position of unilateral market power over their inactive customers, resulting in detriment of around £1.4 billion per year, and recommended remedies including a temporary price cap for customers on prepayment meters. The subsequent legislative price cap for default tariffs, described in section 9.7, went further than the CMA's recommendations.

In Nigeria, the Federal Competition and Consumer Protection Act 2018 established the Federal Competition and Consumer Protection Commission

[4] *European Commission v DEI*, Case C-553/12 P, EU:C:2014:2083.

(FCCPC) with powers over anti-competitive conduct and consumer protection across the economy, while the Electricity Act 2023 confers on NERC responsibility for promoting competition and preventing abuse of market power in the electricity industry. The coordination of sector regulators and competition authorities, examined in Chapter 4, is a recurrent challenge.

Regulatory Enforcement Powers

Effective regulation depends on effective enforcement. Electricity regulators typically possess powers to require information, to investigate suspected breaches of licence conditions, codes, and legislation, to issue compliance orders, to impose financial penalties, to require the payment of compensation to consumers, and to modify, suspend, or revoke licences. In Great Britain, Ofgem may impose penalties of up to ten per cent of a licensee's turnover for breaches of licence conditions or relevant requirements, and has frequently agreed settlements under which licensees make payments to consumer redress funds. In Nigeria, the Electricity Act 2023 empowers NERC to impose fines and sanctions on licensees for breaches of the Act, regulations, codes, and licence conditions, and NERC has periodically imposed fines on DisCos for violations such as failure to meet metering obligations, overbilling, and failure to deliver required hours of supply.

The exercise of enforcement powers is subject to administrative law principles of legality, procedural fairness, rationality, and proportionality, and enforcement decisions are subject to appeal or judicial review. The design of enforcement — the balance between deterrence and compliance-oriented approaches, the use of settlements and consumer redress, and the transparency of enforcement decisions — has been the subject of extensive regulatory scholarship, including the theory of "responsive regulation" developed by Ian Ayres and John Braithwaite, which advocates an "enforcement pyramid" in which regulators escalate from persuasion to increasingly severe sanctions in response to non-compliance. Regulatory enforcement and judicial review are examined further in Chapter 27.

LEADING AUTHORITY

The California Electricity Crisis (2000–2001) and the Law of Market Manipulation

Background. California's restructuring legislation of 1996 (Assembly Bill 1890) required the state's investor-owned utilities to divest much of their generation and to buy electricity in a spot market operated by the California Power Exchange, while capping retail prices for a transitional period and discouraging long-term contracts. In 2000, a combination of rising demand, drought reducing hydroelectric imports, rising gas prices, and capacity withdrawals caused wholesale prices to rise to many times their historical levels, while retail rates remained frozen.

Manipulation. Investigations by FERC and others subsequently revealed that traders, notably at Enron, had exploited flaws in market rules through strategies given names such as "Death Star", "Get Shorty", and "Ricochet", which involved scheduling fictitious congestion to collect congestion relief payments, selling ancillary services they did not have, and moving electricity out of and back into California to evade price caps. Generators were also found to have withheld capacity to raise prices.

Consequences. Pacific Gas and Electric filed for bankruptcy in April 2001, Southern California Edison came close to insolvency, and the state government purchased power on behalf of the utilities at very high prices under long-term contracts. Rolling blackouts affected millions of customers. FERC ordered refunds, and extensive litigation followed. Enron collapsed in December 2001 amid accounting fraud, and several of its executives and traders were convicted of criminal offences.

Legal significance. The crisis led directly to the anti-manipulation provisions of the Energy Policy Act of 2005, to reforms of market design in California and elsewhere, including the introduction of locational marginal pricing and market power mitigation measures, and to a lasting scepticism about retail price caps combined with spot market exposure. Its lessons informed the design of REMIT in the European Union. It remains the paradigmatic illustration of the importance of market design and enforcement in electricity markets.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

Legislative drafting: a state electricity market law (hypothetical). *The state and facts are fictitious, although the exercise is set within the actual framework of Nigeria's Constitution and Electricity Act 2023.* The fictional Nigerian state of Oroke, with a population of five million, receives electricity from a single DisCo whose licence covers Oroke and two neighbouring states. The state government wishes to establish its own electricity market under the Electricity Act 2023, with the objectives of increasing metering, attracting investment in embedded generation and mini-grids, protecting vulnerable consumers, and ensuring a smooth transition of regulatory oversight from NERC.

Prepare (a) a policy memorandum (1,500 words) identifying the constitutional and statutory basis for the state's competence, the matters that remain within federal competence, and the principal transitional issues; (b) the draft text of the key provisions of an "Oroke State Electricity Market Law" (approximately 2,000 words), including provisions establishing a state regulatory commission, licensing, tariff methodology, consumer protection, and the transition of existing licences; and (c) a short note (500 words) assessing the risks of regulatory fragmentation and proposing mechanisms for coordination with NERC and other states.

CHAPTER SUMMARY

- The physical characteristics of electricity — the need for continuous balancing, flows governed by physics rather than contract, inelastic demand, and natural monopoly networks — shape its legal structure.
- Electricity industries range from vertically integrated monopolies through single-buyer and wholesale competition models to full retail competition; Nigeria's industry has moved from the NEPA monopoly through privatisation to constitutional and statutory decentralisation under the Electricity Act 2023.
- Regulation of generation, transmission, and distribution involves licensing, unbundling, tariff regulation, planning, system operation, and reliability standards, with significant federal–state allocation issues in the United States and now Nigeria.
- Liberalisation has produced diverse market designs; the EU's 2024 reform retains marginal pricing while promoting long-term contracts and two-way contracts for difference.
- Grid access law governs third-party access, connection procedures, cost allocation, priority dispatch, and access for distributed resources; connection queues have become a major bottleneck for the energy transition.
- Power purchase agreements allocate risk through capacity payments, currency indexation, change in law, force majeure, and termination provisions, supported by sovereign and multilateral credit enhancement; their renegotiation under economic pressure is a recurrent source of disputes.
- Cross-border trade is governed by the EU internal market, the WAPP and ERERA in West Africa, other regional power pools, and the emerging African Single Electricity Market.
- Consumer protection addresses billing, metering, switching, price regulation, disconnection, and energy poverty, with constitutional litigation emerging as a means of enforcement.
- Market abuse is addressed through REMIT, United States anti-manipulation law, competition law, and regulatory enforcement powers.

DOCTORAL RESEARCH QUESTIONS

1. Is the standard model of electricity liberalisation appropriate for developing countries, or should reform pursue "hybrid" models adapted to local conditions? Evaluate with reference to Nigeria's experience since 2013.
2. Analyse the constitutional and legal implications of the decentralisation of electricity regulation in Nigeria under the 2023 constitutional amendments and the Electricity Act 2023, drawing comparisons with the federal–state division in the United States.
3. Does marginal pricing remain an appropriate foundation for wholesale electricity markets in systems dominated by zero-marginal-cost renewable generation?

4. To what extent can regional electricity regulators such as ERERA exercise effective authority over national utilities, and what legal reforms would strengthen regional market integration in West Africa?
5. How should the law balance the sanctity of long-term PPAs against the need to renegotiate contracts that have become unaffordable as a result of macroeconomic shocks?
6. Is there a justiciable right to electricity? Evaluate South African and other jurisprudence and its potential application in Nigeria.

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10 CHAPTER

Renewable Energy Law and Sustainable Power Development

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the international, regional, and domestic legal foundations of renewable energy development, including targets, definitions, and institutional frameworks.
2. Analyse the licensing and land rights issues associated with utility-scale and distributed solar energy, with particular reference to customary land tenure.
3. Evaluate the permitting and environmental compliance requirements for onshore and offshore wind energy, including the protection of wildlife and the rights of indigenous peoples.
4. Assess the legal framework for hydropower development and the allocation of water resources at the project level.
5. Identify the distinctive legal issues raised by geothermal and biomass energy, including resource classification and sustainability criteria.
6. Compare the principal renewable energy support mechanisms and evaluate their legal design and compatibility with State aid and trade law.
7. Analyse the law governing renewable energy certificates and attribute markets, including guarantees of origin and corporate claims.
8. Critically assess the jurisprudence of renewable energy investment disputes, including the Spanish and Italian arbitrations and their enforcement.

Introduction

Renewable energy has moved from the margins to the centre of energy law. In the space of two decades, wind and solar power have been transformed from expensive niche technologies dependent on generous subsidies into the cheapest sources of new electricity generation in most of the world. Renewable sources now account for the large majority of new generating capacity installed globally each year, and at the 2023 United Nations Climate Change Conference (COP28) in Dubai, the

parties to the Paris Agreement called for a tripling of global renewable energy capacity and a doubling of the global average annual rate of energy efficiency improvements by 2030. Renewable energy is also central to the achievement of universal energy access: decentralised solar systems and mini-grids offer the most cost-effective means of electrifying many of the rural communities in sub-Saharan Africa that the grid has not reached.

The law has been both a driver and a constraint of this transformation. Legal instruments — renewable energy targets, feed-in tariffs, renewable portfolio standards, auctions, tax credits, priority grid access, and permitting regimes — created the markets that drove down costs through deployment and learning. At the same time, legal obstacles — lengthy and uncertain permitting, conflicts over land and resources, inadequate grid access, regulatory instability, and opposition from affected communities — have slowed deployment. And the retroactive reduction of renewable support schemes by several European governments in the aftermath of the financial crisis generated one of the largest waves of investment treaty arbitration in history, testing the limits of investor protection against regulatory change.

The legal issues raised by renewable energy differ from those of fossil fuels in important ways. Renewable resources — sunlight, wind, flowing water, geothermal heat, and biomass — are not "owned" in the same way as subsurface minerals, and their legal status has required new conceptual frameworks. Renewable projects have low operating costs but high upfront capital costs, so their economics depend critically on the cost of capital and therefore on regulatory stability. Many renewable resources are variable, requiring new approaches to grid integration and market design. And the land-intensive character of solar and wind energy, the impacts of hydropower on rivers, and the sustainability concerns surrounding bioenergy generate conflicts with other land uses, biodiversity, and the rights of local and indigenous communities.

This chapter examines the legal foundations of renewable energy development (section 10.1), the licensing and land rights issues of solar energy (section 10.2), wind energy permitting and environmental compliance (section 10.3), hydropower law and water resource allocation (section 10.4), geothermal and biomass energy regulation (section 10.5), renewable energy incentives and support mechanisms (section 10.6), renewable energy certificates and attribute markets (section 10.7), and renewable energy investment disputes (section 10.8). The integration of renewable energy into electricity markets is addressed in Chapter 9, the emerging technologies of storage, hydrogen, and offshore renewables in Chapter 12, and the transboundary dimensions of hydropower in Chapter 19.

10.1 Legal Foundations of Renewable Energy Development

Defining Renewable Energy

Legal definitions of renewable energy matter because they determine eligibility for support schemes, targets, priority access, and certificates. Most definitions enumerate the sources considered renewable. Article 2(1) of the EU Renewable Energy Directive (Directive (EU) 2018/2001, as amended by Directive (EU) 2023/2413) defines "energy from renewable sources" as energy from renewable non-fossil sources, namely wind, solar (solar thermal and solar photovoltaic) and geothermal energy, osmotic energy, ambient energy, tide, wave and other ocean energy, hydropower, biomass, landfill gas, sewage treatment plant gas, and biogas. The Statute of the International Renewable Energy Agency (IRENA), adopted in 2009, defines renewable energy in Article III to include all forms of energy produced from renewable sources in a sustainable manner, including bioenergy, geothermal energy, hydropower, ocean energy (including tidal, wave, and ocean thermal energy), solar energy, and wind energy.

Definitions raise contested questions at the margins. Is large hydropower renewable, given its environmental and social impacts? Is biomass renewable if its harvesting causes deforestation or releases more carbon than it saves over relevant time horizons? Is waste incineration renewable? Should nuclear energy, which is low-carbon but not renewable, be included in "clean energy" targets? Different legal systems answer these questions differently, and the trend has been to supplement definitions with sustainability criteria, as in the EU's sustainability and greenhouse gas saving criteria for biomass, examined in section 10.5, and to adopt broader "clean" or "low-carbon" categories alongside "renewable" categories.

International Foundations

There is no comprehensive international treaty on renewable energy. The international legal foundations of renewable energy are found in several instruments.

Climate change law. The UNFCCC, the Kyoto Protocol, and the Paris Agreement, examined in Chapter 18, provide the overarching framework. Although the Paris Agreement does not specifically mention renewable energy in its operative provisions, the nationally determined contributions of most parties include renewable energy targets or measures, and the COP28 decision on the first global stocktake called on parties to contribute to tripling renewable energy capacity globally by 2030. Article 4(1)(c) of the UNFCCC commits all parties to promote and cooperate in the development, application, and diffusion of technologies that control, reduce, or prevent anthropogenic emissions of greenhouse gases in all relevant sectors, including energy.

The Sustainable Development Goals. SDG 7, adopted by the United Nations General Assembly in 2015, calls for access to affordable, reliable, sustainable,

and modern energy for all by 2030, with target 7.2 calling for a substantial increase in the share of renewable energy in the global energy mix.

IRENA. The Statute of the International Renewable Energy Agency, which entered into force in 2010, established IRENA as an intergovernmental organisation to promote the widespread and increased adoption and sustainable use of all forms of renewable energy. IRENA's institutional role is examined in Chapter 4.

The International Solar Alliance. The Framework Agreement on the Establishment of the International Solar Alliance, launched by India and France at COP21 in 2015 and in force since 2017, establishes a treaty-based organisation of countries committed to promoting solar energy, particularly in countries located between the Tropics of Cancer and Capricorn, through the aggregation of demand, the mobilisation of finance, and capacity building. Nigeria is a party.

The Energy Charter Treaty. Article 19 of the ECT requires contracting parties to strive to minimise harmful environmental impacts of energy activities, including by "promoting and using renewable energy sources", and the ECT's investment protection provisions have been the basis of most renewable energy investment arbitrations, examined in section 10.8.

Regional and Domestic Targets

Renewable energy targets are the most widespread legal instrument of renewable energy policy, adopted by almost every country. Targets may be expressed as shares of final energy consumption, electricity generation, or installed capacity, and may be binding or indicative.

The European Union has adopted binding targets for renewable energy since the Renewable Energy Directive of 2009 (Directive 2009/28/EC), which set a binding target of twenty per cent of gross final energy consumption by 2020, broken down into binding national targets. The recast Directive of 2018 set a binding Union-level target of at least thirty-two per cent by 2030, without binding national targets, relying on national energy and climate plans and a governance mechanism. Directive (EU) 2023/2413 ("RED III") raised the binding Union target to at least 42.5 per cent by 2030, with an indicative aspiration of forty-five per cent, and introduced sub-targets for transport, buildings, industry, and district heating and cooling, including a binding target for the use of renewable fuels of non-biological origin (principally renewable hydrogen) in industry.

In Africa, the ECOWAS Renewable Energy Policy, adopted in 2013 and implemented through the ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE) in Praia, Cape Verde, set targets for increasing the share of renewable energy in the regional electricity mix, including large hydropower, and member states adopted National Renewable Energy Action Plans. The African Union's Agenda 2063 and the Africa Renewable Energy Initiative (AREI),

launched at COP21, aimed to install substantial new renewable capacity across the continent.

The Nigerian Framework

Nigeria's legal and policy framework for renewable energy has developed incrementally. The **National Renewable Energy and Energy Efficiency Policy** (NREEEP), approved in 2015, set out the government's objectives for renewable energy, and the **Nigeria Renewable Energy Master Plan** set targets for the contribution of renewables to electricity supply. NERC's **Feed-in Tariff Regulations for Renewable Energy Sourced Electricity** (2015) provided for feed-in tariffs for small hydropower, wind, solar, and biomass, and required DisCos to procure a proportion of their electricity from renewable sources, although implementation was limited. In 2016, NBET signed PPAs with fourteen solar IPPs for a combined capacity of around 1,125 MW, but most of these projects failed to reach financial close, owing largely to disagreements over tariffs, the absence of adequate payment security, and the liquidity crisis in the electricity market.

The **Rural Electrification Agency** (REA), established under EPSRA and continued under the Electricity Act 2023, administers the Rural Electrification Fund and implements programmes, including the World Bank-supported Nigeria Electrification Project and the Distributed Access through Renewable Energy Scale-up (DARES) programme, that have supported the deployment of solar mini-grids and stand-alone solar systems. The **Climate Change Act 2021** set a target of net zero emissions between 2050 and 2070 and established the National Council on Climate Change, and the **Energy Transition Plan** of 2022 set out a pathway to net zero by 2060, with a substantial expansion of solar capacity. The **Electricity Act 2023** contains specific provisions on renewable energy, including obligations on the regulator to promote renewable energy, provision for renewable purchase obligations, incentives for renewable energy, and the treatment of renewable energy in integrated resource planning. These developments are examined further in Chapter 24.

The Legal Status of Renewable Resources

A distinctive conceptual question is whether, and by whom, renewable resources can be owned. Wind and sunlight are generally regarded as *res communes* — things common to all, incapable of ownership in their natural state. The legal question therefore concerns not ownership of the resource itself, but rights to access and use it: rights to use land on which solar panels and wind turbines are located, protection against interference with access to the resource (for example, by structures that shade solar panels or obstruct wind flows), and the allocation of offshore areas for renewable development. Some jurisdictions have legislated to protect access to solar or wind resources: several states of the United States have enacted

"solar access" laws permitting the creation of solar easements or protecting solar installations from shading, and the law of nuisance and easements may be invoked to protect access.

Offshore, the coastal state's sovereign rights in the exclusive economic zone under Article 56(1)(a) of UNCLOS extend expressly to "the production of energy from the water, currents and winds", providing the international legal basis for offshore wind and ocean energy development, examined in Chapters 12 and 20. Water used for hydropower and geothermal resources are, by contrast, subject to well-developed legal regimes of public ownership and allocation, examined in sections 10.4 and 10.5.

10.2 Solar Energy Licensing and Land Rights

The Solar Revolution

Solar photovoltaic (PV) energy has experienced the most dramatic cost reductions of any energy technology: the cost of solar modules has fallen by around ninety per cent over the past decade, and utility-scale solar is now among the cheapest sources of new electricity in most regions. Solar energy is deployed at every scale: utility-scale solar farms of hundreds or thousands of megawatts; commercial and industrial installations on rooftops and land; residential rooftop systems; solar mini-grids serving rural communities; and stand-alone solar home systems and appliances. Each scale raises different legal issues.

Licensing and Permitting of Utility-Scale Solar

Utility-scale solar projects typically require a generation licence (where the capacity exceeds the licensing threshold), land rights, planning or land use permission, environmental assessment, grid connection, and, for projects selling to a utility, a PPA. The environmental impacts of solar farms are generally modest compared with other generation technologies, but they include land use change, habitat loss, visual impacts, glare, and the use of water for cleaning panels in arid areas. Large solar projects typically require an EIA, and in some jurisdictions they benefit from streamlined permitting. Under the EU's RED III, member states must designate "renewables acceleration areas" in which renewable projects benefit from simplified and accelerated permitting, including exemption from project-level environmental assessment where a strategic environmental assessment has been carried out, and must ensure that permit-granting procedures for projects in such areas do not exceed twelve months for onshore projects. RED III also provides that the planning, construction, and operation of renewable energy plants are presumed to be in the overriding public interest for the purposes of certain environmental assessments under EU nature conservation law, until climate neutrality is achieved.

In Nigeria, utility-scale solar projects above the licensing threshold require a generation licence from NERC (or, where applicable, a state regulator), an EIA approved by the Federal Ministry of Environment, land rights secured under the Land Use Act, and a grid connection agreement with TCN or a DisCo. Projects intended to sell to the grid require a PPA with a buyer — historically NBET, and increasingly DisCos, eligible customers, or traders.

Land Rights for Solar Projects

Utility-scale solar projects are land-intensive, requiring around one to two hectares per megawatt, depending on technology and location. Developers must secure long-term rights to the land for the life of the project (typically twenty-five to thirty-five years), sufficient to satisfy lenders. The form of these rights depends on the land tenure system: freehold purchase, long-term leasehold, rights of occupancy, easements, or concessions over public land.

In jurisdictions with customary land tenure, such as most of sub-Saharan Africa, securing land rights for solar projects raises distinctive challenges. Land may be held communally by families, clans, or communities under customary law, often without formal registration; rights may be overlapping and contested; and the identification of the persons entitled to consent to a transfer may be uncertain. In Nigeria, the **Land Use Act 1978** vests all land in each state in the governor, who may grant statutory rights of occupancy over urban land, while local governments may grant customary rights of occupancy over non-urban land. Holders of customary rights may transfer their interests only with the consent of the governor (for statutory rights) or the local government (for customary rights), and the requirement of governor's consent has been a notorious source of delay and cost. Developers of solar projects in Nigeria must therefore negotiate with customary landholders and communities, obtain consent to the transfer or grant of rights of occupancy, and register their interests, while managing the risk of competing claims.

The **Voluntary Guidelines on the Responsible Governance of Tenure of Land, Fisheries and Forests** (VGGT), endorsed by the Committee on World Food Security in 2012, and **IFC Performance Standard 5** on land acquisition and involuntary resettlement provide international standards for responsible land acquisition, emphasising the recognition of legitimate tenure rights (including customary rights), consultation, the avoidance of involuntary resettlement where possible, and fair compensation. Large solar projects in Africa, such as the Noor Ouarzazate complex in Morocco, which was built on land acquired from a local collective of the Ait Oukrour Toundout community under Moroccan law on collective lands, have attracted scrutiny of the adequacy of consultation and compensation. These issues are examined further in Chapter 21.

Agrivoltaics and Competing Land Uses

The land requirements of solar energy create potential conflicts with agriculture, particularly in densely populated areas or regions with high-quality agricultural land. Agrivoltaics — the co-location of solar panels and agricultural production, for example by mounting panels on elevated structures over crops or grazing land — offers a means of reducing conflict, and some jurisdictions have adopted specific legal frameworks for agrivoltaic projects. France's Law No 2023-175 on the acceleration of renewable energy production defined agrivoltaic installations and established conditions under which they may be located on agricultural land, requiring that they provide a direct service to agricultural production, such as improving agronomic potential or protection against hazards, and do not significantly reduce agricultural output.

Distributed Solar and Mini-Grids

Distributed solar — rooftop systems and small ground-mounted installations — raises different legal issues: the right of consumers to install generation and connect it to the grid; net metering and compensation for exports; building regulations and fire safety standards; and the treatment of distributed generation in network tariffs. In Nigeria, NERC's Regulations on Embedded Generation (2012) and its regulatory instruments on net metering govern the connection of distributed generation to distribution networks and the treatment of exports.

Solar mini-grids are particularly important in Nigeria and other countries with low rates of electrification. The NERC **Mini-Grid Regulations** (2016, revised 2023) distinguish between isolated mini-grids (not connected to the distribution network) and interconnected mini-grids (connected to an existing distribution network, often to serve underserved customers), set thresholds for permits and registration, allow mini-grid operators to charge cost-reflective tariffs approved by the regulator, and address the consequences of the arrival of the main grid, including options for the mini-grid to convert to a small power distributor or to be compensated for its assets. These provisions are critical to investor confidence, because the risk of grid arrival rendering the mini-grid uneconomic is one of the principal risks of mini-grid investment.

Stand-alone solar systems and solar home systems, often sold on "pay-as-you-go" terms using mobile money, have brought electricity to millions of households in sub-Saharan Africa. Their legal issues include consumer protection (product quality standards, warranties, transparent pricing, and responsible debt collection), the regulation of mobile payments, import duties and taxes (several countries have exempted solar products from import duties and value added tax), and the management of electronic waste from end-of-life batteries and panels.

The quality standards developed by the World Bank Group's Lighting Global programme (now VeraSol) have been widely adopted as a reference.

10.3 Wind Energy Permitting and Environmental Compliance

Onshore and Offshore Wind

Wind energy is the second largest source of renewable electricity after hydropower and a cornerstone of decarbonisation strategies in Europe, China, the United States, and increasingly elsewhere. Onshore wind is a mature and low-cost technology; offshore wind, which benefits from stronger and more consistent winds and fewer conflicts with land uses, has grown rapidly in northern Europe and China and is expanding to the United States, Asia, and other regions, although rising costs, supply chain constraints, and interest rates led to the cancellation or delay of several projects in 2023 and 2024. Wind energy raises distinctive permitting issues relating to its visual and landscape impacts, noise, impacts on birds and bats, interference with radar and aviation, and, offshore, interactions with fisheries, shipping, marine ecosystems, and military activities.

Permitting of Onshore Wind

Onshore wind projects typically require planning permission or development consent, environmental assessment, a generation licence, grid connection, and, depending on the jurisdiction, consents relating to aviation, radar, and wildlife. Permitting has been widely identified as the principal bottleneck for onshore wind deployment. In Europe, permitting procedures for onshore wind projects historically took many years, and a substantial pipeline of projects was blocked in permitting or litigation. The EU's Council Regulation (EU) 2022/2577, adopted as an emergency measure in December 2022, laid down temporary rules to accelerate the deployment of renewable energy, including a presumption that renewable energy projects are in the overriding public interest; RED III made many of these measures permanent, requiring maximum permitting periods and the designation of renewables acceleration areas.

Germany's experience illustrates the significance of legal reform. Onshore wind deployment in Germany slowed dramatically after 2017, owing to the introduction of auctions, restrictive distance rules (notably Bavaria's "10H rule", which required a distance of ten times the height of the turbine from residential buildings), and litigation. The 2023 amendments to the Renewable Energy Sources Act (EEG 2023) declared that the construction and operation of renewable energy installations are in the overriding public interest and serve public security (§ 2 EEG), and the Wind-on-Land Act (Wind-an-Land-Gesetz) of 2022 required the federal states to designate two per cent of the national land area for onshore wind by 2032, overriding restrictive distance rules where states fail

to meet their area targets. Deployment and permitting accelerated substantially thereafter.

Wildlife Protection

The protection of birds and bats is a recurrent issue in wind energy permitting. In the European Union, the **Birds Directive** (Directive 2009/147/EC) and the **Habitats Directive** (Council Directive 92/43/EEC) require the protection of wild birds and of species and habitats of Community interest. Article 6(3) of the Habitats Directive requires an "appropriate assessment" of any plan or project likely to have a significant effect on a Natura 2000 site, and permits approval only if the project will not adversely affect the integrity of the site, unless, under Article 6(4), there are imperative reasons of overriding public interest, no alternative solutions, and compensatory measures are taken. Article 12 of the Habitats Directive and Article 5 of the Birds Directive prohibit the deliberate killing or disturbance of protected species. The Court of Justice has interpreted these prohibitions strictly; in *Föreningen Skydda Skogen* (Joined Cases C-473/19 and C-474/19, 2021), it held that the prohibitions apply irrespective of the conservation status of the species concerned and are not limited to activities aimed at killing or disturbing. The presumption of overriding public interest introduced by RED III is intended to facilitate the derogations necessary for renewable energy projects.

In the United States, the Migratory Bird Treaty Act of 1918, the Bald and Golden Eagle Protection Act of 1940, and the Endangered Species Act of 1973 protect birds and other wildlife, and wind energy operators may obtain incidental take permits for eagles. In 2022, a wind energy company pleaded guilty to violations of the Migratory Bird Treaty Act in connection with the deaths of eagles at its facilities, illustrating the criminal liability risks of wildlife impacts.

Wind Energy and Indigenous Rights: The Fosen Case

The most significant judicial decision on the relationship between wind energy and indigenous rights is the judgment of the Supreme Court of Norway in the *Fosen* case (2021). The Norwegian government had granted licences and expropriation permits for the Storheia and Roan wind farms on the Fosen peninsula, among the largest onshore wind developments in Europe, located in an area used by Sámi reindeer herders for winter grazing. The reindeer herders argued that the wind farms would interfere with their ability to practise their culture, in violation of Article 27 of the International Covenant on Civil and Political Rights, which protects the right of persons belonging to minorities, in community with other members of their group, to enjoy their own culture. In October 2021, the Supreme Court held unanimously that the licences and expropriation decisions were invalid because the wind farms would have a significant negative effect on the

herders' ability to practise reindeer husbandry, in violation of Article 27, and that the mitigation measures proposed were insufficient. The Court rejected the argument that the interference could be justified by the public interest in renewable energy, noting that the green transition could be achieved through other means that did not infringe Sámi rights.

The judgment did not specify the consequences for the turbines, which had been constructed and were operating. Following protests by Sámi activists and environmentalists in 2023, which drew attention to the continued operation of the turbines more than five hundred days after the judgment, the Norwegian government apologised and negotiated settlements with the affected reindeer herding groups, concluded in 2023 and 2024, providing for compensation, additional winter grazing land, and other measures allowing the wind farms to continue operating. The *Fosen* case is a landmark in the emerging law of "green" or "just" transition, demonstrating that renewable energy projects are subject to the same human rights constraints as fossil fuel projects, and that courts may invalidate projects that disproportionately burden indigenous peoples. It is examined further in Chapter 21.

Comparable conflicts have arisen elsewhere. In Mexico, wind farm development in the Isthmus of Tehuantepec in Oaxaca, one of the windiest regions in the world, has been contested by Zapotec and Ikoots indigenous communities, and the Supreme Court of Justice of the Nation in 2018 addressed the adequacy of the consultation conducted for the Eólica del Sur project, upholding the consultation in that case while affirming the right to prior consultation. In Kenya, the Lake Turkana Wind Power project, the largest wind farm in Africa when commissioned in 2019, faced litigation by local communities challenging the acquisition of the land on which it was built; in 2021, the Environment and Land Court at Meru found that the process by which the land had been set aside was irregular and ordered the regularisation of the land acquisition in accordance with the Community Land Act, without ordering the removal of the project.

Offshore Wind Leasing and Permitting

Offshore wind development requires a legal framework for the allocation of rights to the seabed and water column, environmental assessment, and coordination with other marine users. Coastal states allocate offshore wind rights through several models. In the **leasing model** used in the United Kingdom, the Crown Estate (for England, Wales, and Northern Ireland) and Crown Estate Scotland grant seabed leases through competitive leasing rounds, and developers must separately obtain development consent and a route to market (typically a contract for difference). In the **centralised tender model** used in Denmark, the Netherlands, and Germany for certain sites, the state pre-investigates and pre-consents sites and tenders them, together with grid connection, to developers, in some cases through

"negative bidding" or criteria-based tenders. In the United States, the Bureau of Ocean Energy Management (BOEM) leases areas of the outer continental shelf under the Outer Continental Shelf Lands Act, as amended by the Energy Policy Act of 2005, through competitive lease sales, and developers must obtain approval of construction and operations plans following environmental review under the National Environmental Policy Act. In 2025, the federal government suspended new offshore wind leasing and halted work on certain projects under construction, giving rise to litigation and illustrating the regulatory risks of offshore wind in the United States.

Offshore wind permitting addresses impacts on marine mammals (particularly from pile-driving noise), seabirds, fish and benthic habitats, fisheries, navigation, aviation, defence, and cultural heritage, and requires coordination through marine spatial planning, which allocates marine space among competing uses. The EU's Maritime Spatial Planning Directive (Directive 2014/89/EU) requires member states to establish maritime spatial plans. Offshore wind and the law of the sea are examined in Chapters 12 and 20.

Decommissioning and Repowering

Wind turbines have operating lives of twenty to thirty years, and the first generation of wind farms in Europe and the United States is reaching the end of its life. Decommissioning raises issues of financial security (analogous to those in mining and petroleum, examined in Chapters 5 and 7), the recycling of turbine blades (which are made of composite materials that are difficult to recycle, and which several European countries have banned from landfill), and site restoration. **Repowering** — the replacement of old turbines with fewer, larger, and more efficient turbines on the same site — can substantially increase output, and RED III requires member states to facilitate repowering through simplified permitting.

10.4 Hydropower Law and Water Resource Allocation

The Role and Impacts of Hydropower

Hydropower is the largest source of renewable electricity globally and the backbone of electricity systems in many countries, including Norway, Brazil, Canada, Paraguay, Ethiopia, the Democratic Republic of the Congo, and Zambia. Reservoir hydropower provides dispatchable generation and storage that can balance variable wind and solar output, and pumped storage hydropower is the largest form of grid-scale electricity storage. In Nigeria, the Kainji, Jebba, and Shiroro hydropower plants on the Niger and Kaduna rivers, and the more recent Zungeru plant, contribute a significant share of grid generation.

At the same time, large dams have serious environmental and social impacts: displacement of communities (the World Commission on Dams estimated in 2000 that large dams had displaced between forty and eighty million people

worldwide), loss of livelihoods dependent on rivers, disruption of fisheries and sediment flows, inundation of forests and cultural heritage, greenhouse gas emissions from reservoirs in tropical areas, and risks of catastrophic dam failure. The report of the **World Commission on Dams**, *Dams and Development: A New Framework for Decision-Making* (2000), proposed a rights-and-risks approach to dam decision-making, emphasising gaining public acceptance, comprehensive options assessment, addressing existing dams, sustaining rivers and livelihoods, recognising entitlements and sharing benefits, ensuring compliance, and sharing rivers for peace, development, and security. Although its recommendations are not binding, they have influenced the policies of development institutions and the industry's **Hydropower Sustainability Standard**, launched in 2021 by the Hydropower Sustainability Council, which provides for independent certification of hydropower projects against environmental, social, and governance criteria.

This section examines the domestic legal framework for hydropower projects — water rights, licensing, concessions, and dam safety. The international law of transboundary watercourses, and the basin-level management of hydropower on shared rivers, are examined in Chapter 19.

Water Rights and Allocation for Hydropower

Hydropower depends on the right to use water. Legal systems allocate water rights according to several models: **riparian rights**, under which owners of land bordering a watercourse are entitled to reasonable use of the water (the traditional common law position); **prior appropriation**, under which rights are allocated on a "first in time, first in right" basis to those who put water to beneficial use (the position in the western United States); and **public ownership and administrative allocation**, under which water is owned or controlled by the state and allocated through permits or licences (the position in most modern water codes, including those of most developing countries). In most jurisdictions, hydropower developers require a water use permit or licence specifying the quantity of water that may be abstracted or diverted, the conditions of use (including minimum environmental flows), and the duration of the right.

In Nigeria, the **Water Resources Act 1993** (Cap W2, Laws of the Federation of Nigeria 2004) vests the right to the use and control of all surface water and groundwater, and of all water in any watercourse affecting more than one state, in the Government of the Federation, and provides for the licensing of water use, including for hydropower. The **Minerals and Mining Act 2007** separately provides for water use permits for mining. The River Basin Development Authorities, established by statute, are responsible for the development of water resources within their basins, and the **Nigeria Hydrological Services Agency** monitors water resources. A National Water Resources Bill, intended to modernise the framework, has been the subject of controversy, in part because of concerns from

states about the concentration of control over water resources in the federal government.

Environmental Flows

The concept of **environmental flows** — the quantity, timing, and quality of water flows required to sustain freshwater ecosystems and the livelihoods that depend on them — has become a central feature of modern water and hydropower law. Water licences for hydropower increasingly specify minimum flows to be released downstream, and require operators to manage releases to mimic natural flow patterns. The EU Water Framework Directive (Directive 2000/60/EC) requires member states to achieve "good ecological status" of water bodies, which has required the revision of hydropower licences to provide adequate flows and fish passage. In *Commission v Austria* (Case C-346/14, 2016), the Court of Justice upheld Austria's authorisation of a hydropower plant on the Schwarze Sulm river, notwithstanding the deterioration of the river's status, because the national authorities had properly applied the derogation for projects of overriding public interest under Article 4(7) of the Directive. In *Bund für Umwelt und Naturschutz Deutschland* (Case C-461/13, 2015, the "Weser" case), the Court held that member states must refuse authorisation for projects that may cause deterioration of the status of a body of surface water, unless a derogation applies, interpreting "deterioration" to include deterioration of a single quality element.

Hydropower Concessions and Licences

Hydropower plants are typically developed under concessions or licences granted by the state for long periods — thirty to seventy-five years or more — reflecting the long life of dams. Concessions specify the rights and obligations of the concessionaire, including water use, operation, maintenance, dam safety, environmental obligations, payment of royalties or fees, and the reversion of assets to the state at the end of the concession. In Norway, the Industrial Licensing Act of 1917 introduced the principle of "reversion" (*hjemfall*), under which hydropower plants owned by private companies revert to the state without compensation at the end of the licence period; the principle was upheld by the EFTA Court in *EFTA Surveillance Authority v Norway* (Case E-2/06, 2007) in so far as it applied to public as well as private owners, after Norway amended its legislation to remove discrimination against private and foreign owners.

In France, the renewal of expiring hydropower concessions, many of them held by EDF, has been the subject of a long-running dispute with the European Commission, which considers that EU public procurement and concession law requires competitive tendering, while successive French governments have resisted opening the concessions to competition. In the United States, non-federal hydropower plants are licensed by FERC under the Federal Power Act for

terms of thirty to fifty years, and relicensing proceedings involve consideration of environmental conditions imposed by federal and state agencies.

In Nigeria, the Kainji and Jebba hydropower plants were concessioned in 2013 to Mainstream Energy Solutions Ltd, and the Shiroro plant to North South Power Company Ltd, under thirty-year concession agreements as part of the privatisation of the power sector. The 700 MW Zungeru hydropower plant, constructed by a Chinese consortium with Chinese financing, was concessioned in 2023 to Mainstream Energy Solutions for a period of thirty years. The proposed **Mambilla hydropower project**, a planned 3,050 MW project in Taraba State, has been delayed for decades and became the subject of an ICC arbitration brought by Sunrise Power and Transmission Company Ltd against the Federal Government of Nigeria, alleging breach of a contract to develop the project — an illustration of the legal risks associated with large hydropower projects.

Dam Safety

The failure of dams can cause catastrophic loss of life and property, as the failures of the Derna dams in Libya in September 2023, which killed thousands of people, demonstrated. Dam safety legislation requires owners and operators of dams to design, construct, operate, maintain, and inspect dams according to specified standards, to prepare emergency action plans, and to coordinate releases with downstream authorities. In Nigeria, the release of water from the Lagdo Dam in neighbouring Cameroon has been associated with severe flooding in Nigeria, notably in 2012 and 2022, raising questions about transboundary coordination examined in Chapter 19. The International Commission on Large Dams (ICOLD) publishes technical bulletins on dam safety that are widely used as standards.

10.5 Geothermal and Biomass Energy Regulation

Geothermal Energy

Geothermal energy — heat derived from the earth's interior — can be used to generate electricity, where high-temperature resources are available, typically in tectonically active regions, and for direct heating and cooling. Geothermal power provides firm, dispatchable, low-carbon electricity, and is a major source of supply in Kenya, Iceland, El Salvador, New Zealand, the Philippines, and Indonesia. Kenya, which has developed the Olkaria geothermal fields in the East African Rift, generates a substantial share of its electricity from geothermal sources and is among the world's leading geothermal producers. Emerging technologies, including enhanced geothermal systems and closed-loop systems, may extend geothermal development beyond traditional high-temperature regions.

Legal classification. The legal classification of geothermal resources is a threshold question with significant consequences. Geothermal resources may

be classified as minerals (and governed by mining law), as water (and governed by water law), or as a sui generis resource (governed by specific geothermal legislation). Classification determines who owns the resource, which authority regulates it, and which legal regime applies. In the United States, the Geothermal Steam Act of 1970 governs geothermal leasing on federal lands, and state laws variously classify geothermal resources as minerals, water, or a distinct resource. In Indonesia, geothermal resources were historically regulated as a mining activity, which meant that geothermal exploration was prohibited in conservation forests under the forestry laws governing mining; Law No 21 of 2014 on Geothermal reclassified geothermal energy as a non-mining activity, enabling development in conservation forest areas subject to environmental permits, and transferred licensing authority for geothermal power from regional governments to the central government. Kenya's Energy Act 2019, which replaced the Geothermal Resources Act 1982, vests geothermal resources in the national government and provides for licensing by the Cabinet Secretary.

Licensing and risk. Geothermal development involves high upfront exploration risk: drilling exploratory wells is expensive, and the characteristics of the resource cannot be confirmed until wells are drilled. Many governments therefore undertake or finance early-stage exploration through state geothermal companies (such as Kenya's Geothermal Development Company) or risk mitigation facilities (such as the Geothermal Risk Mitigation Facility for Eastern Africa, supported by the African Union Commission and development partners), and then offer proven steam fields to private developers. Geothermal licences typically follow a sequence of exploration and exploitation permits similar to that in mining and petroleum law.

Environmental issues. Geothermal development can cause induced seismicity (particularly with enhanced geothermal systems), land subsidence, emissions of hydrogen sulphide and other gases, and impacts on surface thermal features. The suspension of an enhanced geothermal project in Pohang, South Korea, after it was found by a government commission to have triggered a magnitude 5.4 earthquake in 2017, and the abandonment of a geothermal project in Basel, Switzerland, after induced earthquakes in 2006, illustrate the risks. Geothermal projects in Kenya's Rift Valley have also raised issues of land acquisition and resettlement affecting Maasai communities, which were the subject of complaints to the World Bank's Inspection Panel concerning the Olkaria IV project.

Biomass and Bioenergy

Bioenergy — energy derived from biomass, including wood, agricultural residues, energy crops, and organic waste — is the largest source of renewable energy globally when traditional uses of biomass for cooking and heating are included. Modern bioenergy includes the combustion of biomass for electricity and heat,

the production of liquid biofuels (ethanol and biodiesel) for transport, and the production of biogas and biomethane from anaerobic digestion. Traditional biomass use — the burning of wood, charcoal, crop residues, and dung for cooking in inefficient stoves — remains the primary source of cooking energy for around two billion people, predominantly in sub-Saharan Africa and South Asia, causing household air pollution that the World Health Organization estimates contributes to millions of premature deaths annually, and contributing to deforestation. The transition to clean cooking is examined in Chapter 23.

Sustainability concerns. The renewable and low-carbon credentials of bioenergy depend critically on how the biomass is produced. The combustion of wood releases carbon dioxide at the point of combustion, and the carbon is reabsorbed only as replacement trees grow, which may take decades; if biomass is sourced from primary forests or causes deforestation, its net climate impact may be negative over relevant time horizons. Biofuels produced from food crops may compete with food production and drive land use change, including the conversion of forests and grasslands to agriculture elsewhere ("indirect land use change", or ILUC), which may negate their emissions savings. These concerns have led to the development of sustainability criteria in law.

The EU sustainability criteria. The EU Renewable Energy Directive establishes sustainability and greenhouse gas emissions saving criteria that biofuels, bioliquids, and biomass fuels must satisfy in order to count towards renewable energy targets, to be eligible for financial support, and to count towards renewable energy obligations. Under Article 29 of Directive (EU) 2018/2001, as amended by RED III, the criteria prohibit the sourcing of agricultural biomass from land with high biodiversity value (primary forests, highly biodiverse grasslands, and protected areas) or high carbon stock (wetlands, continuously forested areas, and peatland); impose requirements on forest biomass to ensure legal harvesting, regeneration, protection of biodiversity and soil, and maintenance of carbon stocks; and require minimum greenhouse gas emissions savings compared with fossil fuels. The ILUC Directive (Directive (EU) 2015/1513) capped the contribution of biofuels produced from food and feed crops towards transport targets, and the Commission's Delegated Regulation (EU) 2019/807 identified palm oil as a high-ILUC-risk feedstock, to be phased out from counting towards targets by 2030. The phase-out of palm oil biofuels was challenged by Malaysia and Indonesia before the WTO in *EU — Palm Oil (Malaysia)* (DS600) and *EU — Palm Oil (Indonesia)* (DS593); in March 2024, the panel in the Malaysian case accepted the legitimacy of the EU's objective of addressing ILUC, while finding deficiencies in certain aspects of the design and application of the measures, and a panel report in the Indonesian case followed in early 2025.

Biofuel mandates. Many countries mandate the blending of biofuels with petrol and diesel. The United States Renewable Fuel Standard, established under the Energy Policy Act of 2005 and expanded by the Energy Independence and Security Act of 2007, requires the blending of specified volumes of renewable fuels, administered through a system of tradable Renewable Identification Numbers. Brazil's long-standing ethanol programme and blending mandates have made it a world leader in sugarcane ethanol. Nigeria's **Official Gazette of the Nigerian Bio-fuel Policy and Incentives** (2007) set targets for blending ethanol and biodiesel with petroleum products, but implementation has been limited.

Waste-to-energy and biogas. The conversion of municipal solid waste, agricultural residues, and sewage into energy through incineration, gasification, or anaerobic digestion is regulated by waste law, air quality law, and energy law. EU law establishes a waste hierarchy, under Directive 2008/98/EC, that prioritises prevention, reuse, and recycling over energy recovery, and waste-to-energy plants are subject to emission limits under the Industrial Emissions Directive.

10.6 Renewable Energy Incentives and Support Mechanisms

The Rationale for Support

Public support for renewable energy has been justified on several grounds: the failure of markets to price the environmental costs of fossil fuels, particularly greenhouse gas emissions, which places renewable energy at an artificial disadvantage; the learning-by-doing benefits of deployment, which drive down costs for all future users but cannot be fully captured by early adopters; energy security and diversification; industrial policy and job creation; and the need to accelerate the transition to meet climate targets. As the costs of wind and solar have fallen, the rationale for direct subsidies has weakened in many markets, and the focus of support has shifted from subsidising generation costs to reducing investment risk and the cost of capital, which is the principal determinant of the cost of renewable electricity.

	What the state fixes	Who bears price risk	Typical instruments / examples
Feed-in tariff	Price per MWh	Consumers / state	German EEG 2000; Nigeria FiT Regs 2015
Feed-in premium	Premium over market	Shared	EU schemes after 2014 State aid guidelines
Quota + certificates	Quantity (obligation)	Generator	US RPS; UK Renewables Obligation
Auction + PPA / CfD	Quantity; price by bid	Shared (two-way CfD)	South Africa REIPPPP; UK CfD
Fiscal incentives	Tax credit or relief	Taxpayers	US ITC/PTC; 45Y/48E; duty waivers

Figure 10.1 Principal renewable energy support mechanisms and the allocation of price and volume risk. *Source:* prepared for this edition.

Feed-in Tariffs

A **feed-in tariff** (FiT) guarantees eligible renewable generators a fixed price per unit of electricity fed into the grid for a long period (typically fifteen to twenty years), often combined with an obligation on network operators or suppliers to purchase the electricity. The German Electricity Feed-in Act (*Stromeinspeisungsgesetz*) of 1990 and Renewable Energy Sources Act (*Erneuerbare-Energien-Gesetz*, EEG) of 2000 established the model, which was adopted by numerous countries and is credited with driving the early growth of wind and solar energy. FiTs provide high investment certainty, but they require the regulator to set the tariff administratively, with the risk of overcompensation if costs fall faster than anticipated — as occurred with solar PV in Spain, Italy, the Czech Republic, and Germany in the late 2000s, leading to "solar booms" and escalating costs for consumers, and ultimately to retroactive tariff reductions that generated the investment disputes examined in section 10.8.

A **feed-in premium** (FiP) pays generators a premium on top of the market price, either fixed or sliding (varying with the market price to provide a target level of revenue). Feed-in premiums expose generators to market signals while providing revenue support, and EU State aid guidelines from 2014 required member states to move from feed-in tariffs to market-based premiums for all but small installations.

Quota Obligations and Tradable Certificates

Under a **quota obligation** — known in the United States as a **renewable portfolio standard** (RPS) and in the United Kingdom, until 2017, as the **Renewables Obligation** — electricity suppliers or other obligated parties must source a specified and generally rising proportion of their electricity from renewable sources, demonstrating compliance by surrendering tradable green certificates issued to renewable generators. The price of certificates is determined by the market, and generators receive revenue from the sale of both electricity and certificates. Quota systems leave the quantity of renewable energy to be set by the regulator and the price to be determined by the market, inverting the logic of feed-in tariffs. They have been criticised for exposing generators to volatile certificate prices, increasing the cost of capital. More than half of the states of the United States, together with the District of Columbia, have adopted RPS or clean energy standards, many with targets of one hundred per cent clean electricity by 2040 or 2050.

Auctions and Contracts for Difference

Competitive auctions (or tenders) have become the dominant mechanism for allocating support to utility-scale renewable energy worldwide. Under an auction, the government or a designated agency invites bids for the right to build a specified quantity of renewable capacity or generation, and awards support — usually a long-term PPA or contract for difference — to the bidders offering the lowest price, sometimes combined with other criteria. Auctions combine the investment certainty of long-term contracts with competitive price discovery, and have driven dramatic reductions in the prices of wind and solar electricity in countries including India, Brazil, Mexico, the United Arab Emirates, Saudi Arabia, and Morocco.

South Africa's **Renewable Energy Independent Power Producer Procurement Programme** (REIPPPP), launched in 2011, is widely regarded as one of the most successful renewable procurement programmes in a developing country. Through successive bid windows, it procured several gigawatts of wind, solar, and other renewable capacity at falling prices, combining price competition with economic development criteria (local content, job creation, and community ownership) and a robust contractual framework of twenty-year PPAs with Eskom backed by government support agreements. Its success was attributed to transparent procurement, clear rules, credible institutions (the IPP Office, a unit established jointly by the Department of Energy, the National Treasury, and the Development Bank of Southern Africa), and bankable documentation, although its momentum was interrupted between 2015 and 2018 when Eskom delayed signing PPAs.

In the United Kingdom, the **Contracts for Difference** (CfD) scheme, established under the Energy Act 2013, awards fifteen-year (now twenty-year for some

technologies) private law contracts between generators and the Low Carbon Contracts Company, a government-owned counterparty, through competitive allocation rounds. Under a two-way CfD, the generator receives the difference between a "strike price" and a market reference price when the market price is below the strike price, and pays the difference when the market price is above it. During the 2022 crisis, generators with CfDs paid substantial sums back to the counterparty, reducing consumer bills. As noted in Chapter 9, the EU's 2024 electricity market design reform requires direct price support for new renewable and nuclear investment to take the form of two-way CfDs, subject to exceptions.

Fiscal Incentives

Tax incentives are a major instrument of renewable energy support, particularly in the United States, where the federal **investment tax credit** (ITC) and **production tax credit** (PTC) were the principal instruments of support for decades. The Inflation Reduction Act of 2022 extended and restructured these credits, replacing them from 2025 with technology-neutral clean electricity investment and production credits (sections 48E and 45Y of the Internal Revenue Code), with bonus credits for meeting domestic content, energy community, and prevailing wage and apprenticeship requirements, and introduced "transferability", permitting credits to be sold to unrelated taxpayers. The One Big Beautiful Bill Act of July 2025 significantly curtailed these incentives for wind and solar energy, terminating the credits for wind and solar facilities placed in service after 2027 unless construction began within a short period after enactment, and imposing restrictions related to "prohibited foreign entities" — an illustration of the vulnerability of fiscal incentives to political change. Developing countries commonly use import duty and VAT exemptions for renewable energy equipment, tax holidays, and accelerated depreciation; Nigeria has granted pioneer status incentives and import duty waivers for certain renewable energy equipment.

State Aid Control in the European Union

In the European Union, renewable energy support schemes funded through state resources are subject to the State aid rules of Articles 107 and 108 TFEU. The Court of Justice's jurisprudence on whether support schemes constitute State aid has been complex. In *PreussenElektra* (Case C-379/98, 2001), the Court held that the German obligation on electricity suppliers to purchase renewable electricity at minimum prices did not involve the transfer of state resources and therefore did not constitute State aid. In *Vent De Colère* (Case C-262/12, 2013), it held that the French scheme, under which the additional costs of purchase obligations were financed by a levy on consumers collected and administered under state control, involved state resources. In *Germany v Commission* (Case C-405/16 P, 2019), the Court set aside the General Court's judgment and annulled the Commission's

decision classifying the EEG 2012 as State aid, holding that the sums generated by the EEG surcharge were not under public control and therefore did not constitute state resources. Where schemes do constitute State aid, they must be notified and approved by the Commission under the **Guidelines on State aid for climate, environmental protection and energy** (CEEAG), adopted in 2022, which require, among other things, competitive allocation of aid and a demonstrated incentive effect.

The interaction of national support schemes with the free movement of goods was considered in *Ålands Vindkraft AB v Energimyndigheten* (Case C-573/12, 2014), in which the Court held that a Swedish green certificate scheme that awarded certificates only to producers located in Sweden constituted a measure having equivalent effect to a quantitative restriction on imports, but was justified by the objective of promoting renewable energy, given that the Renewable Energy Directive expressly permitted member states to limit support to domestic production. The WTO compatibility of renewable support schemes with local content requirements, considered in *Canada — Renewable Energy/Feed-in Tariff Program* (2013) and *India — Solar Cells* (2016), is examined in Chapter 13.

10.7 Renewable Energy Certificates and Attribute Markets

The Concept of Environmental Attributes

Once electricity enters the grid, electrons from renewable and fossil sources are physically indistinguishable, and it is impossible to trace the electricity consumed by a particular customer to a particular generator. Renewable energy certificates address this problem by separating the "environmental attributes" of renewable generation — its renewable origin and associated characteristics — from the physical electricity, and embodying them in a tradable instrument. A certificate typically represents one megawatt-hour of electricity generated from a renewable source and records information about the source, location, technology, date of generation, and the installation. Certificates enable consumers, suppliers, and companies to substantiate claims that their electricity is renewable, to comply with disclosure or quota obligations, and to account for the emissions associated with their electricity consumption.

Certificates serve different legal functions in different systems, and it is important to distinguish between them. **Compliance certificates** are used to demonstrate compliance with quota obligations (such as renewable portfolio standards and the former UK Renewables Obligation, examined in section 10.6). **Disclosure certificates** — notably guarantees of origin in the European Union — are used to substantiate the information that suppliers must give consumers about the sources of their electricity. **Voluntary certificates** are purchased by

companies and individuals to support voluntary claims of renewable energy use, for example under corporate sustainability commitments.

Guarantees of Origin in the European Union

Article 19 of the Renewable Energy Directive requires member states to ensure that the origin of electricity from renewable sources can be guaranteed, for the purpose of proving to final customers the share or quantity of renewable energy in a supplier's energy mix, by issuing **guarantees of origin** (GOs) in response to a request from a producer. Each guarantee of origin has a standard size of one megawatt-hour, may be used only once, and must be used within twelve months of the production of the corresponding unit of energy; GOs that have not been cancelled expire no later than eighteen months after production. Member states must recognise GOs issued by other member states, and may refuse recognition only where they have well-founded doubts about their accuracy, reliability, or veracity. RED III extended the GO system to renewable gases, including hydrogen. Under the Electricity Directive, suppliers must disclose the share of each energy source in their overall fuel mix, and must use GOs to demonstrate the share of renewable energy. The technical standard for GOs is the European Energy Certificate System (EECS), administered by the Association of Issuing Bodies, which links national registries through a hub enabling cross-border transfers.

Article 19(2) of the Directive, as amended, also provides that member states may decide not to issue GOs to producers that receive financial support from a support scheme, or to issue them and transfer their value to the market, to avoid double compensation — an issue that has generated debate about whether certificates from subsidised plants should be sold to consumers who might believe that their purchases contribute to new renewable investment.

Certificate Systems Elsewhere

In the United States, **renewable energy certificates** (RECs) are used both for compliance with state renewable portfolio standards and for voluntary claims, tracked through regional tracking systems. Australia's large-scale generation certificates, created under the Renewable Energy (Electricity) Act 2000, are used for compliance with the Renewable Energy Target. In countries without national certificate systems, including most of Africa, Asia, and Latin America, the **I-REC Standard**, administered by a non-profit foundation, provides an international framework for issuing and tracking certificates through local issuers, and has been used in a growing number of African countries. The development of certificate systems in developing countries enables domestic and foreign companies to procure renewable electricity credibly, and may support corporate renewable procurement and investment, but requires robust registries, independent verification, and safeguards against double counting.

Corporate Renewable Procurement and Claims

Corporate demand has become an important driver of renewable energy investment. Companies with commitments to use renewable electricity — many of them members of the **RE100** initiative, which brings together companies committed to sourcing one hundred per cent renewable electricity — procure renewable electricity through several means: on-site generation; **corporate PPAs**, under which the company contracts directly with a renewable generator for its output and associated certificates; green tariffs offered by utilities; and the purchase of "unbundled" certificates separately from electricity. Corporate PPAs may be **physical** (involving the delivery of electricity to the buyer through the grid) or **virtual** or **financial** (a contract for difference settled financially against the market price, with the transfer of certificates). RED II required member states to remove unjustified barriers to corporate renewable PPAs, and the EU's 2024 electricity market reform requires member states to facilitate PPAs, including through state-backed guarantee schemes addressing the credit risk of buyers.

The accounting rules for corporate claims are set by the **Greenhouse Gas Protocol**, a private standard-setting initiative of the World Resources Institute and the World Business Council for Sustainable Development. Its Scope 2 Guidance (2015) requires companies to report emissions from purchased electricity using two methods: a **location-based method**, reflecting the average emissions intensity of the grid in the locations where consumption occurs, and a **market-based method**, reflecting the emissions associated with contractual instruments such as certificates and PPAs, subject to quality criteria. The market-based method has been criticised for allowing companies to claim zero emissions from electricity by purchasing cheap unbundled certificates from existing plants, which may have no effect on actual emissions or on investment in new renewable capacity ("additionality"), and from distant locations or different times of day from the company's consumption. In response, some companies have adopted "24/7 carbon-free energy" commitments, matching their consumption on an hourly basis with carbon-free generation in the same grid region, and the Greenhouse Gas Protocol has consulted on revisions to the Scope 2 Guidance, including proposals for hourly matching and requirements that certificates be sourced from the same market as consumption.

The integrity of corporate claims has become a legal as well as a reputational issue. In the European Union, Directive (EU) 2024/825 on empowering consumers for the green transition, which applies from September 2026, prohibits generic environmental claims that cannot be substantiated and claims, based on the offsetting of greenhouse gas emissions, that a product has a neutral, reduced, or positive impact on the environment. Consumer protection authorities and courts in several jurisdictions have taken action against misleading "green" claims,

and the use of renewable energy certificates to support claims may be scrutinised under consumer protection and unfair competition law. The relationship between renewable certificates and carbon credits, which are distinct instruments representing emission reductions or removals and are governed by the carbon market rules examined in Chapter 18, is a recurrent source of confusion: a renewable energy certificate attests to the renewable origin of a unit of electricity, while a carbon credit represents a tonne of carbon dioxide equivalent reduced or removed relative to a baseline, and the same project should not generate both instruments for the same attribute without safeguards against double counting.

10.8 Renewable Energy Investment Disputes

The Context: Retroactive Changes to Support Schemes

The renewable energy investment disputes arose from a common pattern. In the mid-2000s, several European states — notably Spain, Italy, and the Czech Republic — introduced generous feed-in tariffs to attract investment in renewable energy, particularly solar PV. The tariffs, combined with rapidly falling costs of solar panels, generated investment booms far exceeding government projections, and the cost of support, borne by consumers or the state, escalated rapidly. Following the global financial crisis of 2008, and facing large deficits in their electricity systems, these governments enacted measures reducing the support available to existing installations, including caps on the hours of production eligible for support, levies on revenues, and, in Spain's case, the replacement of the feed-in tariff regime by a new regime guaranteeing a "reasonable return" on investment calculated over the regulatory life of installations. Investors, many of them foreign, brought claims under the Energy Charter Treaty and bilateral investment treaties, arguing that the measures breached the obligation to accord fair and equitable treatment (FET), including by frustrating their legitimate expectations of regulatory stability, and constituted indirect expropriation.

The Spanish Arbitrations

Spain's original regime, set out principally in Royal Decree 661/2007, offered renewable installations registered by specified deadlines a regulated tariff for their entire operational life, with tariffs for solar PV substantially above market prices. In 2010, Spain introduced initial measures limiting the hours of production eligible for the tariff and imposing a grid access charge. Between 2012 and 2014, it enacted more radical measures: a seven per cent tax on the value of electricity production (Law 15/2012); the elimination of the inflation adjustment; and, by Royal Decree-law 9/2013, Law 24/2013, Royal Decree 413/2014, and Ministerial Order IET/1045/2014, the abolition of the RD 661/2007 regime and its replacement by a "specific remuneration" regime designed to provide a "reasonable return" for a "standard installation", initially set at around 7.4 per cent pre-tax,

over the regulatory life of the installation, taking into account the remuneration already received.

Spain faced around fifty arbitration claims, making it one of the most frequently sued states in the world. The outcomes varied, but investors prevailed in the majority of cases decided on the merits in respect of the 2013–2014 measures.

Claims dismissed. In *Charanne BV and Construction Investments SARL v Spain* (SCC, 2016), the first award, the tribunal dismissed claims concerning the 2010 measures, holding that the investors could not have had a legitimate expectation that the regulatory framework would remain unchanged in the absence of a specific commitment, that RD 661/2007 was a general regulation, and that the 2010 measures were not unreasonable or disproportionate. In *Isolux Infrastructure Netherlands BV v Spain* (SCC, 2016), the tribunal dismissed claims concerning the 2013–2014 measures, holding that the investor, which had invested in 2012 after the first changes and when further reform was foreseeable, could not have had a legitimate expectation that the regime would remain unchanged. In *Stadtwerke München GmbH and others v Spain* (ICSID, 2019), the tribunal similarly found that a diligent investor would have known that the regime could be modified, and that the new regime was not unreasonable.

Claims upheld. In *Eiser Infrastructure Ltd and Energía Solar Luxembourg SÀRL v Spain* (ICSID, 2017), examined in the case box below, the tribunal found that Spain had breached the FET standard by fundamentally and abruptly altering the regulatory regime, and awarded around EUR 128 million. Subsequent tribunals, including in *Novenergia II v Spain* (SCC, 2018), *Masdar Solar & Wind Cooperatief UA v Spain* (ICSID, 2018), *Antin Infrastructure Services Luxembourg SARL v Spain* (ICSID, 2018), *RREEF Infrastructure v Spain* (ICSID, 2019), *NextEra Energy Global Holdings BV v Spain* (ICSID, 2019), and *Cube Infrastructure Fund SICAV v Spain* (ICSID, 2019), found breaches of the FET standard, with awards ranging from tens to hundreds of millions of euros. The reasoning varied. Some tribunals, such as *Masdar*, found that specific assurances — including letters from the Spanish authorities confirming that the installations qualified for the RD 661/2007 tariff for their operational life — created legitimate expectations. Others, such as *RREEF*, found no legitimate expectation that the specific tariff would be preserved, but held that the investors were entitled to expect a reasonable return, and that Spain had breached the FET standard to the extent that the new regime failed to provide it, or applied retroactively by "clawing back" past remuneration.

The tax carve-out. Tribunals generally held that the seven per cent tax on the value of electricity production was a "taxation measure" within the meaning of Article 21 of the ECT, which excludes taxation measures from the scope of the treaty's protections (subject to exceptions for expropriation and certain non-dis-

crimination obligations), and therefore declined jurisdiction over claims relating to it, notwithstanding the investors' argument that the tax was in substance a disguised reduction of the tariff.

::: casebox {title="Eiser Infrastructure Ltd and Energía Solar Luxembourg SÀRL v Kingdom of Spain (ICSID, 2017)"} **Facts.** The claimants, a United Kingdom investment fund and its Luxembourg affiliate, invested in three concentrated solar power plants in Spain in 2007, relying on the regulated tariff regime of Royal Decree 661/2007. Following the 2013–2014 reforms, the plants' revenues fell dramatically, and the claimants alleged that the plants could no longer service their project finance debt.

Decision. The tribunal held that Article 10(1) of the ECT, which requires contracting parties to accord fair and equitable treatment, did not give investors a right to regulatory stability in the sense of a regulatory regime frozen in time; states retain the right to modify their regulatory regimes to meet changing circumstances in the public interest. However, the FET obligation protected investors against the "total and unreasonable" change of the regime on which they had relied in making long-term investments. Spain's new regime had been applied to existing installations, on the basis of a "standard installation" model that disregarded the claimants' actual costs and the capital-intensive character of their investment, and had eliminated the essential characteristics of the regime on which they had relied, depriving them of essentially all of the value of their investment. The tribunal awarded around EUR 128 million, using a discounted cash flow valuation.

Annulment. In 2020, an ICSID *ad hoc* committee annulled the award in its entirety because one of the arbitrators, appointed by the claimants, had failed to disclose long-standing professional connections with the firm of damages experts that had given evidence for the claimants, which the committee found could have had a material effect on the award. The claimants subsequently resubmitted their claims.

Significance. *Eiser* was the first award against Spain and set the template for many subsequent awards. Its distinction between permissible regulatory change and the "radical" or "fundamental" alteration of a regime on which investors had relied influenced the jurisprudence on regulatory stability under the ECT. Its annulment is a leading illustration of the importance of arbitrator disclosure, a theme examined in Chapter 26. :::

The Italian and Czech Arbitrations

Italy introduced a series of *Conto Energia* decrees offering feed-in tariffs for solar PV, and in 2014, by the so-called *Spalma-incentivi* decree, required existing installations to choose between reduced tariffs spread over a longer period and

other options reducing their incentives. Investors brought claims under the ECT. In *Blusun SA and others v Italy* (ICSID, 2016), the tribunal dismissed the claims, holding that the investors had not demonstrated that Italy had made specific commitments that the incentives would not be altered and that, in any event, the failure of the project was not caused by the measures. Other tribunals reached different results: in *Greentech Energy Systems A/S and others v Italy* (SCC, 2018) and *CEF Energia BV v Italy* (SCC, 2019), tribunals found that Italy had breached the FET standard by reducing incentives that had been specifically guaranteed to the investors' installations by agreements with the state agency responsible for administering the incentives, while in *Eskosol SpA in liquidazione v Italy* (ICSID, 2020), the tribunal dismissed the claim.

The Czech Republic introduced a levy on revenues from solar installations commissioned in 2009 and 2010 (the "solar levy") and withdrew certain tax exemptions. Claims under the ECT and BITs were largely unsuccessful: in *Antaris Solar GmbH and Dr Michael Göde v Czech Republic* (PCA, 2018) and in *JSW Solar (zwei) GmbH and others v Czech Republic* (PCA, 2019), the tribunals dismissed the claims, finding that the investors could not legitimately expect the incentives to remain unchanged, that the Czech Republic had acted reasonably in response to the solar boom, and that the investors, many of whom invested late in the boom, had been on notice of possible changes.

Beyond Europe, *Mesa Power Group LLC v Canada* (UNCITRAL, NAFTA, 2016) concerned the allocation of feed-in tariff contracts under Ontario's Green Energy Act, which included domestic content requirements found inconsistent with WTO law in *Canada — Renewable Energy*. The tribunal dismissed the claims, holding, among other things, that the domestic content requirements fell within the procurement exception of NAFTA Article 1108.

Doctrinal Themes

The renewable energy arbitrations have produced a substantial body of jurisprudence on several doctrinal questions.

Legitimate expectations and general legislation. A central question is whether general legislation — as distinct from specific commitments made to the investor — can create legitimate expectations protected by the FET standard. The prevailing view, reflected in *Charanne*, *Blusun*, and many subsequent awards, is that general legislation does not, by itself, create a legitimate expectation that it will not be changed; states retain the right to regulate. However, tribunals have differed as to whether legislation that specifically targets a defined class of investors and promises them particular treatment for a defined period (such as RD 661/2007's promise of tariffs for the operational life of registered installations)

can create legitimate expectations, and as to the significance of registration and other individualised acts.

Due diligence and foreseeability. Tribunals have consistently held that investors must conduct reasonable due diligence into the legal framework and its stability, including judicial decisions (for example, Spanish Supreme Court judgments upholding earlier regulatory changes), and that expectations of stability are not legitimate if regulatory changes were reasonably foreseeable at the time of investment. The timing of investment has therefore been critical: investors who invested late, after warnings of reform, have generally fared worse.

Proportionality and the reasonable return. Many tribunals have assessed whether the measures were reasonable and proportionate, considering the public interest in addressing the tariff deficit and the impact on investors. Spain's argument that its new regime satisfied investors' legitimate expectations by guaranteeing a reasonable return was accepted by some tribunals and rejected by others, depending on whether the reasonable return was in fact achieved and whether the regime clawed back past remuneration.

Valuation of damages. The arbitrations generated extensive debate over the valuation of damages, including the use of discounted cash flow methods for operating assets, the appropriate discount rate, the useful life of installations, and the treatment of regulatory risk. These questions are examined in Chapter 26.

The Intra-EU Objection and Enforcement

Most of the claimants against Spain and Italy were investors from other EU member states, and the disputes became entangled in the controversy over intra-EU investment arbitration. In *Slovak Republic v Achmea BV* (Case C-284/16, 2018), the Court of Justice held that investor–state arbitration clauses in intra-EU BITs are incompatible with EU law, because they remove disputes that may concern the interpretation or application of EU law from the EU judicial system. In *Republic of Moldova v Komstroy LLC* (Case C-741/19, 2021), the Court held that Article 26 of the ECT must be interpreted as not applicable to disputes between a member state and an investor of another member state concerning an investment in the first member state. In 2017, the Commission adopted a decision concluding that Spain's renewable energy support scheme constituted State aid, and in 2025 it adopted a decision concluding that the payment of the award in *Antin* would constitute unlawful State aid. Arbitral tribunals have, almost without exception, rejected the intra-EU objection, holding that their jurisdiction derives from the ECT as a matter of international law, and that EU law does not override the clear terms of the treaty.

The result has been a conflict between the obligations of EU member states under EU law and their obligations under the ECT and the ICSID Convention,

which has played out in enforcement proceedings outside the European Union. In *Kingdom of Spain v Infrastructure Services Luxembourg Sàrl* [2023] HCA 11, the High Court of Australia held that Spain's agreement to Article 54 of the ICSID Convention, which requires contracting states to recognise ICSID awards as binding and enforce the pecuniary obligations imposed by them as if they were final judgments of their own courts, amounted to a waiver of immunity from the jurisdiction of Australian courts to recognise and enforce the award, although not of immunity from execution against particular assets. Courts in the United Kingdom and the United States have also rejected Spain's immunity objections at various stages of enforcement proceedings; in *NextEra Energy Global Holdings BV v Kingdom of Spain* (2024), the United States Court of Appeals for the District of Columbia Circuit held that federal courts had jurisdiction to enforce intra-EU ECT awards against Spain under the arbitration exception to the Foreign Sovereign Immunities Act, and reversed injunctions that had been granted to prevent award creditors from pursuing proceedings in Europe. Further proceedings have continued, and the conflict between EU law and the international obligations of member states remains unresolved.

In response to the arbitrations, Spain adopted Royal Decree-law 17/2019, which offered installations whose owners renounced or discontinued arbitration claims a fixed reasonable return of around 7.4 per cent until 2031. The European Union and many of its member states have since withdrawn from the ECT, a development examined in Chapter 14.

Lessons for Regulatory Design

The renewable energy arbitrations offer important lessons for the design of renewable energy support. First, support schemes should be designed with mechanisms that allow costs to be controlled — such as volume caps, automatic degression of tariffs, and competitive allocation — so that governments are not forced into retroactive changes. Second, governments should distinguish clearly between general policy statements and specific commitments, and should avoid individualised assurances that they may not be able to honour. Third, changes to support schemes should, as far as possible, apply prospectively to new installations, with transitional arrangements for existing installations. Fourth, investors should conduct thorough due diligence into the stability of the legal framework, and price regulatory risk accordingly. These lessons are relevant to developing countries, including Nigeria, that seek to attract private investment in renewable energy through long-term contracts and support schemes.

::: exercise {title="Practical Exercise: The Savania Solar Support Reform and the Lake Ilera Mini-Grid (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic of Savania, the Lake Ilera community, and all companies*

named are fictional, and any resemblance to real states, communities, or companies is coincidental.

Part A — Retroactive reform. In 2017, the Republic of Savania, a party to the Energy Charter Treaty, enacted the Solar Development Law, which guaranteed solar installations registered before 31 December 2019 a fixed tariff of USD 0.22 per kWh "for twenty years from commissioning". The Ministry of Energy issued registration certificates to each installation stating that it was "entitled to the regulated tariff for twenty years". The law contained no mechanism for controlling the volume of installations. Over 3,000 MW was registered against a forecast of 800 MW. By 2023, the cost of support had created a deficit equal to two per cent of GDP. In 2024, Savania enacted the Electricity Sustainability Law, which (i) replaced the fixed tariff for all installations, including existing ones, with a regime providing a "reasonable return" of 6.5 per cent pre-tax over the regulatory life of a "standard installation", calculated by taking into account revenues received since commissioning; and (ii) imposed a five per cent levy on the gross revenues of all generators.

Helios Infrastructure Partners GmbH (HYPOTHETICAL), incorporated in a non-EU ECT contracting party, invested in 2018 in a 150 MW solar plant financed with seventy per cent debt. Its due diligence report noted "political discussion of tariff costs" but concluded that the risk of retroactive change was low given the registration certificate.

1. Advise Helios on the strength of a claim for breach of Article 10(1) of the ECT, considering the case law on legitimate expectations, specific commitments, due diligence, and proportionality.
2. Advise Savania on its defences, including the argument that the reasonable-return regime satisfies any legitimate expectation, and the significance of the "clawback" of past revenues.
3. Can Helios challenge the five per cent levy under the ECT? Consider Article 21.
4. Propose an alternative reform that Savania could have adopted to control costs while minimising its exposure to claims.

Part B — Mini-grid and grid arrival. In 2021, Lumen Mini-Grids Ltd (HYPOTHETICAL) obtained a permit from the Savania Electricity Regulatory Commission to build and operate an isolated 500 kW solar-hybrid mini-grid serving 2,000 households in the Lake Ilera community, under regulations allowing a cost-reflective tariff. The permit stated that if the main grid reached the community, Lumen could elect either to become a small distributor purchasing bulk power from the distribution company, or to sell its distribution assets to the distribution company "at a fair valuation determined by the Commission". In 2025, the distribution company extended its network to Lake Ilera, began

offering connections at a tariff subsidised to forty per cent below Lumen's tariff, and refused to negotiate either option.

5. Identify Lumen's legal remedies under the regulatory framework, including the role of the Commission.
6. What principles should govern the "fair valuation" of Lumen's assets?
7. How should mini-grid regulations be designed to protect both investors and consumers when the main grid arrives? Draft a model clause of no more than 250 words. :::

CHAPTER SUMMARY

- Renewable energy has become the dominant source of new generating capacity worldwide. The law has driven its growth through targets, support mechanisms, and grid access, and has constrained it through permitting delays, land conflicts, and regulatory instability.
- There is no comprehensive international treaty on renewable energy. Its international foundations lie in climate law, SDG 7, the IRENA Statute, the International Solar Alliance, and the ECT. The EU's Renewable Energy Directive, as amended in 2023, sets a binding Union target of at least 42.5 per cent by 2030.
- Wind and sunlight are generally regarded as things common to all. The law therefore concerns rights to use land and to access the resource, together with the allocation of offshore areas under the coastal state's EEZ rights.
- Utility-scale solar raises land rights issues, particularly where land is held under customary tenure. Mini-grid regulations, such as Nigeria's, must address cost-reflective tariffs and the consequences of the main grid's arrival.
- Wind permitting addresses landscape, wildlife, and indigenous rights. The *Fosen* judgment of the Supreme Court of Norway demonstrates that renewable projects are subject to human rights constraints.
- Hydropower law governs water rights, environmental flows, long-term concessions, reversion, and dam safety. Geothermal law turns on the classification of the resource, and bioenergy is subject to sustainability criteria addressing land use change.
- Support mechanisms include feed-in tariffs and premiums, quota obligations with certificates, auctions with long-term contracts or CfDs, and fiscal incentives. Auctions have become dominant, and EU support schemes are subject to State aid control.
- Certificates separate environmental attributes from physical electricity. They serve compliance, disclosure, and voluntary functions, and the integrity of corporate claims is increasingly subject to legal scrutiny.
- The Spanish, Italian, and Czech arbitrations established that general legislation does not, by itself, freeze the regulatory framework. However, fundamental and retroactive changes that defeat specific commitments or a reasonable

return may breach the FET standard. Their enforcement has exposed a conflict between EU law and the ECT and ICSID Convention.

DOCTORAL RESEARCH QUESTIONS

1. Does the *Fosen* judgment imply a general principle that the public interest in decarbonisation cannot justify interference with indigenous peoples' cultural rights? How should courts balance climate objectives against the rights of affected communities in the energy transition?
2. To what extent did the design of European feed-in tariff schemes, rather than the subsequent reforms, cause the renewable energy investment disputes? What does this imply for the allocation of responsibility between states and investors?
3. Evaluate the coherence of arbitral jurisprudence on legitimate expectations arising from general legislation in the Spanish, Italian, and Czech renewable energy cases. Is a consistent doctrine discernible?
4. Should corporate claims to use renewable electricity be regulated by law, and if so, should the law require additionality, hourly matching, or geographic deliverability?
5. Critically assess the adequacy of the Nigerian legal framework for mini-grids and distributed solar in achieving universal electricity access. What reforms would better align investor protection, consumer protection, and the expansion of the main grid?
6. Is the classification of geothermal resources as minerals, water, or a sui generis resource merely technical, or does it reflect deeper choices about ownership and the distribution of resource benefits?

FURTHER READING

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11 CHAPTER

Nuclear Energy Law, Safety, and International Responsibility

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Explain the legal basis of the right to the peaceful uses of nuclear energy and its relationship with non-proliferation obligations.
2. Describe the architecture of international nuclear law, including the IAEA Statute, the principal conventions, and the role of IAEA safety standards as soft law.
3. Analyse the legal requirements of nuclear safety, with particular attention to the independence of the regulatory body and the lessons of Chernobyl and Fukushima.
4. Explain the system of IAEA safeguards, including comprehensive safeguards agreements and the Additional Protocol, and evaluate its effectiveness in contested cases.
5. Assess the international and domestic law of nuclear security and physical protection.
6. Analyse the principles of the international nuclear liability regimes and evaluate their adequacy in the light of major accidents.
7. Evaluate the legal frameworks for radioactive waste management and decommissioning, including their financing.
8. Identify the emerging legal issues raised by small modular and advanced reactors, and the legal preparations required of nuclear newcomer states, including those in Africa.

Introduction

Nuclear energy occupies a unique position in energy law. It is a low-carbon source of firm electricity that provides around a tenth of the world's electricity and a much larger share of its low-carbon electricity. It is also the only energy technology whose materials and know-how can be diverted to the production of weapons

of mass destruction, and whose accidents can cause radioactive contamination across national borders and over generations. These characteristics have produced a body of law that is more internationalised, more technically detailed, and more closely bound up with questions of international peace and security than any other branch of energy law.

The legal framework of nuclear energy is commonly described by reference to "three Ss" — safety, security, and safeguards — to which a fourth pillar, liability, is usually added, together with the management of radioactive waste and decommissioning. **Safety** concerns the protection of workers, the public, and the environment from the harmful effects of ionising radiation, including the prevention and mitigation of accidents. **Security** concerns the prevention and detection of, and response to, theft, sabotage, unauthorised access, and other malicious acts involving nuclear material and facilities. **Safeguards** concern the verification that nuclear material is not diverted from peaceful uses to nuclear weapons. **Liability** concerns compensation for damage caused by nuclear incidents. Each pillar has its own international instruments, institutions, and domestic legal requirements.

After decades of stagnation following the accidents at Three Mile Island (1979) and Chernobyl (1986), and renewed doubt after the Fukushima Daiichi accident (2011), nuclear energy has experienced a revival of interest driven by climate commitments, energy security concerns following the Russian invasion of Ukraine, and the electricity demand of data centres. At COP28 in 2023, a group of states launched a declaration aiming to triple global nuclear energy capacity by 2050; the European Union included nuclear energy, subject to conditions, in its taxonomy of sustainable activities; the World Bank announced in 2025 that it would resume financing nuclear energy projects; and technology companies have concluded long-term contracts to purchase nuclear electricity, including from the restart of a closed reactor. At the same time, several countries, including Germany, have phased out nuclear power, and the costs and delays of large nuclear construction projects in Europe and the United States have been severe. Several African states, including Egypt, Ghana, Kenya, Nigeria, and Uganda, have pursued nuclear power programmes, and Egypt is constructing the continent's second nuclear power plant, after South Africa's Koeberg.

This chapter examines the peaceful uses of nuclear energy (section 11.1), the international nuclear legal framework (section 11.2), nuclear safety and regulatory independence (section 11.3), safeguards and non-proliferation (section 11.4), nuclear security (section 11.5), nuclear liability (section 11.6), radioactive waste and decommissioning (section 11.7), and small modular reactors and emerging legal issues (section 11.8).

	Objective	Principal instruments	Institutions / review
Safety	Protect people & environment from radiation; prevent accidents	CNS 1994; Joint Convention 1997; IAEA Safety Standards	National regulator; CNS peer review; IRRS
Security	Prevent theft, sabotage & malicious acts	CPPNM & 2005 Amendment; ICSANT; UNSCR 1540	Competent authority; IPPAS
Safeguards	Verify non-diversion to weapons	NPT Art III; INFCIRC/153; Additional Protocol	IAEA Board; UN Security Council
Liability	Compensate victims of nuclear damage	Paris/Brussels; Vienna; Joint Protocol; CSC	Courts of installation state
Waste & decommissioning	Avoid undue burdens on future generations	Joint Convention; Euratom Dir. 2011/70	Waste agencies; segregated funds

Figure 11.1 The pillars of nuclear energy law: objectives, principal instruments, and institutions. *Source:* prepared for this edition.

11.1 Peaceful Uses of Nuclear Energy

From Atoms for Peace to the Right to Peaceful Uses

The international law of nuclear energy originated in the effort, after the bombings of Hiroshima and Nagasaki in 1945, to reconcile the promise of civil nuclear technology with the danger of nuclear weapons proliferation. The first resolution of the United Nations General Assembly, adopted in January 1946, established the United Nations Atomic Energy Commission to make proposals for the control of atomic energy "to the extent necessary to ensure its use only for peaceful purposes". The proposals for international ownership and control of nuclear activities presented by the United States (the Baruch Plan) failed amid the onset of the Cold War. In his "Atoms for Peace" address to the General Assembly in December 1953, President Eisenhower proposed an international agency to which states would contribute fissile material for peaceful uses, leading to the adoption of the Statute of the International Atomic Energy Agency in 1956 and its entry into force in 1957.

The bargain at the heart of the international nuclear order was codified in the **Treaty on the Non-Proliferation of Nuclear Weapons** (NPT), opened for signature in 1968 and in force since 1970. The NPT distinguishes between

"nuclear-weapon States" — defined in Article IX(3) as states that manufactured and exploded a nuclear weapon or other nuclear explosive device before 1 January 1967 (the United States, the Soviet Union (now Russia), the United Kingdom, France, and China) — and "non-nuclear-weapon States". Under Articles I and II, nuclear-weapon states undertake not to transfer nuclear weapons or assist non-nuclear-weapon states to acquire them, and non-nuclear-weapon states undertake not to receive, manufacture, or otherwise acquire nuclear weapons. Under Article III, non-nuclear-weapon states undertake to accept IAEA safeguards on all source and special fissionable material in all their peaceful nuclear activities. Under Article VI, all parties undertake to pursue negotiations in good faith on effective measures relating to cessation of the nuclear arms race and to nuclear disarmament.

Article IV of the NPT

Article IV(1) of the NPT provides that nothing in the Treaty shall be interpreted as affecting "the inalienable right of all the Parties to the Treaty to develop research, production and use of nuclear energy for peaceful purposes without discrimination and in conformity with articles I and II of this Treaty". Article IV(2) commits parties to facilitate, and grants them the right to participate in, the fullest possible exchange of equipment, materials, and scientific and technological information for the peaceful uses of nuclear energy, and requires parties in a position to do so to cooperate in contributing to the further development of applications of nuclear energy for peaceful purposes, "especially in the territories of non-nuclear-weapon States Party to the Treaty, with due consideration for the needs of the developing areas of the world".

The interpretation of Article IV has been contested. Many developing states regard the "inalienable right" as encompassing the entire nuclear fuel cycle, including uranium enrichment and spent fuel reprocessing, which are "sensitive" technologies because they can produce weapons-usable material. Supplier states have sought to restrict the spread of enrichment and reprocessing through export controls, coordinated through the **Nuclear Suppliers Group** (NSG), an informal arrangement established in 1974–1975 after India's first nuclear test, whose guidelines restrict the transfer of sensitive technologies and, since 1992, require full-scope safeguards as a condition of the supply of nuclear materials and equipment to non-nuclear-weapon states. The controversy over Iran's enrichment programme, examined in section 11.4, was in significant part a dispute over the scope of Article IV. The qualification that the right must be exercised "in conformity with articles I and II", and, by implication, with the safeguards obligations of Article III, means that the right is not unconditional.

Peaceful Applications Beyond Electricity

Although this chapter focuses on nuclear power, the peaceful uses of nuclear technology extend far beyond electricity generation. Nuclear and radiation technologies are used in medicine (diagnostic imaging and cancer radiotherapy), agriculture (the sterile insect technique and plant mutation breeding), industry (non-destructive testing and sterilisation), water resource management (isotope hydrology), and scientific research. The IAEA's technical cooperation programme supports these applications in developing states, and they constitute a substantial part of the nuclear activities regulated in most countries, including Nigeria. The legal frameworks for the safety and security of radioactive sources used in these applications, including the IAEA's Code of Conduct on the Safety and Security of Radioactive Sources (2003), are an important component of nuclear law.

The Decision to Pursue Nuclear Power

A state's decision to introduce nuclear power is a long-term commitment requiring the development of a comprehensive legal and institutional infrastructure. The IAEA's **Milestones Approach**, set out in its publication *Milestones in the Development of a National Infrastructure for Nuclear Power*, identifies nineteen infrastructure issues — including national position, nuclear safety, management, funding and financing, legal framework, safeguards, regulatory framework, radiation protection, electrical grid, human resource development, stakeholder involvement, site and supporting facilities, environmental protection, emergency planning, nuclear security, nuclear fuel cycle, radioactive waste management, industrial involvement, and procurement — to be addressed in three phases, from the consideration of the nuclear option through the preparation for construction to the activities to implement the first plant. The IAEA conducts Integrated Nuclear Infrastructure Review (INIR) missions to assess newcomer states' progress, and several African states, including Nigeria, Ghana, Kenya, Egypt, Uganda, and Morocco, have hosted such missions.

The decision to pursue nuclear power raises legal questions beyond technical regulation: democratic legitimacy and public participation (the Aarhus Convention's Compliance Committee and the parties' Meeting have addressed the application of public participation requirements to decisions on nuclear power, including decisions to extend the lifetime of reactors); transboundary consultation under the Espoo Convention on Environmental Impact Assessment in a Transboundary Context (1991), whose Implementation Committee and parties have considered the application of the Convention to the lifetime extension of nuclear power plants; and the compatibility of nuclear power with constitutional rights and environmental principles. The Court of Justice's judgment in *Inter-Environnement Wallonie and Bond Beter Leefmilieu Vlaanderen* (Case C-411/17, 2019) held that the extension of the operating life of Belgian nuclear

reactors required an environmental impact assessment under the EIA Directive and an appropriate assessment under the Habitats Directive, illustrating the application of general environmental law to nuclear decisions.

11.2 International Nuclear Legal Frameworks

The International Atomic Energy Agency

The IAEA, an autonomous international organisation within the United Nations family with its headquarters in Vienna, is the central institution of international nuclear law. Its objectives, set out in Article II of its Statute, are to "seek to accelerate and enlarge the contribution of atomic energy to peace, health and prosperity throughout the world" and to "ensure, so far as it is able, that assistance provided by it or at its request or under its supervision or control is not used in such a way as to further any military purpose". The dual mandate of promotion and control has been a source of tension, and critics have questioned whether an organisation charged with promoting nuclear energy can effectively regulate it.

The IAEA's statutory functions under Article III include encouraging research and development, facilitating the exchange of information, establishing and administering safeguards, and establishing or adopting "standards of safety for protection of health and minimization of danger to life and property", and providing for the application of these standards. The IAEA's governing organs are the General Conference of all member states and the Board of Governors of thirty-five member states, which has the power under Article XII(C) of the Statute to report a member state's non-compliance with its safeguards obligations to the Security Council and the General Assembly. The institutional structure of the IAEA is examined further in Chapter 4.

The Treaty Framework

International nuclear law consists of a network of treaties, most adopted in response to specific events, together with a large body of non-binding standards and guidance. The principal treaties may be grouped by pillar.

Safety. The *Convention on Nuclear Safety* (CNS, 1994; in force 1996) applies to land-based civil nuclear power plants and requires parties to establish a legislative and regulatory framework, an independent regulatory body, and safety requirements for siting, design, construction, and operation. The *Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management* (1997; in force 2001) applies analogous obligations to spent fuel and radioactive waste. Both are "incentive" conventions, relying on national reporting and peer review at review meetings of the parties rather than on enforcement.

Emergency preparedness. The *Convention on Early Notification of a Nuclear Accident* (1986) and the *Convention on Assistance in the Case of a Nuclear Accident*

or Radiological Emergency (1986), both adopted within months of the Chernobyl accident, require states to notify potentially affected states and the IAEA of nuclear accidents with transboundary effects and establish a framework for international assistance.

Security. The *Convention on the Physical Protection of Nuclear Material* (CPPNM, 1980; in force 1987), as amended in 2005 (the amendment in force since 2016), and the *International Convention for the Suppression of Acts of Nuclear Terrorism* (ICSANT, 2005; in force 2007), examined in section 11.5.

Safeguards and non-proliferation. The NPT, safeguards agreements concluded between states and the IAEA, regional treaties establishing nuclear-weapon-free zones (including the African Nuclear-Weapon-Free Zone Treaty, or Treaty of Pelindaba, signed in 1996 and in force since 2009), and the Treaty on the Prohibition of Nuclear Weapons (2017; in force 2021), examined in section 11.4.

Liability. The *Paris Convention on Third Party Liability in the Field of Nuclear Energy* (1960), the *Brussels Supplementary Convention* (1963), the *Vienna Convention on Civil Liability for Nuclear Damage* (1963), the *Joint Protocol* (1988), the protocols amending the Vienna (1997) and Paris and Brussels (2004) Conventions, and the *Convention on Supplementary Compensation for Nuclear Damage* (1997; in force 2015), examined in section 11.6.

IAEA Safety Standards as Soft Law

The most detailed and practically important rules of international nuclear safety law are found not in treaties but in the **IAEA Safety Standards Series**, adopted under Article III(A)(6) of the IAEA Statute. The standards are arranged in a hierarchy: the *Fundamental Safety Principles* (Safety Fundamentals, SF-1, 2006), which set out the fundamental safety objective — "to protect people and the environment from harmful effects of ionizing radiation" — and ten safety principles; *Safety Requirements*, which establish the requirements that must be met to ensure safety, expressed as "shall" statements; and *Safety Guides*, which provide recommendations on how to comply with the requirements. The ten principles of SF-1 address responsibility for safety (which rests with the licensee), the role of government (including an independent regulatory body), leadership and management for safety, justification of facilities and activities, optimisation of protection, limitation of risks to individuals, protection of present and future generations, prevention of accidents, emergency preparedness and response, and protective actions to reduce existing or unregulated radiation risks.

The Safety Standards are not binding on states as such. However, they are binding on the IAEA in relation to its own operations, and on states in relation to operations assisted by the IAEA under project and supply agreements. They acquire legal effect through incorporation into national law, which is widespread,

and through reference in treaties: the CNS and the Joint Convention refer to internationally endorsed safety principles, and the review process under those conventions uses the standards as benchmarks. In the European Union, the Euratom Nuclear Safety Directive (Council Directive 2009/71/Euratom, as amended by Council Directive 2014/87/Euratom) gives binding effect to principles derived from the standards. The IAEA standards are therefore a leading example of the transformation of soft law into hard obligations through domestic and regional incorporation, a theme examined in Chapter 2.

Euratom

The *Treaty establishing the European Atomic Energy Community* (Euratom Treaty), signed in Rome in 1957 alongside the Treaty establishing the European Economic Community, established a distinct regional legal order for nuclear energy among the member states of the European Union. The Euratom Treaty provides for the promotion of research, the establishment of uniform safety standards for the protection of workers and the public (Articles 30–39), the supply of nuclear materials through the Euratom Supply Agency, which has a right of option on ores, source materials, and special fissile materials produced in member states and an exclusive right to conclude supply contracts (Article 52), a regional safeguards system administered by the Commission (Chapter VII), and the ownership by the Community of special fissile materials (Article 86). On the basis of Articles 31 and 32 of the Treaty, the Council has adopted the Basic Safety Standards Directive (Council Directive 2013/59/Euratom), the Nuclear Safety Directive, and the Radioactive Waste and Spent Fuel Directive (Council Directive 2011/70/Euratom). In *Commission v Council* (Case C-29/99, 2002), the Court of Justice held that the Community had competence in the field of nuclear installation safety under the Euratom Treaty, opening the way for the adoption of the Nuclear Safety Directive.

The Euratom Treaty also requires member states to notify the Commission of investment projects in the nuclear field (Article 41) and to provide general data on plans for the disposal of radioactive waste (Article 37). In *Austria v Commission* (Case C-594/18 P, 2020), the Court of Justice dismissed Austria's appeal against the General Court's judgment upholding the Commission's approval, under the State aid rules, of United Kingdom support for the Hinkley Point C nuclear power plant, holding that the promotion of nuclear energy was a legitimate objective of common interest under the Euratom Treaty and that State aid for nuclear power need not pursue an objective of common interest beyond that. In September 2025, the General Court dismissed Austria's action challenging the inclusion of nuclear energy and gas in the EU taxonomy of sustainable activities by Commission Delegated Regulation (EU) 2022/1214.

The Nigerian Framework

Nigeria's nuclear legal framework is based principally on the **Nigeria Atomic Energy Commission Act 1976**, which established the Nigeria Atomic Energy Commission (NAEC) as the body responsible for the promotion and development of atomic energy, including nuclear power, and the **Nuclear Safety and Radiation Protection Act 1995**, which established the **Nigerian Nuclear Regulatory Authority** (NNRA) as the regulatory body responsible for nuclear safety, radiation protection, and the regulation of nuclear and radiation activities. The NNRA has adopted regulations on matters including radiation safety, the security of radioactive sources, and the licensing of nuclear installations. Nigeria operates a research reactor, the Nigeria Research Reactor-1 (NIRR-1), a miniature neutron source reactor at the Centre for Energy Research and Training of Ahmadu Bello University, Zaria, which was converted from highly enriched to low enriched uranium fuel in 2018 with the assistance of the IAEA, China, and the United States — an achievement in nuclear security that removed weapons-usable material from Nigeria. Nigeria has concluded intergovernmental agreements with the Russian Federation on cooperation in the construction of nuclear power plants, and has hosted IAEA missions to review its nuclear infrastructure. Nigeria is a party to the NPT, the Treaty of Pelindaba, and several of the IAEA conventions. The separation of the promotional functions of NAEC from the regulatory functions of the NNRA reflects the requirement of regulatory independence examined in section 11.3.

11.3 Nuclear Safety and Regulatory Independence

The Responsibility for Safety

The first principle of nuclear safety law is that the **prime responsibility for safety rests with the person or organisation responsible for facilities and activities that give rise to radiation risks** — that is, the licensee or operator. This principle, set out in Principle 1 of the IAEA Safety Fundamentals and in Article 9 of the Convention on Nuclear Safety, means that the regulator does not assume responsibility for safety by licensing and inspecting a facility; the operator remains responsible at all times, and must establish a management system that gives safety overriding priority. The principle is reinforced by the concept of **safety culture**, defined by the International Nuclear Safety Advisory Group after Chernobyl as the assembly of characteristics and attitudes in organisations and individuals which establishes that, as an overriding priority, nuclear plant safety issues receive the attention warranted by their significance.

The Requirements of the Convention on Nuclear Safety

The CNS imposes obligations on contracting parties concerning the legislative and regulatory framework (Article 7), the regulatory body (Article 8), responsi-

bility of the licence holder (Article 9), priority to safety (Article 10), financial and human resources (Article 11), human factors (Article 12), quality assurance (Article 13), assessment and verification of safety (Article 14), radiation protection (Article 15), emergency preparedness (Article 16), siting (Article 17), design and construction (Article 18), and operation (Article 19). The legislative and regulatory framework must provide for national safety requirements, a system of licensing, a system of regulatory inspection and assessment, and the enforcement of applicable regulations and licence terms, including suspension, modification, or revocation.

Article 18 requires that the design and construction of a nuclear installation provide for several reliable levels and methods of protection — the concept of **defence in depth** — against the release of radioactive materials, with a view to preventing the occurrence of accidents and mitigating their radiological consequences. Defence in depth is the foundational concept of nuclear safety engineering: it requires multiple independent and redundant layers of protection, so that the failure of one layer is compensated by another, ranging from conservative design and high-quality operation, through control of abnormal operation and engineered safety features, to the control of severe accidents and off-site emergency response.

The CNS is implemented through a peer review process. Each contracting party submits a national report on the measures taken to implement its obligations, which is reviewed by the other parties at review meetings held every three years. Following the Fukushima accident, the parties adopted the **Vienna Declaration on Nuclear Safety** (2015), which set out principles for the implementation of the CNS's objective of preventing accidents and mitigating radiological consequences: new nuclear power plants are to be designed, sited, and constructed with the objective of preventing accidents and, should an accident occur, mitigating possible releases of radionuclides causing long-term off-site contamination and avoiding early radioactive releases or releases large enough to require long-term protective measures and actions; comprehensive and systematic safety assessments are to be carried out periodically for existing installations, and reasonably practicable safety improvements implemented in a timely manner; and national requirements are to take into account the relevant IAEA safety standards. The Declaration was the compromise outcome of a Swiss proposal to amend the CNS, which did not attract sufficient support.

Regulatory Independence

The independence of the regulatory body is the cornerstone of nuclear safety governance. Article 8(2) of the CNS requires each contracting party to take the appropriate steps to ensure "an effective separation between the functions of the regulatory body and those of any other body or organization concerned with

the promotion or utilization of nuclear energy". Principle 2 of the IAEA Safety Fundamentals requires an "effectively independent" regulatory body, and the IAEA's General Safety Requirements Part 1, *Governmental, Legal and Regulatory Framework for Safety*, elaborates the elements of independence: the regulatory body must be functionally separate from entities having responsibilities or interests that could unduly influence its decision-making; it must have adequate legal authority, competence, and financial and human resources; and it must be able to make regulatory judgments and decisions free from undue influence, including political influence and pressure from licensees.

In the European Union, Article 5 of the Euratom Nuclear Safety Directive, as amended in 2014, requires member states to ensure that the competent regulatory authority is functionally separate from any other body or organisation concerned with the promotion or utilisation of nuclear energy, and that it does not seek or take instructions from any such body or organisation when carrying out its regulatory tasks; that it makes regulatory decisions founded on robust and transparent safety-related requirements; that it is given dedicated and appropriate budget allocations; and that it employs an appropriate number of staff with the necessary qualifications.

Independence is a matter of substance, not merely form. Regulatory independence may be compromised by structural arrangements (such as the location of the regulator within a ministry responsible for promoting nuclear energy), by financial dependence, by the movement of personnel between regulator and industry, and by a culture of deference. The **IAEA Integrated Regulatory Review Service** (IRRS) provides peer reviews of national regulatory infrastructure against IAEA standards, and IRRS missions frequently comment on the independence of regulators.

::: casebox {title="The Fukushima Daiichi Accident (2011) and Regulatory Capture"} **Facts.** On 11 March 2011, the Great East Japan Earthquake and the tsunami that followed it struck the Fukushima Daiichi Nuclear Power Station, operated by the Tokyo Electric Power Company (TEPCO). The tsunami, which exceeded the height of the plant's sea wall, flooded the plant and disabled the emergency diesel generators and electrical systems. The resulting loss of cooling led to core meltdowns in three reactors, hydrogen explosions, and the release of large quantities of radioactive material. More than one hundred thousand people were evacuated from the surrounding area. The accident was rated at Level 7, the highest level, on the International Nuclear and Radiological Event Scale — the only accident other than Chernobyl to receive that rating.

Findings. The Fukushima Nuclear Accident Independent Investigation Commission established by the National Diet of Japan concluded in its 2012 report that the accident was "a profoundly manmade disaster — that could

and should have been foreseen and prevented". It found that TEPCO and the regulators had been aware of the risk of a tsunami exceeding the design basis but had failed to take action, and that the relationship between the regulators and TEPCO amounted to "regulatory capture": the regulators had lacked independence from the industry and from the ministry that promoted nuclear power, and had deferred to the operator's expertise and interests. The Nuclear and Industrial Safety Agency, the principal regulator, had been located within the Ministry of Economy, Trade and Industry, which was responsible for promoting nuclear energy.

Legal reforms. Japan abolished the Nuclear and Industrial Safety Agency and in 2012 established the **Nuclear Regulation Authority** (NRA) as an independent commission attached to the Ministry of the Environment, adopted new regulatory requirements incorporating severe accident measures and stricter tsunami and seismic standards, and required all reactors to undergo review against the new standards before restarting. Internationally, the accident led to the IAEA Action Plan on Nuclear Safety (2011), "stress tests" of nuclear power plants in the European Union, the 2014 amendment of the Euratom Nuclear Safety Directive, and the Vienna Declaration on Nuclear Safety.

Litigation. The accident generated extensive litigation in Japan. In June 2022, the Supreme Court of Japan held in cases brought by evacuees that the State was not liable for damages, because the tsunami was far larger than anticipated and the accident could not have been prevented even if the State had exercised its regulatory powers to require TEPCO to take countermeasures. In July 2022, the Tokyo District Court, in a shareholders' derivative action, ordered four former TEPCO executives to pay the company around 13 trillion yen in damages for failing to take measures against the foreseeable risk of a tsunami, a judgment subject to appeal. Compensation for victims under the Act on Compensation for Nuclear Damage is examined in section 11.6.

Significance. Fukushima demonstrated that regulatory independence is a substantive safety requirement and not a formality, and that the assumptions underlying the design basis of nuclear installations must be continuously reassessed in the light of new knowledge about external hazards. :::

Licensing and Safety Assessment

National nuclear safety law establishes a system of licensing that governs each stage in the life of a nuclear installation: site evaluation, design, construction, commissioning, operation, and decommissioning. Licensing decisions are based on the regulator's review of the applicant's safety case — a comprehensive set of documents and analyses demonstrating that the installation will meet safety requirements, including a deterministic safety analysis of design basis accidents

and a probabilistic safety assessment estimating the frequency of core damage and large releases. Many legal systems also require **periodic safety reviews**, typically every ten years, in which the operator reassesses the safety of the installation against current standards and identifies reasonably practicable improvements.

Courts generally defer to the technical judgments of nuclear regulators, while scrutinising the legality of their procedures. In *Vermont Yankee Nuclear Power Corp v Natural Resources Defense Council* 435 US 519 (1978), the United States Supreme Court held that courts may not impose procedural requirements on agencies, including the Nuclear Regulatory Commission, beyond those required by the Administrative Procedure Act and other statutes. In *Baltimore Gas & Electric Co v Natural Resources Defense Council* 462 US 87 (1983), it upheld the Commission's assumption, for the purposes of licensing decisions, that the permanent storage of nuclear waste would have no significant environmental impact, emphasising that a reviewing court must be at its most deferential when an agency makes predictions within its area of special expertise at the frontiers of science. In India, the Supreme Court in *G Sundarrajan v Union of India* (2013) declined to halt the commissioning of the Kudankulam nuclear power plant, holding that the safety of the plant was a matter for the expert regulator, but issued directions on safety, the disposal of spent fuel, and the welfare of the local population. In Japan, district courts have on several occasions issued injunctions suspending the operation of reactors on safety grounds, which have generally been overturned on appeal, illustrating both the potential and the limits of judicial supervision of nuclear safety.

The law of nuclear safety also intersects with the law of nuisance and transboundary harm. In *Land Oberösterreich v ČEZ as* (Case C-115/08, 2009), the Court of Justice held that Austria could not refuse to take into account, in proceedings brought by the Austrian province of Upper Austria seeking an injunction against nuisance from the Temelín nuclear power plant in the Czech Republic, the official authorisation granted to the plant in another member state, because to do so would discriminate on grounds of nationality contrary to the Euratom Treaty.

Nuclear Safety in Armed Conflict: Zaporizhzhia

The Russian occupation of the Zaporizhzhia Nuclear Power Plant in Ukraine — the largest nuclear power plant in Europe — in March 2022 confronted nuclear safety law with an unprecedented situation: a nuclear power plant operated under military occupation in an active war zone, subject to shelling, repeated losses of off-site power, and the detention and intimidation of its staff. The IAEA Director General articulated "seven indispensable pillars" for ensuring nuclear safety and security during armed conflict, derived from IAEA safety standards and security guidance, including physical integrity, the functioning of safety and

security systems, the ability of staff to fulfil their duties without undue pressure, secure off-site power supply, uninterrupted supply chains, effective radiation monitoring and emergency arrangements, and reliable communication with the regulator. In May 2023, he set out to the United Nations Security Council five concrete principles for the protection of the plant, including that there should be no attack from or against the plant and that it should not be used as a base for heavy weapons. The IAEA established a continuous presence of its experts at the plant from September 2022. Under international humanitarian law, Article 56 of Additional Protocol I to the Geneva Conventions (1977) provides special protection for works and installations containing dangerous forces, including nuclear electrical generating stations, which are not to be made the object of attack, even where they are military objectives, if such attack may cause the release of dangerous forces and consequent severe losses among the civilian population, subject to limited exceptions. The protection of energy infrastructure in armed conflict is examined further in Chapter 25.

11.4 Safeguards and Non-Proliferation Obligations

The Concept of Safeguards

IAEA safeguards are a system of technical measures through which the IAEA verifies that states are honouring their international legal obligations to use nuclear material and technology only for peaceful purposes. Safeguards comprise nuclear material accountancy (the maintenance of records and reports of nuclear material quantities and movements), containment and surveillance (seals and cameras), and inspections and other verification activities at facilities and other locations. The IAEA's safeguards conclusions, reported annually in the Safeguards Implementation Report, state whether the IAEA has found any indication of the diversion of declared nuclear material and, for states with additional protocols, whether it has found any indication of undeclared nuclear material or activities.

Types of Safeguards Agreements

Safeguards are implemented under agreements concluded between the IAEA and states. There are three principal types.

Comprehensive safeguards agreements (CSAs), concluded by non-nuclear-weapon states party to the NPT in accordance with Article III, are based on the IAEA document *The Structure and Content of Agreements between the Agency and States Required in Connection with the Treaty on the Non-Proliferation of Nuclear Weapons* (INFCIRC/153 (Corrected), 1972). Under a CSA, the state undertakes to accept safeguards on all source or special fissionable material in all peaceful nuclear activities within its territory, under its jurisdiction, or carried out under its control anywhere, for the exclusive purpose of verifying that such material is not diverted to nuclear weapons or other nuclear explosive devices.

The state must establish a State System of Accounting for and Control of nuclear material (SSAC), declare its nuclear material and facilities, and accept inspections. Many states with minimal or no nuclear material have concluded **small quantities protocols** suspending most of the detailed provisions of their CSAs; in 2005, the IAEA Board modified the standard text of the protocol to reinstate certain requirements, including the provision of design information on new facilities and the possibility of inspections.

Item-specific safeguards agreements, based on INFCIRC/66/Rev.2, apply to specified nuclear material, facilities, or equipment, and are concluded with states not party to the NPT — India, Pakistan, and Israel.

Voluntary offer agreements have been concluded by the five nuclear-weapon states, under which they offer some or all of their civil nuclear facilities for safeguards.

The Additional Protocol

The discovery after the 1991 Gulf War of an extensive clandestine nuclear weapons programme in Iraq — a party to the NPT with a CSA in force — revealed that safeguards focused on the verification of declared material were inadequate to detect undeclared activities. The IAEA responded with a programme of strengthened safeguards, culminating in the **Model Additional Protocol** (INFCIRC/540 (Corrected)), approved by the Board of Governors in 1997. Under an additional protocol, a state must provide an expanded declaration of its nuclear-related activities, including nuclear fuel cycle-related research and development, uranium mines and concentration plants, the manufacture of specified equipment, and the export of specified equipment and material; and must grant the IAEA "complementary access" to any location on the site of a facility, to other locations listed in the declaration, and to other locations specified by the IAEA to resolve questions or inconsistencies, on short notice. The additional protocol enables the IAEA to provide credible assurance of the absence of undeclared nuclear material and activities. Although a large majority of states with CSAs have brought additional protocols into force, the additional protocol is not legally required by the NPT, and some states have declined to conclude one. The NSG and some supplier states have debated whether the additional protocol should be a condition of supply.

Non-Compliance and the Security Council

Where the IAEA Board finds that a state is in non-compliance with its safeguards agreement, it may, under Article XII(C) of the Statute, report the non-compliance to the Security Council and the General Assembly. The Security Council may then take action under Chapter VII of the United Nations Charter, as it did in respect of Iraq, the Democratic People's Republic of Korea (DPRK), and Iran.

The DPRK announced its withdrawal from the NPT in 2003 under Article X, which permits withdrawal on three months' notice if extraordinary events related to the subject matter of the Treaty have jeopardised the supreme interests of the withdrawing state, and subsequently conducted nuclear weapons tests; the legal validity of the withdrawal remains contested, and the Security Council has imposed sanctions under Chapter VII.

Iran and the Joint Comprehensive Plan of Action

The case of Iran is the most extensively litigated and negotiated safeguards controversy. In 2003, the IAEA reported that Iran had failed to declare certain nuclear material and activities, including enrichment-related activities, in breach of its CSA. In 2005, the Board found Iran in non-compliance and in 2006 reported the matter to the Security Council, which adopted a series of resolutions under Chapter VII demanding that Iran suspend enrichment and imposing sanctions. In 2015, Iran concluded the **Joint Comprehensive Plan of Action** (JCPOA) with China, France, Germany, Russia, the United Kingdom, the United States, and the European Union, under which Iran accepted limits on its enrichment activities and stockpile, provisional application of its additional protocol, and enhanced monitoring, in exchange for the lifting of nuclear-related sanctions. The Security Council endorsed the JCPOA in Resolution 2231 (2015), which terminated the earlier resolutions while providing a "snapback" mechanism by which a JCPOA participant could trigger the reimposition of the previous sanctions.

The United States withdrew from the JCPOA in 2018 and reimposed its own sanctions, and Iran subsequently exceeded the JCPOA's limits, enriched uranium to sixty per cent, and reduced its cooperation with the IAEA. In June 2025, the IAEA Board found that Iran was in non-compliance with its safeguards obligations, and Israel and the United States carried out military strikes on Iranian nuclear facilities, raising grave questions under the law on the use of force; Iran subsequently suspended cooperation with the IAEA. In August 2025, France, Germany, and the United Kingdom invoked the snapback mechanism, and the previous United Nations sanctions were reimposed at the end of September 2025, although Russia and China contested the legality of the snapback. The Iranian case illustrates the dependence of the safeguards system on political support and the limits of verification without cooperation, and it is a case study in the interaction between technical verification, treaty interpretation, and the use of force.

Nuclear-Weapon-Free Zones and the Prohibition Treaty

Regional nuclear-weapon-free zones complement the NPT. The **Treaty of Pelindaba**, which established the African Nuclear-Weapon-Free Zone, prohibits the research, development, manufacture, stockpiling, acquisition, possession, and testing of nuclear explosive devices in the zone and the dumping of radioactive

waste; requires parties to conclude CSAs with the IAEA; requires the maintenance of the highest standards of physical protection; and prohibits armed attack on nuclear installations. It established the **African Commission on Nuclear Energy** (AFCONE), headquartered in South Africa, to ensure compliance and promote the peaceful uses of nuclear energy in Africa. Other zones cover Latin America and the Caribbean (Treaty of Tlatelolco, 1967), the South Pacific (Treaty of Rarotonga, 1985), South-East Asia (Treaty of Bangkok, 1995), and Central Asia (Treaty of Semipalatinsk, 2006).

The **Treaty on the Prohibition of Nuclear Weapons** (TPNW), adopted in 2017 and in force since January 2021, prohibits states parties from developing, testing, producing, acquiring, possessing, stockpiling, using, or threatening to use nuclear weapons. None of the nuclear-armed states is a party, and the relationship between the TPNW and the NPT is contested. Article 3 of the TPNW requires states parties to maintain, at a minimum, their existing safeguards obligations, and Article 4 provides for safeguards for states that eliminate their nuclear weapons. The legality of the threat or use of nuclear weapons under general international law was addressed by the International Court of Justice in its Advisory Opinion on the *Legality of the Threat or Use of Nuclear Weapons* (1996), in which it held, among other things, that there exists an obligation to pursue in good faith and bring to a conclusion negotiations leading to nuclear disarmament in all its aspects under strict and effective international control.

11.5 Nuclear Security and Physical Protection

The Concept of Nuclear Security

Nuclear security is defined in IAEA guidance as the prevention and detection of, and response to, criminal or intentional unauthorised acts involving or directed at nuclear material, other radioactive material, associated facilities, or associated activities. Its concerns include the theft of nuclear material for the purpose of constructing a nuclear explosive device; the theft of radioactive sources for use in a radiological dispersal device ("dirty bomb"); the sabotage of nuclear facilities or transports with a view to causing radiological release; and cyberattacks on the digital systems of nuclear facilities. Unlike safety, which addresses accidental events, security addresses deliberate acts by adversaries, and the two are complementary: measures designed for safety and for security must be coordinated so that each supports rather than undermines the other — for example, safety requires easy egress in an emergency, while security requires controlled access.

Nuclear security rose to the top of the international agenda after the terrorist attacks of 11 September 2001, which raised concerns about the potential for terrorists to acquire nuclear or radiological material or to attack nuclear facilities. The **Nuclear Security Summits** convened between 2010 and 2016 produced na-

tional commitments to minimise the use of highly enriched uranium, strengthen physical protection, and ratify the relevant conventions, and contributed to the entry into force of the 2005 Amendment to the CPPNM.

The Convention on the Physical Protection of Nuclear Material

The **Convention on the Physical Protection of Nuclear Material** (CPPNM), adopted in 1980 and in force since 1987, originally applied only to nuclear material used for peaceful purposes in international transport. It required parties to ensure that nuclear material in international transport within their territory, or on board ships or aircraft under their jurisdiction, was protected at specified levels; prohibited the export or import of nuclear material unless assurances of such protection had been received; and required parties to criminalise specified offences, including theft, embezzlement, and the threat to use nuclear material to cause death or serious injury or substantial property damage, and to establish jurisdiction over and either prosecute or extradite alleged offenders.

The **Amendment to the CPPNM**, adopted in 2005 and in force since 8 May 2016, substantially expanded the scope of the Convention. The amended Convention, sometimes called the Convention on the Physical Protection of Nuclear Material and Nuclear Facilities, extends to nuclear material in domestic use, storage, and transport, and to the protection of nuclear facilities against sabotage. It requires each party to establish, implement, and maintain an appropriate physical protection regime, consisting of a legislative and regulatory framework, a competent authority, and the responsibility of licence holders, and sets out twelve "fundamental principles" of physical protection, including the responsibility of the state, the responsibility of licence holders, legislative and regulatory framework, competent authority, security culture, threat evaluation (the "design basis threat"), a graded approach, defence in depth, quality assurance, contingency plans, and confidentiality. It adds new offences, including sabotage of a nuclear facility and the smuggling of nuclear material, and requires parties to cooperate in locating and recovering stolen or smuggled material.

The International Convention for the Suppression of Acts of Nuclear Terrorism

The **International Convention for the Suppression of Acts of Nuclear Terrorism** (ICSANT), adopted by the United Nations General Assembly in 2005 and in force since 2007, is one of the sectoral counter-terrorism conventions negotiated under United Nations auspices. It requires parties to criminalise the possession of radioactive material or the making or possession of a device with the intent to cause death or serious bodily injury, substantial damage to property or the environment, or to compel a natural or legal person, an international organisation, or a state to do or refrain from doing an act; the use of radioactive material or

a device, or the use of or damage to a nuclear facility in a manner which releases or risks the release of radioactive material, with such intent; and threats and demands relating to such acts. It applies the "prosecute or extradite" principle and provides for international cooperation. ICSANT applies to radioactive material in general, not only nuclear material, and to acts by any person, but excludes the activities of armed forces during an armed conflict, which are governed by international humanitarian law.

United Nations Security Council Resolution 1540

Security Council Resolution 1540 (2004), adopted under Chapter VII of the United Nations Charter, decided that all states shall refrain from providing any form of support to non-state actors that attempt to develop, acquire, manufacture, possess, transport, transfer, or use nuclear, chemical, or biological weapons and their means of delivery, and shall adopt and enforce appropriate effective laws prohibiting such activities, and appropriate effective measures to account for and secure such items, develop effective border controls, and establish export and trans-shipment controls. Resolution 1540 is notable as an instance of the Security Council imposing general legislative obligations on all states, rather than responding to a specific situation, and it has been described as "legislative" in character. Its implementation is monitored by a committee of the Security Council.

The Security of Radioactive Sources

Radioactive sources used in medicine, industry, and research present significant security risks because they are widely distributed, often poorly controlled, and potentially usable in radiological dispersal devices. Incidents involving lost or abandoned sources — such as the incident at Goiânia, Brazil, in 1987, in which an abandoned radiotherapy source was opened by scrap metal scavengers, causing several deaths and widespread contamination — illustrate the dangers. The IAEA's **Code of Conduct on the Safety and Security of Radioactive Sources** (2003), together with its supplementary Guidance on the Import and Export of Radioactive Sources and Guidance on the Management of Disused Radioactive Sources, provides non-binding guidance that a large number of states have committed politically to implement, including through national registers of sources, licensing, and the safe and secure management of disused sources.

In Nigeria, the NNRA regulates the security of radioactive sources, which are widely used in the oil and gas industry for well logging and in industrial radiography, as well as in medicine. The theft of radioactive sources from oil and gas operations has occurred in Nigeria and elsewhere, illustrating the intersection of nuclear security with the regulation of the extractive industries examined in Chapter 5.

Nuclear Security Law in Practice

The IAEA's **Nuclear Security Series** provides guidance on the implementation of nuclear security, including the *Nuclear Security Fundamentals* (2013) and *Nuclear Security Recommendations* on physical protection, on radioactive material and associated facilities, and on material out of regulatory control. Unlike the Safety Standards, nuclear security guidance is formally treated as guidance rather than standards, reflecting the sensitivity of security matters as an aspect of national sovereignty. The IAEA's **International Physical Protection Advisory Service** (IPPAS) offers peer reviews of national physical protection regimes. The confidentiality of security information means that nuclear security law is less transparent than safety law, which complicates public scrutiny and accountability. Cybersecurity has become a growing dimension of nuclear security, as nuclear facilities increasingly rely on digital instrumentation and control systems, and is examined in the context of energy infrastructure more generally in Chapter 12.

11.6 Nuclear Liability and Compensation

The Rationale for a Special Liability Regime

The potential for nuclear accidents to cause catastrophic and transboundary damage, the difficulty of proving causation for radiation-induced illnesses that may manifest decades later, and the reluctance of suppliers and insurers to participate in the nuclear industry without protection from unlimited liability led, from the late 1950s, to the development of special liability regimes for nuclear damage. These regimes were designed to serve two objectives: to ensure adequate compensation for victims of nuclear incidents, and to enable the development of the nuclear industry by limiting and channelling liability so that it could be insured.

The Principles of Nuclear Liability Law

The international nuclear liability conventions are based on a set of common principles.

Exclusive liability of the operator (channelling). Liability for nuclear damage is channelled exclusively to the operator of the nuclear installation in which the nuclear incident occurs, or from which the nuclear material involved originated. No other person — including suppliers of equipment, designers, and constructors — is liable, and the operator's right of recourse against other persons is limited to cases where recourse is expressly provided for by contract or where the damage results from an act or omission done with intent to cause damage. Channelling simplifies claims for victims, who need only identify the operator, and protects suppliers, avoiding the need for every participant in the nuclear industry to insure against the same risk.

Strict liability. The operator's liability is strict (absolute), in the sense that it does not depend on proof of fault or negligence. Exonerations are limited — principally to damage caused by acts of armed conflict, hostilities, civil war, or insurrection, and, under the older conventions, grave natural disasters of an exceptional character (an exoneration removed by the revised conventions).

Limitation of liability in amount. The operator's liability may be limited in amount, in exchange for strict and exclusive liability, with the possibility of state or collective funding beyond the operator's liability.

Limitation in time. Claims must be brought within specified time limits, which the revised conventions have extended to thirty years from the incident for claims in respect of loss of life and personal injury, recognising the latency of radiation-induced illness.

Compulsory financial security. The operator must maintain insurance or other financial security covering its liability.

Exclusive jurisdiction. Jurisdiction over claims lies exclusively with the courts of the state in whose territory the nuclear incident occurred, ensuring a single forum for the equitable distribution of compensation among all victims.

Non-discrimination. Compensation must be distributed without discrimination based on nationality, domicile, or residence.

The Paris and Vienna Systems

The **Paris Convention on Third Party Liability in the Field of Nuclear Energy** (1960), adopted under the auspices of the OECD Nuclear Energy Agency, applies principally among Western European states. It was supplemented by the **Brussels Supplementary Convention** (1963), which provides additional tiers of compensation from public funds of the installation state and from collective contributions of the parties. The **Protocols of 2004** amending the Paris and Brussels Conventions, which entered into force on 1 January 2022, raised the minimum liability of the operator to EUR 700 million, increased total compensation available under the Brussels tiers to EUR 1.5 billion, broadened the definition of nuclear damage to include economic loss, the costs of measures of reinstatement of impaired environment, loss of income deriving from an economic interest in the use or enjoyment of the environment, and the costs of preventive measures, and extended the time limit for personal injury claims to thirty years.

The **Vienna Convention on Civil Liability for Nuclear Damage** (1963), adopted under the auspices of the IAEA, was intended to be of worldwide application and applies principally among states in Central and Eastern Europe, Latin America, and elsewhere. The **1997 Protocol** amending the Vienna Convention, in force since 2003, raised the minimum liability amount to not less than 300 million Special Drawing Rights, broadened the definition of nuclear damage in

similar terms to the 2004 Paris Protocol, and extended the geographical scope of the Convention to damage suffered wherever, subject to exceptions.

The **Joint Protocol relating to the Application of the Vienna Convention and the Paris Convention** (1988), in force since 1992 and adopted in response to the Chernobyl accident, links the two systems, so that victims in a state party to one convention may claim against an operator in a state party to the other.

The Convention on Supplementary Compensation

The **Convention on Supplementary Compensation for Nuclear Damage** (CSC), adopted in 1997 and in force since 2015, was intended to establish a global liability regime capable of attracting states that are not party to either the Paris or the Vienna Convention, including the United States. It may be joined by parties to the Paris or Vienna Conventions, and by states whose national law complies with the provisions of its Annex, a "grandfather clause" designed to accommodate the economic channelling of the United States Price-Anderson Act. It requires installation states to ensure the availability of at least 300 million SDRs and provides for an international fund of contributions from all parties, calculated principally by reference to installed nuclear capacity. The CSC's parties include the United States, Japan, India, Canada, and the United Arab Emirates, as well as several other states. Japan's accession in 2015 was motivated in part by the desire to facilitate the participation of international companies in the decommissioning of Fukushima Daiichi by providing them with channelling protection.

National Liability Regimes

The United States pioneered nuclear liability legislation with the **Price-Anderson Act** of 1957, which requires operators of nuclear power plants to maintain the maximum available private insurance and to participate in a retrospective premium scheme under which each operator contributes to the compensation of damage caused by an incident at any plant, creating a pool of more than USD 15 billion. Price-Anderson employs "economic channelling" — liability is not legally channelled to the operator, but the insurance covers any person liable — and provides for the consolidation of claims in federal courts. In *Duke Power Co v Carolina Environmental Study Group* 438 US 59 (1978), the Supreme Court upheld the constitutionality of the Act's limitation of liability, holding that it was rationally related to the legitimate objective of encouraging private participation in nuclear energy and that it provided a reasonably just substitute for the common law remedies it replaced.

Japan's **Act on Compensation for Nuclear Damage** (1961) imposes unlimited, strict, and exclusive liability on operators, with government support. Following Fukushima, TEPCO's compensation liabilities far exceeded its capac-

ity, and the **Nuclear Damage Compensation Facilitation Corporation** (later the Nuclear Damage Compensation and Decommissioning Facilitation Corporation) was established in 2011 to provide financial assistance to TEPCO, funded by government bonds and contributions from the nuclear industry. Compensation was paid through direct claims to TEPCO, through the government's Dispute Reconciliation Committee for Nuclear Damage Compensation, which issued guidelines on the categories of compensable damage and conducted alternative dispute resolution, and through litigation. The total compensation paid by TEPCO exceeded ten trillion yen.

India's **Civil Liability for Nuclear Damage Act 2010** departed from the principle of exclusive channelling by granting the operator, in section 17(b), a right of recourse against the supplier where the nuclear incident resulted from the supply of equipment or material with patent or latent defects or substandard services. This provision, a response to the Bhopal gas disaster of 1984 and the inadequacy of the compensation it produced, deterred foreign suppliers from participating in the Indian nuclear market. India ratified the CSC in 2016, and the government issued explanations of the provision's scope, but the compatibility of section 17(b) with the CSC and its implications for suppliers remained controversial.

Adequacy of the Liability Regimes

The international nuclear liability regime has been criticised on several grounds. The liability limits, even as revised, are modest in comparison with the costs of major accidents: the damage caused by Chernobyl and Fukushima has been estimated in the hundreds of billions of US dollars. Participation in the conventions is fragmented, and many states with nuclear power programmes, including China and some newcomer states, are party to none of them; the absence of a treaty relationship means that victims in one state may be unable to claim against an operator in another. Channelling has been criticised for removing the incentive for suppliers to ensure the safety of their products. And the conventions' exclusive jurisdiction provisions may leave victims in neighbouring states dependent on the courts of the installation state. For newcomer states, including those in Africa, adherence to a liability convention — commonly the revised Vienna Convention or the CSC — is a practical prerequisite for attracting foreign suppliers and financing, and the IAEA's International Expert Group on Nuclear Liability (INLEX) has recommended that states establish treaty relations among themselves through adherence to the revised conventions and the CSC.

11.7 Radioactive Waste and Decommissioning

The Challenge of Radioactive Waste

Radioactive waste is generated at every stage of the nuclear fuel cycle and by the use of radioactive materials in medicine, industry, and research. It is classified according to its radioactivity and the duration of its hazard: very low level and low level waste, which may be disposed of in near-surface facilities; intermediate level waste, which requires greater containment; and high level waste, principally spent nuclear fuel (where it is treated as waste rather than reprocessed) and the residues of reprocessing, which generates heat and remains hazardous for many thousands of years. The management of high level waste is the most intractable problem of nuclear energy, both technically and politically, and the long timescales involved raise distinctive questions of intergenerational equity.

The International Framework

The **Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management** (1997) is the only legally binding international instrument addressing the safety of spent fuel and radioactive waste management on a global scale. It applies to spent fuel and radioactive waste resulting from civilian applications, and requires contracting parties to ensure that at all stages of management individuals, society, and the environment are adequately protected against radiological hazards, including by taking into account interdependencies among the different steps of management, providing for effective protection of individuals, society, and the environment, taking into account biological, chemical, and other hazards, striving to avoid actions that impose reasonably predictable impacts on future generations greater than those permitted for the current generation, and aiming to avoid imposing undue burdens on future generations (Articles 4 and 11). The preamble affirms that radioactive waste should, as far as is compatible with the safety of the management of such material, be disposed of in the state in which it was generated, while recognising that in certain circumstances safe and efficient management might be fostered through agreements among parties to use facilities in one of them.

Article 27 of the Joint Convention regulates transboundary movements of spent fuel and radioactive waste, requiring prior notification to and consent of the state of destination, and prohibiting shipments to destinations south of latitude 60 degrees South for storage or disposal. The Bamako Convention on the Ban of the Import into Africa and the Control of Transboundary Movement and Management of Hazardous Wastes within Africa (1991) prohibits the import of hazardous wastes, including radioactive wastes, into Africa from non-parties, and the Treaty of Pelindaba prohibits the dumping of radioactive wastes in the African Nuclear-Weapon-Free Zone. The dumping of radioactive waste at sea is

prohibited by the 1996 Protocol to the London Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter (1972). The Joint Convention reflects the principle of national responsibility for radioactive waste, which is also expressed in Article 4 of the Euratom Radioactive Waste and Spent Fuel Directive (Council Directive 2011/70/Euratom), under which each member state has ultimate responsibility for management of the spent fuel and radioactive waste generated in it, and must establish a national policy and national programme. Article 4(4) of the Directive provides that radioactive waste shall be disposed of in the member state in which it was generated, unless an agreement has been concluded for the use of a disposal facility in another member state or a third country, subject to strict conditions.

Deep Geological Disposal

There is a broad scientific consensus that deep geological disposal — the emplacement of high level waste and spent fuel in engineered repositories several hundred metres underground in stable geological formations, relying on a combination of engineered and natural barriers to isolate the waste for the required period — is the most appropriate long-term solution for high level waste. Progress in implementing geological disposal has nevertheless been slow, owing principally to public opposition and the difficulty of securing the consent of host communities.

Finland is the first country to construct a deep geological repository for spent fuel. The Onkalo repository at Olkiluoto, developed by Posiva, received a construction licence in 2015 following a decision-in-principle ratified by Parliament in 2001, and commenced trial operations in 2024. The Finnish process has been regarded as a model because it combined a staged decision-making process, the right of the host municipality (Eurajoki) to veto the decision-in-principle, an independent and trusted regulator (the Radiation and Nuclear Safety Authority, STUK), and the long-term commitment of the waste producers. **Sweden** approved the construction of a repository at Forsmark in 2022, following a process in which the host municipality consented. **France** has pursued the Cigéo project at Bure under legislation adopted in 1991, 2006, and 2016, which introduced the concept of reversibility — the ability to retrieve waste for a period after emplacement — into the legal framework. The emphasis in these processes on voluntarism, consent-based siting, and staged decision-making reflects lessons learned from earlier failures.

The United States illustrates the consequences of a failed siting process. The **Nuclear Waste Policy Act** of 1982 established a federal responsibility for the disposal of spent fuel and required utilities to pay a fee into a Nuclear Waste Fund in exchange for a federal commitment to begin accepting spent fuel by 31 January 1998. The 1987 amendments to the Act designated Yucca Mountain in Nevada as the sole site to be characterised, over Nevada's opposition. The

Department of Energy applied for a construction licence in 2008, but the Obama administration sought to withdraw the application in 2010 and ceased funding. In *In re Aiken County* 725 F.3d 255 (DC Cir 2013), the Court of Appeals for the District of Columbia Circuit issued a writ of mandamus requiring the Nuclear Regulatory Commission to continue the licensing process with the funds available, holding that the Commission could not decline to comply with the statutory mandate on policy grounds. The project nevertheless remained stalled. The federal government's failure to accept spent fuel by 1998 has resulted in extensive litigation by utilities for breach of contract, in which the government has been held liable for billions of dollars in damages for the costs of on-site storage, and the fee was suspended in 2014 following a ruling of the D.C. Circuit. Efforts to license private interim storage facilities were addressed by the Supreme Court in *Nuclear Regulatory Commission v Texas* (2025), which held that the State of Texas and other challengers, which had not been parties to the licensing proceeding, were not entitled to seek judicial review of a licence granted for an interim storage facility.

Decommissioning

Decommissioning comprises the administrative and technical actions taken to allow the removal of some or all regulatory controls from a nuclear facility at the end of its life, including decontamination, dismantling, and the management of resulting waste. Strategies include **immediate dismantling**, in which decommissioning commences shortly after permanent shutdown, and **deferred dismantling** (safe enclosure), in which the facility is maintained in a safe condition for several decades to allow radioactive decay before dismantling. IAEA safety standards and the Euratom Directives increasingly favour immediate dismantling, on grounds including the availability of staff familiar with the facility and the avoidance of transferring burdens to future generations.

Financing. The legal regulation of decommissioning financing is critical, because decommissioning costs are incurred decades after the revenue from operation has been earned, and there is a risk that operators will become insolvent or that funds will be inadequate. The principal models are segregated external funds, in which operators make contributions during operation to an independently managed fund; internal provisions on the operator's balance sheet; and state funding. Article 9 of the Euratom Radioactive Waste Directive requires member states to ensure that the national framework requires that adequate financial resources be available for the implementation of national programmes, with due regard to the responsibility of waste generators. The United Kingdom's arrangements for new nuclear power stations, under the Energy Act 2008, require operators to have an approved Funded Decommissioning Programme before construction may begin.

The German Nuclear Phase-Out

Germany's phase-out of nuclear energy illustrates the legal consequences of the political decision to end nuclear power. In 2002, the government and the nuclear operators reached a consensus, implemented by legislation, under which each plant was allocated a residual quantity of electricity it could produce. In 2010, the government extended the operating lives of the plants, but following the Fukushima accident in 2011, the Thirteenth Amendment to the Atomic Energy Act revoked the extension and required all plants to close by the end of 2022. The operators challenged the amendment before the Federal Constitutional Court, and the Swedish state-owned company Vattenfall also brought a claim under the Energy Charter Treaty (*Vattenfall AB and others v Germany*, ICSID Case No ARB/12/12).

In its judgment of 6 December 2016, the Federal Constitutional Court held that the accelerated phase-out was largely compatible with the Basic Law: it did not constitute an expropriation, and it was, in principle, a permissible determination of the content and limits of property under Article 14 of the Basic Law, justified by the legislature's reassessment of the risks of nuclear energy after Fukushima. However, the Court held that the amendment was unconstitutional in so far as it failed to provide compensation for the residual electricity quantities allocated in 2002 that certain operators could not use, and for investments made in reliance on the 2010 extension. Germany enacted compensation legislation, and in 2021 reached a settlement with the operators, under which they received approximately EUR 2.4 billion in compensation, including around EUR 1.4 billion for Vattenfall, which discontinued its ECT claim. The last three German nuclear power plants closed in April 2023, after a brief extension due to the energy crisis.

Separately, the **Act on the Reorganisation of Responsibility in Nuclear Waste Management** (2017) transferred responsibility for the interim and final storage of radioactive waste from the operators to the state, in exchange for the payment by the operators of approximately EUR 24 billion into a public fund (KENFO), while the operators remained responsible for decommissioning and dismantling their plants and for packaging the waste. This arrangement illustrates a model for the allocation of long-term liability between private operators and the state.

11.8 Small Modular Reactors and Emerging Legal Issues

The Promise of Small Modular and Advanced Reactors

Small modular reactors (SMRs) are nuclear reactors with a power capacity of up to around 300 MW(e) per unit, designed to be manufactured in factories and transported to sites for installation, in contrast to large reactors constructed

on-site. Proponents claim that SMRs offer lower upfront capital costs, shorter construction times, enhanced safety through passive safety features and smaller radioactive inventories, siting flexibility (including at the sites of retiring coal plants and in remote locations), and suitability for smaller electricity grids, including those of many developing countries. **Advanced reactors** — including high-temperature gas-cooled reactors, molten salt reactors, sodium-cooled fast reactors, and microreactors of a few megawatts — employ non-light-water technologies with potential advantages in safety, efficiency, waste management, and industrial heat applications. Several SMR designs have progressed towards deployment: the BWRX-300 at Darlington in Ontario, Canada, received a licence to construct from the Canadian Nuclear Safety Commission in 2025; the United Kingdom selected a Rolls-Royce SMR design for deployment in 2025; and the floating nuclear power plant *Akademik Lomonosov*, comprising two small reactors mounted on a barge, has operated in the Russian Arctic since 2019–2020. Technology companies have concluded agreements to purchase electricity from SMRs and other advanced reactors to power data centres.

Licensing Challenges

Existing licensing frameworks were developed for large light-water reactors and may not be well suited to SMRs and advanced reactors, whose designs, safety features, and fuel cycles differ substantially. Regulators have responded with technology-inclusive and risk-informed approaches. In the United States, the **Nuclear Energy Innovation and Modernization Act** of 2019 required the Nuclear Regulatory Commission to develop a technology-inclusive regulatory framework for advanced reactors, and the **ADVANCE Act** of 2024 directed the Commission to modernise its mission and licensing processes, reduce fees for advanced reactor applicants, and expedite reviews. The NRC approved the design of NuScale's SMR in 2020 and an uprated version in 2025, the first SMR designs certified or approved in the United States. In the United Kingdom, the Generic Design Assessment process permits the assessment of reactor designs independently of specific sites.

Key licensing issues include the scope and size of **emergency planning zones** (which proponents argue can be reduced for SMRs, given their smaller inventories and passive safety, enabling siting closer to population centres and industrial users); **staffing requirements** and remote or autonomous operation; the treatment of multi-module plants; **security requirements** proportionate to the risk; and the qualification of **novel fuels**, such as high-assay low-enriched uranium (HALEU), which raises safeguards and security considerations.

International Harmonisation

The economic case for SMRs depends on serial factory production of standardised designs, which in turn depends on the ability to deploy the same design in multiple countries without extensive redesign to meet different national requirements. Licensing has traditionally been an exclusively national prerogative, and differences in national requirements are a significant barrier to SMR deployment. The IAEA launched the **Nuclear Harmonization and Standardization Initiative** (NHSI) in 2022 to promote regulatory collaboration and industrial standardisation for SMRs, and regulators have entered into cooperation arrangements, including agreements between the United States and Canadian regulators on joint reviews of advanced reactor designs. Proposals for mutual recognition of design assessments, or for an international design certification, raise questions of sovereignty and regulatory accountability: under the CNS and IAEA standards, responsibility for safety cannot be delegated by the host state, which must retain an independent and competent regulatory body capable of making its own judgments.

Transportable and Floating Reactors

Transportable reactors — including floating nuclear power plants and microreactors designed to be moved between sites — raise novel legal questions under international nuclear law, which was developed for stationary installations. These include the determination of the "installation state" for the purposes of safety, safeguards, and liability; the application of the liability conventions to reactors in transit or on the high seas or in the territorial sea of a foreign state; the application of the law of the sea to floating reactors (including innocent passage and the rights of coastal states); the allocation of safeguards responsibilities where a reactor is fuelled in one state and operated in another; and the security of reactors in transit. The IAEA has studied these questions, but the international legal framework remains incomplete.

Safeguards by Design and Proliferation Considerations

SMRs and advanced reactors raise safeguards considerations relating to the use of higher-enriched fuels, novel fuel forms (such as pebbles or molten salt fuels, which complicate item counting), long-life cores that are not refuelled for many years, and deployment in large numbers in many locations, including in states with limited nuclear infrastructure. The IAEA promotes "safeguards by design", the integration of safeguards considerations into the design of nuclear facilities from the earliest stages, and the participation of the IAEA in the design process.

Newcomer States and Nuclear Governance in Africa

SMRs are frequently marketed to newcomer states, including in Africa, as a means of introducing nuclear power at a scale compatible with smaller grids

and financing capacities. However, the IAEA Milestones Approach applies equally to SMRs: a newcomer state must establish a comprehensive legal framework covering safety, security, safeguards, and liability; an independent and competent regulator; human resources; and arrangements for waste management, decommissioning, and financing, regardless of the size of the reactor. The vendor-financed "build-own-operate" model, under which a foreign state-owned vendor finances, builds, owns, and operates a nuclear plant in the host state, as in Rosatom's Akkuyu project in Turkey, may reduce the burden on the host state but raises questions of long-term dependence, regulatory independence from a powerful foreign operator, and the allocation of liability for waste and decommissioning. Egypt's El Dabaa project, under construction with Rosatom since 2022 with four VVER-1200 reactors and financed largely by a Russian state loan, will be the largest nuclear project in Africa. South Africa's Koeberg plant, the only operating nuclear power plant in Africa, received a twenty-year extension of the operating licence of its first unit in 2024. The development of nuclear governance in Africa, including the role of AFCONE and regional cooperation in regulation, is a significant subject for doctoral research.

Financing, Taxonomy, and Market Design

The cost and financing of nuclear power are as important as its regulation. Large nuclear projects have high capital costs, long construction times, and a history of cost overruns, which make them difficult to finance in liberalised markets without state support. Governments have adopted a range of support mechanisms: CfDs, as used for Hinkley Point C in the United Kingdom; the **regulated asset base** (RAB) model, introduced in the United Kingdom by the Nuclear Energy (Financing) Act 2022 and applied to the Sizewell C project, under which investors receive a regulated return during construction funded by consumers; direct state ownership, as in France's full nationalisation of EDF in 2023; and state loans and guarantees. The inclusion of nuclear energy in the EU taxonomy, the removal of the World Bank's longstanding restriction on financing nuclear projects in 2025, and the pledges made at COP28 to triple nuclear capacity have improved the prospects of access to sustainable finance.

::: comparative {title="Comparative Perspective: Regulatory Models for Nuclear Newcomers"} Newcomer states have adopted different institutional models for nuclear regulation, each with distinctive strengths and risks.

- **The United Arab Emirates** enacted a comprehensive Federal Law by Decree No 6 of 2009 on the Peaceful Uses of Nuclear Energy, establishing the Federal Authority for Nuclear Regulation as an independent regulator before contracting for its first plant. It renounced domestic enrichment and reprocessing in a cooperation agreement with the United States, brought

an additional protocol into force, and acceded to the CSC and the Vienna Convention. The Barakah plant, built by a Korean consortium, entered commercial operation between 2021 and 2024. The UAE is frequently cited as a model of sequencing, establishing law and institutions before construction.

- **Turkey**, under the Akkuyu intergovernmental agreement with Russia (2010), adopted a build-own-operate model in which a Russian-owned project company owns and operates the plant. This model reduces the host state's financial burden but places heavy demands on the regulator's capacity to oversee a powerful foreign operator.
- **Bangladesh**, constructing the Rooppur plant with Russian financing and technology, enacted the Bangladesh Atomic Energy Regulatory Act 2012 to separate regulation from promotion. It has relied heavily on IAEA support and the vendor's regulatory documentation.
- **Ghana** created an independent Nuclear Regulatory Authority by the Nuclear Regulatory Authority Act 2015 (Act 895), separating regulation from the Ghana Atomic Energy Commission. It has established Nuclear Power Ghana as an owner-operator entity and has considered both large reactors and SMRs.
- **Nigeria** separated promotion (NAEC) from regulation (the NNRA) by statute in 1995, well before any decision to construct a power reactor. Commentators have identified the modernisation of the 1995 Act, the strengthening of the NNRA's financial independence, and adherence to a liability convention as priorities before any power reactor project proceeds.

The common lesson is that the legal framework must precede, rather than follow, the commitment to construct. The regulator's independence and competence also depend as much on funding, staffing, and culture as on statutory form. :::

::: exercise {title="Practical Exercise: The Republic of Kesena SMR Programme (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic of Kesena, the Kesena Atomic Energy Agency, NordReactor Corporation, and all other entities named are fictional.*

The Republic of Kesena is a West African state with an electricity grid of 4,500 MW peak demand. It is a party to the NPT and the Treaty of Pelindaba, and has a comprehensive safeguards agreement with a small quantities protocol but no additional protocol. It is not a party to any nuclear liability convention. Its Atomic Energy Act 1988 establishes the Kesena Atomic Energy Agency (KAEA). The KAEA is responsible for both "the promotion of nuclear energy" and "the licensing and inspection of nuclear facilities", and its Director General reports to the Minister of Energy, who chairs the national nuclear power programme.

The government has signed a memorandum of understanding with NordReactor Corporation (HYPOTHETICAL), a state-owned vendor from a non-African state. Under the memorandum, NordReactor will build, own, and operate two 150 MW(e) SMR units at Port Abeni, 40 km from the capital, and will sell electricity to the state utility under a twenty-five-year PPA. The draft intergovernmental agreement includes the following provisions:

- (a) "The Republic shall recognise the design certification issued by the vendor state's regulator as satisfying all Kesenan licensing requirements relating to reactor design."
- (b) "The Republic shall indemnify NordReactor and its suppliers against all claims for nuclear damage in excess of USD 50 million."
- (c) "Spent fuel shall be stored at the site for the life of the plant. Responsibility for its final disposal shall be agreed by the Parties in due course."
- (d) "NordReactor may, with the consent of the Minister, relocate a third floating SMR unit to Kesenan territorial waters to supply the Port Abeni industrial zone."

Answer the following:

1. Conduct a gap analysis of Kesena's legal framework against the Convention on Nuclear Safety, the IAEA Safety Fundamentals, and the IAEA Milestones Approach. What legislative and institutional reforms should be made before a construction licence is issued?
2. Is clause (a) compatible with the principle that the host state retains responsibility for safety and must maintain an independent and competent regulator? Redraft it to permit reliance on the vendor state's assessment without abdicating Kesena's regulatory responsibility.
3. Advise Kesena on whether to accede to the revised Vienna Convention, the CSC, or both. Evaluate clause (b) in the light of the principles of channelling and the position of potential victims in neighbouring Pelindaba states.
4. Evaluate clause (c) against the Joint Convention, the Treaty of Pelindaba, and the Bamako Convention. Draft an alternative clause addressing spent fuel management, decommissioning, and their financing.
5. What safeguards, security, and law of the sea issues are raised by clause (d)? Should Kesena bring an additional protocol into force and rescind its small quantities protocol before proceeding? :::

CHAPTER SUMMARY

- The international nuclear order rests on the bargain of the NPT. Non-nuclear-weapon states renounce nuclear weapons and accept safeguards, in exchange for the inalienable right to the peaceful uses of nuclear energy under Article IV, which must be exercised in conformity with non-proliferation obligations.

- International nuclear law is organised around safety, security, safeguards, and liability, together with waste management and decommissioning. Each pillar has its own conventions, and the IAEA, with its dual mandate of promotion and control, is the central institution.
- The detailed rules of nuclear safety are found in the IAEA Safety Standards. These are not binding as such but acquire legal effect through incorporation into national law, treaty review processes, and Euratom directives.
- The prime responsibility for safety rests with the operator. The Convention on Nuclear Safety requires an effective separation between the regulator and bodies promoting nuclear energy. Fukushima demonstrated that regulatory independence and the continuous reassessment of external hazards are substantive safety requirements.
- Safeguards agreements, strengthened by the Additional Protocol, verify the non-diversion of nuclear material. The cases of Iraq, the DPRK, and Iran show both the importance and the political limits of verification.
- The amended CPPNM, ICSANT, and Security Council Resolution 1540 form the core of nuclear security law. The security of radioactive sources is a significant practical concern, including in the oil and gas industry.
- The nuclear liability conventions channel strict, limited, and insured liability to the operator in a single forum. Their limits, fragmented participation, and the Indian departure from channelling are central issues, and newcomers need liability treaty relations to attract suppliers.
- Deep geological disposal is the accepted solution for high level waste, and Finland leads its implementation. Consent-based siting and secure financing of waste management and decommissioning are critical, as the United States and German experiences demonstrate.
- SMRs and advanced reactors raise issues of licensing modernisation, international harmonisation, emergency planning, transportable reactors, and safeguards by design. They do not relieve newcomer states of the obligation to establish a comprehensive legal and institutional infrastructure.

DOCTORAL RESEARCH QUESTIONS

1. Is the IAEA's dual mandate to promote and to regulate nuclear energy compatible with effective international oversight of nuclear safety? Should international nuclear safety standards become legally binding?
2. Evaluate the legal significance of the IAEA Board's findings of non-compliance and the invocation of the JCPOA snapback mechanism. What do the Iranian and North Korean cases reveal about the enforcement of non-proliferation obligations?

3. Does the principle of legal channelling in the nuclear liability conventions remain justified, or should suppliers bear liability for defects in the equipment and services they provide? Assess India's approach in this light.
4. How should international law respond to the operation of nuclear power plants in zones of armed conflict? Is there a need for a new instrument protecting nuclear installations in armed conflict?
5. Critically assess the readiness of the legal and institutional frameworks of one or more African states to host nuclear power, with reference to the IAEA Milestones Approach and the vendor-financed build-own-operate model.
6. Can international nuclear law accommodate transportable and floating reactors without amendment of the principal conventions? Identify the key gaps and propose solutions.

FURTHER READING

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12 CHAPTER

Hydrogen, Energy Storage, and Emerging Energy Technologies

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Analyse the regulatory classification of hydrogen and the legal frameworks governing its production, transport, and use.
2. Distinguish the principal hydrogen production pathways and evaluate the legal criteria — including additionality, temporal correlation, and carbon intensity thresholds — by which "renewable" and "low-carbon" hydrogen are defined.
3. Assess the law of hydrogen certification and the emerging framework for international trade in hydrogen and its derivatives.
4. Explain the legal treatment of electricity storage, including its definition, ownership restrictions, market participation, and network charging.
5. Evaluate the legal frameworks for carbon capture, utilisation, and storage, including the London Protocol, storage permitting, pore space rights, and long-term liability.
6. Analyse the international and domestic law applicable to offshore renewable energy technologies, including floating wind, wave and tidal energy, and offshore grids.
7. Assess the legal issues raised by smart grids and the digitalisation of energy systems, including data protection and cybersecurity.
8. Evaluate the use of regulatory sandboxes and other instruments for accommodating technological innovation in energy regulation.

Introduction

The decarbonisation of the energy system requires not only the replacement of fossil-fuelled electricity generation with renewable and other low-carbon sources, examined in Chapters 10 and 11, but also the transformation of the infrastructure, markets, and technologies through which energy is converted, stored, transported, and used. Electricity from variable wind and solar resources must be

stored or balanced; sectors that are difficult to electrify — steel, chemicals, fertilisers, shipping, and aviation — require low-carbon fuels and feedstocks; residual emissions from industrial processes must be captured and stored; offshore resources must be harnessed and connected; and an increasingly decentralised and digitalised energy system must be managed securely. Hydrogen, electricity storage, carbon capture and storage, offshore renewable technologies, and smart grids are the principal technological responses to these challenges.

These emerging technologies pose a common legal problem: the law was designed for an energy system with different technologies, actors, and boundaries. Electricity law presupposed a clear separation between generation, transmission, distribution, and consumption, which storage, which can act as each in turn, does not respect. Gas law was designed for natural gas, and its application to hydrogen raises questions of classification, network regulation, and safety. Mining and petroleum law regulates the extraction of substances from the subsurface, and must be adapted to regulate the injection of carbon dioxide into it. The law of the sea was developed for navigation, fishing, and seabed resources, and must accommodate large-scale offshore energy infrastructure. And energy regulation was designed for a system of large, centrally controlled assets, not a system of millions of connected devices exchanging data. The legal response has been a combination of the adaptation of existing frameworks, the creation of new regimes, and the use of experimental instruments such as regulatory sandboxes.

This chapter examines hydrogen production and regulatory classification (section 12.1), hydrogen production pathways and their legal definition (section 12.2), hydrogen certification and cross-border trade (section 12.3), battery storage and grid integration (section 12.4), carbon capture, utilisation, and storage (section 12.5), offshore renewable energy technologies (section 12.6), smart grids, digital infrastructure, and cybersecurity (section 12.7), and regulatory sandboxes and technological innovation (section 12.8).



Figure 12.1 The hydrogen value chain and the principal legal questions arising at each stage. *Source:* prepared for this edition.

12.1 Hydrogen Production and Regulatory Classification

The Role of Hydrogen in the Energy Transition

Hydrogen is the most abundant element in the universe, but it occurs on Earth almost entirely in compounds, principally water and hydrocarbons, from which it must be separated using energy. It is therefore an energy carrier rather than a primary energy source. Hydrogen has been produced and used for more than a century as an industrial feedstock, principally in oil refining and in the production of ammonia for fertilisers and methanol, and global production amounts to around one hundred million tonnes per year, almost all of it produced from natural gas and coal without carbon capture, resulting in substantial greenhouse gas emissions.

The interest in hydrogen as an instrument of decarbonisation rests on its potential, when produced with low emissions, to replace fossil fuels in applications that are difficult to electrify directly: as a feedstock and reducing agent in steel

production (replacing coking coal), as a feedstock in ammonia and methanol production, as a fuel for shipping (directly or through derivatives such as ammonia and methanol), aviation (through synthetic fuels), and heavy road transport, and as a medium for the long-duration storage of energy. Its role in other applications, such as residential heating and light passenger vehicles, is contested, because direct electrification is generally more efficient. Many governments have adopted national hydrogen strategies, and several resource-rich states, including Australia, Chile, Morocco, Namibia, Egypt, Oman, and Saudi Arabia, aspire to become exporters of hydrogen and its derivatives.

The legal framework for hydrogen is at an early stage of development. Hydrogen raises legal questions at each stage of its value chain: production (permitting, electricity and water supply, and the definition of "renewable" or "low-carbon" hydrogen); conversion (into ammonia, methanol, or other carriers); transport and storage (by pipeline, ship, or truck, and in underground salt caverns or depleted reservoirs); trade (certification and the treatment of hydrogen under trade law); and end use (safety standards, demand-side mandates, and support schemes).

The Problem of Classification

The threshold legal question is how hydrogen is to be classified. Existing legal categories — "gas", "natural gas", "fuel", "chemical", "dangerous substance", and "energy product" — have been defined with other substances in mind, and the classification of hydrogen determines which regulatory regimes apply to it.

Hydrogen as a gas. In some jurisdictions, hydrogen falls within the definition of "gas" for the purposes of gas legislation, bringing hydrogen networks within existing frameworks of licensing, network regulation, and safety. The United Kingdom's Gas Act 1986 defines "gas" broadly to include any substance in a gaseous state consisting wholly or mainly of methane, ethane, propane, butane, hydrogen, or carbon monoxide, among other things, so that the conveyance of hydrogen is, in principle, within the licensing regime. In other jurisdictions, gas legislation is limited to "natural gas" or "methane", and hydrogen is excluded.

Hydrogen as a chemical and dangerous substance. Hydrogen is a flammable gas with a wide flammability range and low ignition energy, and is subject to the legislation governing dangerous substances, industrial safety, and the prevention of major accidents. In the European Union, the **Seveso III Directive** (Directive 2012/18/EU) on the control of major-accident hazards involving dangerous substances applies to establishments where hydrogen is present above specified thresholds, and the **ATEX Directives** (Directives 2014/34/EU and 1999/92/EC) govern equipment and workplaces in potentially explosive atmospheres. Hydrogen is also classified under chemicals legislation, including the

EU's REACH Regulation, and under the international rules on the transport of dangerous goods.

Hydrogen in the EU gas market framework. The European Union adopted the **Hydrogen and Decarbonised Gas Market Package** in 2024, comprising Directive (EU) 2024/1788 on common rules for the internal markets for renewable gas, natural gas and hydrogen, and Regulation (EU) 2024/1789 on the internal markets for renewable gas, natural gas and hydrogen. The package establishes, for the first time, a dedicated regulatory framework for hydrogen networks, distinct from that for natural gas. It provides for the vertical unbundling of hydrogen network operators from hydrogen production and supply, using models analogous to those for gas and electricity examined in Chapters 6 and 9; for regulated third-party access to hydrogen networks from 2033, with negotiated access permitted until then; for the certification of hydrogen network operators; for the establishment of a European Network of Network Operators for Hydrogen (ENNOH); for integrated network planning; and for rules on the blending of hydrogen into natural gas networks, under which transmission operators must accept flows with up to two per cent hydrogen by volume at interconnection points. The package also defines "low-carbon hydrogen" — hydrogen whose energy content is derived from non-renewable sources and which meets a greenhouse gas emissions reduction threshold of seventy per cent compared with fossil fuel — and provides for its certification.

Hydrogen pipelines in the United States. The regulation of hydrogen pipelines in the United States illustrates the difficulties of fitting hydrogen into existing categories. Hydrogen pipelines are subject to safety regulation by the Pipeline and Hazardous Materials Safety Administration, but the federal authority responsible for their economic regulation is unclear: the Federal Energy Regulatory Commission's jurisdiction under the Natural Gas Act extends to "natural gas", which may not include pure hydrogen, while the Surface Transportation Board's residual jurisdiction over pipelines transporting commodities other than water, gas, or oil may extend to hydrogen. Blends of hydrogen with natural gas transported in natural gas pipelines are generally considered to remain within FERC's jurisdiction.

Production Permitting

The production of hydrogen by electrolysis requires an electrolyser — industrial equipment that uses electricity to split water into hydrogen and oxygen — together with a supply of electricity and water. Electrolysers are typically permitted as industrial installations, subject to land use, environmental, and major-accident hazard requirements. Large electrolysers above specified thresholds may be subject to the EU's Industrial Emissions Directive and environmental impact assessment, and RED III requires member states to ensure that permit-granting procedures

for renewable hydrogen production and for associated infrastructure are streamlined. Production from natural gas with carbon capture requires, in addition, the permitting of capture facilities and carbon dioxide transport and storage, examined in section 12.5.

Water. Electrolysis requires around nine litres of purified water per kilogram of hydrogen, in addition to cooling water. Although the water requirements of hydrogen production are modest relative to agriculture, large-scale production in water-scarce regions — including many of the regions with the best solar resources, such as North Africa, the Middle East, Namibia, and Australia — may require desalination, raising issues of water rights, the disposal of brine, and the allocation of scarce water between export industries and local needs. Water allocation law, examined in Chapter 10, therefore forms part of the legal framework for hydrogen.

Land and benefit-sharing. Export-scale hydrogen projects in developing countries involve the allocation of large areas of land for renewable generation, electrolysers, and port facilities, raising issues of land rights, environmental protection, and the sharing of benefits with host states and communities. Namibia's flagship green hydrogen project in the Tsau //Khaeb National Park near Lüderitz, for which the government concluded a feasibility and implementation agreement with a private developer and acquired an option to take an equity stake, has attracted debate about the location of an industrial project in a national park, the terms of the state's participation, and the distribution of benefits. These questions echo those examined in the context of extractive industries in Chapters 5 and 7, and resource nationalism and local content in the hydrogen sector are emerging themes of energy law in Africa.

Nigeria. Nigeria does not yet have a dedicated legal framework for hydrogen. Hydrogen production from natural gas is undertaken incidentally in refining and in the fertiliser industry, and the PIA's provisions on gas utilisation and midstream licensing may apply to some hydrogen-related activities. Nigeria's abundant natural gas, solar resources, and existing fertiliser and ammonia industries, together with the Energy Transition Plan's identification of hydrogen as a potential decarbonisation pathway, suggest that the development of a hydrogen strategy and legal framework is a subject of growing interest.

12.2 Green, Blue, and Other Hydrogen Production Pathways

The Colour Spectrum

Hydrogen production pathways are commonly described using a spectrum of colours. **Grey hydrogen** is produced from natural gas by steam methane reforming without carbon capture, and accounts for most current production; **brown** or **black hydrogen** is produced from coal by gasification. **Blue hydrogen**

is produced from natural gas with the capture and storage of a large share of the resulting carbon dioxide. **Green hydrogen** is produced by electrolysis using renewable electricity. **Pink** or **purple hydrogen** is produced by electrolysis using nuclear electricity. **Turquoise hydrogen** is produced by methane pyrolysis, which splits methane into hydrogen and solid carbon. **White** or **natural hydrogen** is hydrogen occurring naturally in geological formations, whose potential has attracted recent exploration interest.

The colour terminology is a useful shorthand, but it is legally imprecise and potentially misleading. The climate impact of hydrogen depends not on its colour but on its life-cycle greenhouse gas emissions, which vary within each colour: green hydrogen produced by electrolysers drawing electricity from a grid dominated by fossil fuels may have higher emissions than grey hydrogen, and blue hydrogen's emissions depend on the capture rate and the upstream methane emissions from natural gas production. Law and policy have therefore moved towards **carbon intensity** approaches, which define hydrogen by reference to maximum life-cycle emissions per unit of hydrogen, sometimes combined with criteria relating to the source of energy.

The EU Definition of Renewable Hydrogen

The European Union has adopted the most detailed legal definition of renewable hydrogen. Under RED II and RED III, hydrogen and other fuels produced from renewable electricity are classified as **renewable fuels of non-biological origin** (RFNBOs), which count towards renewable energy targets, including RED III's binding target that at least forty-two per cent of the hydrogen used in industry should be RFNBOs by 2030 and sixty per cent by 2035, and its sub-target for RFNBOs in transport. To qualify as RFNBOs, fuels must satisfy two Commission Delegated Regulations adopted in 2023.

Delegated Regulation (EU) 2023/1184 establishes rules for determining when electricity used to produce RFNBOs may be counted as fully renewable. Electricity taken directly from a renewable installation connected to the electrolyser may be counted as renewable if the installation came into operation no earlier than thirty-six months before the electrolyser. Electricity taken from the grid may be counted as fully renewable if the bidding zone in which the electrolyser is located has a share of renewable electricity above ninety per cent, if the emissions intensity of the electricity is below a specified threshold, or if the producer has concluded renewable PPAs that satisfy three criteria:

- **Additionality**: the renewable installations from which the electricity is sourced must have come into operation no earlier than thirty-six months before the electrolyser, and must not have received operating or investment aid, so that the hydrogen production is matched by new, unsubsidised

renewable capacity rather than diverting existing renewable electricity from other uses. Electrolysers in operation before 1 January 2028 are exempted from the additionality requirement until 1 January 2036.

- **Temporal correlation**: the renewable electricity must be produced in the same calendar month as the hydrogen until 31 December 2029, and in the same one-hour period from 1 January 2030.
- **Geographical correlation**: the renewable installation must be located in the same bidding zone as the electrolyser, in an interconnected bidding zone with higher electricity prices, or in an offshore bidding zone interconnected with the electrolyser's bidding zone.

Delegated Regulation (EU) 2023/1185 establishes the methodology for assessing the greenhouse gas emissions savings of RFNBOs and recycled carbon fuels, and sets a minimum emissions savings threshold of seventy per cent compared with a fossil fuel comparator of 94 grams of carbon dioxide equivalent per megajoule, equivalent to approximately 3.4 kilograms of carbon dioxide equivalent per kilogram of hydrogen.

The additionality, temporal, and geographical correlation requirements are intended to ensure that the production of renewable hydrogen does not increase emissions by diverting renewable electricity from the grid and increasing fossil generation elsewhere. Industry stakeholders have argued that they are excessively strict and increase the cost of renewable hydrogen, while environmental organisations have defended them as necessary to ensure environmental integrity. The rules apply to RFNBOs imported into the European Union as well as those produced within it, with significant implications for exporting states, as examined in section 12.3. In 2025, the Commission adopted a delegated act setting out the methodology for determining the greenhouse gas emissions of low-carbon fuels, including low-carbon hydrogen produced from natural gas with carbon capture or from nuclear electricity, under the Hydrogen and Decarbonised Gas Market Package.

The United States Clean Hydrogen Production Credit

In the United States, the Inflation Reduction Act of 2022 introduced a **clean hydrogen production tax credit** under section 45V of the Internal Revenue Code, with four tiers of credit value according to the life-cycle greenhouse gas emissions of the hydrogen, up to a maximum emissions rate of four kilograms of carbon dioxide equivalent per kilogram of hydrogen, determined by a "well-to-gate" analysis. Final regulations issued by the Treasury Department in January 2025 adopted three requirements for hydrogen produced using grid electricity to be credited with the emissions of the clean electricity procured through energy attribute certificates — incrementality (new clean generation), temporal matching

(annual matching until 2029, and hourly matching thereafter), and deliverability (generation in the same region) — closely analogous to the EU's criteria, while providing flexibilities for existing nuclear plants at risk of retirement and for states with robust clean electricity standards. The One Big Beautiful Bill Act of July 2025 curtailed the credit by limiting it to facilities beginning construction before 1 January 2028, an illustration of the political volatility of hydrogen support.

Other National Approaches

The **United Kingdom's Low Carbon Hydrogen Standard** sets a maximum threshold of twenty grams of carbon dioxide equivalent per megajoule of hydrogen (on a lower heating value basis) for hydrogen to be eligible for government support, applicable to all production pathways, including electrolysis and natural gas reforming with carbon capture. The United Kingdom supports low-carbon hydrogen production through the **Hydrogen Production Business Model**, a contract for difference-style mechanism under which producers receive the difference between a strike price and a reference price, and the first hydrogen allocation round resulted in contracts with eleven electrolytic hydrogen projects in 2023. **Japan's Hydrogen Society Promotion Act** of 2024 provides for fifteen-year price-gap support for low-carbon hydrogen and ammonia supply chains, including imports, meeting carbon intensity requirements. Other states, including India, China, Australia, and Chile, have adopted standards defining green or low-carbon hydrogen, some based on carbon intensity alone and others combining intensity with source-based criteria.

Natural Hydrogen and Legal Classification

The discovery of naturally occurring hydrogen accumulations — notably at Bourakébougou in Mali, where a well drilled for water in the 1980s was found to produce almost pure hydrogen, which has since been used to generate electricity for a village — has generated interest in exploration for natural hydrogen in several countries. Natural hydrogen raises the classification question in acute form: is it a mineral, a petroleum substance, a gas, or a sui generis resource, and which licensing regime applies to its exploration and production? South Australia amended its Petroleum and Geothermal Energy Act in 2021 to include hydrogen as a regulated substance, enabling the grant of exploration licences for natural hydrogen, and France amended its Mining Code to include natural hydrogen among the substances subject to mining titles. In most jurisdictions, including Nigeria, the classification of natural hydrogen is unaddressed, and the definitions of "petroleum" and "minerals" in the PIA and the Nigerian Minerals and Mining Act may not clearly extend to it.

12.3 Hydrogen Certification and Cross-Border Trade

The Function of Certification

Because the climate benefit of hydrogen depends on how it was produced, and because molecules of hydrogen produced by different pathways are physically identical, regulatory and commercial claims about low-carbon hydrogen depend on reliable certification of its origin and emissions. Certification serves several legal functions: demonstrating eligibility for support schemes and tax credits; demonstrating compliance with renewable energy targets and obligations; enabling consumers to make claims under emissions accounting frameworks; and enabling the differentiation of hydrogen in international trade. Certification systems may follow a **book-and-claim** model, analogous to guarantees of origin, under which certificates are traded separately from the physical product, or a **mass balance** model, under which certificates and the physical product move together along the supply chain, with the quantity of certified hydrogen withdrawn not exceeding the quantity injected. EU law requires the mass balance approach for the purposes of demonstrating compliance with the RFNBO and biofuel sustainability criteria under Article 30 of the Renewable Energy Directive, while guarantees of origin, which follow a book-and-claim model, are used for disclosure to consumers.

EU Certification Through Voluntary Schemes

Under Article 30 of the Renewable Energy Directive, economic operators may demonstrate compliance with the RFNBO criteria, as with the biofuel sustainability criteria, through national schemes or voluntary international schemes recognised by the Commission. The Commission has recognised a number of voluntary schemes for the certification of RFNBOs, which audit producers' compliance with the additionality, temporal and geographical correlation, and emissions savings requirements, including for producers located outside the European Union. Implementing Regulation (EU) 2022/996 sets out rules for verifying sustainability, greenhouse gas emissions saving, and low indirect land use change risk criteria, including the requirements for audits and the use of the Union Database, which tracks the transactions of certified fuels to prevent fraud and double counting.

International Standards

The proliferation of national definitions and certification schemes threatens to fragment the international market for hydrogen, increasing the costs of compliance for producers seeking to export to multiple markets. Efforts to develop internationally harmonised methodologies have been led by the **International Partnership for Hydrogen and Fuel Cells in the Economy** (IPHE), whose working paper on a methodology for determining the greenhouse gas emissions associated with the production of hydrogen informed the development of **ISO**

19870:2023, the first international standard on the methodology for determining greenhouse gas emissions associated with the production, conditioning, and transport of hydrogen to the consumption gate. The ISO standard provides a common accounting methodology, but does not set thresholds or determine which hydrogen is "clean", which remain matters for national and regional law. Mutual recognition of certification schemes between importing and exporting states has been proposed as a means of reducing fragmentation, and the European Union has explored cooperation with partner countries on certification.

Trade Arrangements and Support for Imports

Several importing states have developed instruments to support imports of renewable hydrogen and derivatives. Germany's **H2Global** mechanism, managed by an intermediary company (Hintco), uses a double-auction model: the intermediary concludes long-term purchase agreements with producers outside the European Union, selected through competitive tenders on the basis of the lowest price, and sells the products in the European market through short-term auctions to the highest bidder, with the difference between the purchase and sale prices funded by the German government. The first H2Global tender, concluded in 2024, resulted in a contract for the supply of renewable ammonia produced in Egypt. The European Union's **European Hydrogen Bank**, funded by the Innovation Fund, conducts auctions for fixed premium support for renewable hydrogen production in the European Economic Area, with the first auction in 2024 awarding around EUR 720 million to projects selected on the basis of the lowest premium requested, and is developing an international component to support imports.

Hydrogen and International Trade Law

International trade in hydrogen and its derivatives is subject to the WTO agreements examined in Chapter 13. Several questions arise. First, the **classification** of hydrogen and its derivatives in the Harmonized System — hydrogen is classified under heading 28.04 and ammonia under heading 28.14 — does not distinguish between low-carbon and high-carbon production, so that tariff differentiation based on production methods would require new subheadings or other instruments. Second, the **non-discrimination** obligations of the GATT raise the question whether low-carbon and high-carbon hydrogen are "like products" — a question that turns on the debate over whether non-product-related processes and production methods can be a basis for regulatory distinction, examined in Chapter 13. Third, the **TBT Agreement** applies to technical regulations and standards defining low-carbon hydrogen and the conformity assessment procedures by which compliance is certified, requiring them not to be more trade-restrictive than necessary to fulfil a legitimate objective and encouraging the use of international standards, such as ISO 19870. Fourth, **subsidies** for hydrogen

production, such as the US 45V credit and EU support schemes, may be subject to the disciplines of the Agreement on Subsidies and Countervailing Measures, particularly where they are conditioned on domestic content. Fifth, the EU **Carbon Border Adjustment Mechanism** (Regulation (EU) 2023/956) covers hydrogen and ammonia among its initial sectors, so that imports of carbon-intensive hydrogen and ammonia into the European Union will bear a carbon price from the definitive period beginning in 2026, as examined in Chapter 13.

Transport of Hydrogen and Derivatives

International trade in hydrogen requires its transport over long distances, by pipeline or by ship. Pipeline transport of hydrogen within Europe is being developed through the conversion of natural gas pipelines and the construction of new hydrogen pipelines, including the planned "European Hydrogen Backbone" and cross-border corridors such as the H2Med project linking the Iberian Peninsula with France and Germany, and the proposed SoutH2 Corridor linking North Africa and Italy with Austria and Germany. Such cross-border pipelines raise issues of intergovernmental agreements, transit, and regulatory coordination similar to those examined for natural gas pipelines in Chapter 6.

Shipping of hydrogen requires its conversion into a form with higher energy density: liquefied hydrogen (at minus 253 degrees Celsius), ammonia, methanol, or liquid organic hydrogen carriers. The *Suiso Frontier*, the world's first liquefied hydrogen carrier, completed a pilot voyage from Australia to Japan in 2022. The International Maritime Organization adopted interim recommendations for the carriage of liquefied hydrogen in bulk, pending amendment of the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code). Ammonia, which is already traded internationally in large quantities, is likely to be the principal carrier in the near term, but its toxicity raises safety and environmental issues in its handling, transport, and use as a fuel, which the IMO has begun to address through interim guidelines on the safety of ships using ammonia as fuel.

12.4 Battery Storage and Grid Integration

The Role of Electricity Storage

Electricity storage — the conversion of electrical energy into a form that can be stored and reconverted into electricity when needed — is increasingly central to the integration of variable renewable energy. Storage can shift energy from periods of surplus to periods of scarcity, provide fast-responding frequency regulation and other ancillary services, defer or replace investment in network reinforcement, provide backup supply, and enable consumers to increase their self-consumption of rooftop solar. **Pumped storage hydropower** remains the largest form of grid-scale storage globally, but **battery energy storage systems**, principally using

lithium-ion technology, have grown dramatically as their costs have fallen, and are now deployed at grid scale in many markets. Other technologies include compressed air, flywheels, thermal storage, and, for long durations, hydrogen.

The rapid growth of battery storage has been illustrated by projects such as the Hornsdale Power Reserve in South Australia, commissioned in 2017 following a series of blackouts, which demonstrated the capability of large batteries to provide fast frequency response and reduced the costs of frequency control services in the Australian National Electricity Market. In sub-Saharan Africa, batteries are integral to solar mini-grids and stand-alone solar systems, examined in Chapter 10, and to the management of grids subject to frequent outages, including in Nigeria, where behind-the-meter batteries increasingly substitute for diesel generators.

The Legal Characterisation of Storage

The principal legal difficulty presented by storage is that it does not fit within the categories of traditional electricity law. Storage consumes electricity when charging and produces it when discharging; it may be connected at transmission or distribution level or behind the customer's meter; and it may provide services to the market, to network operators, and to consumers. Early regulatory frameworks treated storage as generation (because it produces electricity), as consumption (because it draws electricity), or as both, with the result that storage could be subject to double charges — paying network charges and levies both when charging and when discharging — and to uncertainty about whether network operators could own it.

The European Union addressed these issues in the Clean Energy for All Europeans package. Article 2(59) of Directive (EU) 2019/944 defines "energy storage" in the electricity system as "deferring the final use of electricity to a moment later than when it was generated, or the conversion of electrical energy into a form of energy which can be stored, the storing of such energy, and the subsequent reconversion of such energy into electrical energy or use as another energy carrier". The definition recognises storage as a distinct activity, rather than as generation or consumption. Article 15(5) of the Directive requires member states to ensure that active customers who own energy storage facilities are not subject to double charges, including network charges, for stored electricity remaining within their premises or when providing flexibility services to system operators.

In Great Britain, storage was treated as a form of generation for licensing purposes, and the Energy Act 2023 amended the Electricity Act 1989 to define electricity storage expressly as a distinct subset of generation, providing legal clarity. In the United States, storage has been treated as a distinct resource for the purposes of wholesale market participation, as examined below. In Nigeria, the legal treatment of storage under the Electricity Act 2023 and NERC's regulations

is still developing, and a comprehensive framework for the licensing, market participation, and network charging of storage has yet to be established.

Ownership of Storage by Network Operators

Whether network operators — whose activities are regulated monopolies — may own and operate storage facilities is a significant question. Storage owned by network operators could defer network investment and support system security, but it could also distort competition in markets in which storage competes with generation and demand response to provide energy and flexibility services. The EU's unbundling principles, examined in Chapter 9, apply: Articles 36 and 54 of Directive (EU) 2019/944 provide that distribution and transmission system operators shall not own, develop, manage, or operate energy storage facilities, except where the facilities are "fully integrated network components" (such as capacitors that are used solely to maintain network stability) and the regulator has approved them, or where, following an open, transparent, and non-discriminatory tendering procedure, no other party has expressed interest in providing the storage services at reasonable cost, the facilities are necessary for the system operator's obligations, and the regulator has approved the derogation. The regulator must periodically reassess whether third parties are interested in owning the facilities, and if so, the system operator must phase out its storage activities.

Market Participation

The ability of storage to participate in wholesale electricity markets on equal terms with other resources depends on market rules designed for generators and loads. In the United States, the Federal Energy Regulatory Commission's **Order No 841** (2018) required regional transmission organisations and independent system operators to revise their market rules to establish participation models for electric storage resources, enabling them to provide all capacity, energy, and ancillary services that they are technically capable of providing, to be dispatched and to set market prices as both buyers and sellers, to account for their physical and operational characteristics, and to buy energy for later resale at the wholesale market price. In *National Association of Regulatory Utility Commissioners v FERC* 964 F.3d 1177 (DC Cir 2020), the Court of Appeals upheld Order No 841, rejecting the argument that FERC had exceeded its jurisdiction by preventing states from prohibiting storage resources connected at the distribution level from participating in wholesale markets, holding that the Order regulated practices directly affecting wholesale rates within FERC's jurisdiction under the Federal Power Act while preserving states' authority over the safety and reliability of the distribution system. **Order No 2222** (2020), examined in Chapter 9, extended wholesale market participation to aggregations of distributed energy resources, including behind-the-meter batteries.

Several jurisdictions have adopted **procurement mandates** for storage. California's Assembly Bill 2514 (2010) directed the California Public Utilities Commission to determine whether to adopt storage procurement targets, and the Commission subsequently required the state's investor-owned utilities to procure 1,325 MW of storage by 2020, which was followed by substantially larger procurement orders to maintain reliability as solar generation increased. Capacity markets, examined in Chapter 9, have also become an important source of revenue for storage, subject to "de-rating" factors that reflect the limited duration of battery storage.

Safety, Siting, and End of Life

Large battery installations raise safety issues, principally the risk of thermal runaway leading to fires that are difficult to extinguish and may release toxic gases. A major fire at a large battery storage facility at Moss Landing, California, in January 2025 led to local evacuations and renewed scrutiny of battery safety standards and siting. Safety is addressed through fire codes and standards, including the United States National Fire Protection Association's NFPA 855 standard for the installation of stationary energy storage systems and certification standards such as UL 9540, and through planning and permitting requirements. The end-of-life management of batteries — including their collection, reuse in "second life" applications, and recycling — is governed in the European Union by the Batteries Regulation (Regulation (EU) 2023/1542), examined in Chapter 8, which also imposes carbon footprint, recycled content, due diligence, and battery passport requirements on industrial batteries, including those used for grid storage.

12.5 Carbon Capture, Utilisation, and Storage

The Role of CCUS

Carbon capture, utilisation, and storage (CCUS) comprises technologies that capture carbon dioxide from large point sources — such as power plants, cement and steel works, hydrogen production, and natural gas processing — or directly from the atmosphere, and either use it in products or store it permanently in deep geological formations. The Intergovernmental Panel on Climate Change has identified CCUS as a necessary component of many pathways limiting global warming to 1.5°C or 2°C, particularly for the abatement of emissions from industrial processes, such as cement production, for which there are few alternatives, and for the removal of carbon dioxide from the atmosphere through bioenergy with carbon capture and storage (BECCS) and direct air capture with carbon storage (DACCS). CCUS is nevertheless controversial: critics argue that it has a history of underperformance and cost overruns, that it may prolong the use of fossil fuels, and that much captured carbon dioxide has been used for enhanced oil recovery, which increases oil production.

Commercial-scale geological storage of carbon dioxide has been undertaken since 1996 at the **Sleipner** field in the Norwegian North Sea, where carbon dioxide separated from natural gas has been injected into a saline aquifer beneath the seabed, partly in response to Norway's carbon tax on offshore emissions. The **Northern Lights** project, part of Norway's Longship project, began injecting carbon dioxide transported by ship from industrial emitters in 2025, providing the first commercial cross-border carbon dioxide transport and storage service.

The International Framework: The London Protocol

The injection of carbon dioxide into sub-seabed geological formations raised the question whether it constituted "dumping" prohibited by the international law of marine pollution. The 1996 **Protocol to the Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter** (the London Protocol) adopts a "reverse list" approach, prohibiting the dumping of all wastes or other matter except those listed in Annex 1. In 2006, the parties amended Annex 1 to include "carbon dioxide streams from carbon dioxide capture processes for sequestration", permitting the storage of carbon dioxide in sub-seabed geological formations, provided that the streams consist overwhelmingly of carbon dioxide, no wastes are added for the purpose of disposal, and the storage is subject to a permit and to the specific guidelines adopted by the parties. The **OSPAR Convention** for the protection of the marine environment of the North-East Atlantic was amended in 2007 to permit the storage of carbon dioxide in sub-soil geological formations.

Article 6 of the London Protocol prohibits the export of wastes or other matter to other countries for dumping or incineration at sea, which would have prevented the cross-border transport of carbon dioxide for sub-seabed storage. In 2009, the parties adopted an amendment to Article 6 permitting the export of carbon dioxide streams for storage, provided that an agreement or arrangement is entered into by the countries concerned, allocating permitting responsibilities. Because the 2009 amendment had not received sufficient acceptances to enter into force, the parties adopted a resolution in 2019 permitting its provisional application by parties that so declare, enabling cross-border projects such as Northern Lights to proceed on the basis of bilateral arrangements, such as those concluded between Norway and several European states.

Domestic Storage Regimes

The European Union. Directive 2009/31/EC on the geological storage of carbon dioxide (the CCS Directive) establishes a comprehensive legal framework for storage in the territory of member states, including their exclusive economic zones and continental shelves. Member states retain the right to determine the areas from which storage sites may be selected, including the right not to allow

any storage in parts or the whole of their territory. Exploration requires an exploration permit, and storage requires a storage permit, which may be issued only if the site has been characterised and assessed and found to present no significant risk of leakage or of environmental or health damage. Operators must monitor the storage site and take corrective measures in the event of leakage or significant irregularities, and must surrender emission allowances under the EU ETS for any leaked emissions. Following closure, responsibility for the site is transferred to the competent authority when all available evidence indicates that the stored carbon dioxide will be completely and permanently contained, a minimum period — normally not less than twenty years — has elapsed, a financial contribution has been made, and the site has been sealed and the injection facilities removed. Operators must provide financial security for their obligations. Captured and stored emissions are not counted as emissions under the EU ETS, providing an economic incentive for CCS in sectors covered by the ETS. The **Net-Zero Industry Act** (Regulation (EU) 2024/1735) set an objective of achieving an annual carbon dioxide injection capacity of at least fifty million tonnes in the Union by 2030, and imposes on holders of oil and gas production licences an obligation to contribute to that objective in proportion to their share of production.

The United Kingdom. The Energy Act 2008 established a licensing regime for offshore carbon dioxide storage, administered by the North Sea Transition Authority, and the Energy Act 2023 established a framework for the economic regulation of carbon dioxide transport and storage networks by Ofgem, together with revenue support mechanisms for industrial carbon capture, power with CCS, and hydrogen production.

The United States. The geological storage of carbon dioxide is regulated by the Environmental Protection Agency under the Safe Drinking Water Act's Underground Injection Control programme, which established in 2010 a dedicated "Class VI" well category for geological sequestration, with requirements relating to site characterisation, well construction, monitoring, financial responsibility, and post-injection site care; states may obtain primacy to administer the programme. The **section 45Q** tax credit, substantially enhanced by the Inflation Reduction Act, provides a credit per tonne of carbon oxide captured and stored or used, and has been a principal driver of CCUS investment.

Pore Space and Long-Term Liability

Two legal questions are distinctive to onshore geological storage. The first concerns the **ownership of pore space** — the void spaces in subsurface rock formations in which carbon dioxide is stored. In common law jurisdictions where land ownership extends to the subsurface, the question arises whether pore space belongs to the surface owner or the mineral owner, where these have been severed. Several states of the United States, including North Dakota, Montana,

and Wyoming, have enacted legislation vesting pore space in the surface owner, and provided for the "unitisation" or "amalgamation" of pore space, permitting storage operators to proceed with the consent of a specified proportion of owners, with compensation for non-consenting owners. In jurisdictions where the subsurface is owned by the state, as in most civil law and many developing countries, the state may grant storage rights through licences. In Nigeria, where minerals and petroleum are vested in the Federal Government, the ownership of pore space and the legal basis for storage licensing remain unaddressed, although the PIA's framework for petroleum operations might be adapted to regulate storage in depleted reservoirs.

The second concerns **long-term liability** for stored carbon dioxide, which must remain contained for centuries or millennia, far beyond the lifetime of the companies that inject it. Most regimes provide for the transfer of liability to the state after a period of post-closure monitoring, as in the EU CCS Directive, subject to conditions and financial contributions, and some, such as Australia's Offshore Petroleum and Greenhouse Gas Storage Act 2006, provide for the Commonwealth to assume long-term liability after a closing certificate is issued. The design of liability transfer must balance the need to provide investors with certainty against the risk of socialising the costs of leakage.

CCUS in Climate Law and Carbon Markets

CCS was accepted as an eligible project activity under the Kyoto Protocol's Clean Development Mechanism by a decision of the parties in 2011, subject to modalities and procedures addressing site selection, monitoring, liability, and financial provision. Under the Paris Agreement, carbon dioxide removals, including through BECCS and DACCS, may generate credits under the Article 6.4 mechanism, subject to standards on removals adopted by its Supervisory Body, as examined in Chapter 18. The treatment of CCUS in climate law raises questions about the permanence of storage, the accounting for emissions from enhanced oil recovery, and the risk that reliance on future removals may delay emission reductions.

12.6 Offshore Renewable Energy Technologies

The Expansion of Offshore Energy

The ocean offers vast renewable energy resources: offshore wind, which is stronger and more consistent than onshore wind; wave energy; tidal energy, derived from tidal ranges and tidal currents; ocean thermal energy conversion (OTEC), which exploits temperature differences between surface and deep waters in tropical regions; and salinity gradient energy. Offshore wind is the only one of these technologies that has achieved commercial maturity at scale, with tens of gigawatts installed, predominantly in the North Sea, the Baltic Sea, and the coastal waters

of China. **Floating offshore wind**, in which turbines are mounted on floating platforms moored to the seabed, extends the potential of offshore wind to deep waters, which account for most of the global offshore wind resource, including off the coasts of Japan, Korea, the western United States, Norway, and parts of Africa. Wave and tidal energy remain at the demonstration and early commercial stage, with projects such as the MeyGen tidal stream array in Scotland.

This section examines the legal framework governing offshore renewable technologies — their installation, the allocation of marine space, offshore grids, and decommissioning. The leasing and permitting of offshore wind have been introduced in Chapter 10, and the law of the sea relating to maritime zones, submarine cables, and seabed resources more generally is examined in Chapter 20.

The Law of the Sea Framework

The United Nations Convention on the Law of the Sea (UNCLOS) provides the international legal framework for offshore renewable energy. In the **territorial sea**, extending up to twelve nautical miles from the baselines, the coastal state exercises sovereignty, subject to the right of innocent passage of foreign ships, and may authorise offshore renewable installations under its domestic law. In the **exclusive economic zone**, extending up to two hundred nautical miles, Article 56(1)(a) grants the coastal state sovereign rights for the purpose of exploring and exploiting, conserving, and managing the natural resources of the zone, "and with regard to other activities for the economic exploitation and exploration of the zone, such as the production of energy from the water, currents and winds". Article 60 grants the coastal state the exclusive right to construct and to authorise and regulate the construction, operation, and use of artificial islands, installations, and structures for these purposes, and exclusive jurisdiction over them, including with regard to customs, fiscal, health, safety, and immigration laws and regulations.

Article 60 imposes obligations on the coastal state in respect of installations in the EEZ. Due notice must be given of their construction, and permanent means for giving warning of their presence must be maintained. Installations which are abandoned or disused must be removed to ensure safety of navigation, taking into account generally accepted international standards (the IMO's 1989 *Guidelines and Standards for the Removal of Offshore Installations and Structures on the Continental Shelf and in the Exclusive Economic Zone*). The coastal state may establish reasonable safety zones around installations, which shall not exceed a distance of five hundred metres around them, except as authorised by generally accepted international standards or as recommended by the competent international organisation. Installations may not be established where interference may be caused to the use of recognised sea lanes essential to international navigation. In exercising its rights in the EEZ, the coastal state must have due regard to the

rights and duties of other states, including the freedoms of navigation, overflight, and the laying of submarine cables and pipelines (Article 56(2) and Article 58).

The growth of large offshore wind farms raises questions about the adequacy of the five-hundred-metre limit on safety zones, the effect of large arrays on navigation and on the rights of other states, the extension of offshore renewable development beyond the EEZ onto the extended continental shelf (where the coastal state's rights under Part VI of UNCLOS relate to natural resources of the seabed and subsoil and may not extend to wind energy in the superjacent waters), and the protection of marine biodiversity. The **Agreement under UNCLOS on the Conservation and Sustainable Use of Marine Biological Diversity of Areas beyond National Jurisdiction** (BBNJ Agreement, 2023), which provides for environmental impact assessments of activities with potential effects in areas beyond national jurisdiction, may apply to the transboundary impacts of large offshore energy installations.

Maritime Spatial Planning

The allocation of marine space among competing uses — offshore energy, shipping, fishing, aquaculture, conservation, defence, cables and pipelines, and tourism — is increasingly governed by **maritime spatial planning**. The EU Maritime Spatial Planning Directive (Directive 2014/89/EU) required member states to establish maritime spatial plans by 2021, taking into account land-sea interactions, environmental, economic, social, and safety aspects, and cooperation with neighbouring states. Maritime spatial plans designate areas suitable for offshore renewable energy, and RED III requires member states to designate renewables acceleration areas, including offshore. The integration of offshore wind with conservation objectives has generated debate about the co-location of wind farms with marine protected areas and the potential for offshore wind foundations to act as artificial reefs.

Offshore Grids and Hybrid Projects

The scale of planned offshore wind development, particularly in the North Sea, has led to a shift from the radial connection of individual wind farms to shore towards **offshore grids** — networks of offshore transmission infrastructure that connect multiple wind farms and interconnect national electricity systems. **Hybrid projects** combine offshore wind connection and cross-border interconnection, allowing offshore wind to be exported to more than one country. These projects raise complex legal questions: the classification of hybrid assets as transmission or interconnection; the allocation of cross-border capacity (under EU rules requiring that at least seventy per cent of interconnection capacity be made available for cross-border trade, examined in Chapter 9); the definition of **offshore bidding zones**, in which offshore wind farms connected to a hybrid

asset receive the price of the offshore zone, rather than that of the onshore zone, which may reduce their revenues; the allocation of costs and benefits between states; and the governance of cross-border infrastructure.

In the **Esbjerg Declaration** of May 2022, Denmark, Germany, Belgium, and the Netherlands committed to developing the North Sea as a "green power plant of Europe", with a combined target of at least 65 GW of offshore wind by 2030 and 150 GW by 2050. In the **Ostend Declaration** of April 2023, nine North Seas states set targets of 120 GW by 2030 and 300 GW by 2050. The **North Seas Energy Cooperation**, an intergovernmental cooperation framework, supports coordination of offshore grid planning. The EU's TEN-E Regulation (Regulation (EU) 2022/869) provides for priority offshore grid corridors and for cooperation among member states in the development of offshore network development plans, and the 2024 electricity market reform addresses cost sharing and revenue arrangements for offshore hybrid projects. Denmark's plans for artificial "energy islands" in the North Sea and on Bornholm in the Baltic, intended as hubs for offshore wind and interconnectors, illustrate both the ambition and the difficulties of offshore grid development: Denmark put the North Sea energy island on hold after cost estimates rose substantially.

Wave, Tidal, and Ocean Thermal Energy

Wave and tidal energy technologies require legal frameworks for seabed leasing, consenting, and support. The United Kingdom's Contracts for Difference scheme has included ring-fenced budgets for tidal stream energy in recent allocation rounds, supporting deployment. The proposed Swansea Bay tidal lagoon, which would have used a seawall to harness the tidal range, was rejected by the United Kingdom government in 2018 on the ground that it did not offer value for money, illustrating the significance of support mechanisms and cost for novel technologies. Tidal range and tidal stream projects raise environmental issues relating to marine mammals, fish, and estuarine ecosystems, and require appropriate assessment under nature conservation law. OTEC, which is of interest to tropical island states, raises issues relating to the discharge of cold deep water and the location of installations in deep waters.

Offshore Renewables in Africa

Offshore renewable energy development in Africa is at an early stage. South Africa has identified significant offshore wind potential, and the World Bank's Offshore Wind Development Program has supported assessments of offshore wind resources in several African countries, including Morocco, Mauritania, South Africa, and Nigeria, and the development of roadmaps. Nigeria does not have a legal framework specifically addressing offshore renewable energy; offshore activities in its territorial sea and EEZ are governed principally by petroleum legis-

lation, the **Exclusive Economic Zone Act 1978**, and the **Territorial Waters Act 1967**, as amended, and the allocation of offshore areas for renewable energy would require consideration of the interaction with petroleum licensing under the PIA, coastal state jurisdiction, and federal–state relations. The development of offshore renewable frameworks in African states offers an opportunity to learn from the experience of other jurisdictions in the design of leasing, consenting, local content, and decommissioning rules.

Decommissioning of Offshore Renewable Installations

The decommissioning of offshore renewable installations is governed by Article 60(3) of UNCLOS, the IMO Guidelines of 1989, and, in the North-East Atlantic, the OSPAR Convention, together with domestic law. Domestic regimes typically require developers to prepare decommissioning programmes and to provide financial security. In the United Kingdom, the Energy Act 2004 requires persons responsible for offshore renewable energy installations to submit decommissioning programmes for approval and empowers the Secretary of State to require financial security. The decommissioning of offshore wind farms raises questions similar to those of offshore petroleum installations, examined in Chapter 5, including the extent of removal required (particularly of foundations, scour protection, and cables), the potential for "rigs-to-reefs" approaches, and the recycling of turbine blades.

12.7 Smart Grids, Digital Infrastructure, and Cybersecurity

The Digitalisation of Energy Systems

The electricity system is being transformed by digitalisation: smart meters that measure consumption at short intervals and communicate data to suppliers and network operators; sensors and automated controls on transmission and distribution networks; distributed energy resources — rooftop solar, batteries, electric vehicles, and heat pumps — that can be controlled remotely and aggregated; digital platforms for peer-to-peer trading and flexibility markets; and the use of artificial intelligence to forecast demand and renewable output and to optimise system operation. A **smart grid** is an electricity network that uses digital technologies to monitor and manage the flow of electricity from all generation sources to meet the varying demands of end users efficiently, reliably, and sustainably. At the same time, the rapid growth of electricity demand from data centres, driven by cloud computing and artificial intelligence, has created new pressures on electricity systems and new questions of grid access, cost allocation, and planning, examined in Chapter 9.

Digitalisation raises legal issues in four principal areas: the regulation of smart metering; the protection of energy data and privacy; the governance of data access and interoperability; and cybersecurity.

Smart Metering

The deployment of smart meters has been mandated or encouraged in many jurisdictions. In the European Union, the Third Energy Package required member states to ensure the implementation of intelligent metering systems where a cost-benefit assessment was positive, with a target of rolling out smart meters to at least eighty per cent of consumers by 2020. Articles 19 to 21 of Directive (EU) 2019/944 set out requirements for smart metering systems, including functionality, the right of consumers to access their data, data protection, cybersecurity, and interoperability, and the right of every consumer to request a smart meter where the roll-out is not positively assessed. In Great Britain, energy suppliers were required by licence conditions to take all reasonable steps to roll out smart meters to their domestic and small business customers, and the Data Communications Company, a licensed body, provides a national communications infrastructure for smart meters.

In Nigeria, metering is a critical issue for the electricity sector, because a large proportion of customers have been unmetered and charged through "estimated billing", which has been a major source of consumer grievance and revenue leakage. NERC's **Meter Asset Provider Regulations** (2018, subsequently revised) enabled third-party meter asset providers to finance, supply, and install meters for distribution companies, and the **National Mass Metering Programme**, launched in 2020, sought to provide meters free of charge to unmetered customers. NERC has issued orders capping estimated bills for unmetered customers, and the transition to prepayment and smart meters raises issues of consumer protection, data, and the security of revenue examined in Chapter 9.

Energy Data and Privacy

Smart meters and connected devices generate granular data on energy consumption that can reveal intimate details of private life, including occupancy patterns, daily routines, and the use of particular appliances. Energy data is therefore personal data subject to data protection law where it relates to identified or identifiable individuals. In the European Union, the **General Data Protection Regulation** (Regulation (EU) 2016/679) applies to the processing of energy data, requiring a lawful basis for processing, purpose limitation, data minimisation, and security, and the Electricity Directive requires member states to ensure that the processing of consumers' metering data complies with data protection law. In the Netherlands, the initial proposal for a mandatory smart meter roll-out was withdrawn in 2009 after the Senate expressed concerns that compulsory installation would infringe the right to respect for private life under Article 8 of the European Convention on Human Rights, and the roll-out proceeded on a voluntary basis with privacy safeguards. In Nigeria, the **Nigeria Data Protection Act 2023**, which established the Nigeria Data Protection Commission, governs

the processing of personal data, including by electricity distribution companies and meter providers.

Data Access and Interoperability

The value of energy data for innovation — enabling new services such as demand response, energy efficiency advice, and the aggregation of flexibility — depends on the ability of consumers and authorised third parties to access data in standardised, interoperable formats. Article 23 of Directive (EU) 2019/944 requires member states to specify rules on access to data of the final customer by eligible parties, ensuring efficient and secure data access and exchange, and Article 24 provides for the adoption of interoperability requirements and non-discriminatory and transparent procedures for access to data. The EU's **Data Act** (Regulation (EU) 2023/2854) establishes general rules on access to data generated by connected products, including smart energy devices, giving users the right to access and share such data with third parties. The governance of energy data raises tensions between innovation, competition, privacy, and security.

Cybersecurity

The digitalisation of energy systems has increased their exposure to cyberattacks, which can disrupt supply, damage equipment, and endanger safety. In December 2015, a cyberattack on distribution companies in western Ukraine caused power outages affecting around two hundred and twenty-five thousand customers — the first publicly confirmed cyberattack causing a power outage — and a further attack in December 2016, using malware designed specifically to attack electricity grid control systems, caused an outage in part of Kyiv. In May 2021, a ransomware attack on the information technology systems of **Colonial Pipeline**, which operates the largest refined products pipeline system in the United States, led the company to shut down the pipeline for several days, causing fuel shortages on the east coast; the company paid a ransom, part of which was subsequently recovered by the Department of Justice. In response, the Transportation Security Administration issued security directives imposing mandatory cybersecurity requirements on pipeline operators, illustrating the shift from voluntary to mandatory cybersecurity regulation in the energy sector.

Regulatory frameworks. In the United States, the North American Electric Reliability Corporation, the electric reliability organisation certified by FERC under section 215 of the Federal Power Act (added by the Energy Policy Act of 2005), develops mandatory **Critical Infrastructure Protection** (CIP) reliability standards, approved and enforced by FERC, which require owners and operators of the bulk electric system to identify critical cyber assets and implement security controls, with substantial penalties for violations. In the European Union, the **NIS2 Directive** (Directive (EU) 2022/2555) imposes

cybersecurity risk management and incident reporting obligations on essential and important entities, including in the energy sector (electricity, district heating, oil, gas, and hydrogen); the **Critical Entities Resilience Directive** (Directive (EU) 2022/2557) addresses the physical and other resilience of critical entities; and the **Network Code on Cybersecurity** for cross-border electricity flows (Commission Delegated Regulation (EU) 2024/1366) establishes sector-specific rules for cybersecurity risk assessment and management in the electricity sector. In Nigeria, the **Cybercrimes (Prohibition, Prevention, etc.) Act 2015**, as amended in 2024, provides for the designation and protection of critical national information infrastructure and criminalises attacks on it.

International law. Cyberattacks on energy infrastructure by states or state-sponsored actors raise questions of international law, including the prohibition of the use of force, the principle of non-intervention, state responsibility, and, in armed conflict, international humanitarian law. The reports of the United Nations Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security, endorsed by the General Assembly, affirmed in 2013 that international law, and in particular the United Nations Charter, applies to cyberspace, and in 2015 set out voluntary norms of responsible state behaviour, including that a state should not conduct or knowingly support activity contrary to its obligations under international law that intentionally damages critical infrastructure or otherwise impairs the use and operation of critical infrastructure to provide services to the public. The *Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations* (2017), prepared by an international group of experts, offers a scholarly restatement of the application of international law to cyber operations, including the protection of critical infrastructure. These questions are examined further in Chapter 25.

12.8 Regulatory Sandboxes and Technological Innovation

The Regulatory Challenge of Innovation

Energy regulation faces a recurrent dilemma in responding to technological innovation. Rules designed for existing technologies and business models may inadvertently prohibit, disadvantage, or create uncertainty for innovations — such as peer-to-peer trading, community energy, vehicle-to-grid services, local flexibility markets, and novel tariff structures — that do not fit established categories. Yet regulators cannot simply suspend the rules that protect consumers, ensure system security, and prevent the abuse of market power, and they often lack the information needed to design new rules for technologies whose characteristics and risks are not yet understood. This is a variant of the "Collingridge dilemma":

at an early stage, when a technology can still be shaped, its impacts are difficult to predict; once its impacts are known, it may be entrenched and difficult to regulate.

Several instruments have been developed to address this dilemma, including **experimentation clauses**, which authorise regulators or ministers to permit deviations from statutory requirements for pilot projects; **innovation funding** schemes, such as the network innovation allowances and competitions administered by Ofgem under the RIIO price control framework; **principles-based regulation**, which sets outcomes rather than prescriptive rules; and **regulatory sandboxes**.

The Concept of a Regulatory Sandbox

A regulatory sandbox is a framework established by a regulator that allows innovators to test new products, services, or business models in a real-world environment, for a limited period and subject to safeguards, with temporary relief from specified regulatory requirements or with regulatory guidance on how existing rules apply. The concept originated in financial regulation, with the United Kingdom Financial Conduct Authority's sandbox launched in 2016, and was rapidly adopted in the energy sector and elsewhere. The OECD's **Recommendation of the Council for Agile Regulatory Governance to Harness Innovation** (2021) encourages the use of experimental approaches, including sandboxes, as part of agile regulatory governance.

Energy regulatory sandboxes have been established in many jurisdictions. **Ofgem's Energy Regulation Sandbox**, launched in 2017, offers innovators bespoke guidance, comfort, and, where Ofgem has the power to do so, derogations from licence conditions and industry codes for trials, and has supported trials of peer-to-peer trading, local energy markets, and innovative tariffs. **Singapore's** Energy Market Authority launched a regulatory sandbox in 2017. **Germany** adopted the SINTEG Regulation (2017), which permitted participants in the "Smart Energy Showcases" programme to be compensated for certain economic disadvantages arising from trials, and subsequently developed the concept of "real-world laboratories" (*Reallabore*) for the energy transition. In the European Union, the **Net-Zero Industry Act** (Regulation (EU) 2024/1735) requires member states to establish regulatory sandboxes for net-zero technologies, and the AI Act (Regulation (EU) 2024/1689) requires the establishment of AI regulatory sandboxes, which may encompass energy applications.

::: casebox {title="Regulatory Sandboxing in the Australian National Electricity Market (2022)"} **Background.** In 2019, the Australian Energy Market Commission (AEMC) recommended the establishment of a regulatory sandboxing framework for the National Electricity Market. The Commission concluded that the existing regulatory framework, which is set out in detailed national legis-

lation and rules, contained barriers to the trialling of innovative technologies and business models. It also found that the regulators lacked the powers needed to grant relief for trials.

The framework. Legislation amending the National Energy Laws, enacted in 2022, together with the National Electricity Amendment (Regulatory sandboxing) Rule 2022, established a framework with three components:

- an **innovation enquiry service**, provided by the Australian Energy Regulator (AER), which offers guidance to innovators on how the existing framework applies to their proposals;
- **trial waivers**, which the AER may grant to exempt a trial project from specified provisions of the National Electricity Rules for a limited period, subject to conditions;
- **trial rules**, which the AEMC may make to enable trials that require exemptions beyond the AER's waiver powers, such as exemptions from requirements of the national legislation.

Trial projects must be genuinely innovative and must have the potential to contribute to the achievement of the national energy objective. They must be subject to appropriate consumer protections and must be of limited duration and scale.

Significance. The Australian framework is notable for giving sandboxing an express statutory basis, for distinguishing guidance, waivers, and rule-making, and for linking trials to the possibility of permanent rule changes where they succeed. It addresses the principle of legality by ensuring that regulators have clear authority to grant relief from binding legal requirements. That authority was absent from some earlier sandboxes, which could offer only guidance or comfort. :::

Legal Issues Raised by Sandboxes

Regulatory sandboxes raise several legal questions.

Legal basis. Under the principle of legality, a regulator may waive or modify binding legal requirements only if it has statutory authority to do so. Where a sandbox involves relief from requirements contained in primary legislation, express legislative authorisation is required, as in Australia. Where the regulator lacks such authority, it may be able to offer only guidance, informal comfort, or the non-enforcement of rules — which may provide limited legal certainty for participants and may raise questions about the legitimacy of selective non-enforcement.

Equal treatment and competition. The selection of participants for a sandbox, and the grant of exemptions to some market participants but not others, may confer competitive advantages and raise questions of equal treatment, non-discrimination, and, in the European Union, State aid. Sandboxes therefore

require transparent and objective eligibility criteria, open application processes, and publication of decisions.

Consumer protection. Consumers participating in trials may be exposed to risks — of higher prices, interruption of supply, or loss of the protections they would otherwise enjoy. Sandbox frameworks typically require informed consent, maintenance of core protections (such as protections for vulnerable customers and access to dispute resolution), exit arrangements returning consumers to standard arrangements at the end of a trial, and limits on the number of participating consumers.

From experiment to rule. The value of sandboxes lies not only in enabling individual trials but also in informing permanent regulatory change. Frameworks that provide for the systematic evaluation and publication of trial results, and for the consideration of rule changes where trials succeed, are more likely to achieve systemic reform. Conversely, sandboxes that merely grant temporary exemptions without a pathway to rule change may create a cycle of trials without lasting effect.

Regulatory capacity and capture. Sandboxes demand significant regulatory resources and close engagement between regulators and innovators, which raises concerns about regulatory capture and the regulator's ability to maintain its independence and impartiality.

Innovation and Regulation in Developing Countries

Regulatory sandboxes and innovation-friendly frameworks are of particular relevance to developing countries, where new technologies and business models — such as pay-as-you-go solar, mobile money payment for electricity, mini-grids, productive use appliances, and digital metering — have expanded energy access faster than traditional utility models, often operating in gaps or grey areas of the regulatory framework. In Nigeria, the **Nigeria Startup Act 2022** provides a framework for supporting technology start-ups, including through regulatory sandboxes, and financial regulators have adopted sandbox frameworks for financial technology, relevant to the payment systems used in energy access business models. The NERC Mini-Grid Regulations, examined in Chapter 10, are an example of a regulatory framework designed to accommodate a new delivery model with tailored, lighter-touch requirements. Regulators in developing countries face the challenge of encouraging innovation while protecting consumers, ensuring quality, and integrating new models with the expansion of the main grid, and comparative study of the design of energy sandboxes is a promising area for doctoral research.

::: exercise {title="Practical Exercise: The Daruna Green Ammonia Export Project (HYPOTHETICAL)"} *This exercise is entirely hypothetical. The Republic*

of Daruna, the Port Asaba Special Economic Zone, Solveig Hydrogen AS, and all other entities named are fictional.

The Republic of Daruna is a coastal West African state with excellent solar and wind resources, limited freshwater, and a national grid in which 70 per cent of electricity is generated from natural gas. Solveig Hydrogen AS (HYPOTHETICAL), a European developer, proposes a USD 8 billion project in the Port Asaba Special Economic Zone. The project comprises:

- 3 GW of solar and wind generation on 40,000 hectares of land, part of which is held under customary tenure by the Ilaro and Kemi communities;
- a 1.5 GW electrolyser;
- a desalination plant;
- an ammonia synthesis plant and export terminal.

The ammonia will be sold to European buyers seeking to qualify it as an RFNBO, including under an H2Global-style tender. Daruna has no hydrogen legislation. Its Electricity Act requires all generators above 1 MW to hold a licence and to sell to the national grid company, and its Water Act vests all water in the state.

Solveig proposes the following arrangements:

- (a) the renewable plants will be connected both directly to the electrolyser and to the national grid, so that surplus electricity can be sold to the grid and grid electricity can be used when renewable output is low;
- (b) a twenty-five-year host government agreement will grant Solveig exemption from the Electricity Act, a ten-year tax holiday, stabilisation of the legal framework, and the right to export all of the ammonia;
- (c) the government will acquire a 10 per cent carried interest in the project company.

Community groups and civil society organisations have argued that the project should supply electricity and water to local communities, which have low rates of access, and that the government should obtain a larger share of the benefits.

1. Identify the legislative reforms required to enable the project, including the classification of hydrogen and ammonia, licensing of production, electricity sector rules, water rights, and land acquisition.
2. Advise Solveig on whether the ammonia can qualify as an RFNBO under Delegated Regulation (EU) 2023/1184, given arrangement (a), and on the certification arrangements required. How do the additionality and temporal correlation requirements affect the project's design?
3. Evaluate arrangement (b) from the perspective of Daruna's long-term interests, with reference to the material on stabilisation clauses in Chapter 14 and on resource governance in Chapter 7. Draft alternative provisions that would better balance investor protection and the state's regulatory autonomy.

4. Propose legal mechanisms by which the project could contribute to domestic energy access and water supply and share benefits with the Ilaro and Kemi communities. Consider the risk of "green extractivism" and the requirements of free, prior, and informed consent discussed in Chapter 21.
5. Would a regulatory sandbox be an appropriate instrument for enabling a pilot phase of the project? Explain your answer. :::

CHAPTER SUMMARY

- Hydrogen, storage, CCUS, offshore renewables, and smart grids are essential to decarbonisation. Each strains legal categories designed for an energy system with different technologies and boundaries.
- The classification of hydrogen determines the applicable regimes of gas, chemical, and safety law. The EU's 2024 Hydrogen and Decarbonised Gas Market Package creates a dedicated framework for hydrogen networks, with unbundling and regulated third-party access from 2033.
- The colour terminology for hydrogen is legally imprecise, and law has moved towards carbon intensity criteria. The EU's RFNBO rules require additionality, temporal correlation, and geographical correlation. The US 45V credit adopted analogous requirements before it was curtailed in 2025.
- Certification, through book-and-claim or mass balance systems, underpins regulatory and commercial claims about hydrogen. International harmonisation through ISO 19870 and mutual recognition is incomplete. International trade in hydrogen raises questions under the GATT, the TBT Agreement, the SCM Agreement, and CBAM.
- EU law defines electricity storage as a distinct activity, protects storage owners against double charging, and restricts network operator ownership of storage. FERC Order No 841, upheld in *NARUC v FERC*, opened wholesale markets to storage.
- CCUS is governed by the London Protocol, whose 2006 and 2009 amendments permit sub-seabed storage and its export, and by domestic regimes such as the EU CCS Directive and the US Class VI programme. Pore space ownership and long-term liability are distinctive issues.
- UNCLOS grants coastal states sovereign rights over energy production from winds and currents in the EEZ, subject to obligations regarding installations, safety zones, navigation, and removal. Offshore grids and hybrid projects raise complex cross-border questions.
- Smart grids raise issues of smart metering, data protection, data access, and cybersecurity. Mandatory cybersecurity regulation of energy infrastructure has expanded, as the NERC CIP standards, NIS2, and the response to Colonial Pipeline show.

- Regulatory sandboxes enable controlled experimentation but require a clear legal basis, equal treatment, consumer protection, and a pathway from successful trials to permanent rule change.

DOCTORAL RESEARCH QUESTIONS

1. Is a carbon intensity approach to the definition of low-carbon hydrogen preferable to a source-based approach? Evaluate the EU's RFNBO criteria in the light of environmental integrity, cost, and the development interests of exporting states.
2. To what extent do the EU's rules on renewable hydrogen and their application to imports constitute a form of extraterritorial regulation, and are they compatible with WTO law?
3. Who should own pore space, and who should bear long-term liability for stored carbon dioxide? Compare the approaches of two or more jurisdictions and propose a model for a developing country.
4. Do the rights of coastal states under UNCLOS adequately accommodate large-scale offshore energy development, including beyond the EEZ? Is there a need for new international rules on offshore energy installations?
5. Critically assess whether existing international law adequately protects energy infrastructure against cyberattacks. What role should the norms of responsible state behaviour play?
6. Evaluate the potential of regulatory sandboxes to accelerate energy access in sub-Saharan Africa. What legal design features are necessary to protect consumers and ensure legitimacy?

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